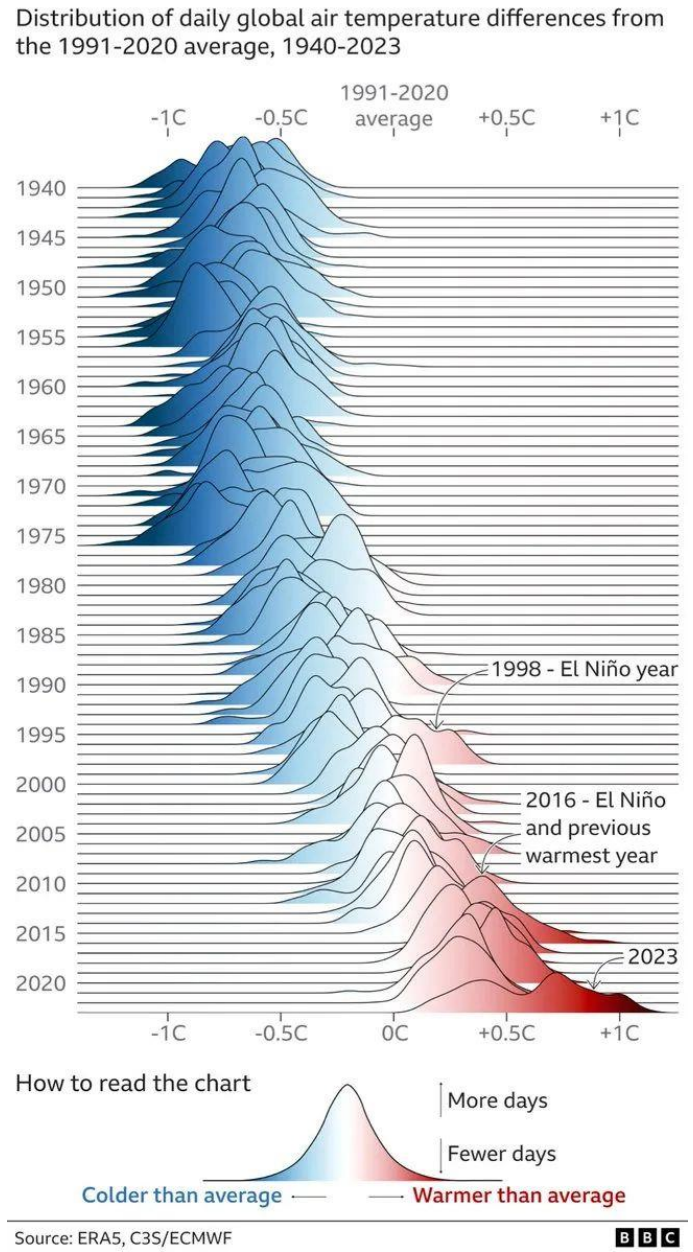
澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

Introduction to Data Visualization

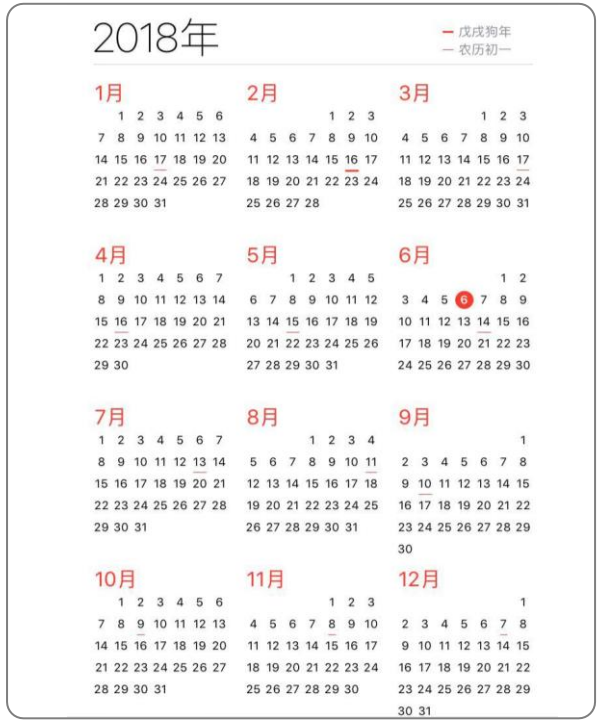
Jie Zhang
Faculty of Applied Sciences, Macao Polytechnic University
Email: jpeter.zhang@mpu.edu.mo
Personal Profile: easy-shu.github.io

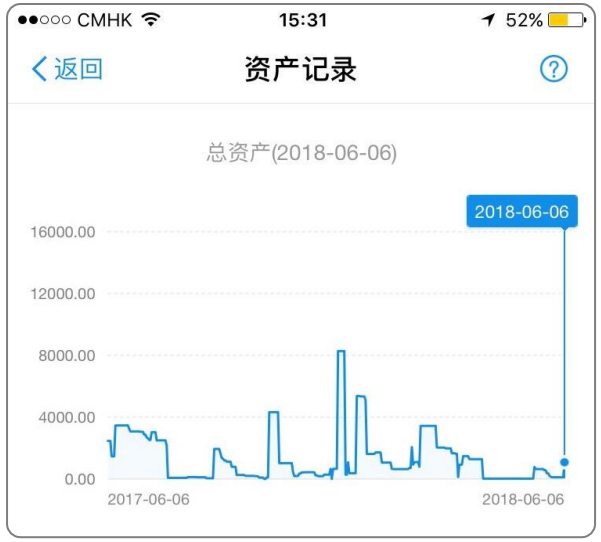


Definition

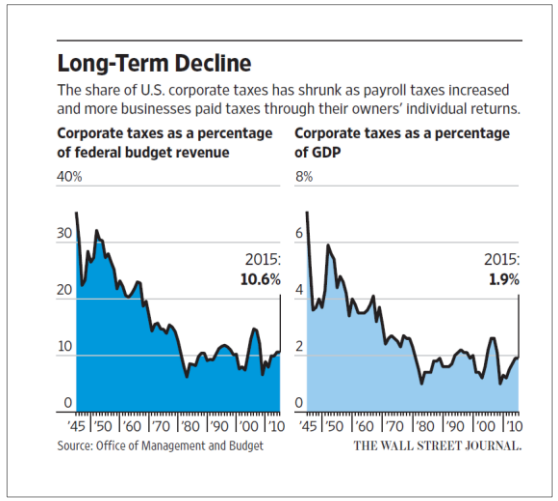

Distribution of daily global air temperature differences from the 1991-2020 average, 1940-2023
1991-2020 average
-1C
-0.5C
+0.5C
+1C
1940
1945
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
1998 - El Niño year
2016 - El Niño and previous warmest year
2023
-1C
-0.5C
0C
+0.5C
+1C
How to read the chart
More days
Fewer days
Colder than average
Warmer than average
Source: ERA5, C3S/ECMWF
BBC

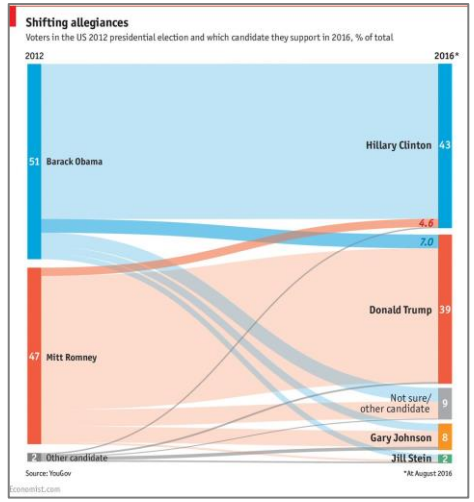

[redacted]
踏足中国 20个省区，47 个城市
超越了97.2%的用户
北京 上海 重庆 香港 澳门 南京 连云港 泰州 南通 常州
无锡 苏州 江阴 杭州 湖州 绍兴 诸暨 成都 乐山 都江堰
峨眉山 昆明 丽江 大理 迪庆 郑州 开封 登封 青岛 保定
太原 荆州 南昌 长春 延吉 图们 珲春 和龙 沈阳 大连
丹东 拉萨 日喀则 山南 西宁 福州 宁德

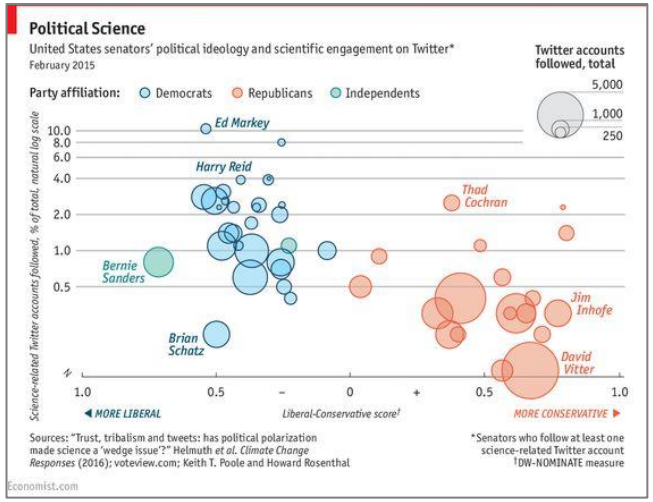

2018年
戊戌狗年
农历初一
1月
2月
3月
4月
5月
6月
7月
8月
9月
10月
11月
12月

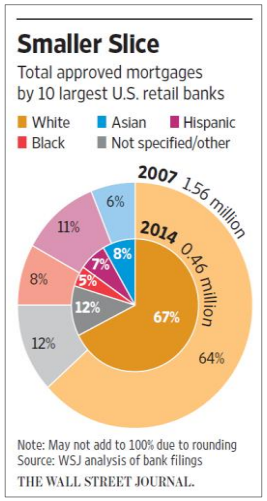

CMHK
15:31
52%
返回
资产记录
总资产(2018-06-06)
2018-06-06
16000.00
12000.00
8000.00
4000.00
0.00
2017-06-06
2018-06-06

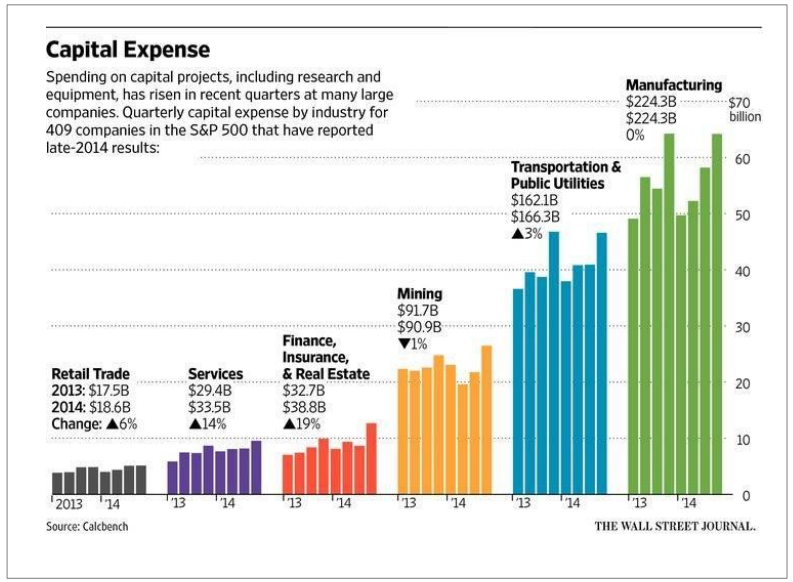

CMHK
15:43
47%
深圳市
广州市
1小时 55
2小时 54
1天
1小时 55分钟
慢
6分钟
东莞市
深圳市
中山市
1小时 55分钟 (136公里)
途经G4京港澳高速/广深高速
具体路线
预览

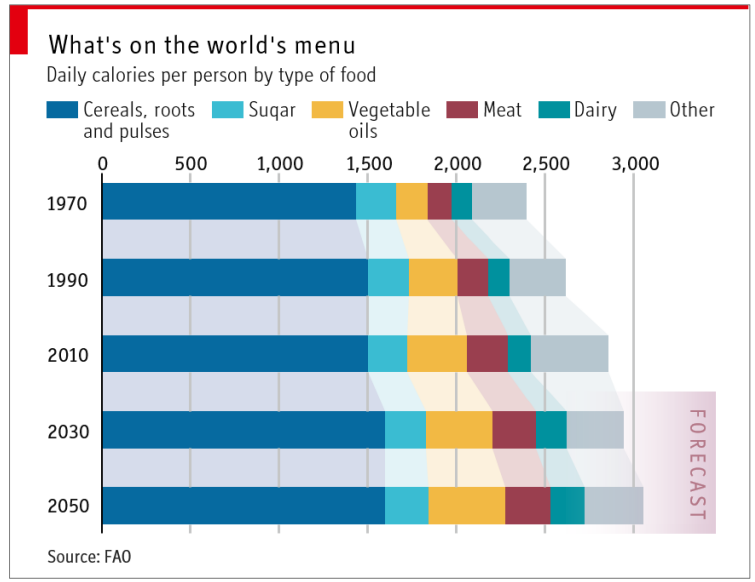

Long-Term Decline
The share of U.S. corporate taxes has shrunk as payroll taxes increased and more businesses paid taxes through their owners' individual returns.
Corporate taxes as a percentage of federal budget revenue
Corporate taxes as a percentage of GDP
40%
30
20
10
0
8%
6
4
2
0
2015: 10.6%
2015: 1.9%
'45 '50 '60 '70 '80 '90 '00 '10
Source: Office of Management and Budget
THE WALL STREET JOURNAL.

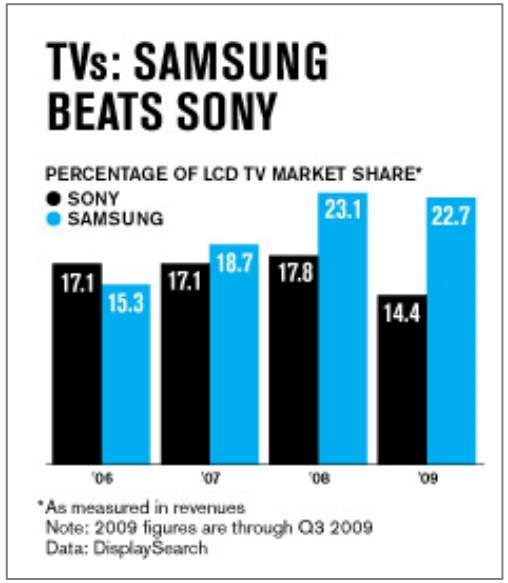

Shifting allegiances
Voters in the US 2012 presidential election and which candidate they support in 2016, % of total
2012
2016*
51 Barack Obama
47 Mitt Romney
Hillary Clinton 43
Donald Trump 39
Not sure/ other candidate 9
Gary Johnson 8
Jill Stein 2
Other candidate 2
4.6
7.0
Source: YouGov
*At August 2016
Economist.com


Political Science
United States senators' political ideology and scientific engagement on Twitter*
February 2015
Party affiliation: Democrats Republicans Independents
Twitter accounts followed, total
5,000
1,000
250
Science-related Twitter accounts followed, % of total, natural log scale
10.0
8.0
6.0
4.0
2.0
1.0
0.5
Ed Markey
Harry Reid
Thad Cochran
Bernie Sanders
Brian Schatz
Jim Inhofe
David Vitter
1.0 0.5 0 0.5 1.0
MORE LIBERAL
Liberal-Conservative score†
MORE CONSERVATIVE
Sources: "Trust, tribalism and tweets: has political polarization made science a 'wedge issue'?" Helmuth et al. Climate Change Responses (2016); voteview.com; Keith T. Poole and Howard Rosenthal
*Senators who follow at least one science-related Twitter account
†DW-NOMINATE measure
Economist.com


Smaller Slice
Total approved mortgages by 10 largest U.S. retail banks
White
Asian
Hispanic
Black
Not specified/other
2007 1.56 million
2014 0.46 million
6%
11%
8%
12%
67%
64%
8%
7%
5%
12%
Note: May not add to 100% due to rounding
Source: WSJ analysis of bank filings
THE WALL STREET JOURNAL.


Capital Expense
Spending on capital projects, including research and equipment, has risen in recent quarters at many large companies. Quarterly capital expense by industry for 409 companies in the S&P 500 that have reported late-2014 results:
Manufacturing
$224.3B
$224.3B
0%
$70 billion
Transportation & Public Utilities
$162.1B
$166.3B
▲3%
Mining
$91.7B
$90.9B
▼1%
Finance, Insurance, & Real Estate
$32.7B
$38.8B
▲19%
Services
$29.4B
$33.5B
▲14%
Retail Trade
2013: $17.5B
2014: $18.6B
Change: ▲6%
60
50
40
30
20
10
0
2013 '14 '13 '14 '13 '14 '13 '14 '13 '14 '13 '14
Source: Calcbench
THE WALL STREET JOURNAL.


What's on the world's menu
Daily calories per person by type of food
Cereals, roots and pulses
Sugar
Vegetable oils
Meat
Dairy
Other
0 500 1,000 1,500 2,000 2,500 3,000
1970
1990
2010
2030
2050
FORECAST
Source: FAO


TVs: SAMSUNG BEATS SONY
PERCENTAGE OF LCD TV MARKET SHARE*
SONY
SAMSUNG
17.1
15.3
17.1
18.7
17.8
23.1
14.4
22.7
'06 '07 '08 '09
*As measured in revenues
Note: 2009 figures are through Q3 2009
Data: DisplaySearch

LETTER

doi:10.1038/nature19309

Cotranslational signal-independent SRP preloading during membrane targeting

Justin W. Chartron[1], Katherine C. L. Hunt[1] & Judith Frydman[1,2]

Ribosome-associated factors must properly decode the limited information available in nascent polypeptides to direct them to their correct cellular fate[1]. It is unclear how the low complexity information exposed by the nascent chain suffices for accurate recognition by the many factors competing for the limited surface near the ribosomal exit site[2,3]. Questions remain even for the well-studied cotranslational targeting cycle to the endoplasmic reticulum, involving recognition of linear hydrophobic signal sequences or transmembrane domains by the signal recognition particle (SRP)[4,5]. Notably, the SRP has low abundance relative to the large number of ribosome–nascent-chain complexes (RNCs), yet it accurately selects those destined for the endoplasmic reticulum[6]. Despite their overlapping specificities, the SRP and the cotranslationally acting Hsp70 display precise mutually exclusive selectivity *in vivo* for their cognate RNCs[7,8]. To understand cotranslational nascent chain recognition *in vivo*, here we investigate the cotranslational membrane-targeting cycle using ribosome profiling[9] in yeast cells coupled with biochemical fractionation of ribosome populations. We show that the SRP preferentially binds secretory RNCs before their targeting signals are translated. Non-coding mRNA elements can promote this signal-independent pre-recruitment of SRP. Our study defines the complex kinetic interaction between elongation in the cytosol and determinants in the polypeptide and mRNA that modulate SRP–substrate selection and membrane targeting.

Secretory proteins are proposed to target to the endoplasmic reticulum (ER) membrane either co- or post-translationally for subsequent translocation[10–12]. Mechanistic models of ER targeting and the role of the SRP derive primarily from cell-free systems using model proteins[10,13], raising the question of how these pathways function in the cell. To investigate membrane targeting *in vivo*, we fractionated soluble and membrane-attached ribosomes from yeast cells, and then used ribosome profiling (termed Ribo-seq)[9] to compare the ribosome-protected mRNA footprints from polysomes obtained from both fractions (Extended Data Fig. 1a). We derived a cotranslational membrane enrichment score for each coding sequence (Methods, Extended Data Fig. 1b and Supplementary Table 1). Transcripts encoding cytosolic or nuclear (cytonuclear) proteins were preferentially translated on cytosolic ribosomes and not enriched on membrane polysomes (Fig. 1a). Tail-anchored proteins, whose single or transmembrane domain (TMD) at the carboxyl terminus is only revealed posttranslationally[14], were also translated on cytosolic ribosomes. By contrast, many nuclear-encoded mitochondrial protein transcripts were enriched in the membrane-bound ribosome fraction, as expected[15]. Transcripts encoding ER-destined secretory proteins were highly enriched on membrane-bound ribosomes. Proteins containing a signal sequence (SS) or TMD had comparable cotranslational membrane enrichment, conflicting with the idea that the targeting signal itself distinguishes which proteins are targeted co- or post-translationally to the ER[11,12] (Fig. 1a).

Ribosome profiling provides a snapshot of the abundance of ribosomes at each codon of each mRNA[9], revealing the dynamics of translation on soluble versus membrane-bound ribosomes. For cytonuclear proteins, soluble ribosome-protected reads were distributed across the entire reading frame, consistent with complete translation in the cytosol (Extended Data Fig. 1c). For secretory proteins, both soluble and membrane-bound polysomes produced protected reads. Cytosolic translation represented only a small fraction of any given secretory transcript, and most of the secretory mRNA pool was membrane anchored. In the classical understanding of cotranslational targeting, secretory protein RNCs bind to the membrane only after exposing a targeting signal[4]. Thus, there should be fewer RNCs found on the membrane translating the portion of transcripts not yet

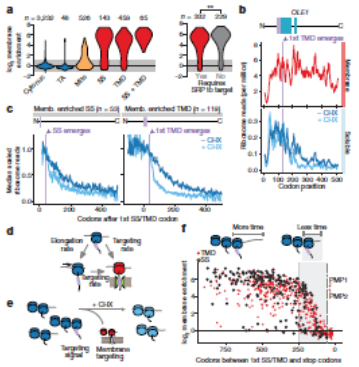


Figure 1 | Cotranslational membrane enrichment. a, Distributions of the open reading frame (ORF) enrichment of ribosome-protected reads in the membrane fraction compared to the soluble fraction. ORFs were alternatively classified by expected SRP dependence[11]. Values are the mean from two biological replicates. *$P < 0.01$, Wilcoxon rank-sum test. **b**, Ribosome-protected reads at each codon of an example transmembrane protein *OLE1*. Membrane topology is indicated above, with the first TMD in lavender. **c**, Metagene analysis of soluble fraction polysome-protected reads from transcripts that were at least twofold membrane enriched. ORFs were aligned at the targeting signal and scaled. **d**, Cotranslational membrane targeting is in competition with elongation. **e**, Elongation inhibitors provide additional time for polysomes exposing a targeting signal to localize to the membrane. **f**, Membrane enrichment was limited by the length of the reading frame remaining after the encoding of targeting signals. The vertical dashed line indicates 50 codons.

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targeted, that is, at codon positions upstream of the first SS or TMD. However, the membrane-bound ribosome-protected reads were evenly distributed across the entire transcript (Fig. 1b and Extended Data Fig. 1c, d). This suggests that once targeted, secretory mRNAs remain associated to the ER and their translation initiates at the membrane. This is consistent with the observed proximity of secretory RNCs to the translocon before synthesis of the targeting signal[16]. The small fraction of secretory mRNA in cytoplasmic pre-targeted RNCs probably represents the pioneer round of targeting.

The positioning of soluble ribosomes along mRNA provides insight into how secretory transcripts are targeted to the membrane. The highest read density for these messages mapped 5′ of the region encoding the first SS or TMD; read density declined after the first targeting signal was exposed by the ribosome, as expected from cotranslational signal-dependent targeting of soluble RNCs to the membrane (Fig. 1b, c and Extended Data Fig. 1d). Surprisingly, the loss of reads after signal emergence was gradual, resulting in many RNCs that remained soluble for hundreds of residues after SS or TMD exposure. This result was inconsistent with the elongation attenuation activity proposed for the SRP[17,18] and suggests that elongation continues on cytosolic RNCs upon exposure of a targeting signal (see Supplementary Discussion).

The idea that there is a kinetic competition between continuing elongation in the cytosol and RNC targeting to the membrane makes two testable predictions (Fig. 1d). First, pharmacological inhibition of elongation with cycloheximide (CHX) should decouple these processes, enhancing targeting of translocation-competent RNCs and promoting their depletion from the soluble fraction (Fig. 1e). Cells were subjected to a brief, two-minute CHX incubation before Ribo-seq analysis of soluble and membrane-bound polysomes. Importantly, such brief incubation did not perturb non-secretory polysomes (Extended Data Fig. 1c). By contrast, CHX treatment markedly reduced the soluble secretory reads, but only after cytosolic RNCs exposed the first SS or TMD, that is, 40 codons after its synthesis (Fig. 1b, c and Extended Data Fig. 1d).

The kinetic competition between targeting and elongation predicts that cotranslational membrane attachment is influenced by translation termination. In the absence of an elongation arrest, the probability of RNCs reaching the membrane cotranslationally will decrease as the first SS or TMD is found closer to the C terminus (Fig. 1f and Extended Data Fig. 1e). Indeed, we observed a decline in the maximum membrane enrichment of secretory RNCs when the first targeting signal is near the C terminus. Thus, secretory proteins with a late targeting signal, SS or TMD, must be targeted to the ER posttranslationally (Supplementary Discussion). Overall, our data suggest that cotranslational targeting to the ER in yeast is accomplished via a pioneer round of translation on soluble ribosomes that establishes a pool of ER-residing mRNA that initiate translation at the membrane (Extended Data Fig. 1f).

We next determined which RNCs are substrates of the SRP *in vivo*. Immunoprecipitation of Srp72p from total soluble RNCs was followed by ribosome profiling of both SRP-associated polysomes and monosomes (Fig. 2a and Extended Data Fig. 2a). Few transcripts encoding cytonuclear or mitochondrial proteins were enriched on SRP, confirming its specificity towards ER-destined transcripts. Notably, the SRP bound to all secretory RNCs that were cotranslationally targeted to the membrane, including SRP-dependent and SRP-independent proteins (Fig. 2b, c).

The number of ribosome-protected reads from soluble, SRP-bound transcripts diminished after ribosome exposure of the first SS or TMD, as expected from its targeting function (Fig. 2d). The loss was gradual and many SRP–RNCs remained soluble well after the targeting signal became fully exposed to the cytosol. This supports the notion that elongation proceeds on cytosolic ribosomes even after SRP binds, in contrast with the expected SRP-induced elongation arrest. Indeed, blocking elongation with CHX for 2 min before lysis caused a marked depletion in SRP-bound reads, but only for RNCs

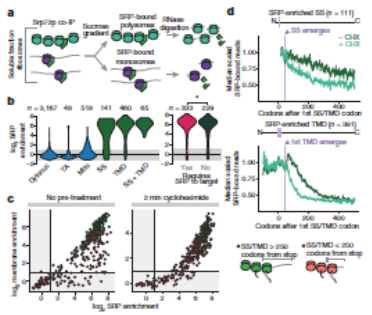


Figure 2 | Cotranslational enrichment of SRP. a, Srp72p-TAP was immunoprecipitated from the total soluble fraction. SRP-bound monosomes and polysomes were separated by sucrose gradient ultracentrifugation. **b**, Distributions of the ORF enrichment of ribosome-protected reads from SRP-bound soluble polysomes over the total soluble polysomes. ORFs were alternatively classified by expected SRP dependence[11]. Values are the mean from two biological replicates. TA, tail-anchored. *$P \leq 0.05$, Wilcoxon rank-sum test. **c**, Cotranslational membrane-fraction enrichment compared to SRP enrichment. **d**, Metagene analysis of soluble SRP-bound polysome-protected reads from transcripts that are at least twofold SRP-enriched. ORFs were aligned at the targeting signal and scaled.

exposing their first targeting signal (Fig. 2d). In principle, the delayed targeting of soluble RNCs to the membrane after SS/TMD emergence could reflect a delay in SRP binding rather than a lack of elongation arrest. Comparing the SRP and membrane enrichment to transcripts indicated that this is not the case. RNCs encoding late targeting signals, that is, near the C terminus, still bound SRP but did not target to the ER membrane (Supplementary Discussion, Fig. 2c and Extended Data Fig. 2b–d). Addition of CHX allowed these late-signal RNCs to enrich at the membrane, indicating the SRP–RNC complexes are competent for ER-targeting. We conclude that the SRP binds the nascent chain quickly, and continued elongation causes termination of late signals before targeting.

Although elongation arrest is not a general consequence of SRP binding *in vivo*, recent work showed that a rare-codon-directed slowdown of elongation facilitates SRP binding[19]. An intrinsic, non-SRP-dependent elongation slowdown should increase ribosome-protected reads at the same codon in both soluble SRP-bound and membrane-bound polysomes. Indeed, several transcripts presented such local increases in ribosome-protected reads at sites corresponding to exposure of a targeting signal on the ribosome (Extended Data Fig. 3a–c). Distinct elongation attenuation mechanisms observed at these sites included clusters of rare codons[19] and stalling polypeptide elements, such as stretches of positively charged amino acids, or proline motifs, positioned within the exit tunnel[20,21]. While most secretory transcripts were not significantly enriched in these attenuator elements compared to the proteome (Extended Data Fig. 3d, e), the few non-secretory proteins that cotranslationally bound to the SRP were enriched in elongation attenuation elements positioned at sites that exposed a near-cognate hydrophobic sequence for SRP binding (Extended Data Fig. 3d, f). We speculate that the presence of such elements enhances SRP recognition of the near-cognate hydrophobic tracts in these non-secretory proteins.

To understand the basis for the specificity of the SRP *in vivo*, we next determined the initial point of SRP recruitment to ribosomes

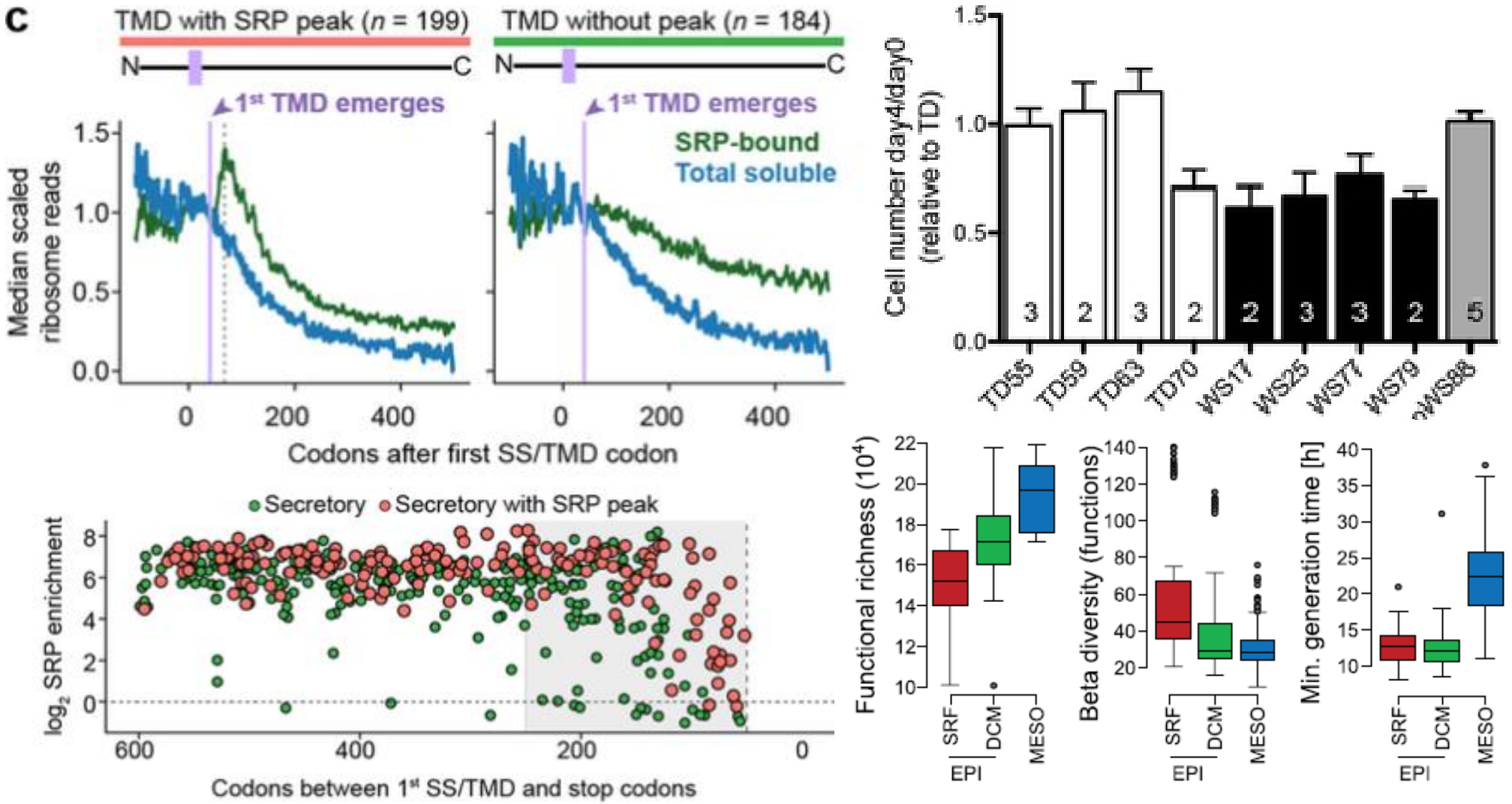
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Source: Chartron et al.. (2016) Nature, **536**, 224.

AIDesign Python-Data Science

澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University





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A picture is worth a thousand words !

LETTER

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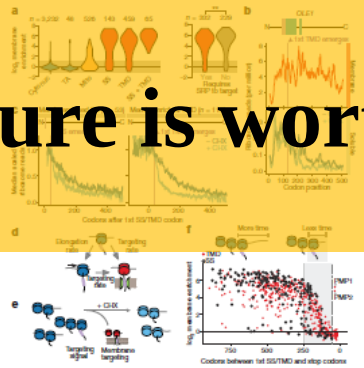


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targeted, that is, at codon positions upstream of the first SS or TMD. However, the membrane-bound ribosome-protected reads were evenly distributed across the entire transcript (Fig. 1b and Extended Data Fig. 1c, d). This suggests that once targeted, secretory mRNAs remain associated to the ER and their translation initiates at the membrane. This is consistent with the observed proximity of secretory RNCs to the translocon before synthesis of the targeting signal[16]. The small fraction of secretory mRNA in cytoplasmic pre-targeted RNCs probably represents the pioneer round of targeting.

The positioning of soluble ribosomes along mRNA provides insight into how secretory transcripts are targeted to the membrane. The highest read density for these messages mapped 5′ of the region encoding the first SS or TMD; read density declined after the first targeting signal was exposed by the ribosome, as expected from cotranslational signal-dependent targeting of soluble RNCs to the membrane (Fig. 1b, c and Extended Data Fig. 1d). Surprisingly, the loss of reads after signal emergence was gradual, resulting in many RNCs that remained soluble for hundreds of residues after SS or TMD exposure. This result was inconsistent with the elongation attenuation activity proposed for the SRP[17,18] and suggests that elongation continues on cytosolic RNCs upon exposure of a targeting signal (see Supplementary Discussion).

The idea that there is a kinetic competition between continuing elongation in the cytosol and RNC targeting to the membrane makes two testable predictions (Fig. 1d). First, pharmacological inhibition of elongation with cycloheximide (CHX) should decouple these processes, enhancing targeting of translocation-competent RNCs and promoting their depletion from the soluble fraction (Fig. 1e). Cells were subjected to a brief, two-minute CHX incubation before Ribo-seq analysis of soluble and membrane-bound polysomes. Importantly, such brief incubation did not perturb non-secretory polysomes (Extended Data Fig. 1c). By contrast, CHX treatment markedly reduced the soluble

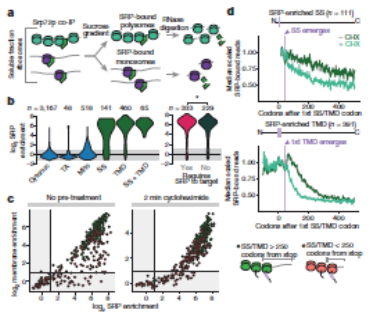


Figure 2 | Cotranslational enrichment of SRP. a, Srp72p-TAP was immunoprecipitated from the total soluble fraction. SRP-bound monosomes and polysomes were separated by sucrose gradient ultracentrifugation. **b**, Distributions of the ORF enrichment of ribosome-protected reads from SRP-bound soluble polysomes over the total soluble polysomes. ORFs were alternatively classified by expected SRP dependence[11]. Values are the mean from two biological replicates. TA, tail-anchored. * $P \leq 0.05$, Wilcoxon rank-sum test. **c**, Cotranslational membrane-fraction enrichment compared to SRP enrichment. **d**, Metagene analysis of soluble SRP-bound polysome-protected reads from transcripts that were at least twofold SRP-enriched. ORFs were aligned at the targeting signal and scaled.

The kinetic competition between targeting and elongation predicts that cotranslational membrane attachment is influenced by translation termination. In the absence of an elongation arrest, the probability of RNCs reaching the membrane cotranslationally will decrease as the first SS or TMD is found closer to the C terminus (Fig. 1f and Extended Data Fig. 1e). Indeed, we observed a decline in the maximum membrane enrichment of secretory RNCs when the first targeting signal is near the C terminus. Thus, secretory proteins with a late targeting signal, SS or TMD, must be targeted to the ER posttranslationally (Supplementary Discussion). Overall, our data suggest that cotranslational targeting to the ER in yeast is accomplished via a pioneer round of translation on soluble ribosomes that establishes a pool of ER-residing mRNA that initiate translation at the membrane (Extended Data Fig. 1f).

We next determined which RNCs are substrates of the SRP *in vivo*. Immunoprecipitation of Srp72p from total soluble RNCs was followed by ribosome profiling of both SRP-associated polysomes and monosomes (Fig. 2a and Extended Data Fig. 2a). Few transcripts encoding cytonuclear or mitochondrial proteins were enriched on SRP, confirming its specificity towards ER-destined transcripts. Notably, the SRP bound to all secretory RNCs that were cotranslationally targeted to the membrane, including SRP-dependent and SRP-independent proteins (Fig. 2b, c).

The number of ribosome-protected reads from soluble, SRP-bound transcripts diminished after ribosome exposure of the first SS or TMD, as expected from its targeting function (Fig. 2d). The loss was gradual and many SRP–RNCs remained soluble well after the targeting signal became fully exposed to the cytosol. This supports the notion that elongation proceeds on cytosolic ribosomes even after SRP binds, in contrast with the expected SRP-induced elongation arrest. Indeed, blocking elongation with CHX for 2 min before lysis caused a marked depletion in SRP-bound reads, but only for RNCs

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Although elongation arrest is not a general consequence of SRP binding *in vivo*, recent work showed that a rare-codon-directed slowdown of elongation facilitates SRP binding[19]. An intrinsic, non-SRP-dependent elongation slowdown should increase ribosome-protected reads at the same codon in both soluble SRP-bound and membrane-bound polysomes. Indeed, several transcripts presented such local increases in ribosome-protected reads at sites corresponding to exposure of a targeting signal on the ribosome (Extended Data Fig. 3a–c). Distinct elongation attenuation mechanisms observed at these sites included clusters of rare codons[19] and stalling polypeptide elements, such as stretches of positively charged amino acids, or proline motifs, positioned within the exit tunnel[20,21]. While most secretory transcripts were not significantly enriched in these attenuator elements compared to the proteome (Extended Data Fig. 3d, e), the few non-secretory proteins that cotranslationally bound to the SRP were enriched in elongation attenuation elements positioned at sites that exposed a near-cognate hydrophobic sequence for SRP binding (Extended Data Fig. 3d, f). We speculate that the presence of such elements enhances SRP recognition of the near-cognate hydrophobic tracts in these non-secretory proteins.

To understand the basis for the specificity of the SRP *in vivo*, we next determined the initial point of SRP recruitment to ribosomes

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Source: Chartron et al.. (2016) Nature, **536**, 224.



Data and Information Visualization



WIKIPEDIA
The Free Encyclopedia

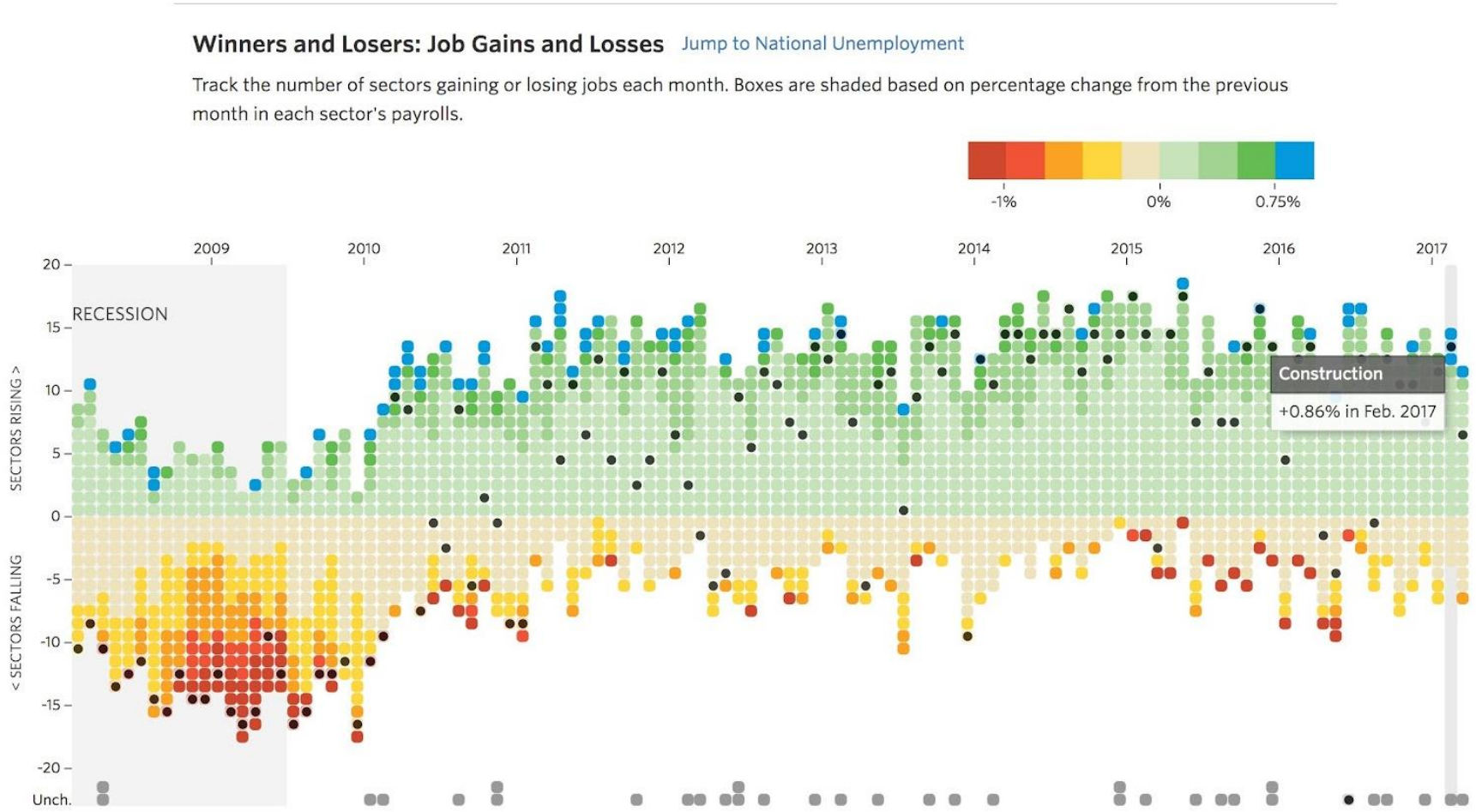
Data and information visualization is the practice of designing and creating easy-to-communicate and easy-to-understand graphic or visual representations of a large amount of complex quantitative and qualitative data and information with the help of static, dynamic or interactive visual items.

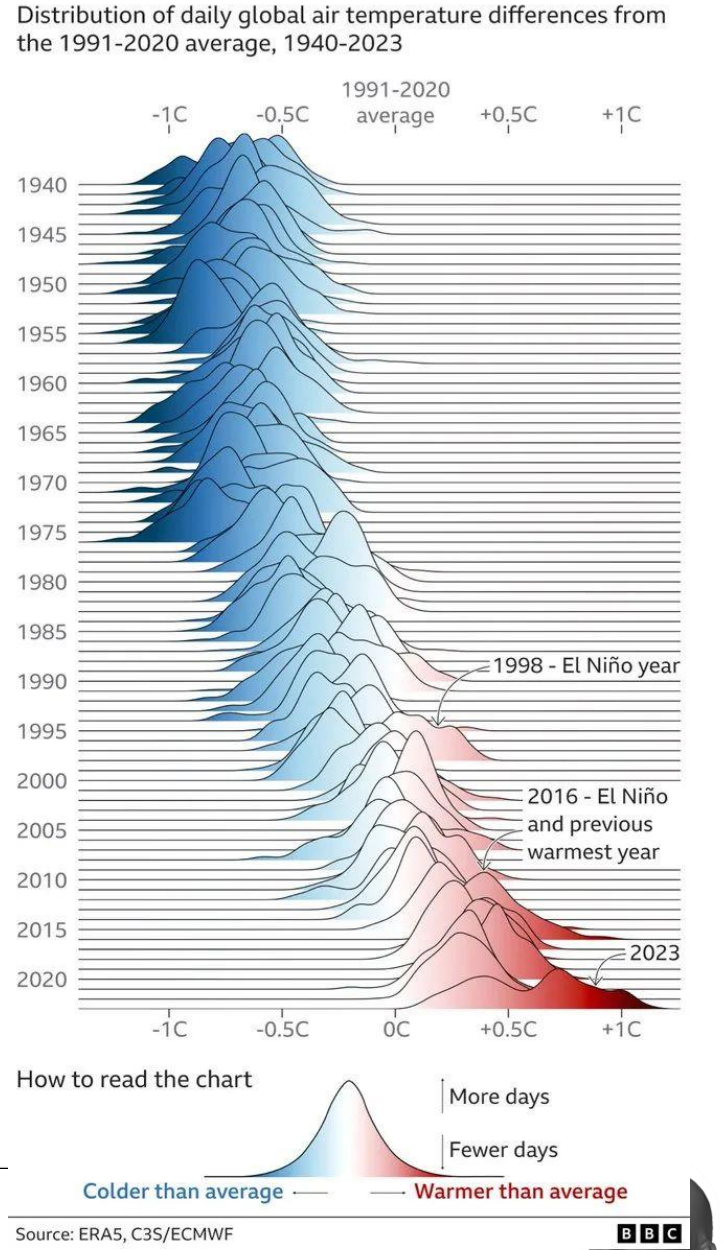
- **Data visualization** is concerned with visually presenting sets of primarily quantitative raw data in a schematic form. The visual formats used in data visualization include tables, charts and graphs (e.g. pie charts, bar charts, line charts, area charts, cone charts, pyramid charts, donut charts, histograms, spectrograms, cohort charts, waterfall charts, funnel charts, bullet graphs, etc.), diagrams, plots (e.g. scatter plots, distribution plots, box-and-whisker plots), geospatial maps (such as proportional symbol maps, choropleth maps, isopleth maps and heat maps), figures, correlation matrices, percentage gauges, etc., which sometimes can be combined in a dashboard.

- **Information visualization**, deals with multiple, large-scale and complicated datasets which contain quantitative (numerical) data as well as qualitative (non-numerical, i.e. verbal or graphical) and primarily abstract information and its goal is to add value to raw data, improve the viewers' comprehension, reinforce their cognition and help them derive insights and make decisions as they navigate and interact with the computer-supported graphical display. Visual tools used in information visualization include maps (such as tree maps), animations, infographics, Sankey diagrams, flow charts, network diagrams, semantic networks, entity-relationship diagrams, venn diagrams, timelines, mind maps, etc.



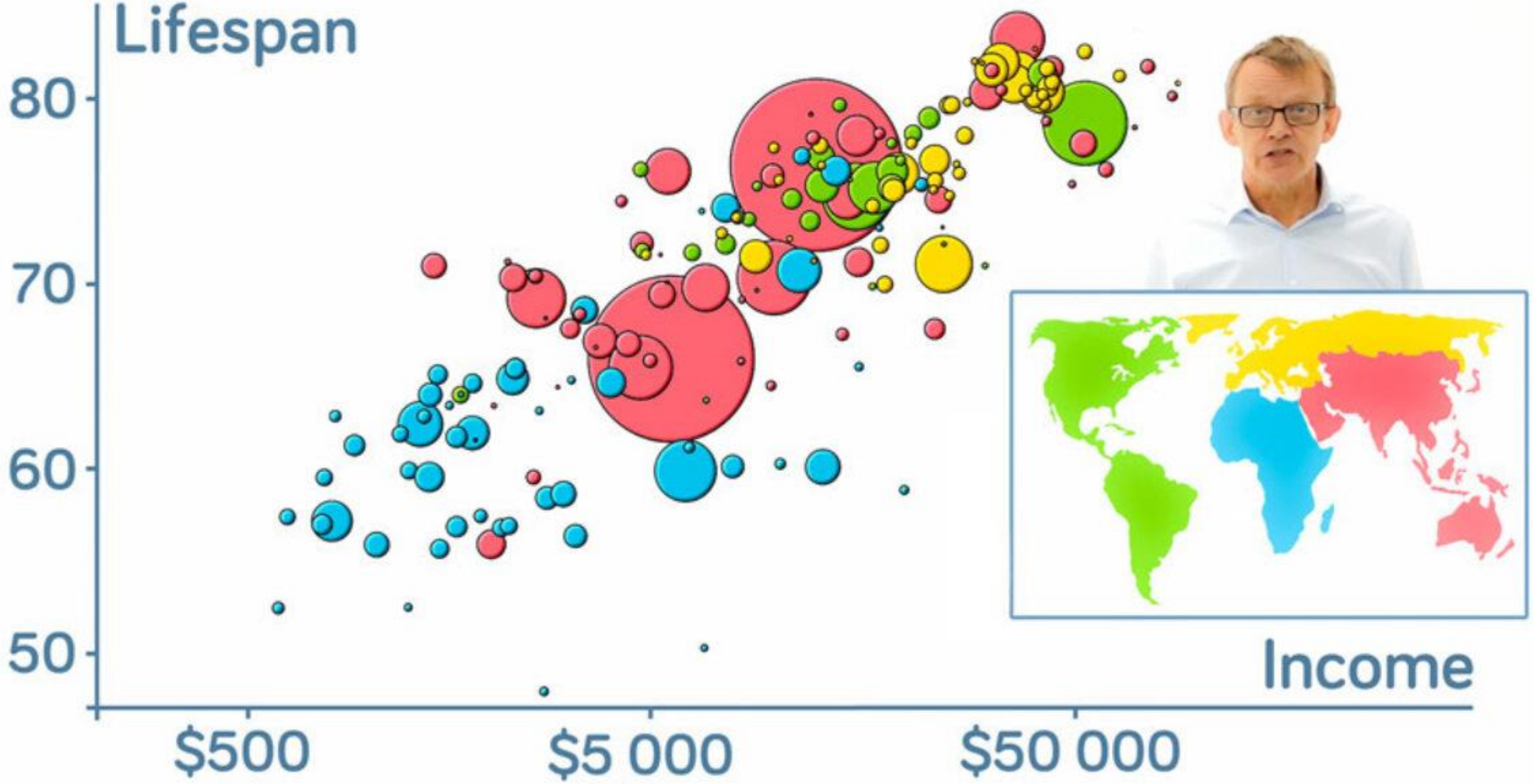
Data Visualization





Hans Rosling: Bubble charts and graph animation

https://www.gapminder.org/





Source: https://www.youtube.com/watch?v=jbkSRLYSojo&t=186s

Information Visualization

Google

6 Phases of a Data Analysis Project

Ask
Objective: Define the problem, understand the pain points, identify the key stakeholders

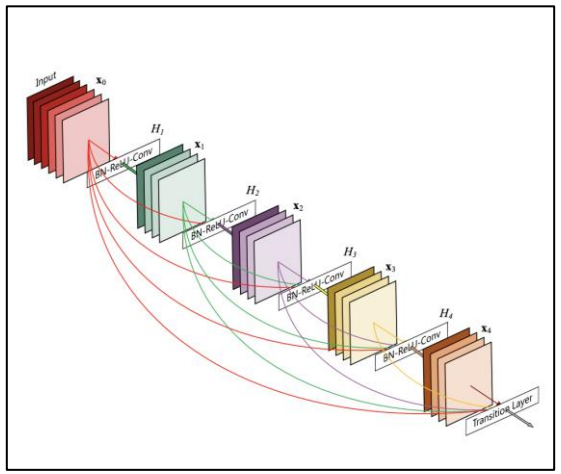
Prepare
Objective: Identify the metrics to measure. Acquire the data

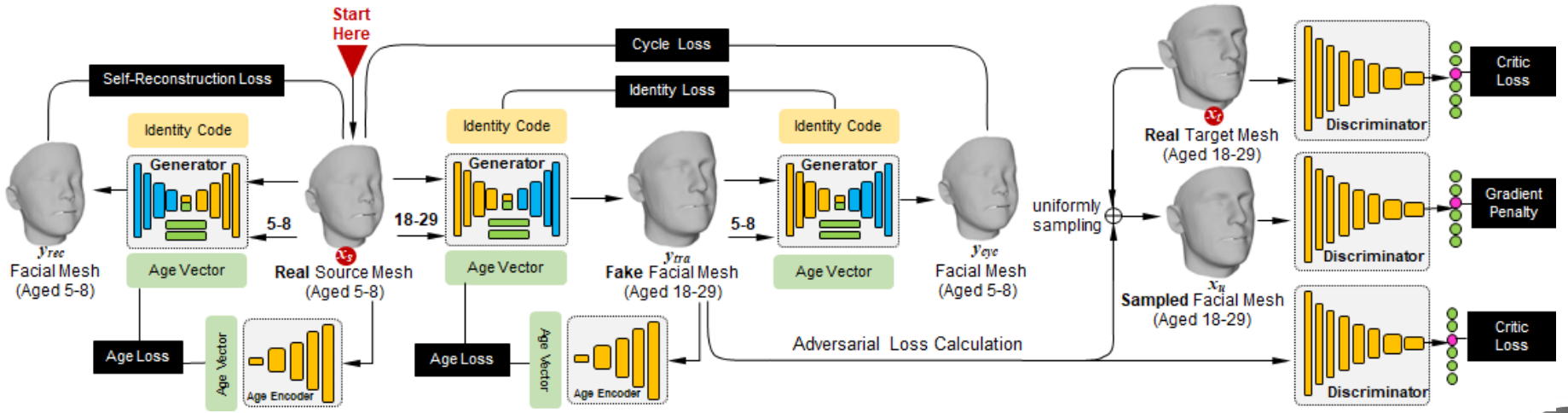
Process
Objective: Data cleaning, understanding the data, feature generation, handling missing values

Analyze
Objective: Extract insights, understand trends, identify gaps and opportunities

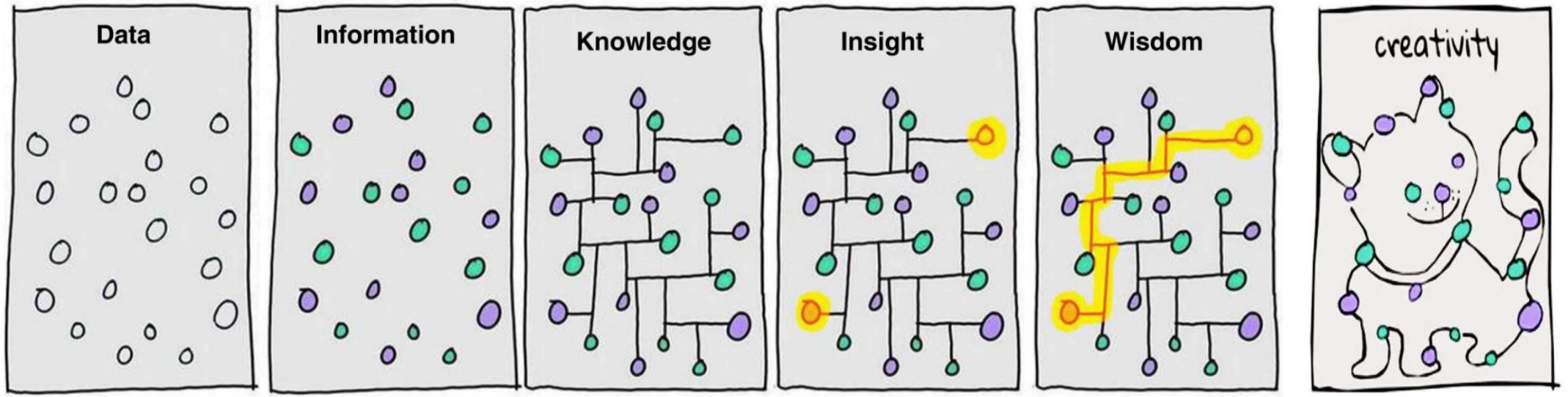
Share
Objective: Craft a data story, communicate findings, data-driven story telling

Act
Objective: Provide recommendations, data-backed business decisions



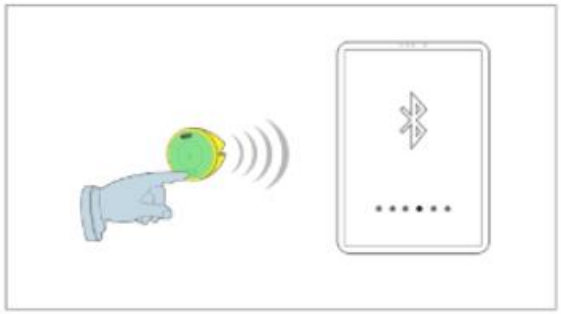


Information Visualization

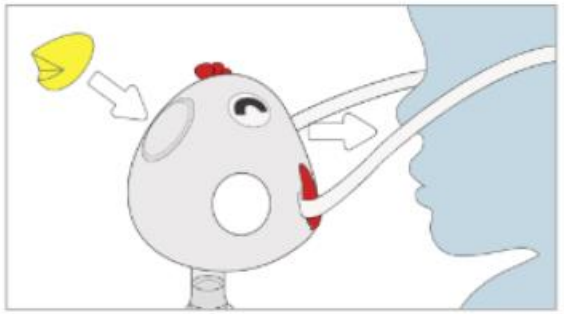


‘Data is not information, information is not knowledge, knowledge is not understanding, understanding is not wisdom’ (CliffordStoll and Gary Schubert).

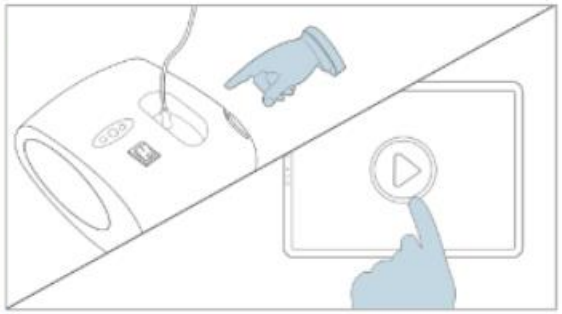
Information Visualization



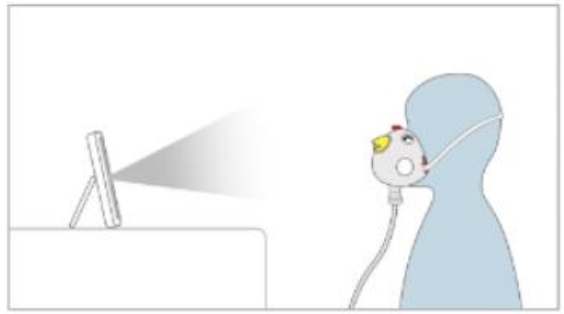
❶ Turn on Bluetooth on smart hardware and connect to the iPad.



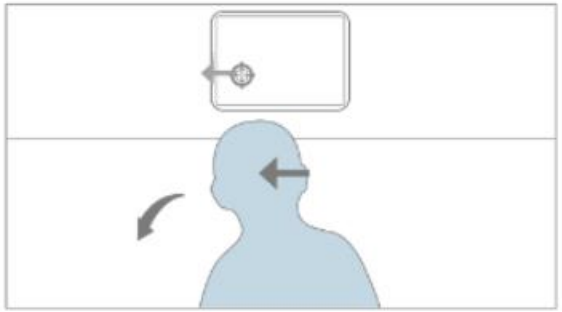
❷ Insert the smart hardware into the mask. User put on the mask.



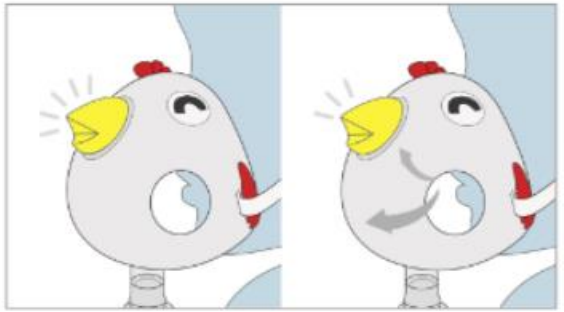
❸ Turn on the nebulizer and prepare for treatment. Also open the game on the iPad.



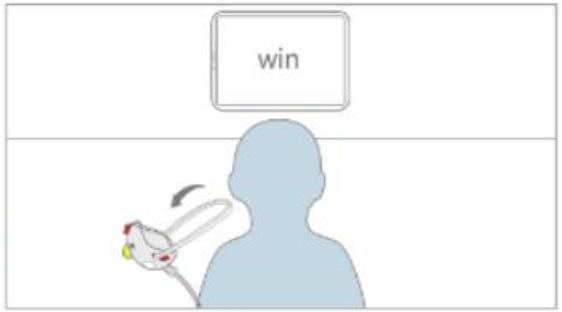
❹ Place the iPad, make sure the user is facing the iPad camera and begin the treatment.



❺ The user controls the direction of the game by turning his head.



❻ The intelligent hardware interacts with sound. The louder the breathing, the brighter the hardware, and also the strength increase in the game.



❼ After complete the game, take off the mask, and the treatment is finished.



❽ Fun Breath will generate a treatment result base on the respiratory number and intensity.

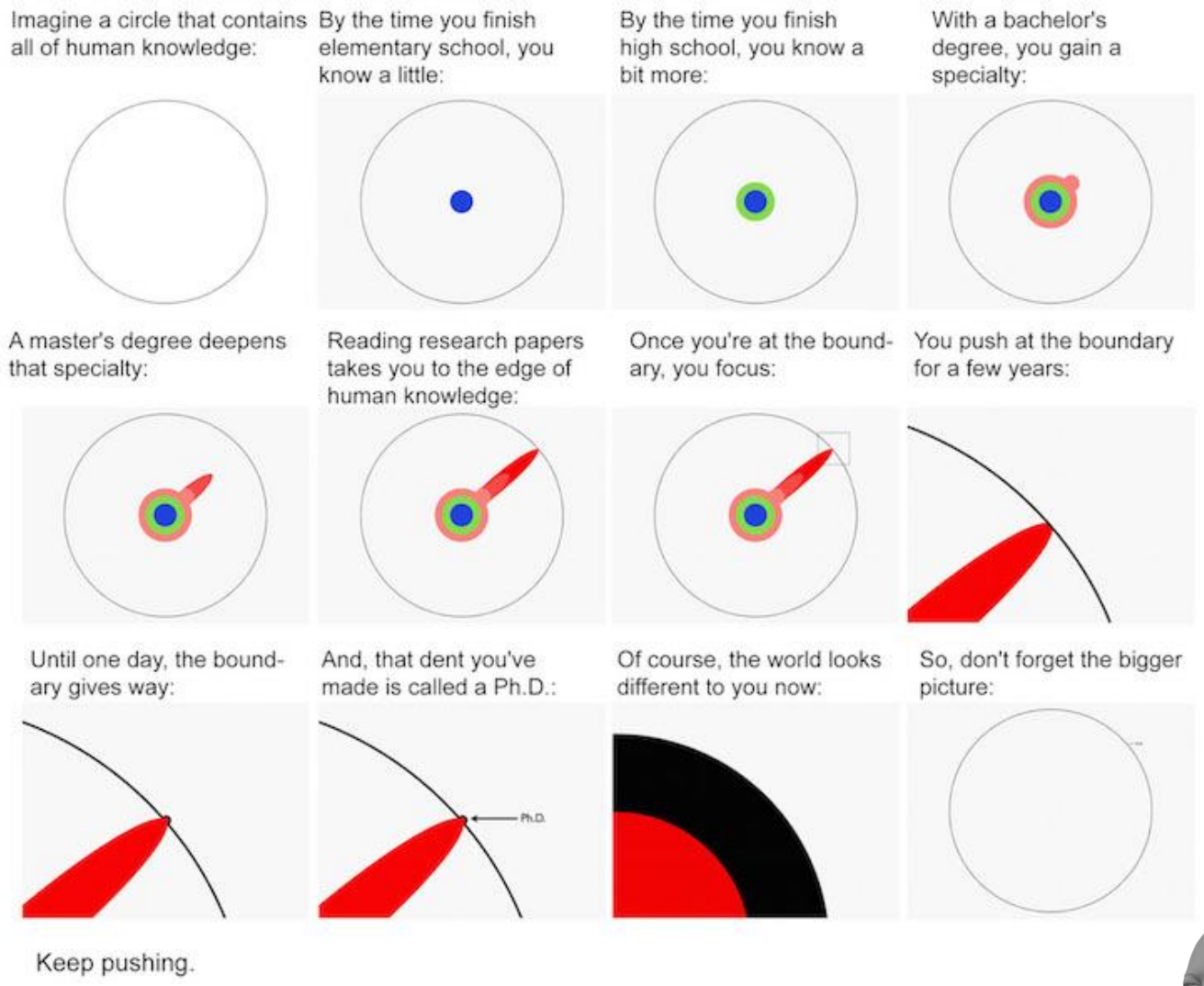
How to use our designed FunBreath (An Intelligent Nebulizer Mask for Children's Happy and Effective Treatment)

Matt Might: Clever 12 Step Infographic Explains the Path of a Ph.D.



In 2012, Matt Might, a professor of computer science at the University of Utah, created The Illustrated Guide to the Ph.D.

This clever data visualization has helped him explain what a Ph.D. is to his incoming doctoral student each year since. The clear, simple, and effective design will ring true for anyone who has gone through the Ph.D. process, and will provide clarity for anyone thinking about taking the plunge.

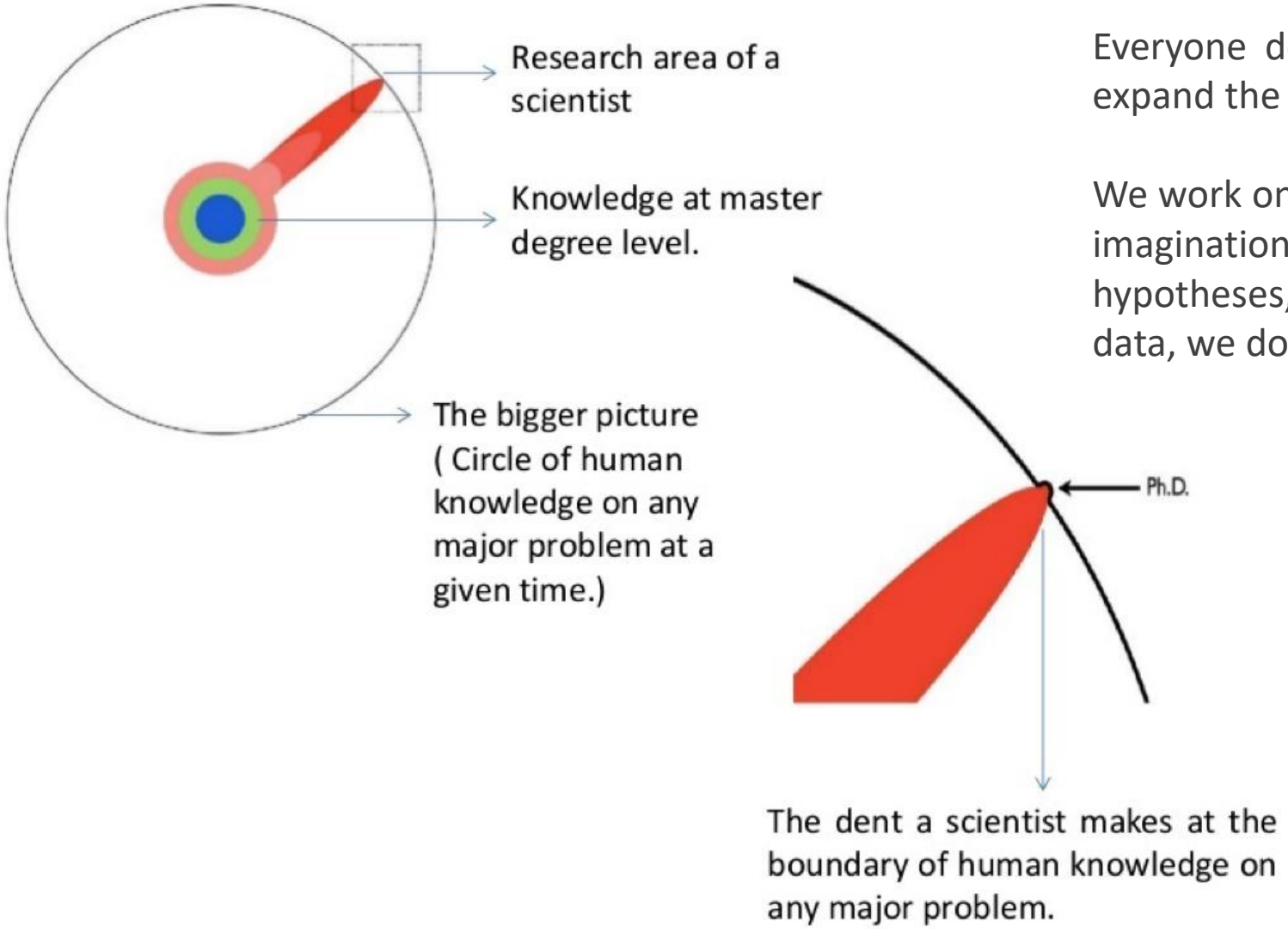




The animated guide to a Ph.D.

Source: https://www.youtube.com/watch?v=We760YM5-iM

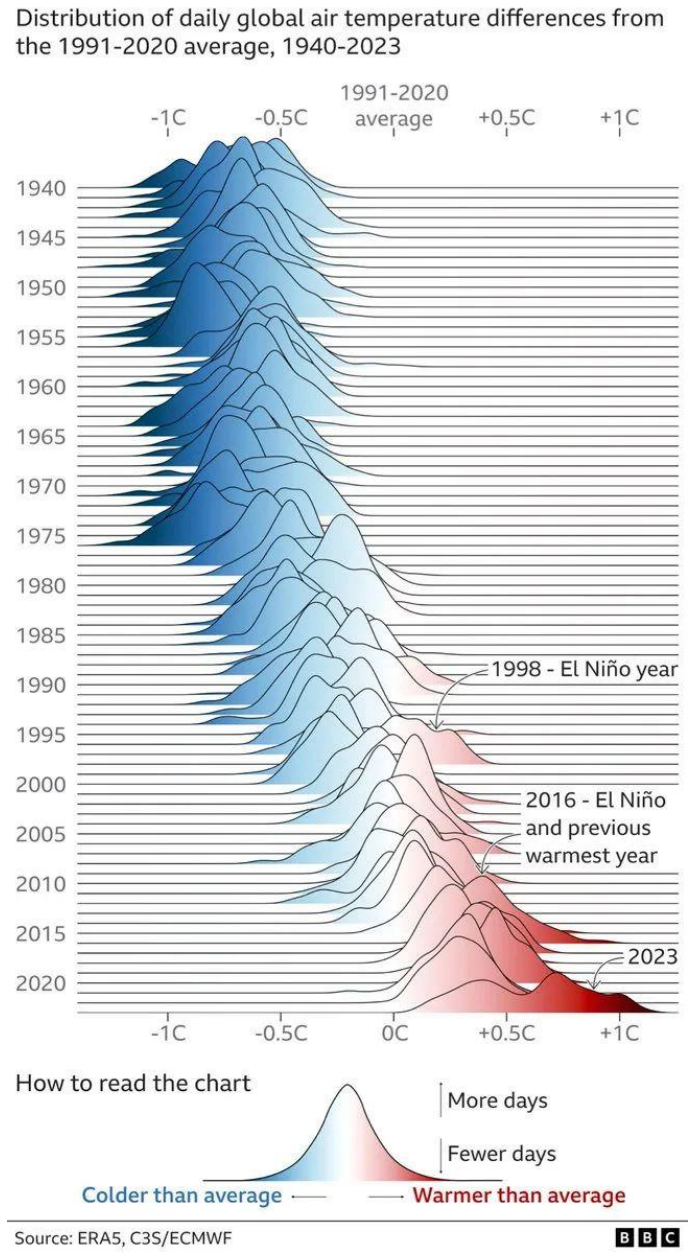
Matt Might: Clever 12 Step Infographic Explains the Path of a Ph.D.



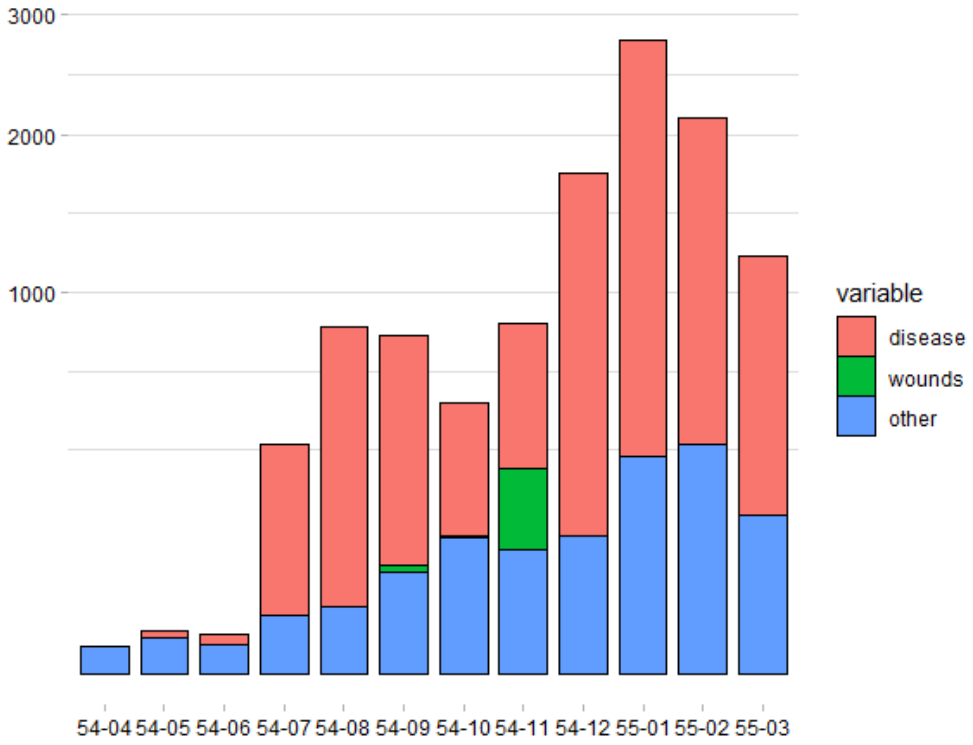
Everyone doing a PhD knows the final task: To expand the limits of knowledge.

We work on it for 3 or 4 years. With responsibility, imagination and enthusiasm, we formulate hypotheses, we make experiments, we analyze data, we do research!

Functions


Distribution of daily global air temperature differences from the 1991-2020 average, 1940-2023
1991-2020 average
-1C
-0.5C
+0.5C
+1C
1940
1945
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
1998 - El Niño year
2016 - El Niño and previous warmest year
2023
-1C
-0.5C
0C
+0.5C
+1C
How to read the chart
More days
Fewer days
Colder than average
Warmer than average
Source: ERA5, C3S/ECMWF
BBC

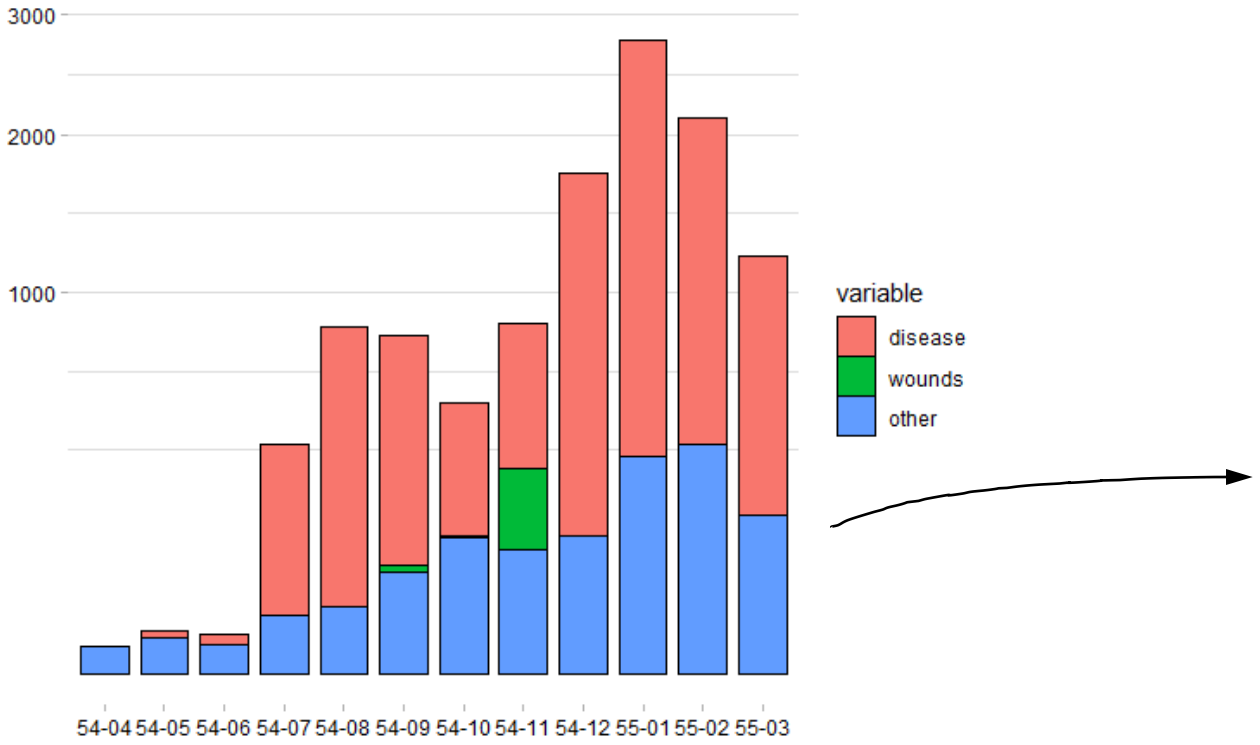
Data Visualization



(a) Rectangular coordinate

Stacked bar chart in different coordinates

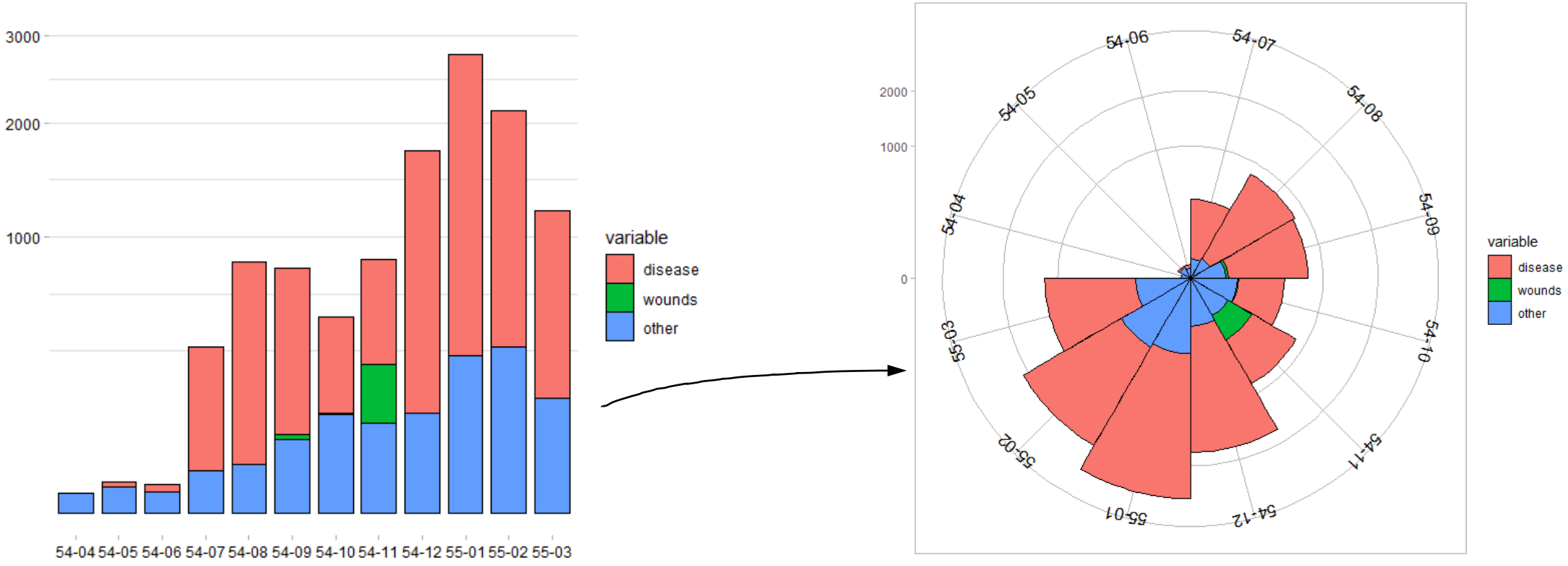
Data Visualization



(a) Rectangular coordinate

Stacked bar chart in different coordinates

Data Visualization



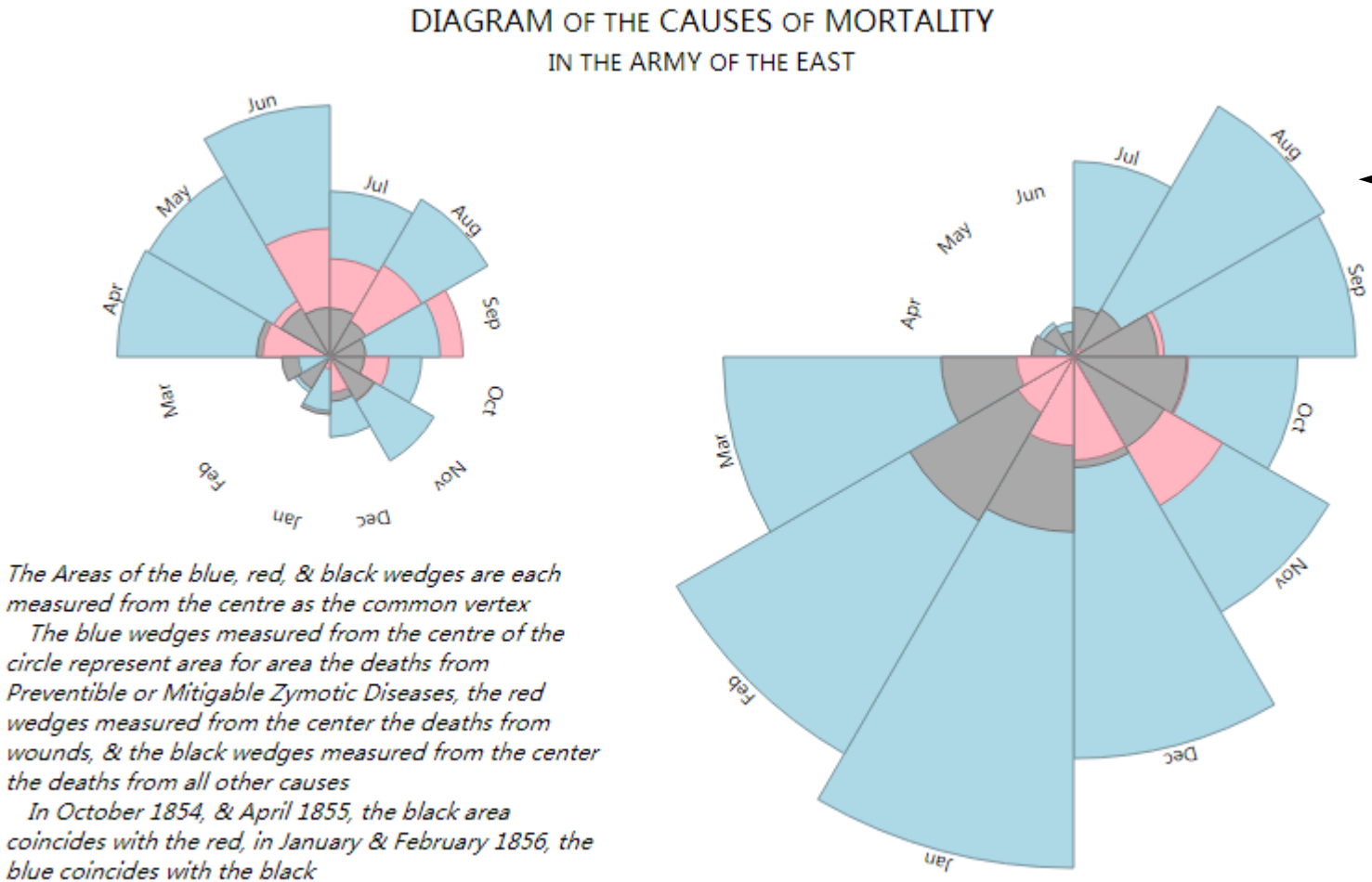
(a) Rectangular coordinate

(b) Polar coordinate

Stacked bar chart in different coordinates



Nightingale’ s Rose

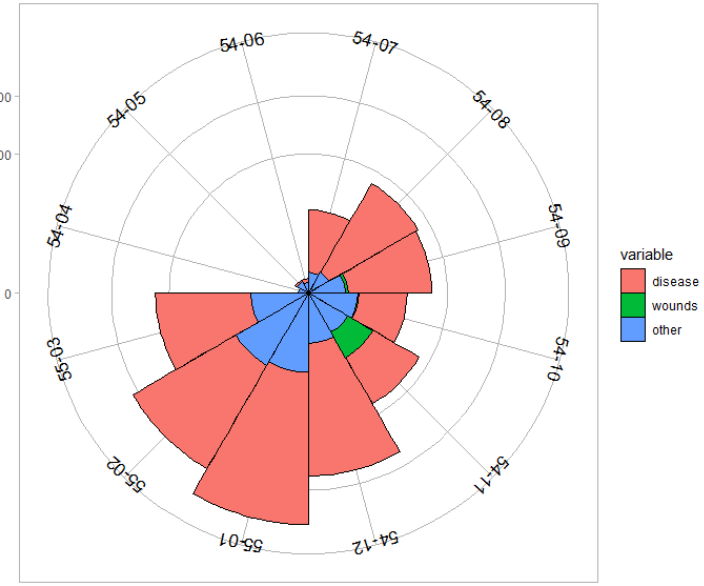


The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex

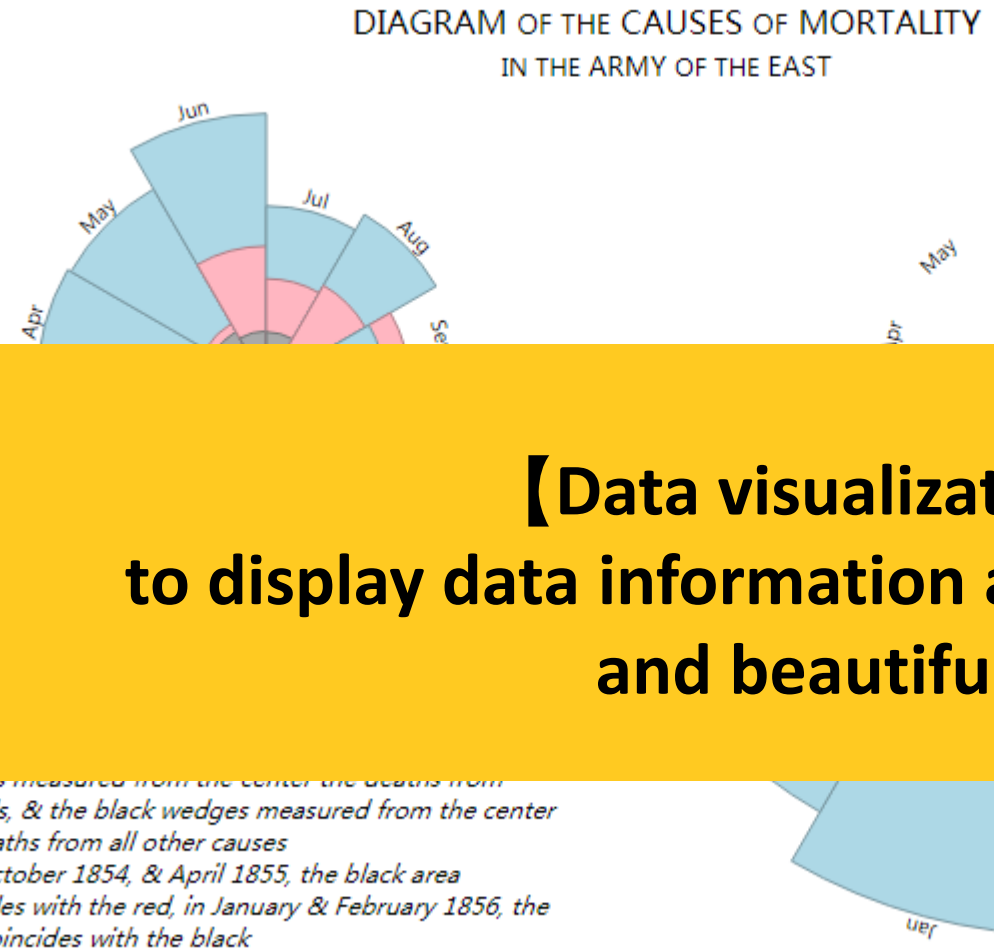
The blue wedges measured from the centre of the circle represent area for area the deaths from Preventible or Mitigable Zymotic Diseases, the red wedges measured from the center the deaths from wounds, & the black wedges measured from the center the deaths from all other causes

In October 1854, & April 1855, the black area coincides with the red, in January & February 1856, the blue coincides with the black

The entire areas may be compared by following the blue, the red & the black lines enclosing them.



Nightingale’ s Rose

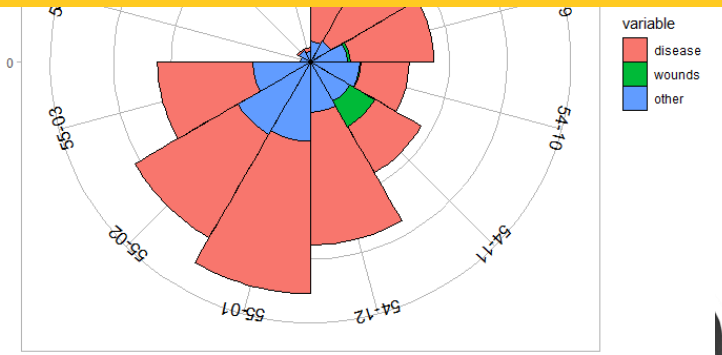


wounds, & the black wedges measured from the center the deaths from all other causes

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The entire areas may be compared by following the blue, the red & the black lines enclosing them.

【Data visualization】
to display data information accurately, simply and beautifully



Data Analysis

Four different datasets with the same x and different y (*y*1, *y*2, *y*3, *y*4)

x	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
*y*1	4.6	5.4	5.2	6.6	5.9	6.1	5.8	6.8	6.5	6.7	6.9	11.1	8.2	10.3	12.8	13
*y*2	6.1	11.6	16.6	19	22.7	31.8	34	33.7	35.6	34.5	39.6	58.3	57.7	72.9	68.4	82.6
*y*3	5.5	31.1	33.1	51.8	55.7	60.7	63.5	75.5	84.4	84.6	76.3	92.4	81.6	91	88.1	93.8
*y*4	1	3	4.9	7.9	9.8	12	18.9	24.7	28.9	28.6	39.3	33.2	42.1	54.4	43.3	90.2

x	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	-
*y*1	20.8	12.4	15.9	15.3	38.8	35.9	24.3	54.5	62.9	43.8	76.9	91	96.9	51.4	100	-
*y*2	84.5	82	89.1	102.1	68.1	96.3	108.5	76.7	107.6	103.4	116.5	106.4	142.5	115.1	110.5	-
*y*3	101.3	103	107.4	104.3	110.7	103.4	113.6	105.1	112.5	119.3	113.7	109.5	108.7	110.1	118.8	-
*y*4	81.2	90.8	70.9	66.8	67.5	88.6	116.9	141.4	104	161.4	101.8	137.1	175.3	119.5	257.3	-

What is the relationship between x and y?

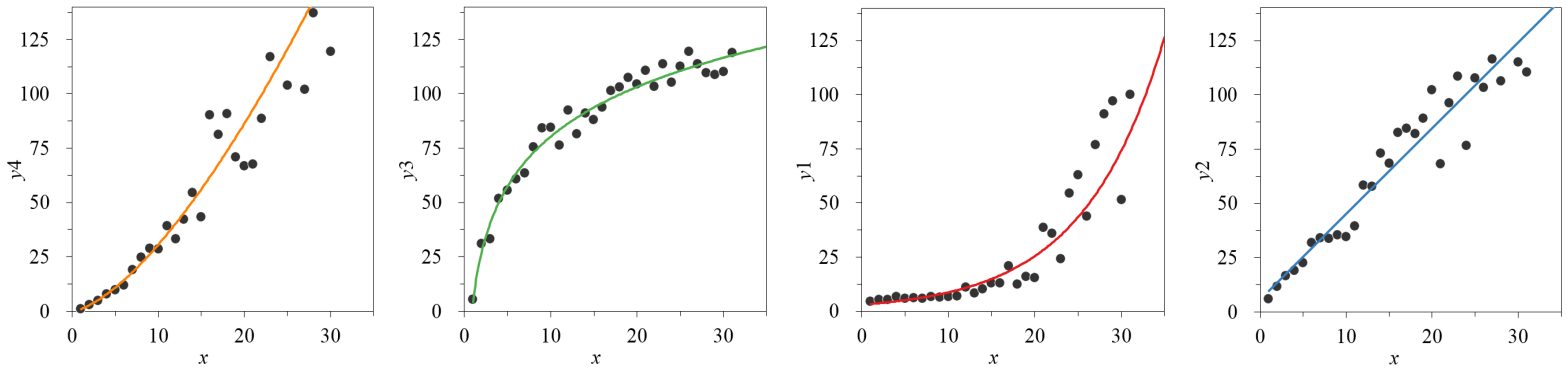


Data Analysis

Four different datasets with the same x and different y (*y*1, *y*2, *y*3, *y*4)

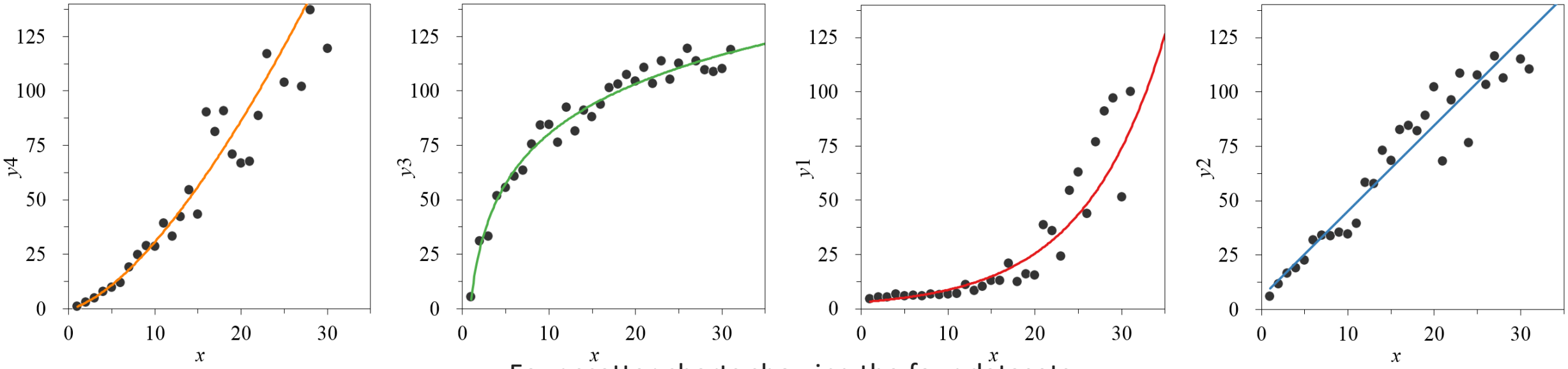
x	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
*y*1	4.6	5.4	5.2	6.6	5.9	6.1	5.8	6.8	6.5	6.7	6.9	11.1	8.2	10.3	12.8	13
*y*2	6.1	11.6	16.6	19	22.7	31.8	34	33.7	35.6	34.5	39.6	58.3	57.7	72.9	68.4	82.6
*y*3	5.5	31.1	33.1	51.8	55.7	60.7	63.5	75.5	84.4	84.6	76.3	92.4	81.6	91	88.1	93.8
*y*4	1	3	4.9	7.9	9.8	12	18.9	24.7	28.9	28.6	39.3	33.2	42.1	54.4	43.3	90.2

x	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	-
*y*1	20.8	12.4	15.9	15.3	38.8	35.9	24.3	54.5	62.9	43.8	76.9	91	96.9	51.4	100	-
*y*2	84.5	82	89.1	102.1	68.1	96.3	108.5	76.7	107.6	103.4	116.5	106.4	142.5	115.1	110.5	-
*y*3	101.3	103	107.4	104.3	110.7	103.4	113.6	105.1	112.5	119.3	113.7	109.5	108.7	110.1	118.8	-
*y*4	81.2	90.8	70.9	66.8	67.5	88.6	116.9	141.4	104	161.4	101.8	137.1	175.3	119.5	257.3	-



Four scatter charts showing the four datasets

Data Analysis



Four scatter charts showing the four datasets

Data Analysis

Four different datasets with the same x and different y (*y*1, *y*2, *y*3, *y*4)

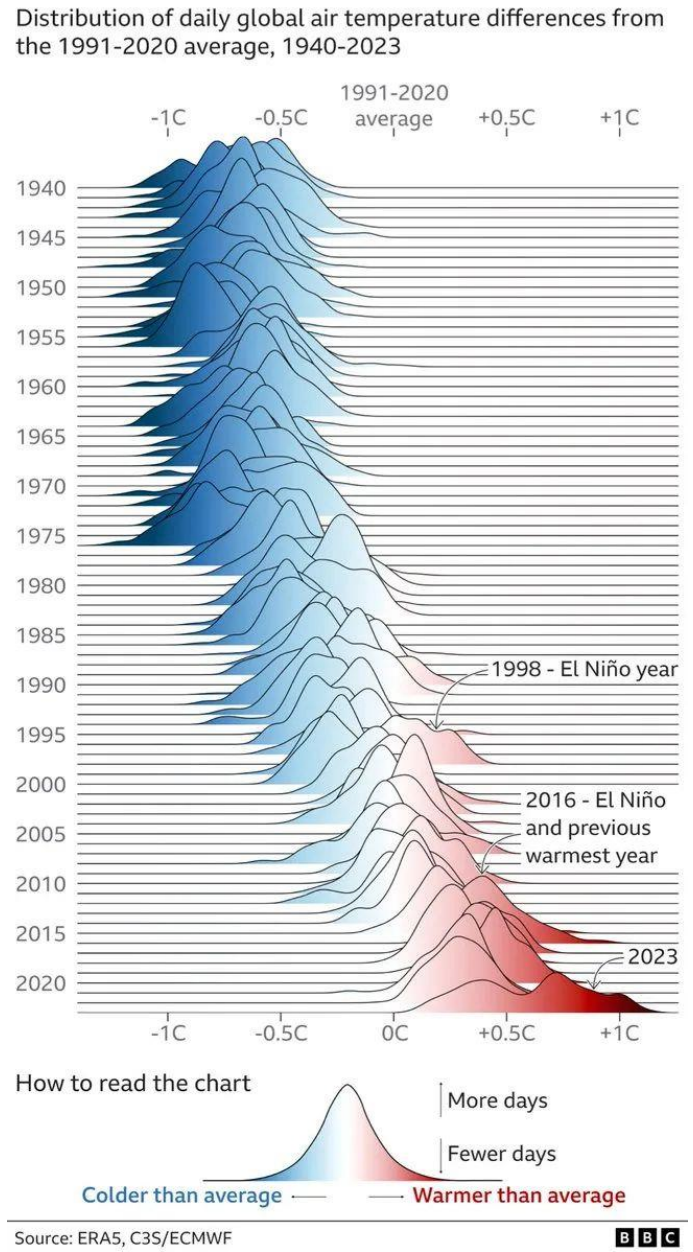
x	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
*y*1	4.6	5.4	5.2	6.6	5.9	6.1	5.8	6.8	6.5	6.7	6.9	11.1	8.2	10.3	12.8	13
*y*2	6.1	11.6	16.6	19	22.7	31.8	34	33.7	35.6	34.5	39.6	58.3	57.7	72.9	68.4	82.6
*y*3	5.5	31.1	33.1	51.8	55.7	60.7	63.5	75.5	84.4	84.6	76.3	92.4	81.6	91	88.1	93.8
*y*4	1	3	4.9	7.9	9.8	12	18.9	24.7	28.9	28.6	39.3	33.2	42.1	54.4	43.3	90.2

x	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	-
*y*1	20.8	12.4	15.9	15.3	38.8	35.9	24.3	54.5	62.9	43.8	76.9	91	96.9	51.4	100	-
*y*2	84.5	82	89.1	102.1	68.1	96.3	108.5	76.7	107.6	103.4	116.5	106.4	142.5	115.1	110.5	-
*y*3	101.3	103	107.4	104.3	110.7	103.4	113.6	105.1	112.5	119.3	113.7	109.5	108.7	110.1	118.8	-
*y*4	81.2	90.8	70.9	66.8	67.5	88.6	116.9	141.4	104	161.4	101.8	137.1	175.3	119.5	257.3	-

【Data visualization】
to find the data relationship and pattern



History

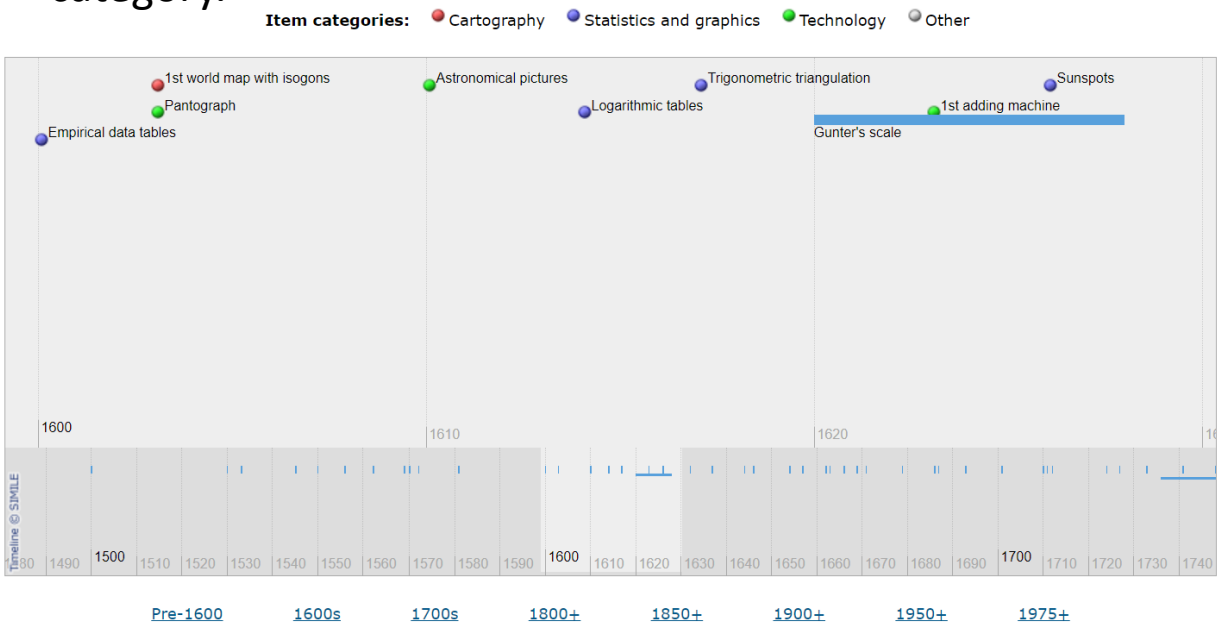

Distribution of daily global air temperature differences from the 1991-2020 average, 1940-2023
1991-2020 average
-1C
-0.5C
+0.5C
+1C
1940
1945
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
1998 - El Niño year
2016 - El Niño and previous warmest year
2023
-1C
-0.5C
0C
+0.5C
+1C
How to read the chart
More days
Fewer days
Colder than average
Warmer than average
Source: ERA5, C3S/ECMWF
BBC

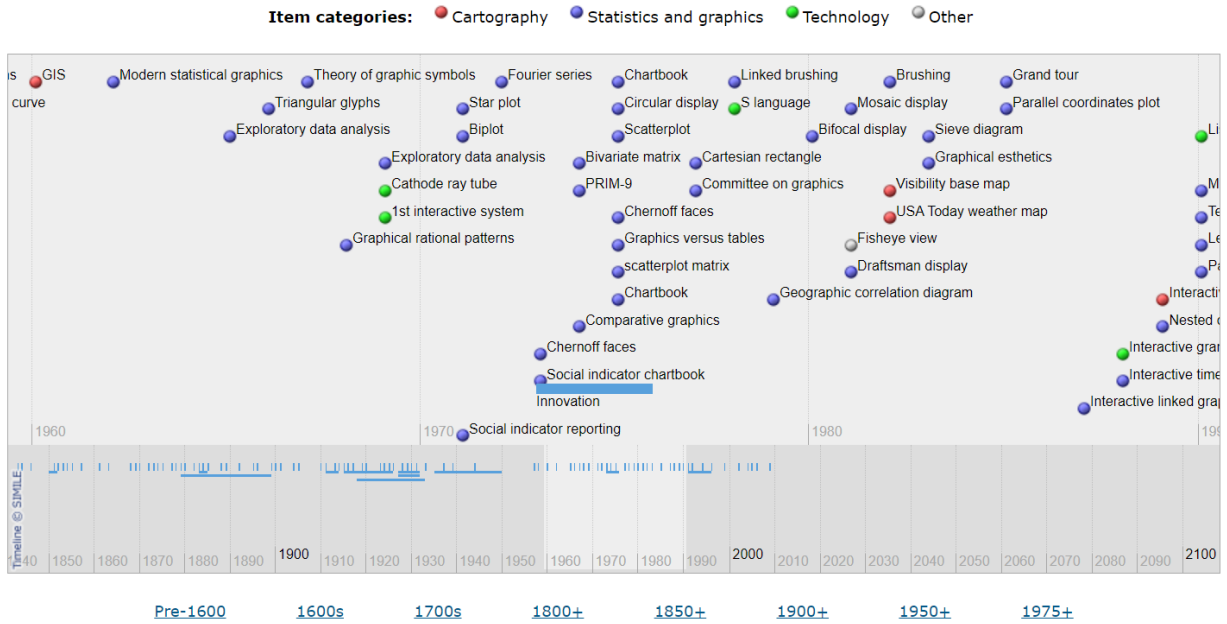
Data Visualization Timeline

Source: https://www.datavis.ca/milestones/

This page provides a graphic overview of the events in the history of data visualization that we call "**milestones**." These milestones are shown below in the form of *an interactive timeline*. The timeline is divided into *two vertical sections*. You can *drag each section left or right* to see milestones of different time periods. You can also click one of the links at the bottom of the timeline to jump to a particular epoch.

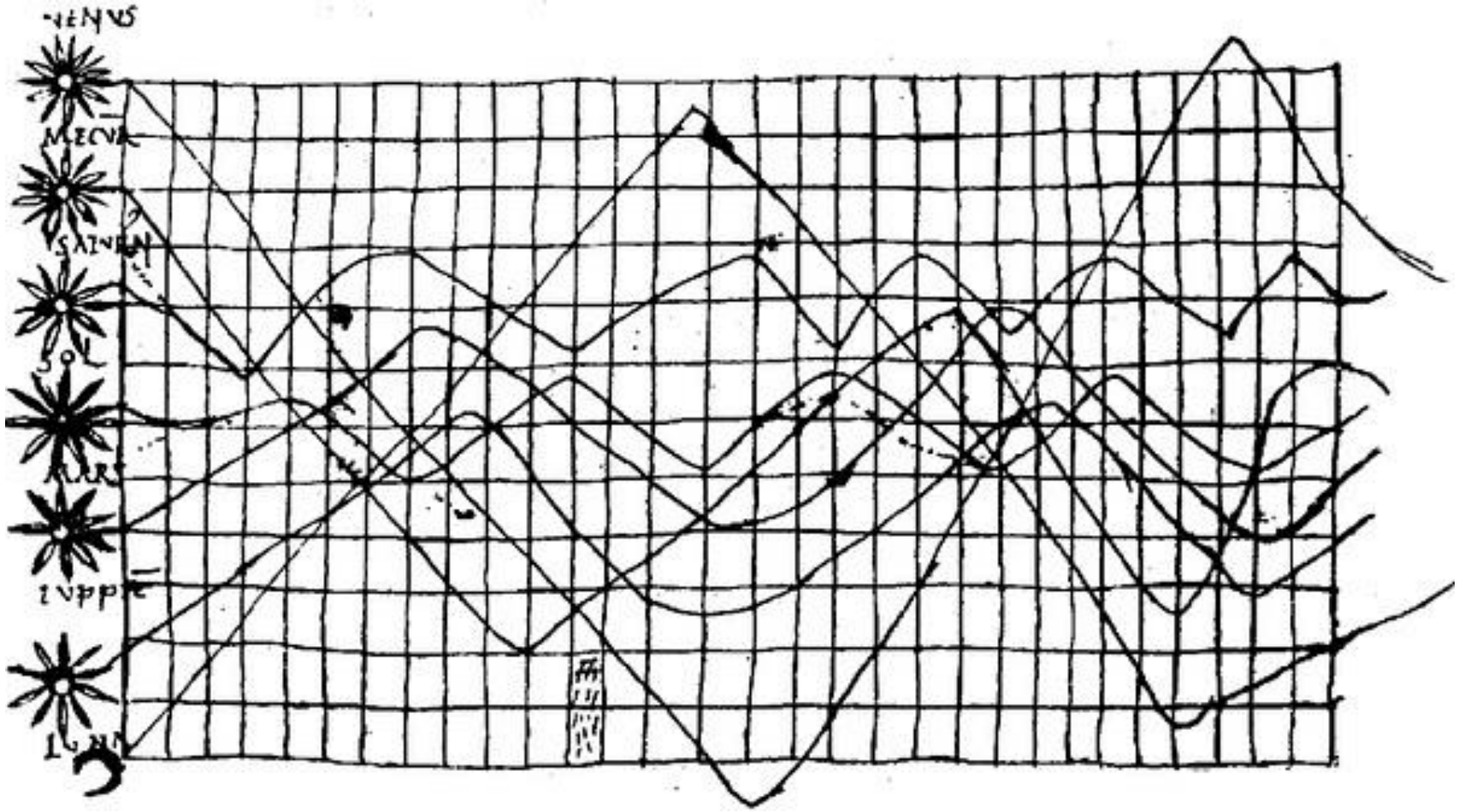
Each of the milestone's in the timeline can be clicked to reveal its summary that includes both a link to its full details and a category to which it belongs. The category can also be clicked to initiate a search of other milestone's based on that category.





950 Unknown: Diagram: planetary movements

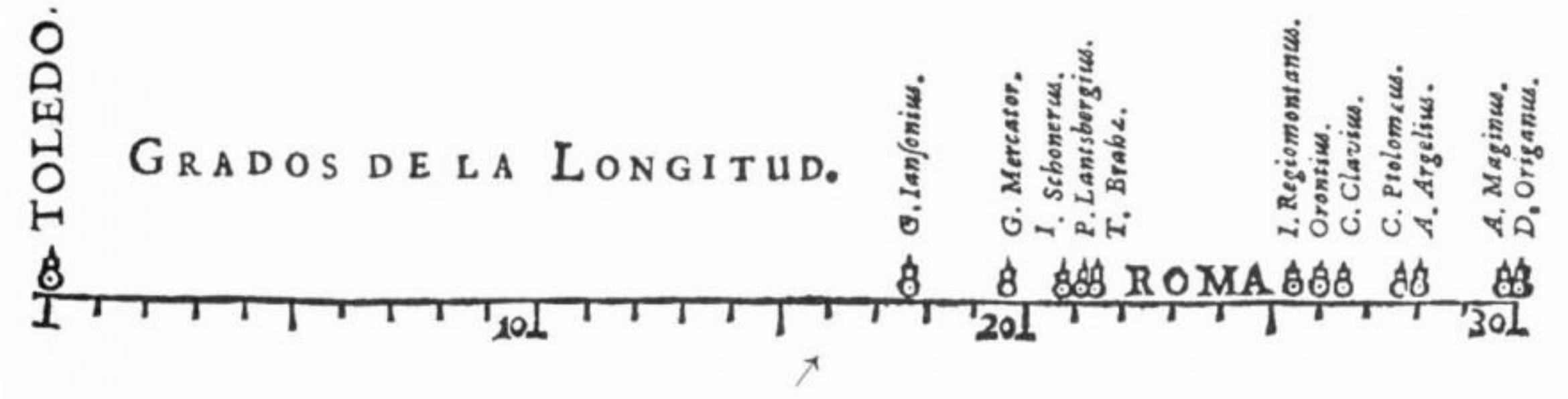
Earliest known attempt to show changing values graphically (positions of the sun, moon, and planets throughout the year)



1644 Michael F. van Langren: 1st data graph

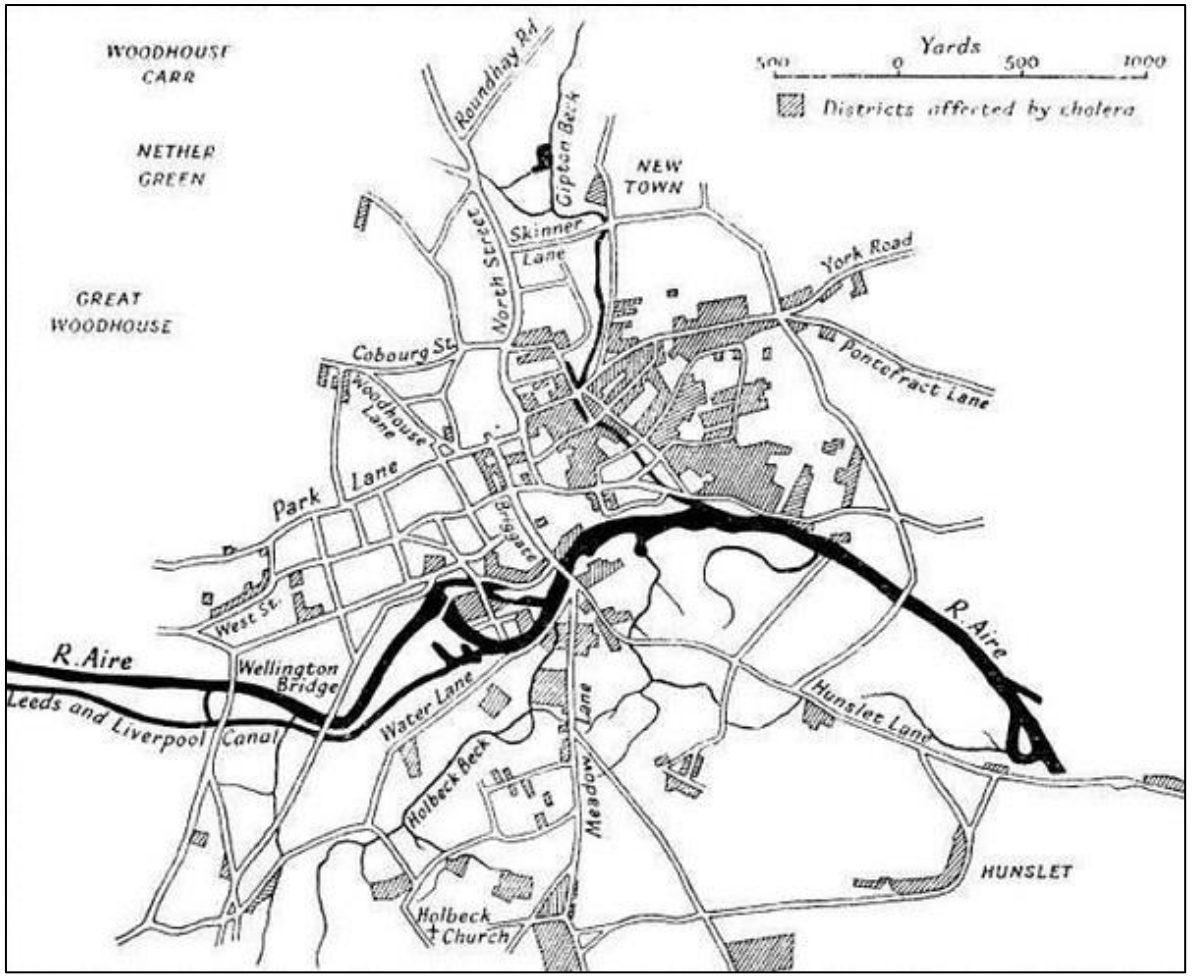
First visual representation of statistical data: variations in determination of longitude between Toledo and Rome

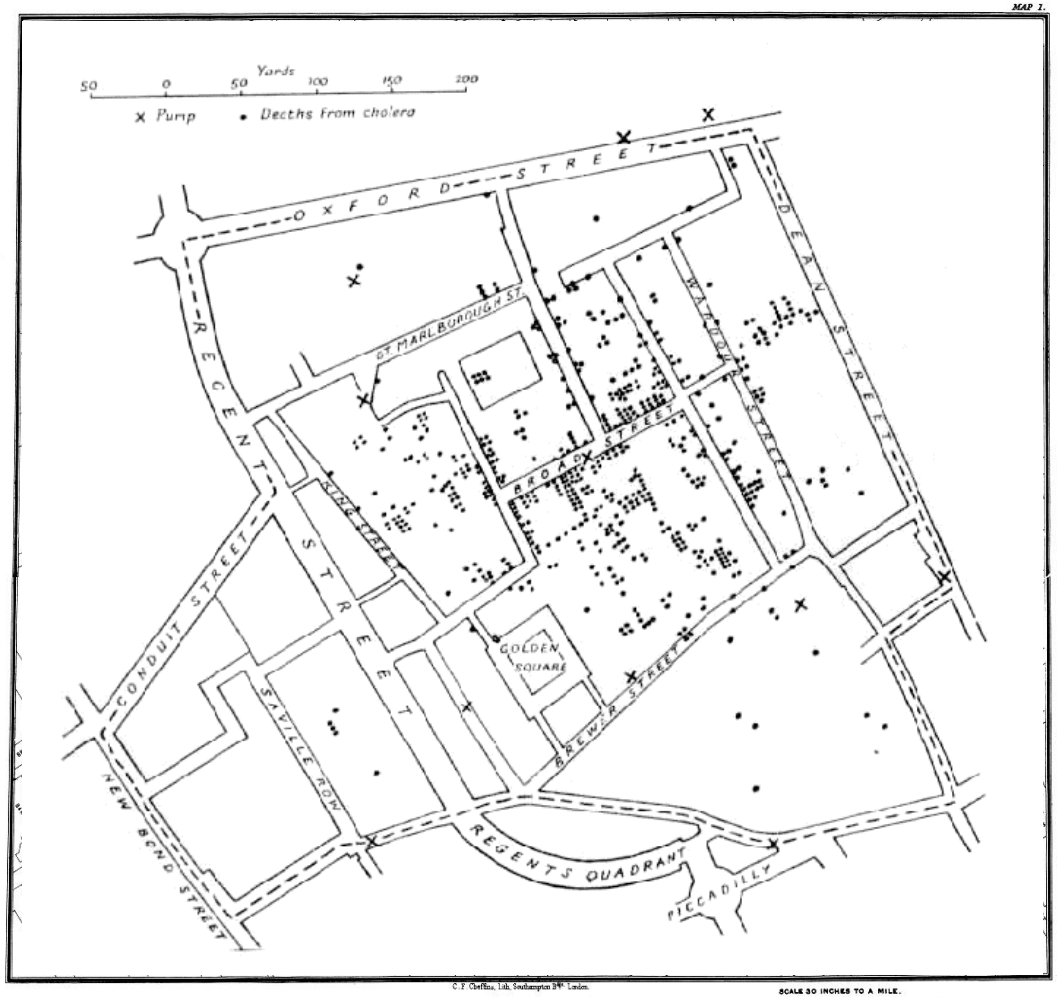
In 1644, Michael Florent Van Langren, a Flemish astronomer, is believed to have provided the first visual representation of statistical data. The one-dimensional line graph below shows the twelve known estimates at the time of the difference in longitude between Toledo and Rome as well as the name of each astronomer who provided the estimate. What is notable here is that while Van Langren could have provided this information in a table, it is the use of the graph that really visually displays the wide variations in estimates.



1855 John Snow: Dot map of disease

Use of a dot map to display epidemiological data, leads to discovery of the source of a cholera epidemic.



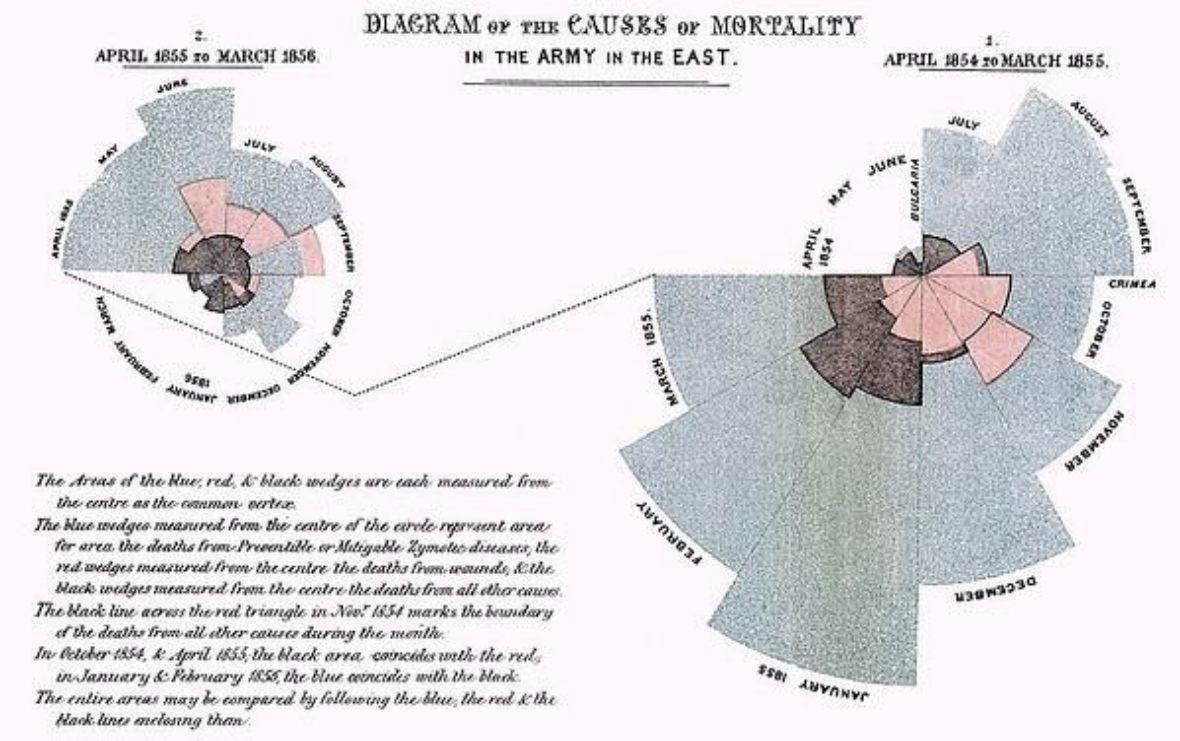


1857 Florence Nightingale: Coxcombs



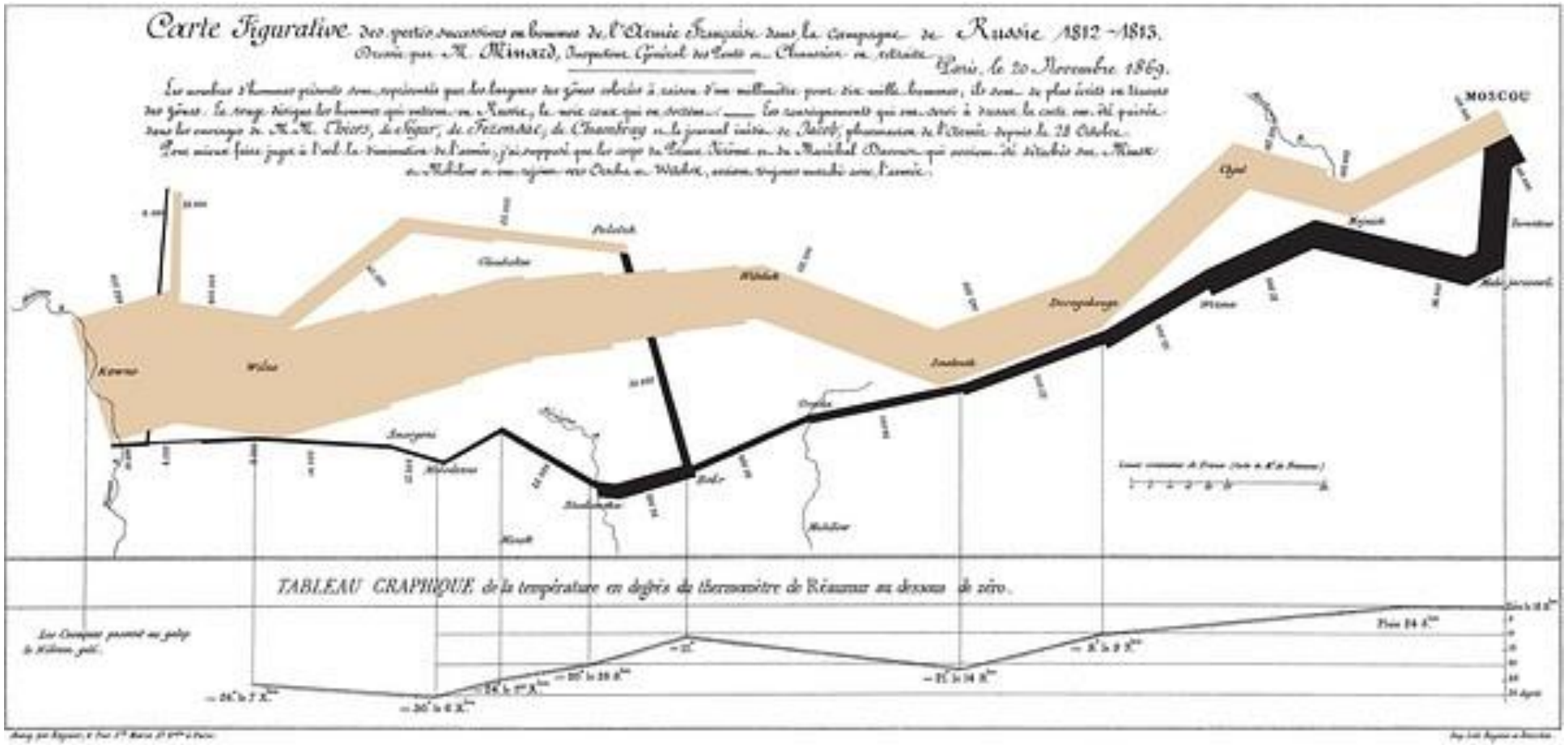
Polar area charts, known as "coxcombs" (used in a campaign to improve sanitary conditions of army)



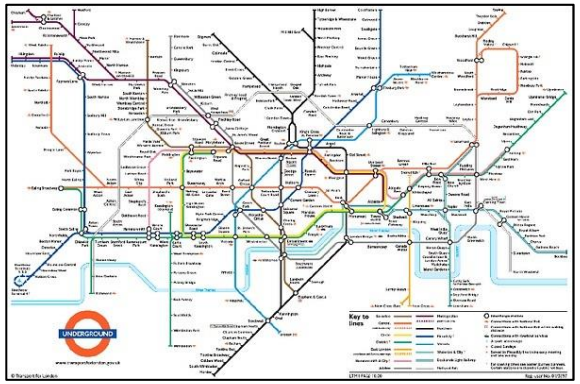


1869 Charles Joseph Minard: Flow map

Minard's flow map graphic of Napoleon's March on Moscow (called "the best graphic ever produced)

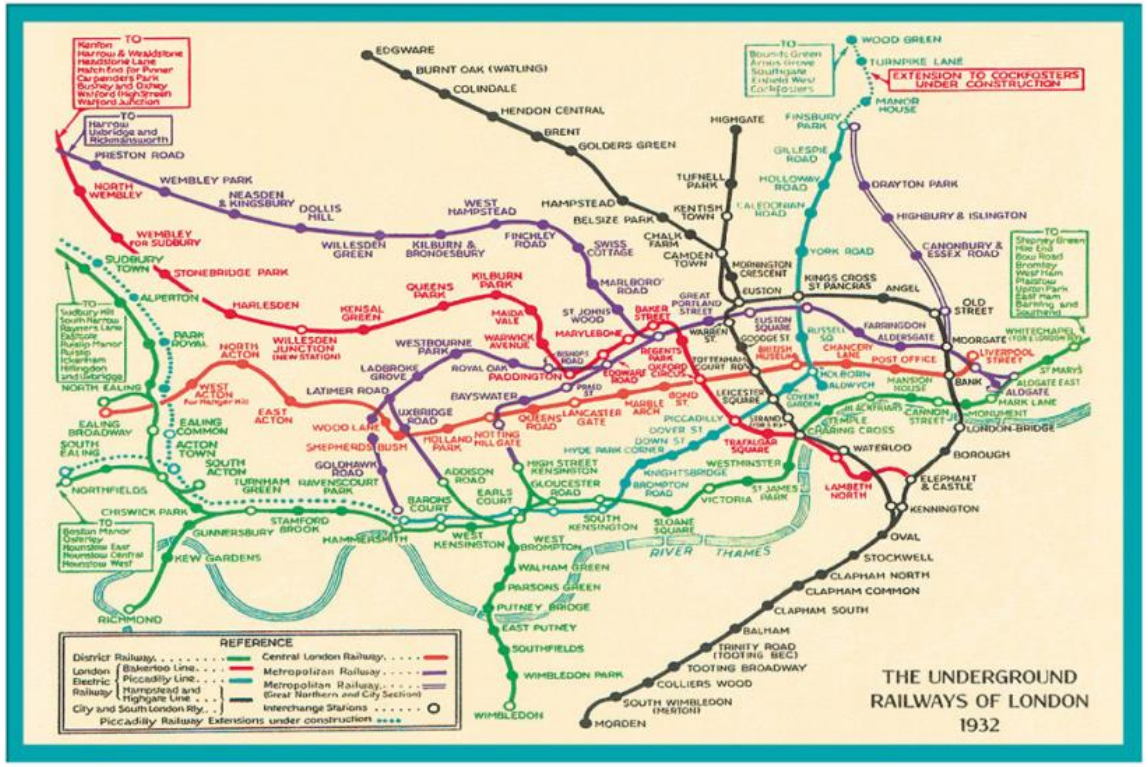


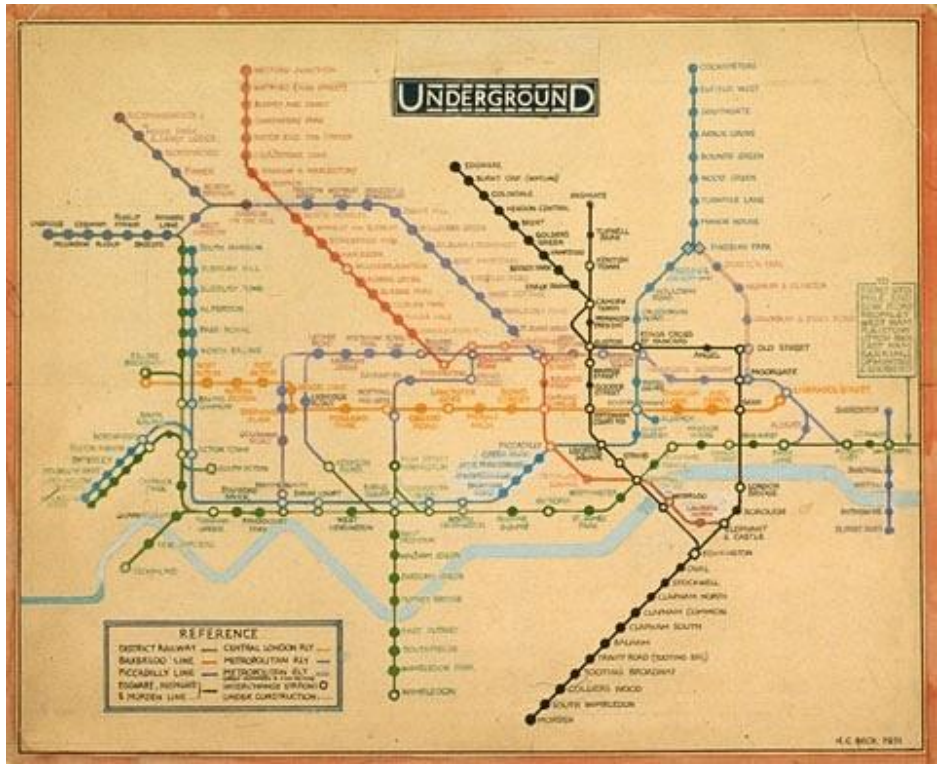
1933 Henry C. Beck: London Underground map



Re-design of the routes of the London underground rail system to favor usability

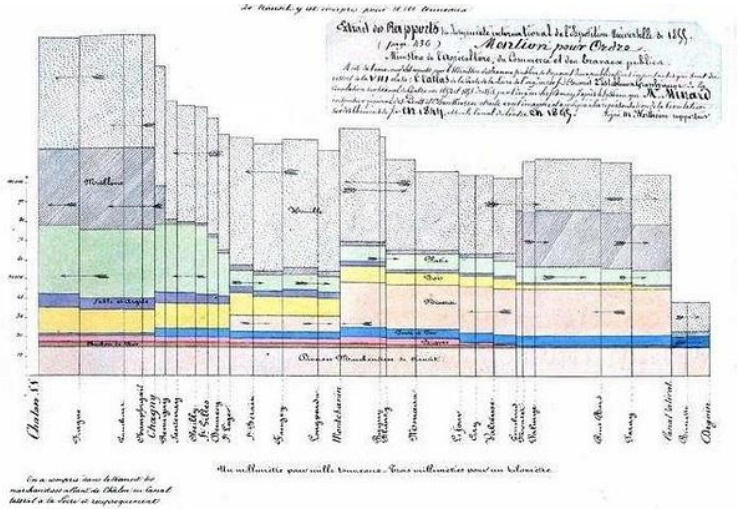
Beck, an engineering draughtsman, designed the map like an electrical circuit board, using only vertical, horizontal and 45 degree angled lines. He located stations according to available space. The resulting map was geographically inaccurate, but easier to use to determine how to get from point A to B. Beck's idea was soon copied by most subway (and bus) companies around the world.

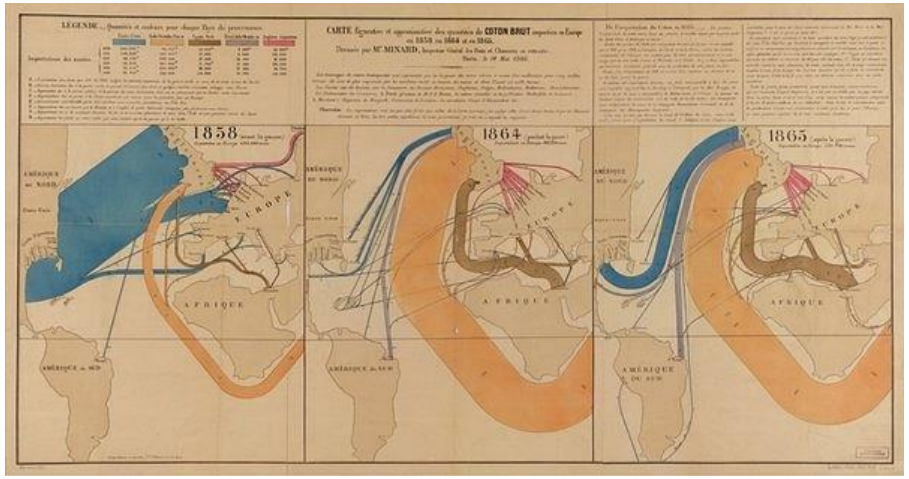


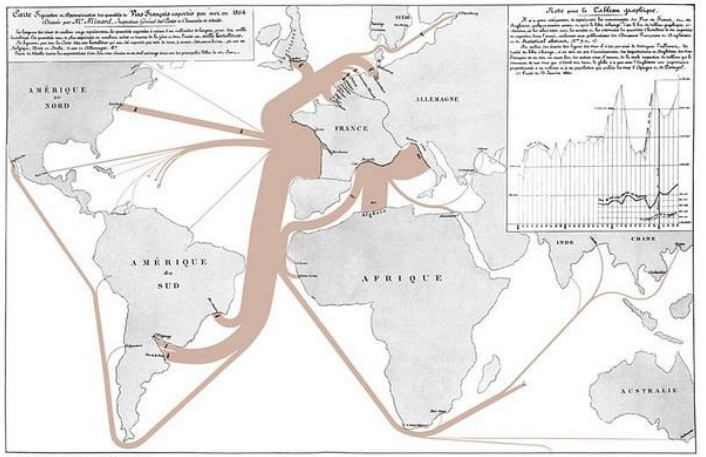


More Beautiful Charts

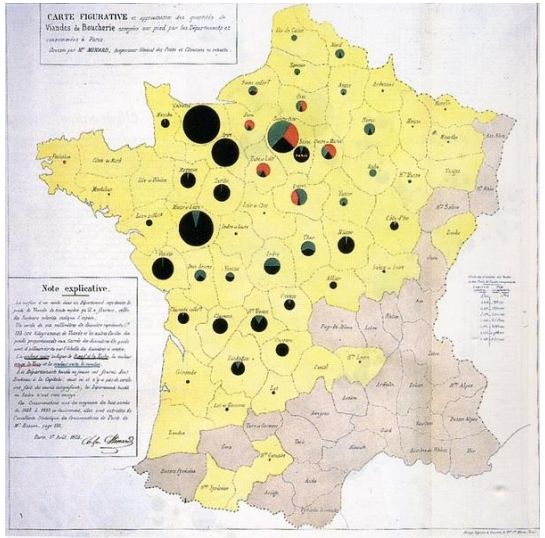
Source: https://www.datavis.ca/milestones/

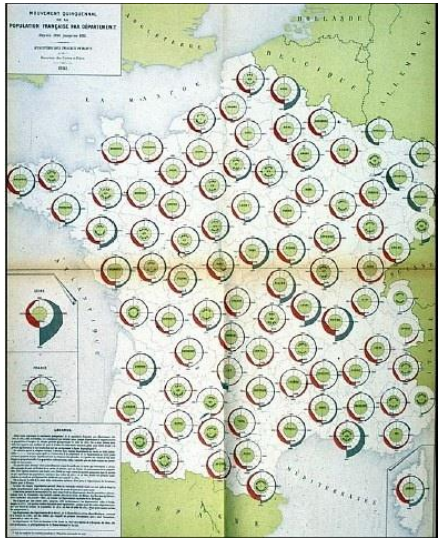


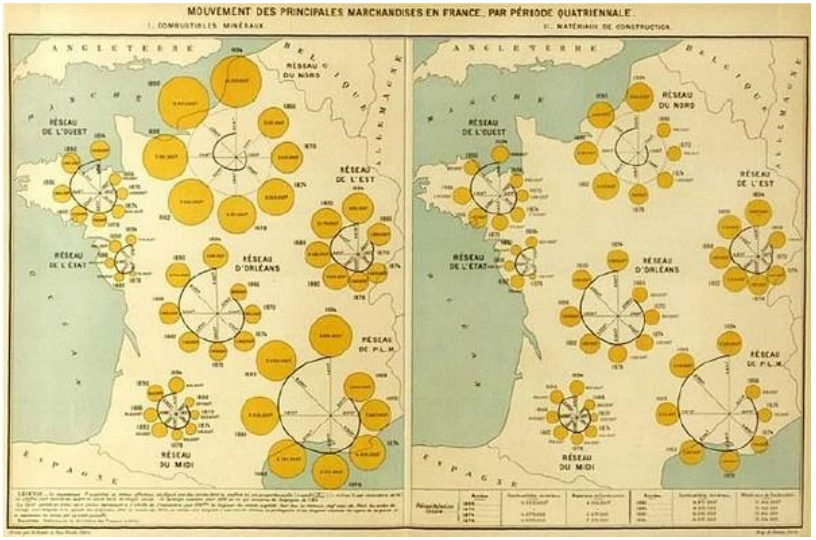


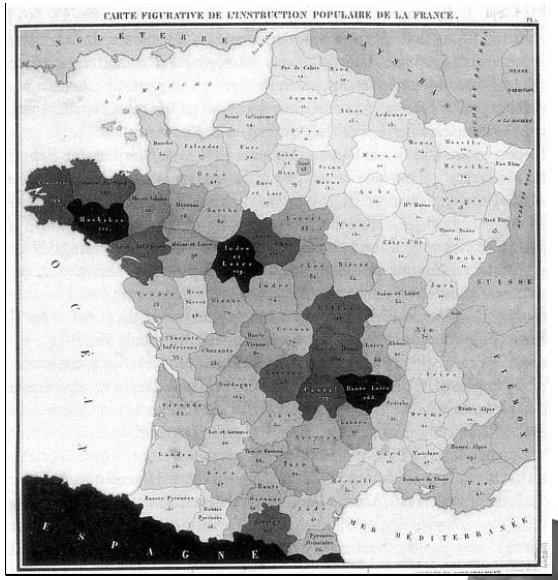


Charles Joseph Minard, *Tableaux Graphiques et Cartes Figuratives de M. Minard*, 1845-1869, a portfolio of his work held by the Bibliothèque de l'École Nationale des Ponts et Chaussées, Paris.









2006 Hadley Wickham: Computational graphics language: ggplot2

An influential, open source implementation of the Grammar of Graphics from Wilkinson (1999) in R, together with other computational tools to make it easier to produce beautiful statistical diagrams

Around this time, software for producing statistical graphs progressed from limited point-and-click interfaces and low-level graphics languages to higher-level computational languages for specifying a graph. This item recognizes the work of Hadley Wickham in the ggplot2, plyr and other R packages, but there are a number of other important contributors to this topic.



Hadley Wickha

Twitter GitHub Email

https://hadley.nz/

Hi! I'm Hadley Wickham, Chief Scientist at Posit, where (among other things) I lead the tidyverse team. I build tools (computational and cognitive) that make data science easier, faster, and more fun. I'm from New Zealand but I currently live in Houston, TX with my husband and dogs.

If you'd like to learn more about what I do, and how to use R effectively, you might enjoy one of my books (which are all free to read on the web):

- R for Data Science, with Mine Çetinkaya-Rundel and Garrett Grolemund, is the place to start if you want to learn how to do data science with R.
- ggplot2: elegant graphics for data analysis shows you how to use ggplot2 to create graphics that help you understand your data.
- Advanced R helps you master R as a programming language, teaching you what makes R tick.
- R packages, with Jenny Bryan, teaches good software engineering practices for R, using packages for bundling, documenting, and testing your code.

Outside of work, I love to bake and make cocktails. My sister Charlotte also works for Posit as a Developer Educator.



2006 Hadley Wickham: Computational graphics language: ggplot2

ggplot2: Elegant Graphics for Data Analysis (3e)

Welcome
Preface to the third edition
Preface to the second edition
Getting Started
1 Introduction
2 First steps
Layers
3 Individual geoms
4 Collective geoms
5 Statistical summaries
6 Maps
7 Networks
8 Annotations
9 Arranging plots
Scales

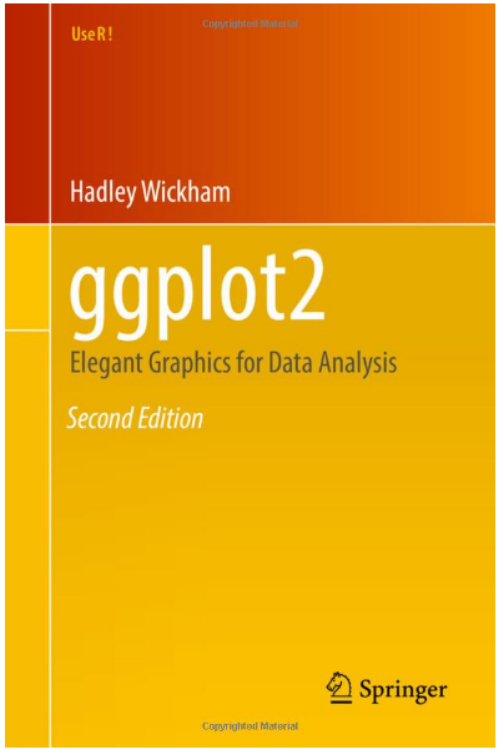
ggplot2: Elegant Graphics for Data Analysis (3e)

Welcome

This is the on-line version of work-in-progress **3rd edition** of "ggplot2: elegant graphics for data analysis" published by Springer. You can learn what's changed from the 2nd edition in the Preface.

While this book gives some details on the basics of ggplot2, its primary focus is explaining the Grammar of Graphics that ggplot2 uses, and describing the full details. It is not a cookbook, and won't necessarily help you create any specific graphic that you need. But it will help you understand the details of the underlying theory, giving you the power to tailor any plot specifically to your needs.

The book is written by Hadley Wickham, Danielle Navarro, and Thomas Lin Pedersen.



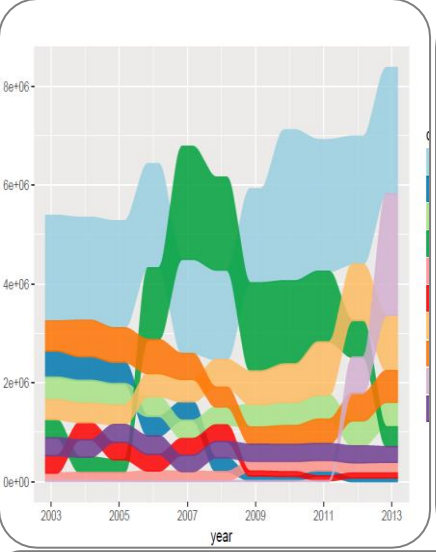
Hadley Wickham: the international 2019 COPSS Presidents' Award

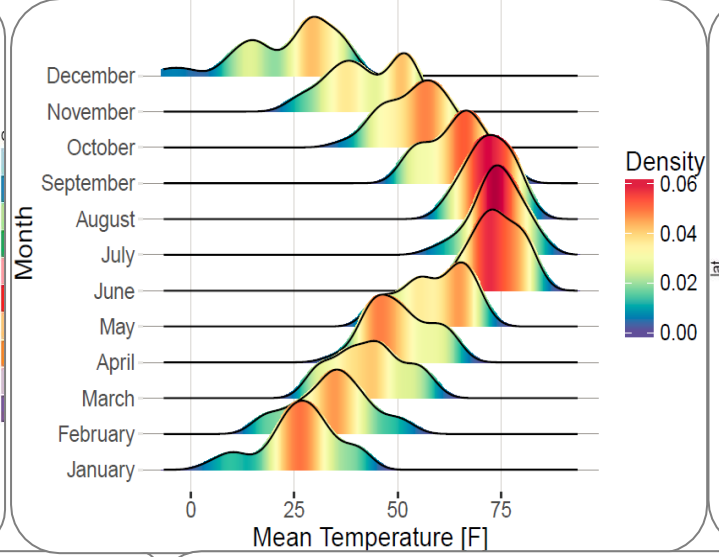


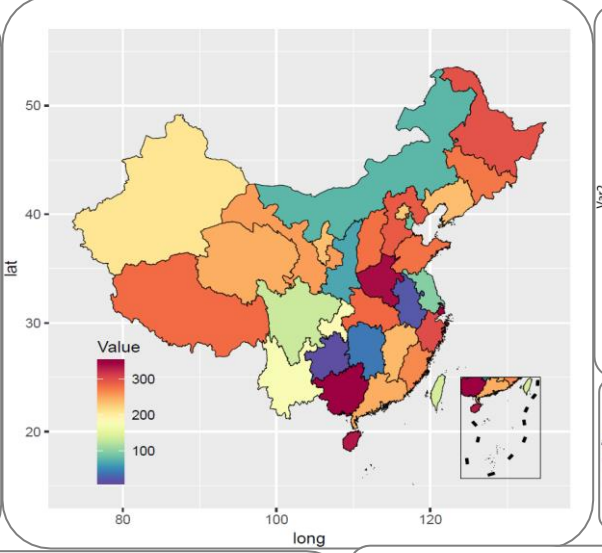
Hadley Wickham, a statistician from Hamilton, has won the international 2019 COPSS Presidents' Award.
Previously the award has primarily recognised highly theoretical contributions to statistics. This year is the first time it has been awarded for practical application.

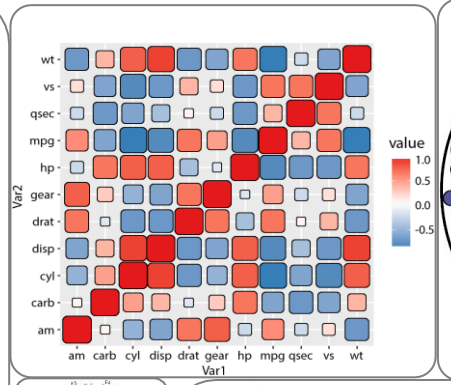
The award citation recognised Wickham's "**influential work in statistical computing, visualisation, graphics, and data analysis**" including "making statistical thinking and computing accessible to a large audience".

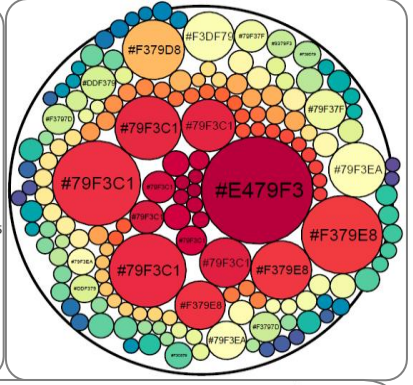
The COPSS Presidents' Award is generally regarded as one of the highest honours in the field of statistics, along with the International Prize in Statistics.

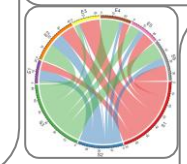

year

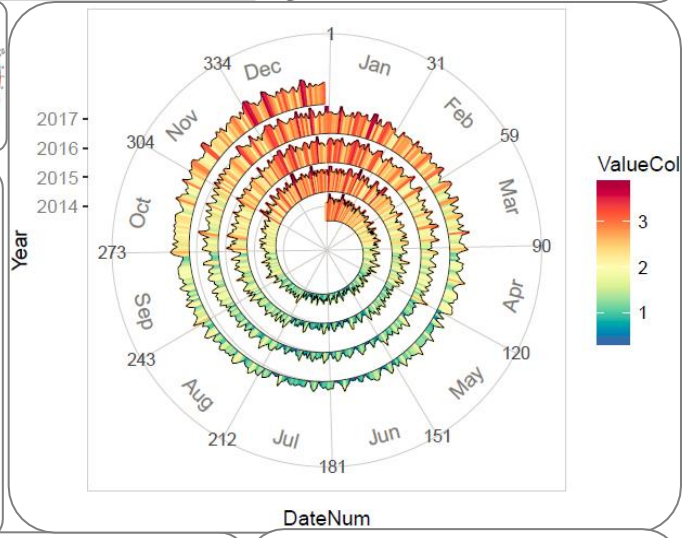

December
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Mean Temperature [F]

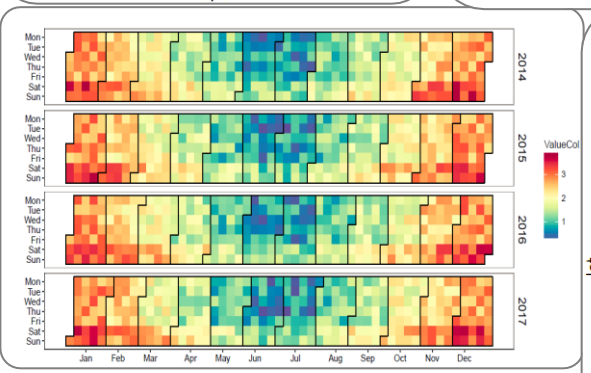

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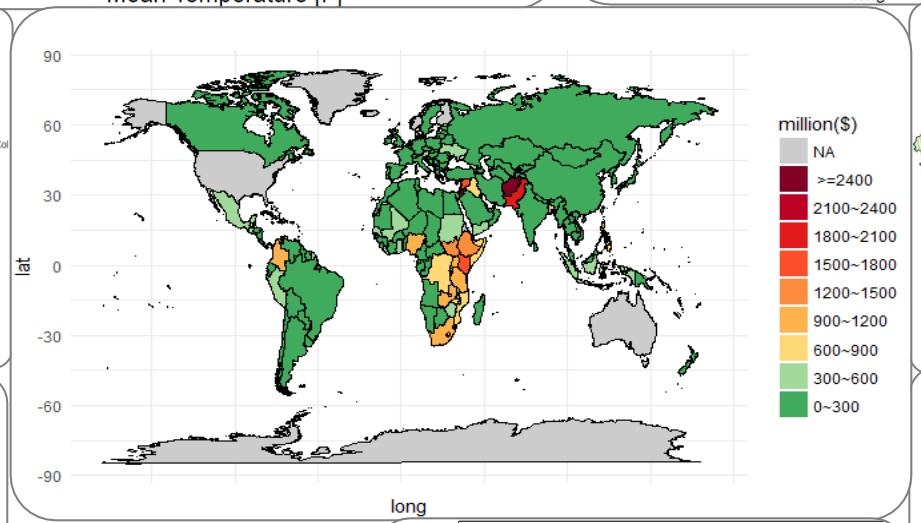

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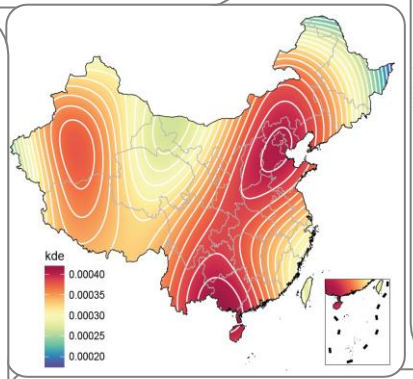

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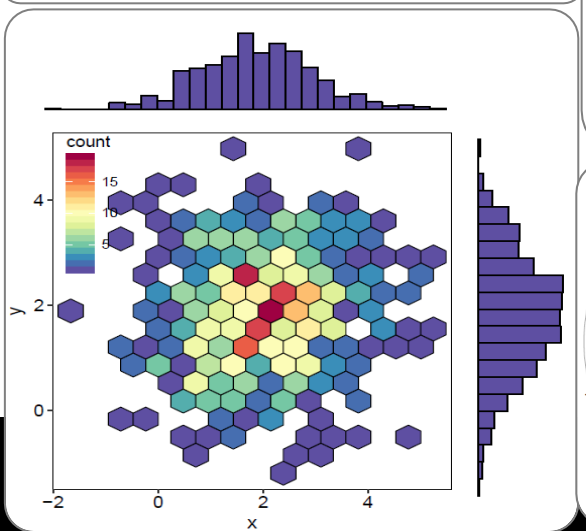


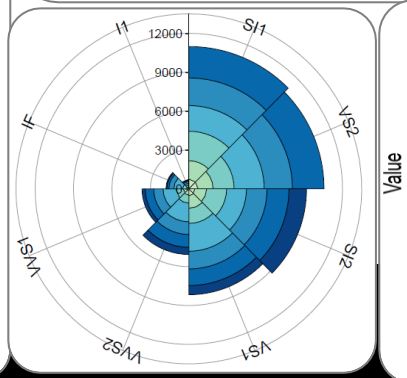

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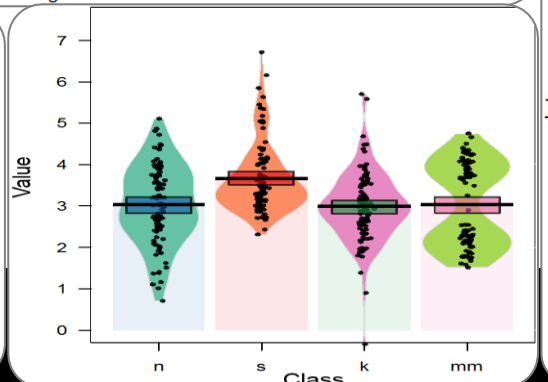

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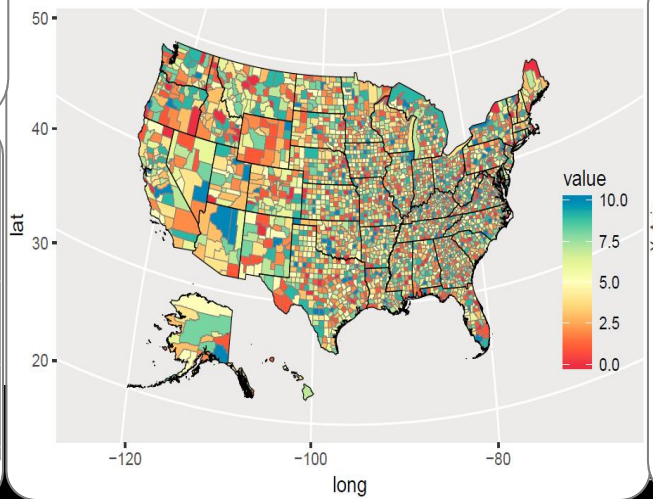

million($)
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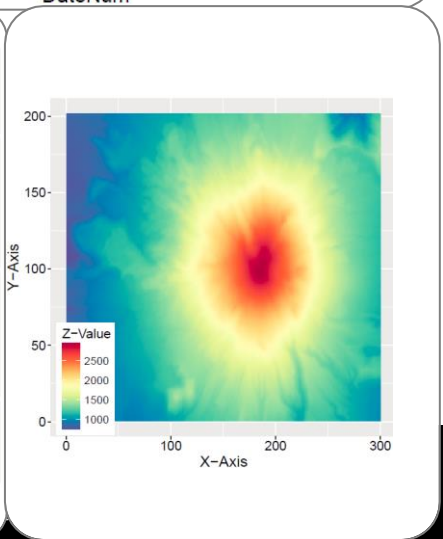

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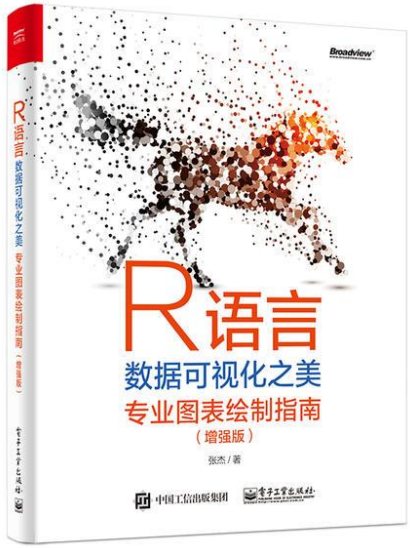

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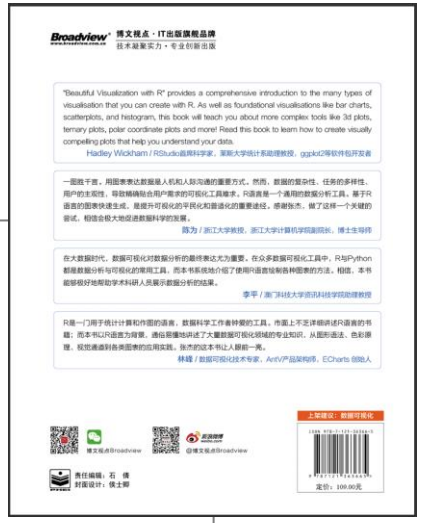

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Z-Value
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Hadley Wickham: Recommendation for Beautiful Visualization with R



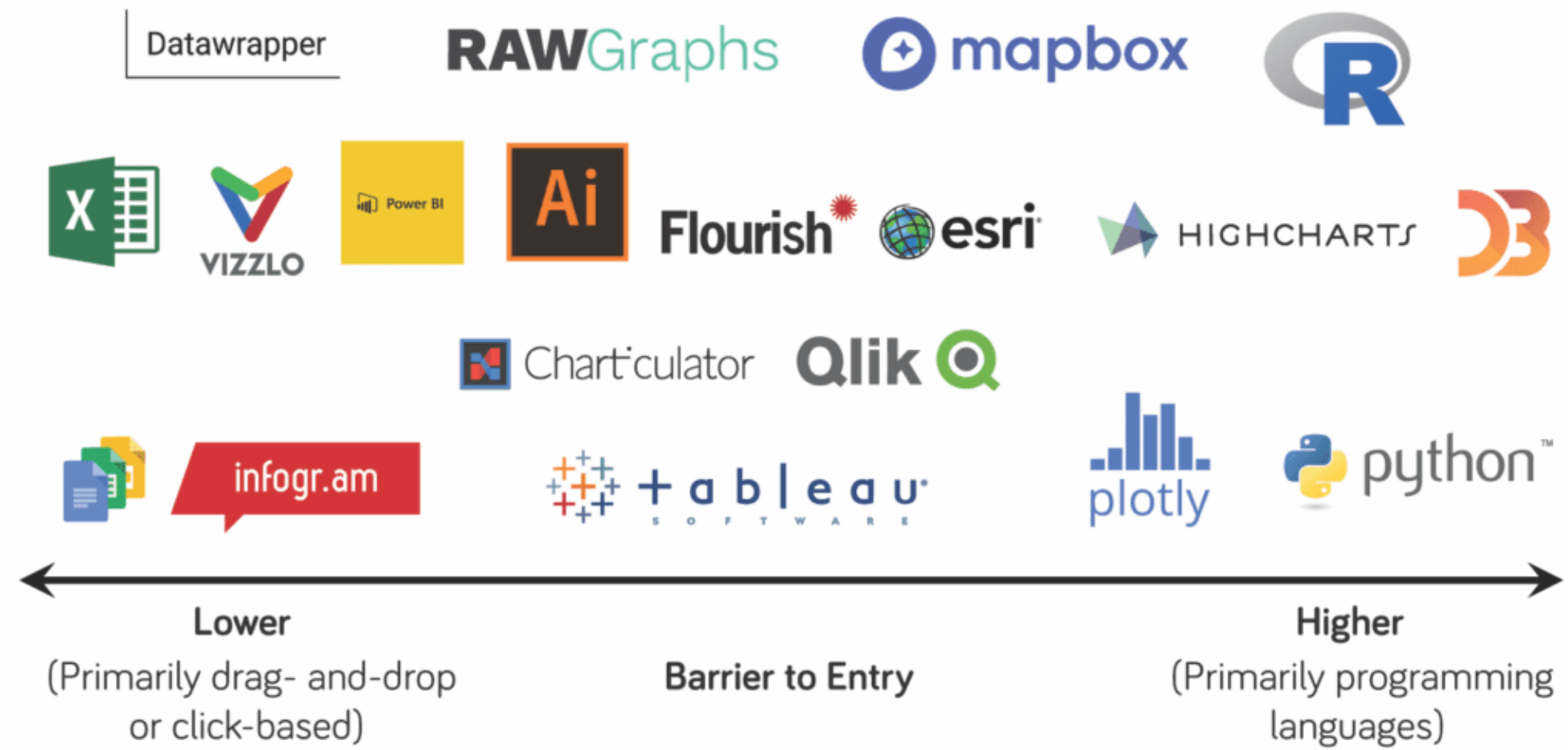


Hadley Wickham <hadley@rstudio.com>
週六 11/5/2019 6:45
收件者: ZHANG, Peter J [ITC] <peter.j.zhang@polyu.edu.hk>
How does this sound?

(BTW I'd suggest using "Beautiful Visualization with R" as the English translation of the title, in case it comes up elsewhere.)

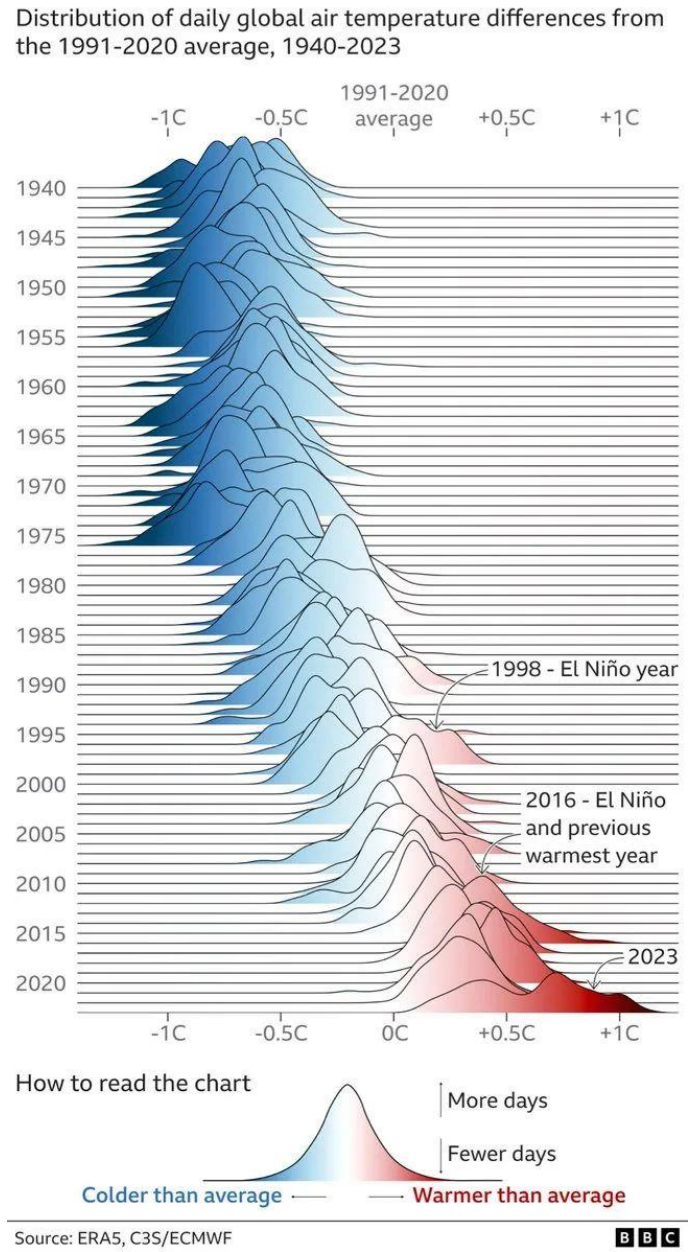
"Beautiful Visualization with R" provides a comprehensive introduction to the many types of visualisation that you can create with R. As well as foundational visualisations like bar charts, scatterplots, and histogram, this book will teach you about more complex tools like 3d plots, ternary plots, polar coordinate plots and more! Read this book to learn how to create visually compelling plots that help you understand your data.

2000s: Data Visualizations

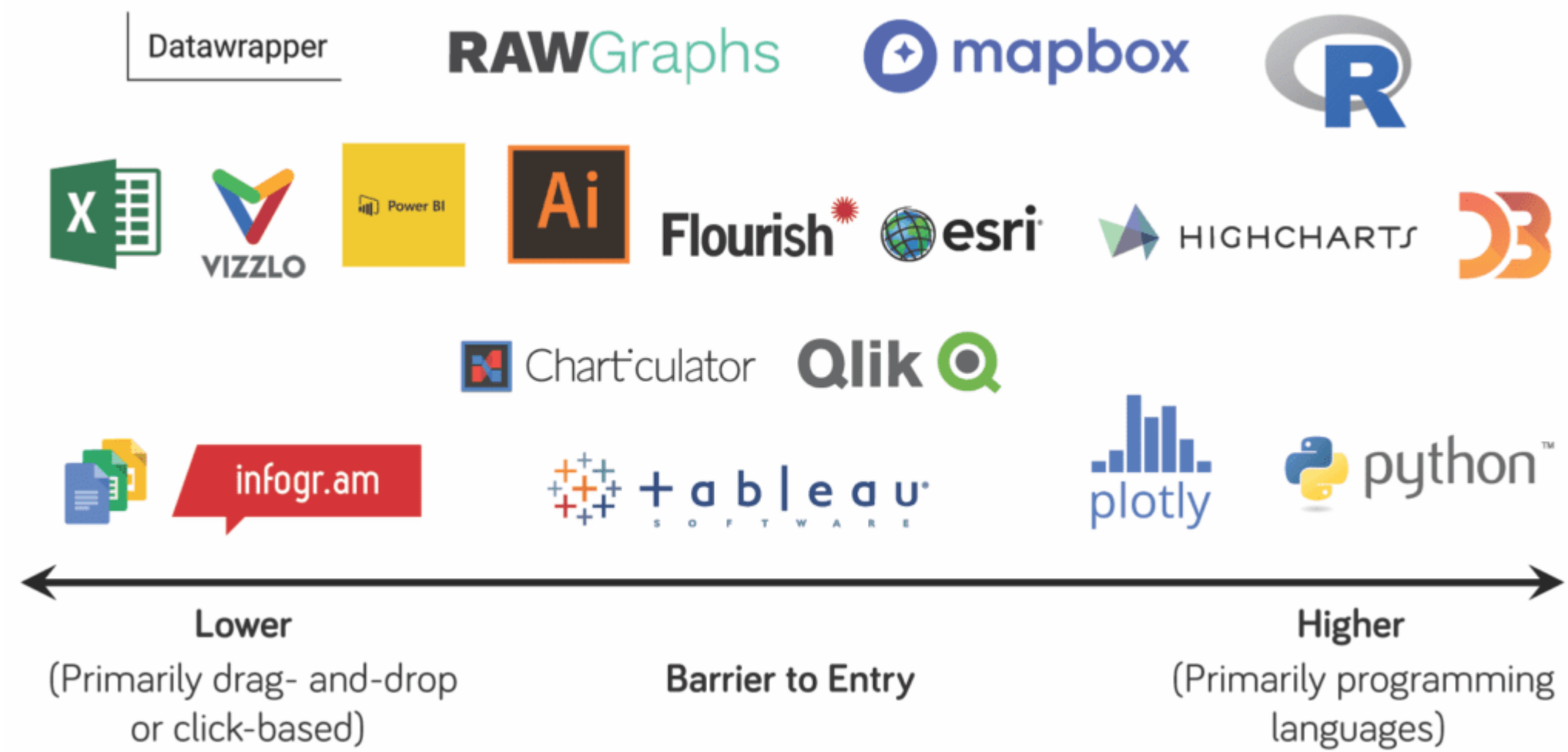


Source: https://policyviz.com/2022/02/01/the-data-visualization-tools-wars/

Tools

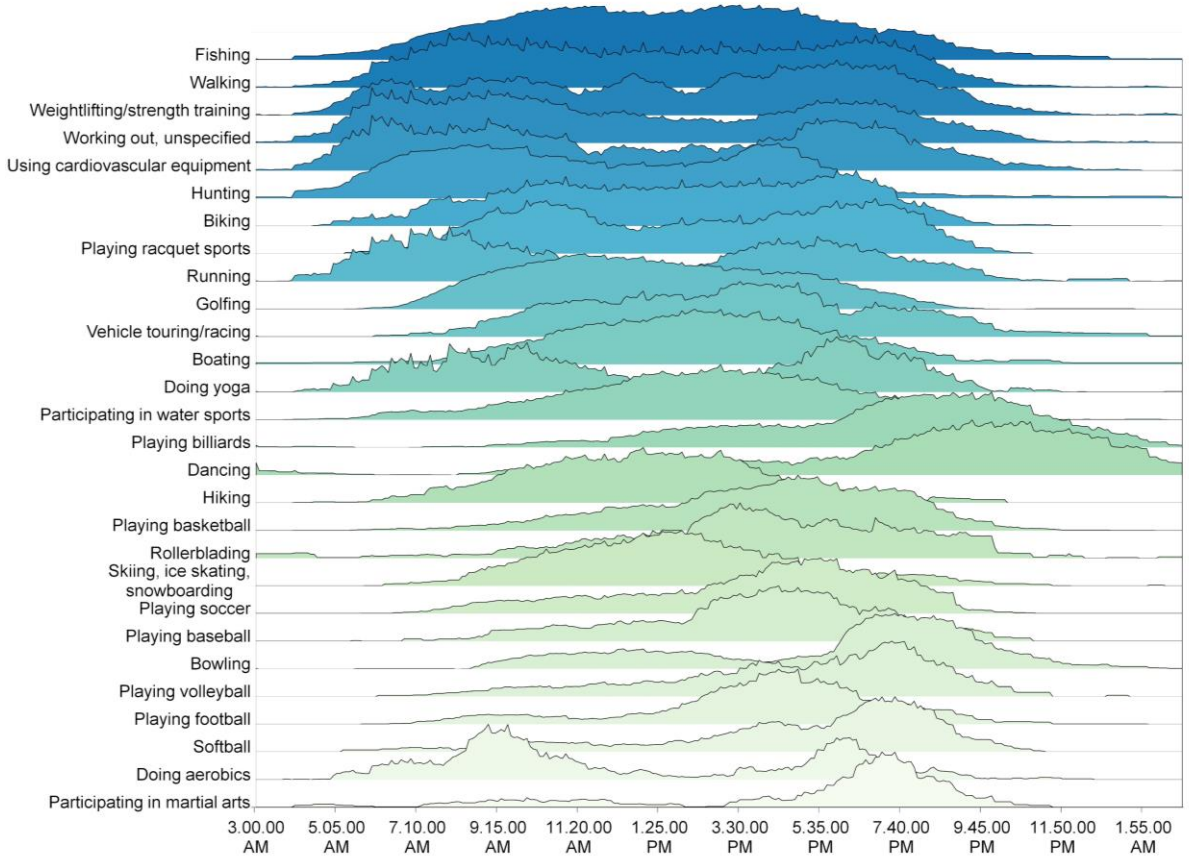

Distribution of daily global air temperature differences from the 1991-2020 average, 1940-2023
1991-2020 average
-1C
-0.5C
+0.5C
+1C
1940
1945
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
1998 - El Niño year
2016 - El Niño and previous warmest year
2023
-1C
-0.5C
0C
+0.5C
+1C
How to read the chart
More days
Fewer days
Colder than average
Warmer than average
Source: ERA5, C3S/ECMWF
BBC

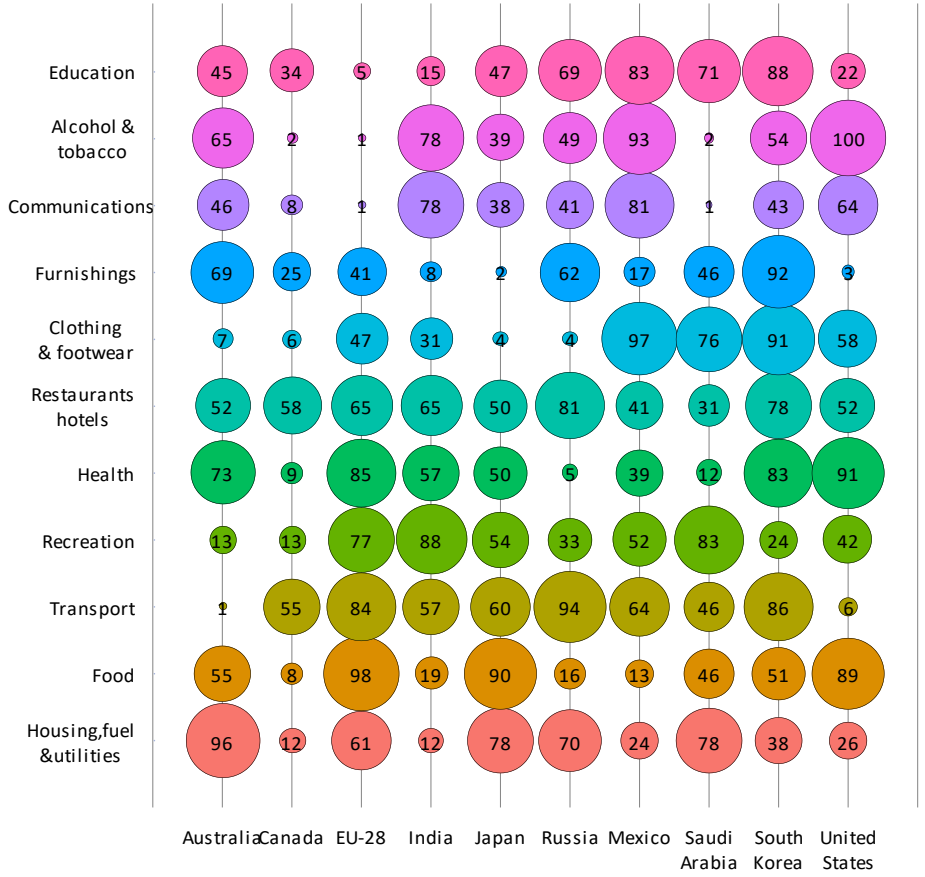
Data Visualization Tools



Excel

Excel is indeed a versatile tool for data visualization, particularly for simpler visualizations and quick exploratory data analysis.





Excel

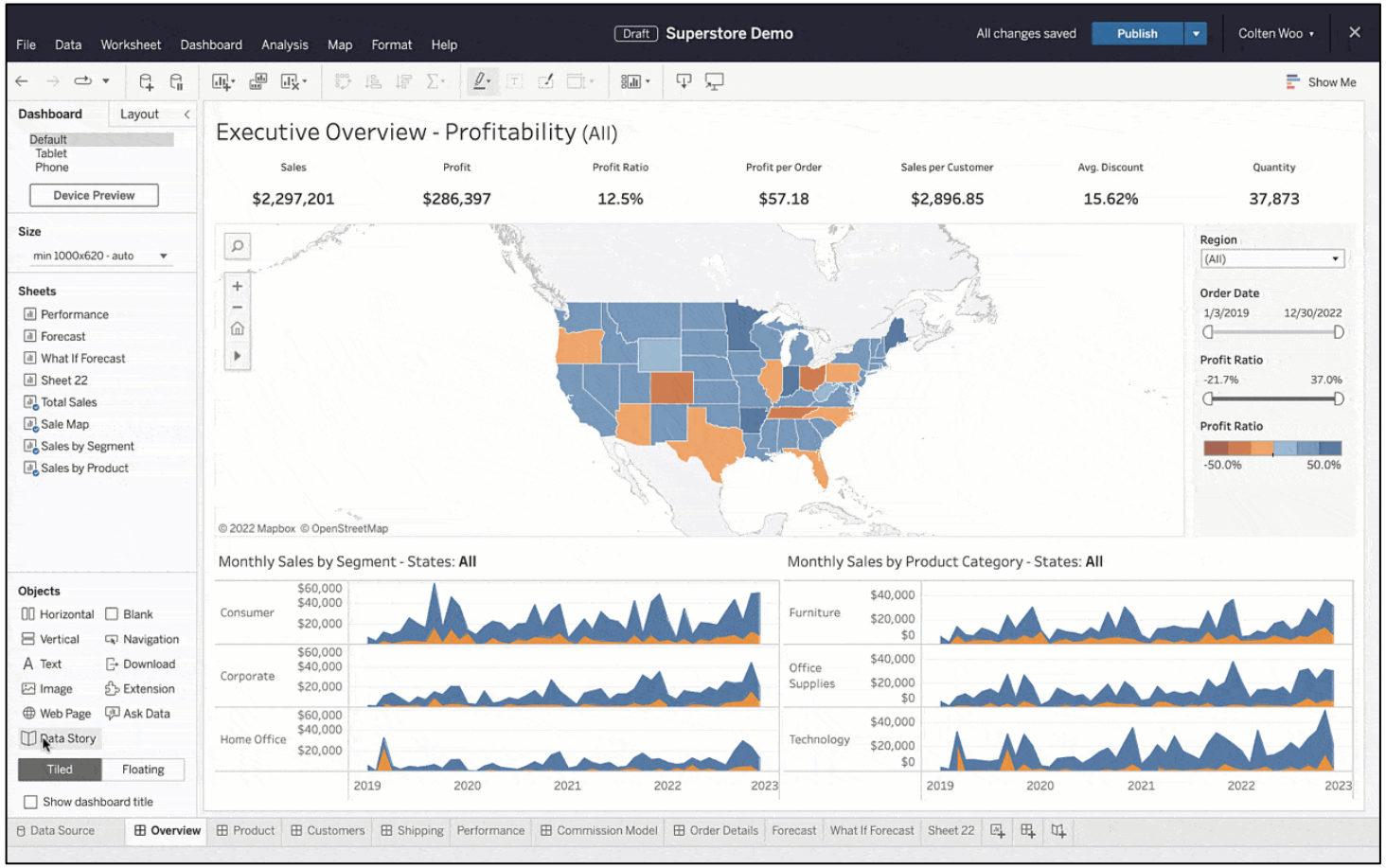
Excel is indeed a versatile tool for data visualization, particularly for simpler visualizations and quick exploratory data analysis.

- **User-Friendly Interface**: Excel provides a user-friendly interface that allows users with varying levels of technical expertise to create visualizations easily.
- **Quick and Easy Chart Creation**: Excel offers a wide range of chart types that can be created with just a few clicks. Users can select their data and create a chart almost instantly.
- **Familiarity**: Many professionals are already familiar with Excel, making it a popular choice for data visualization tasks. This familiarity can reduce the learning curve for creating visualizations.
- **Accessibility**: Excel is widely available and often included in standard office software suites, making it accessible to a large number of users without additional cost.

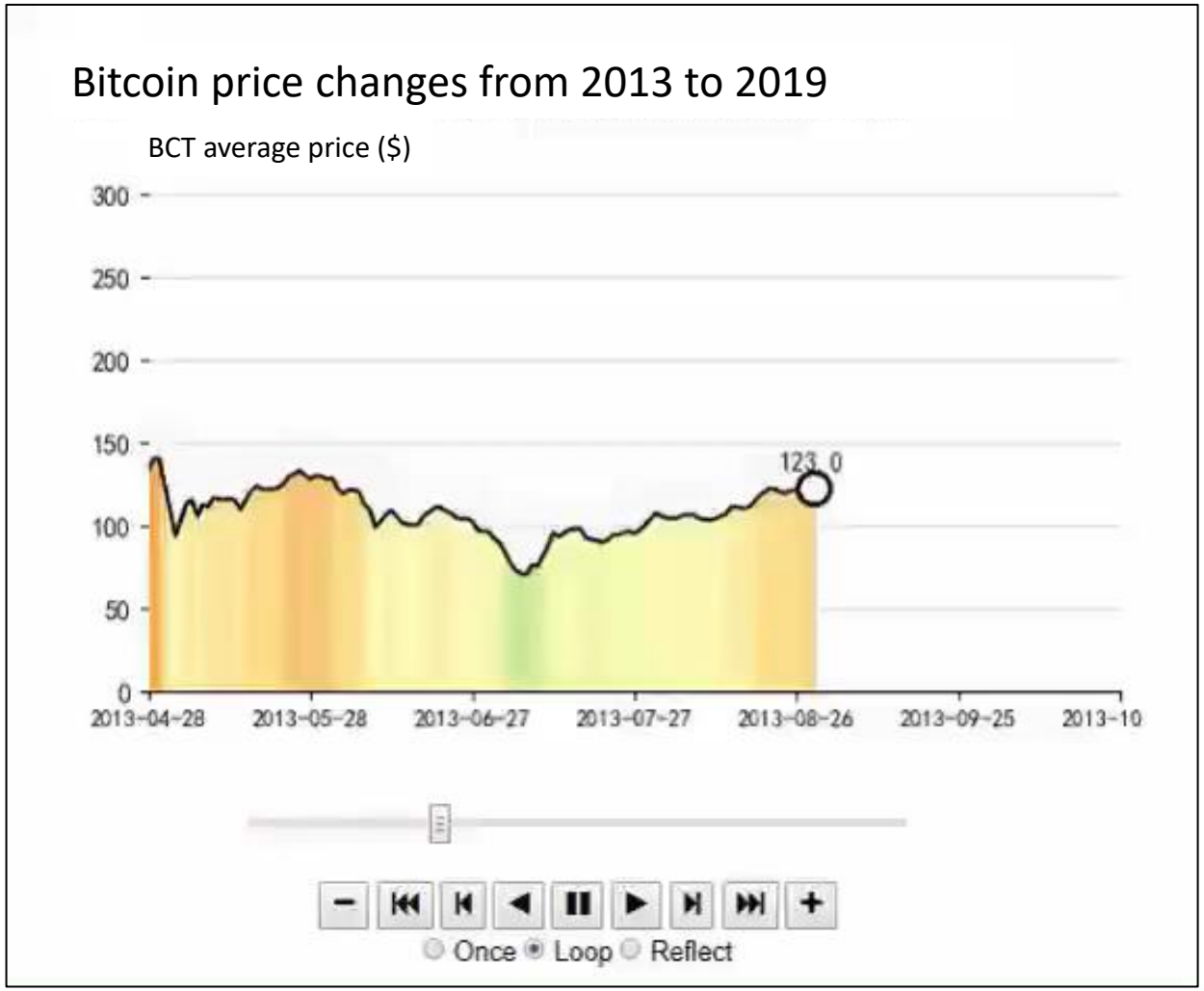


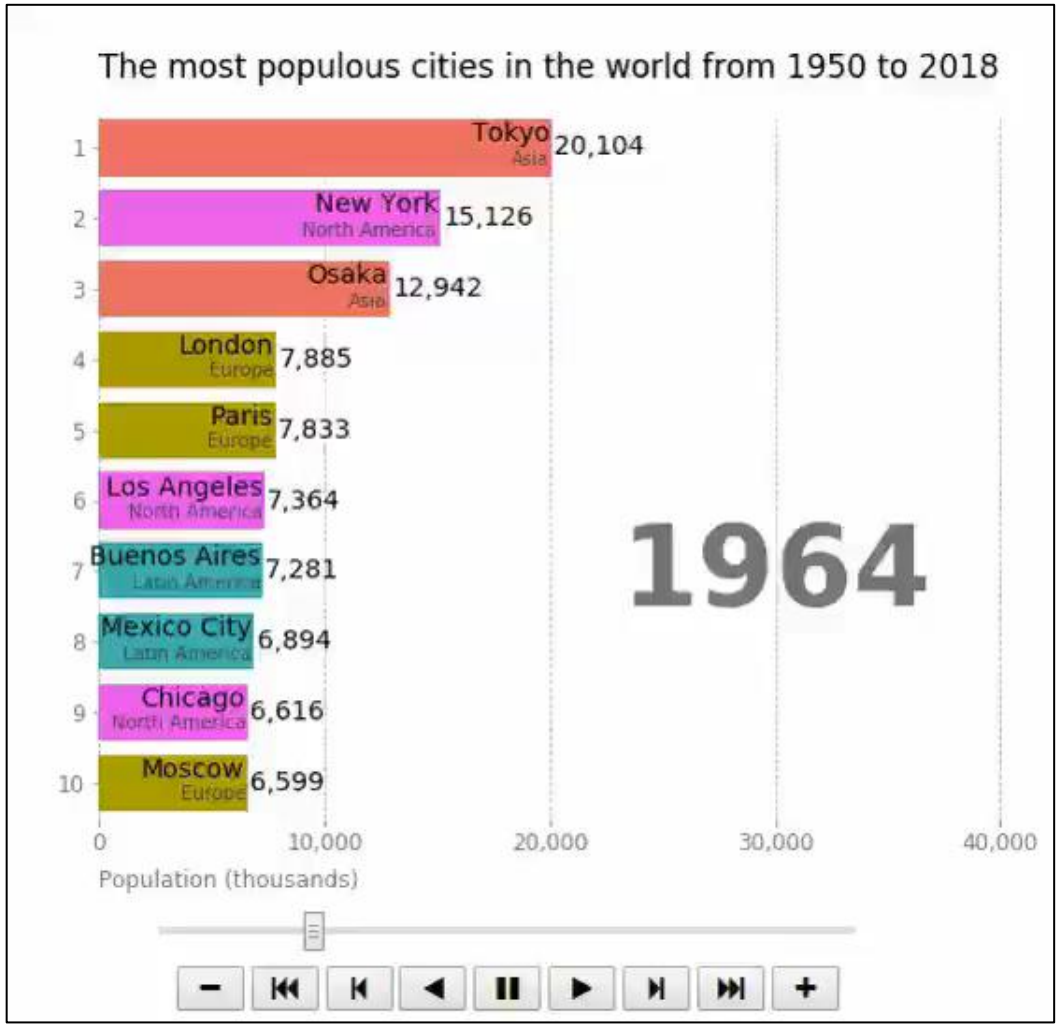
Tableau

Tableau is a widely used data visualization tool that allows users to create interactive and shareable dashboards. It enables users to connect to various data sources, visualize data in different formats, and share insights with others.



Python


Bitcoin price changes from 2013 to 2019
BCT average price ($)
300
250
200
150
100
50
0
123.0
2013-04-28
2013-05-28
2013-06-27
2013-07-27
2013-08-26
2013-09-25
2013-10
Once
Loop
Reflect


The most populous cities in the world from 1950 to 2018
1 Tokyo Asia 20,104
2 New York North America 15,126
3 Osaka Asia 12,942
4 London Europe 7,885
5 Paris Europe 7,833
6 Los Angeles North America 7,364
7 Buenos Aires Latin America 7,281
8 Mexico City Latin America 6,894
9 Chicago North America 6,616
10 Moscow Europe 6,599
1964
0
10,000
20,000
30,000
40,000
Population (thousands)

Python

Python's flexibility, ease of use, extensive library support, and interactivity make it a powerful choice for data visualization tasks, whether for exploratory data analysis, dashboards, reports, or presentations.

matplotlib Version 3.1.1
A Python 2D plotting library

Seaborn
Statistical data visualization

plotnine
A Grammar of Graphics for Python

Cartopy
Geographic information visualization library

Bokeh
An interactive visualization library

folium
Geographic information visualization library-leaflet.js

Pandas' builtin-plotting
Powerful Python data analysis toolkit

plotly
Open Source Graphing Libraries

pyecharts
A Python Echarts Plotting Library



Python

Python's flexibility, ease of use, extensive library support, and interactivity make it a powerful choice for data visualization tasks, whether for exploratory data analysis, dashboards, reports, or presentations.

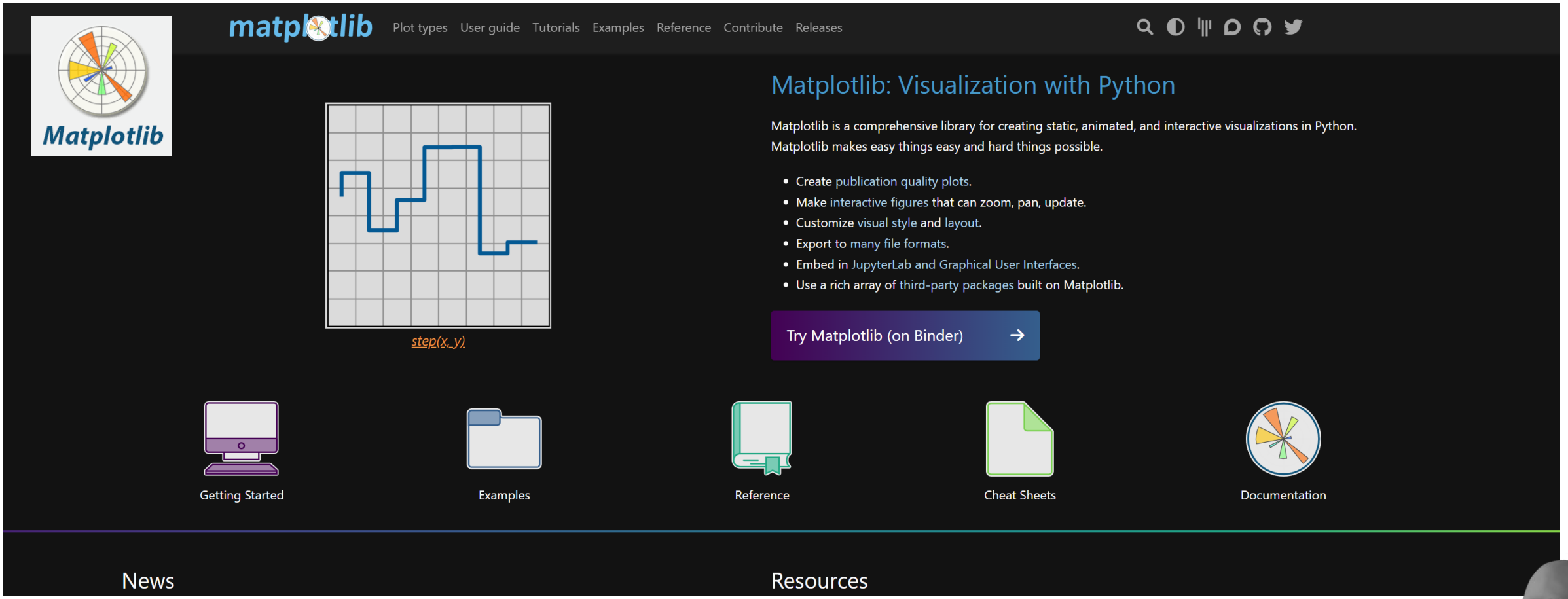
- **Rich Ecosystem of Libraries**: Python offers a vast array of data visualization libraries such as Matplotlib, Seaborn, Plotly, Bokeh, and Altair, each catering to different needs and preferences. These libraries provide diverse options for creating various types of visualizations.
- **Ease of Use**: Python's syntax is straightforward and easy to understand, making it accessible for beginners and experienced programmers alike. This simplicity accelerates the process of creating visualizations.
- **Interactivity**: Libraries like Plotly and Bokeh enable the creation of interactive visualizations, allowing users to explore data dynamically and gain deeper insights.
- **Integration with Data Analysis Tools**: Python seamlessly integrates with popular data analysis libraries like Pandas and NumPy. This integration streamlines the process of data manipulation and visualization.
- **Community Support**: Python has a large and active community of users and developers. This community contributes to the development of new visualization tools, provides support, and shares resources and best practices.
- **Versatility**: Python is a versatile language that can be used for a wide range of tasks beyond data visualization, such as data analysis, machine learning, web development, and more. This versatility makes it a valuable tool in various fields.
- **Customization**: Python visualization libraries offer a high degree of customization. Users can adjust colors, styles, labels, and other elements to create visualizations that suit their specific needs and preferences.



Matplotlib

https://matplotlib.org/

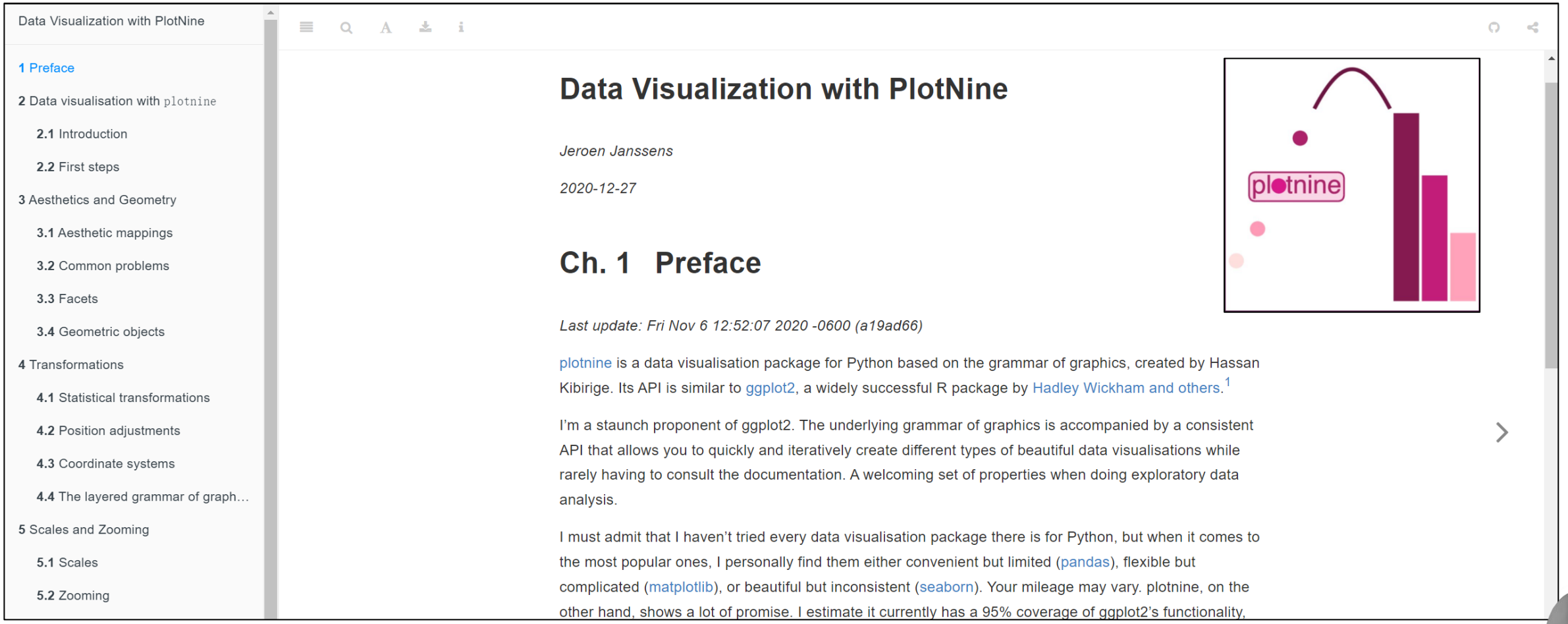
Matplotlib is a popular Python library used for creating static, interactive, and animated plots and visualizations. It is widely used for data visualization due to its flexibility and extensive capabilities.



Plotnine

https://f0nzie.github.io/rmarkdown-python-plotnine/
https://github.com/has2k1/plotnine

Plotnine is a data visualisation package for Python based on the grammar of graphics, created by Hassan Kibirige. Its API is similar to ggplot2, a widely successful R package by Hadley Wickham and others.



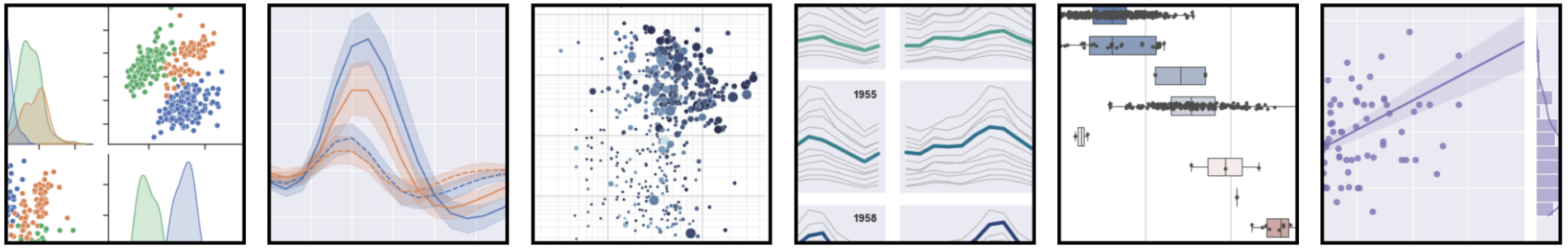
Seaborn

https://seaborn.pydata.org/

Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.

seaborn Installing Gallery Tutorial API Releases Citing FAQ

seaborn: statistical data visualization



Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.

For a brief introduction to the ideas behind the library, you can read the introductory notes or the paper. Visit the installation page to see how you can download the package and get started with it. You can browse the example gallery to see some of the things that you can do with seaborn, and then check out the tutorials or API reference to find out how.

To see the code or report a bug, please visit the GitHub repository. General support questions are most at home on stackoverflow, which has a dedicated channel for seaborn.

Contents

Installing
Gallery
Tutorial
API
Releases
Citing
FAQ

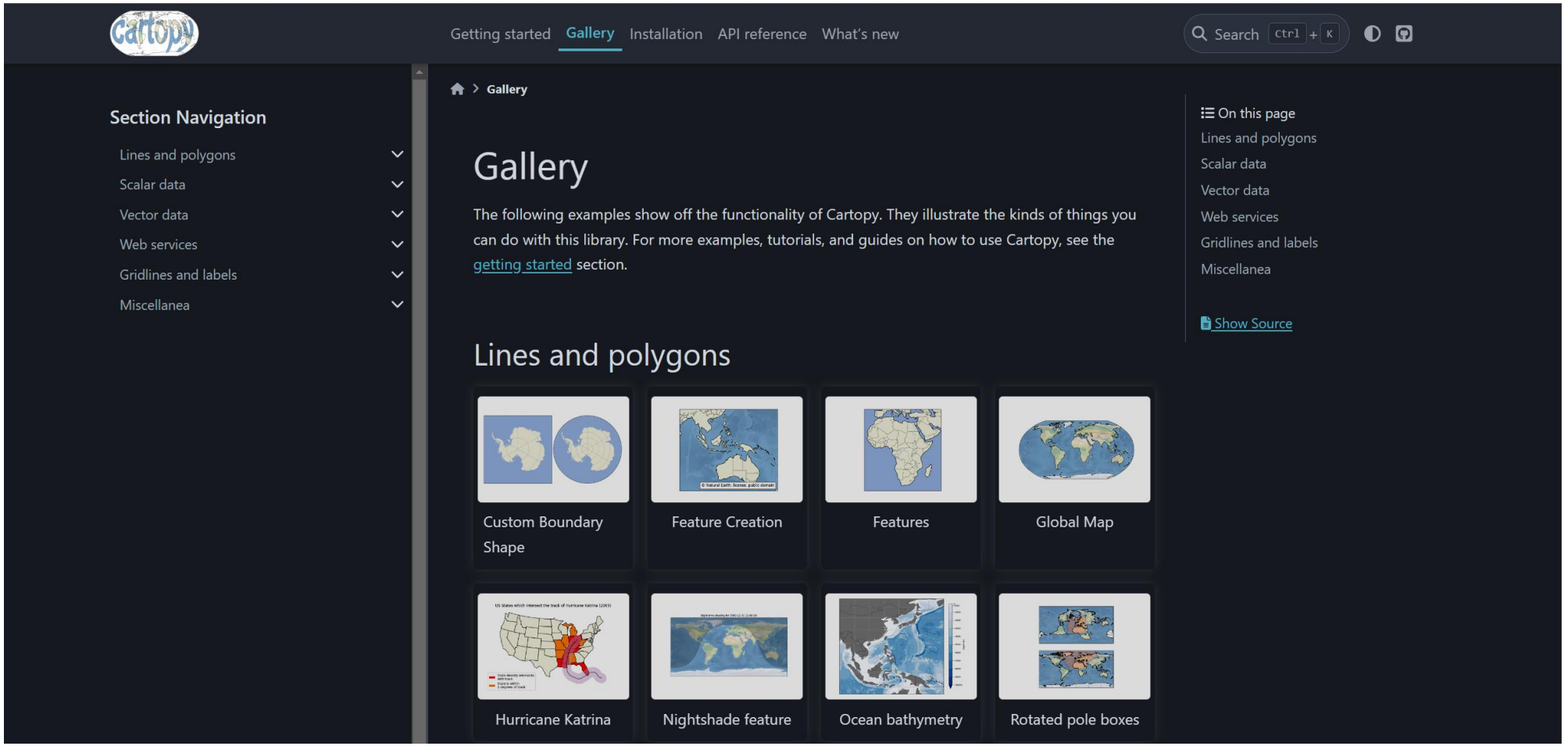
Features

- New Objects: API | Tutorial
- Relational plots: API | Tutorial
- Distribution plots: API | Tutorial
- Categorical plots: API | Tutorial
- Regression plots: API | Tutorial
- Multi-plot grids: API | Tutorial
- Figure theming: API | Tutorial
- Color palettes: API | Tutorial

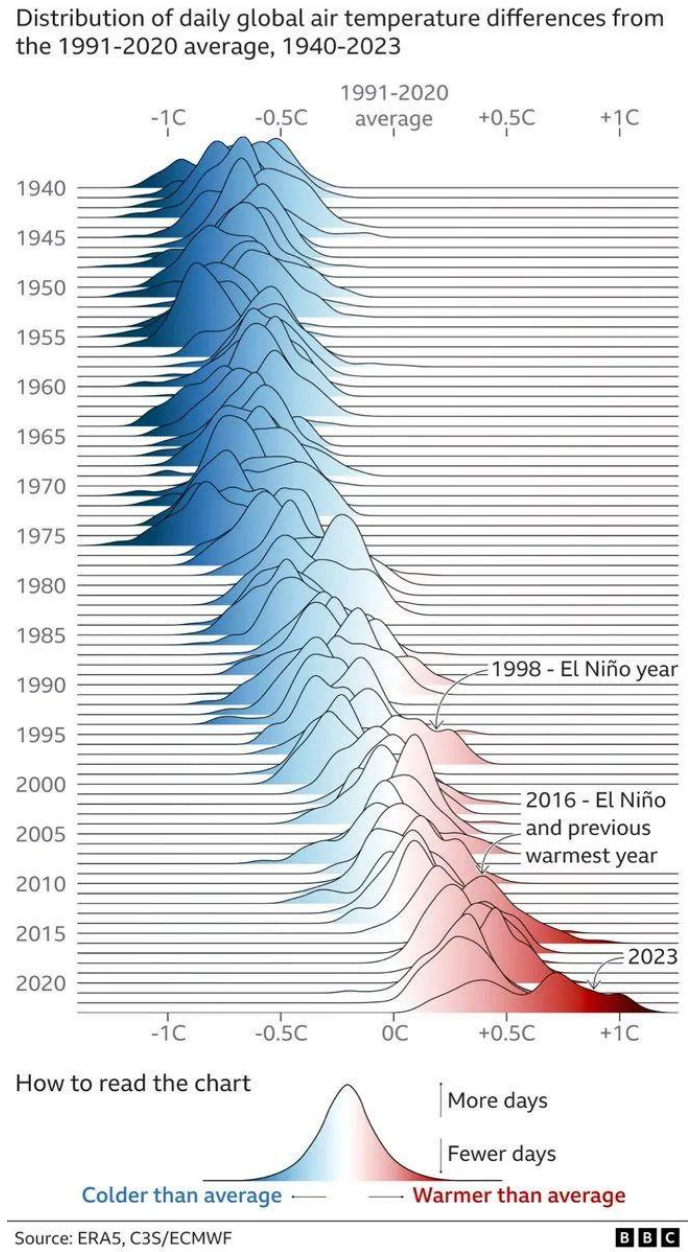
Cartopy

https://scitools.org.uk/cartopy/docs/latest/

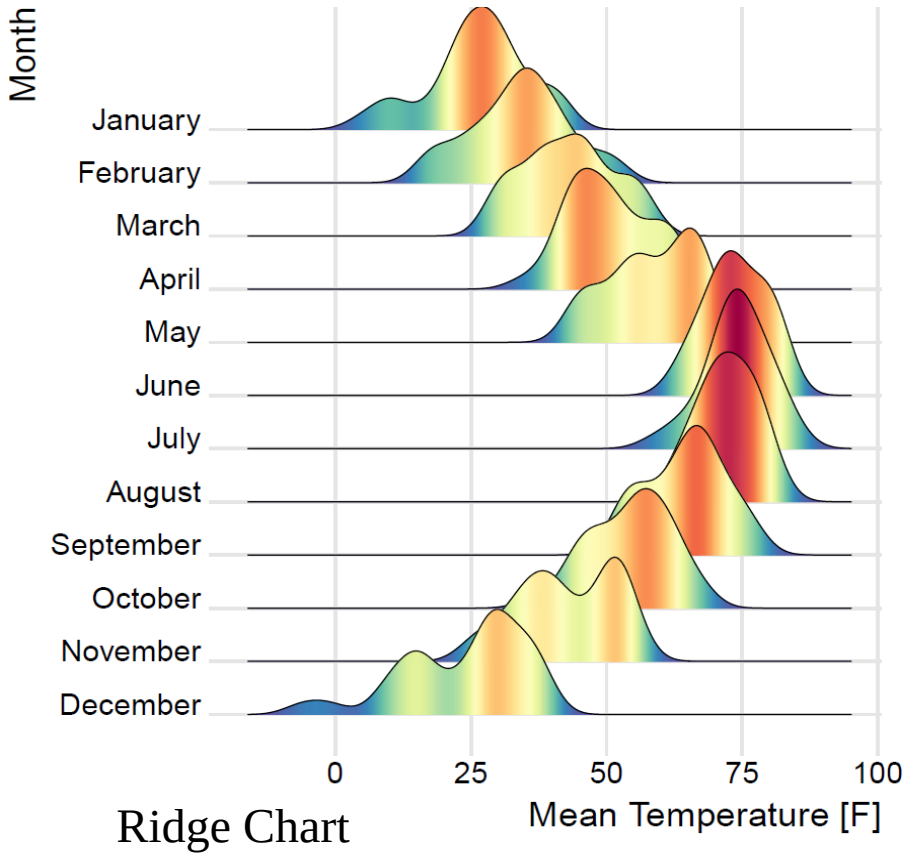
Cartopy is a Python package designed for geospatial data processing in order to produce maps and other geospatial data analyses. Cartopy makes use of the powerful PROJ, NumPy and Shapely libraries and includes a programmatic interface built on top of Matplotlib for the creation of publication quality maps.



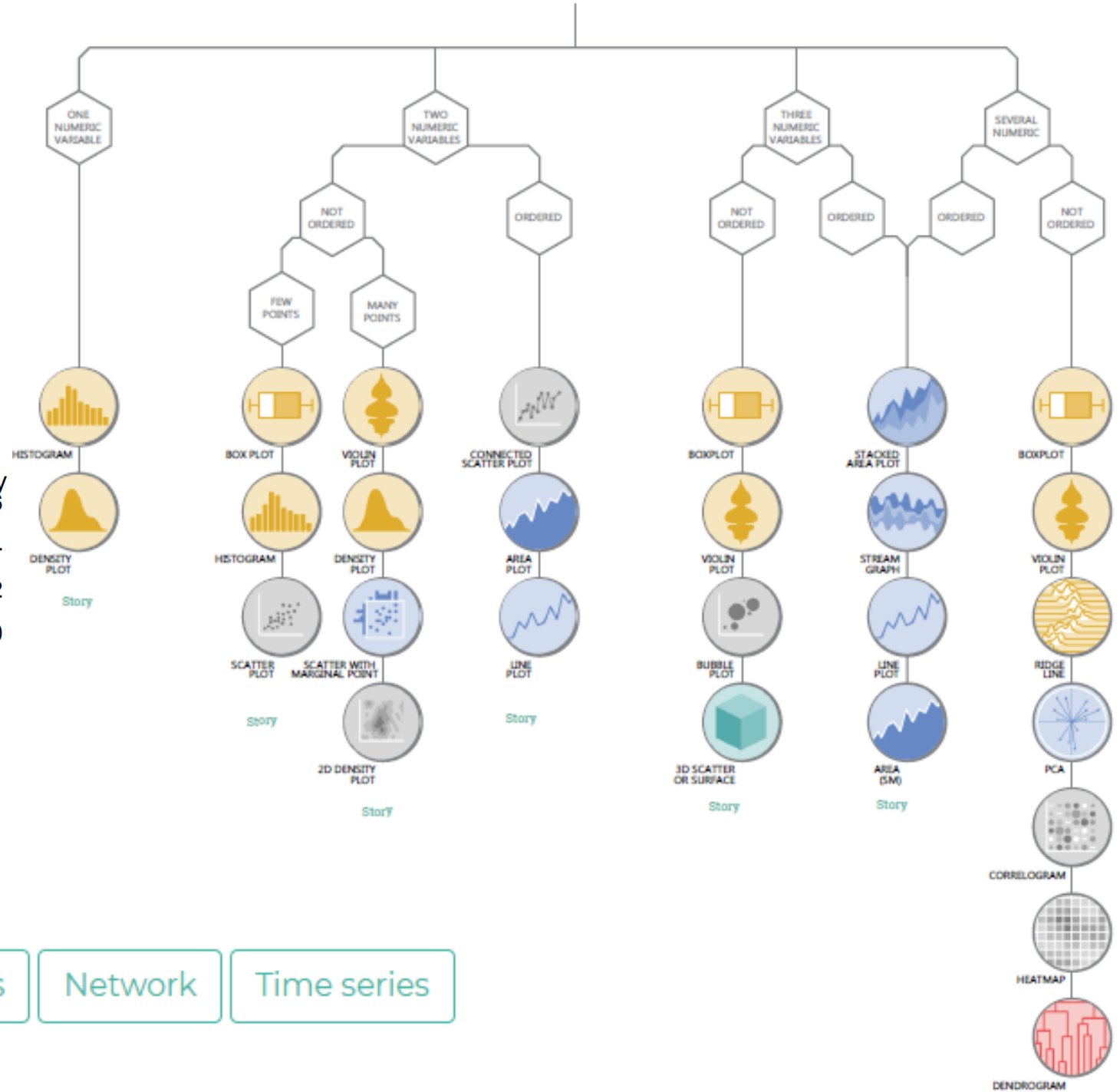
Categories


Distribution of daily global air temperature differences from the 1991-2020 average, 1940-2023
1991-2020 average
-1C
-0.5C
+0.5C
+1C
1940
1945
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
1998 - El Niño year
2016 - El Niño and previous warmest year
2023
-1C
-0.5C
0C
+0.5C
+1C
How to read the chart
More days
Fewer days
Colder than average
Warmer than average
Source: ERA5, C3S/ECMWF
BBC

from Data to Viz



Ridge Chart



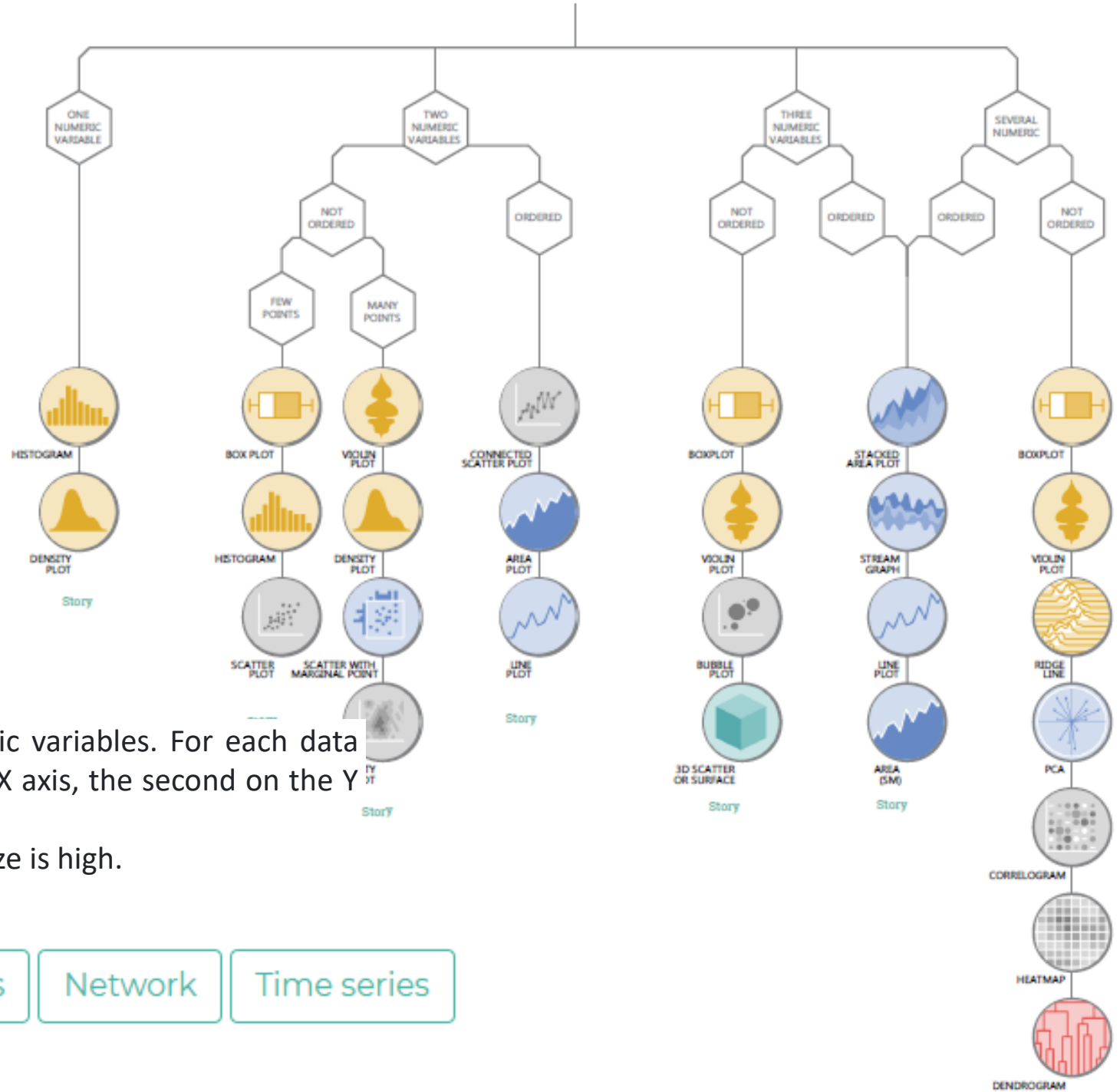
Numeric | Categoric | Num & Cat | Maps | Network | Time series

https://www.data-to-viz.com/#scatter

from Data to Viz

Scatterplot





A scatterplot displays the relationship between 2 numeric variables. For each data point, the value of its first variable is represented on the X axis, the second on the Y axis.

Overplotting is the most common mistake when sample size is high.

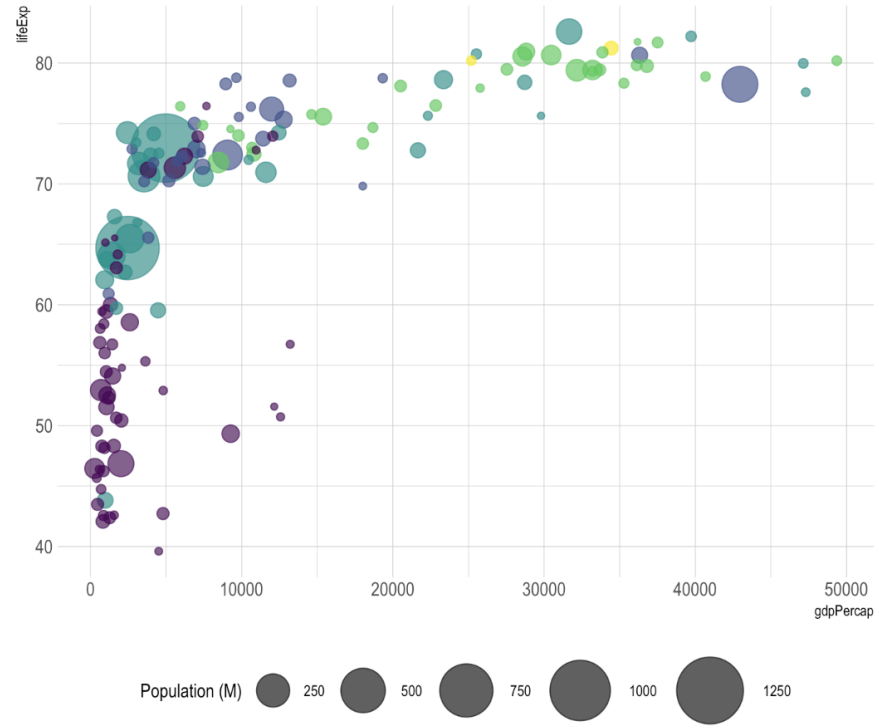
Don't forget to show subgroups if you have some.

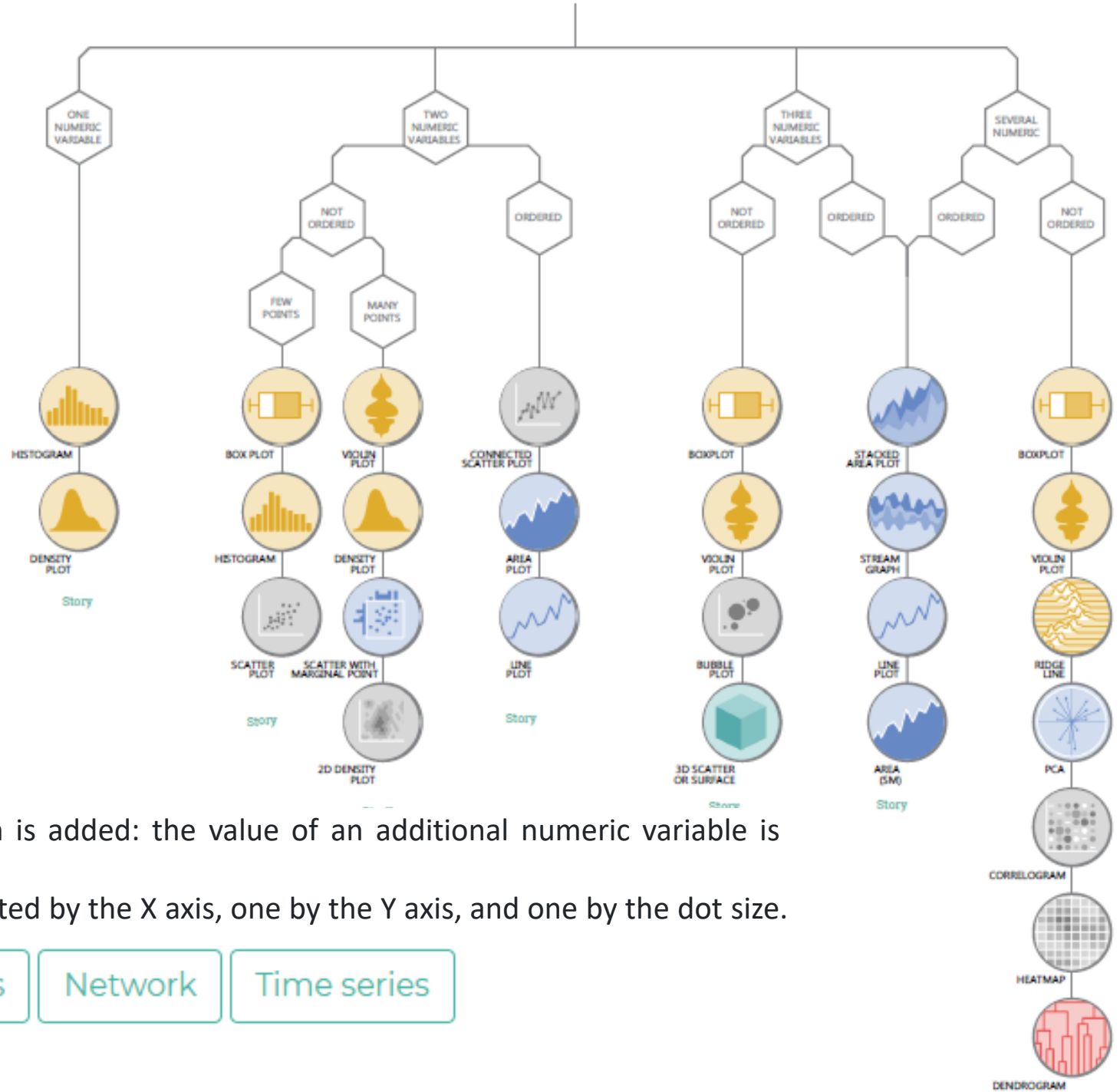
Numeric | Categoric | Num & Cat | Maps | Network | Time series

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from Data to Viz

Bubble Plot





A bubble plot is a scatterplot where a third dimension is added: the value of an additional numeric variable is represented through the size of the dots.

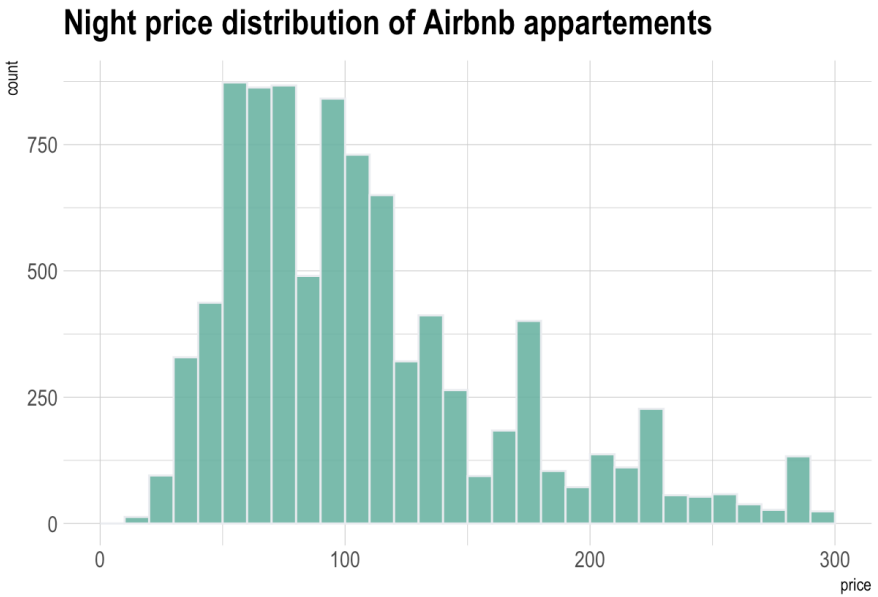
You need 3 numerical variables as input: one is represented by the X axis, one by the Y axis, and one by the dot size.

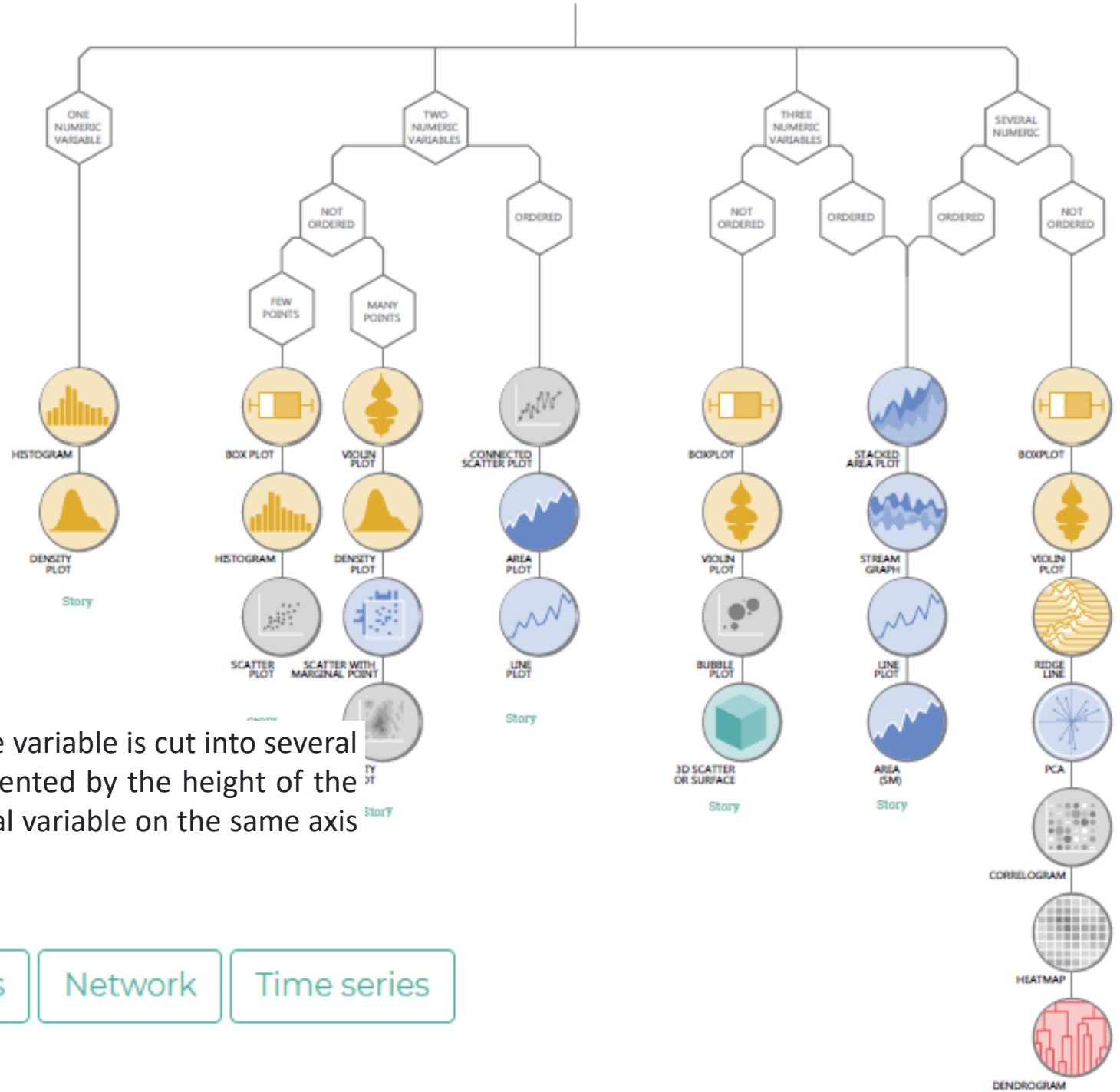
Numeric | Categoric | Num & Cat | Maps | Network | Time series

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Histogram





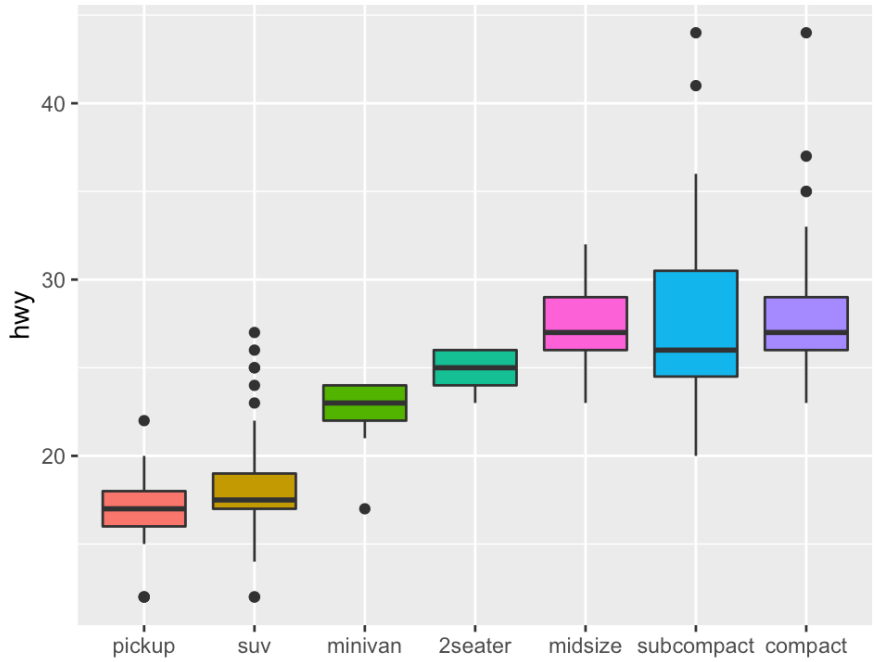
A **histogram** takes as input a numeric variable only. The variable is cut into several bins, and the number of observation per bin is represented by the height of the bar. It is possible to represent the distribution of several variable on the same axis using this technique.

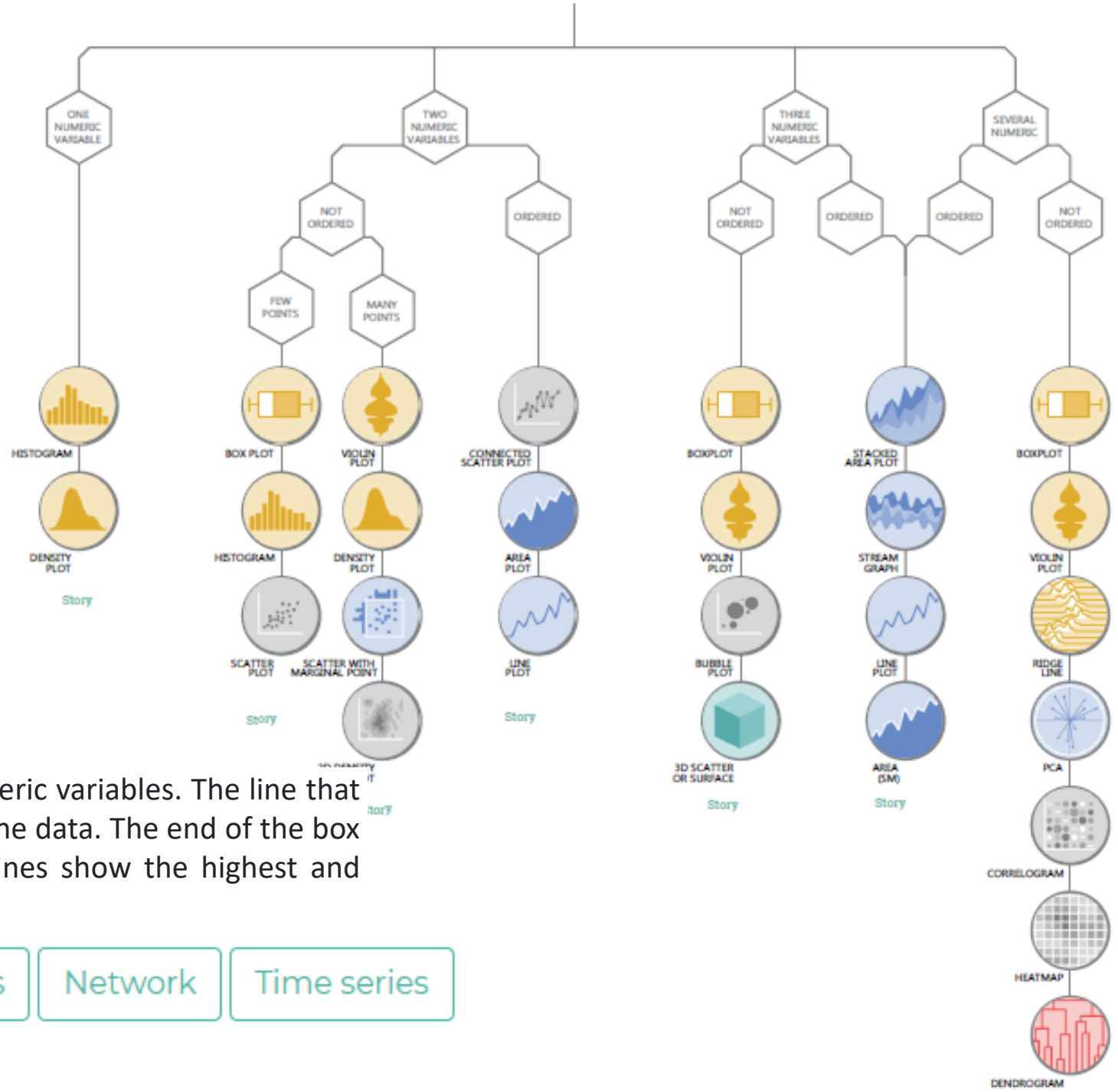
Numeric | Categoric | Num & Cat | Maps | Network | Time series

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Boxplot





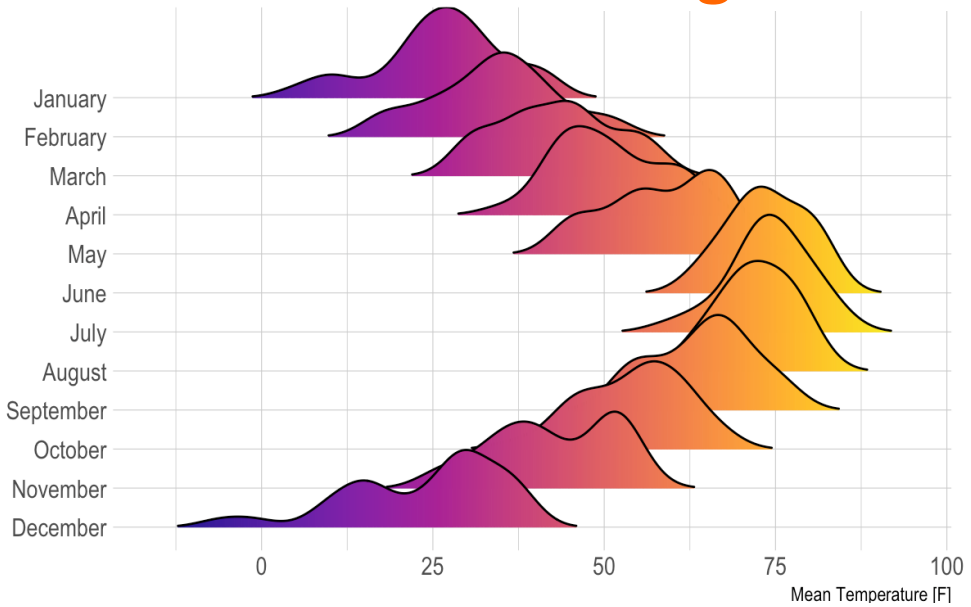
A boxplot gives a nice summary of one or several numeric variables. The line that divides the box into 2 parts represents the median of the data. The end of the box shows the upper and lower quartiles. The extreme lines show the highest and lowest value excluding outliers.

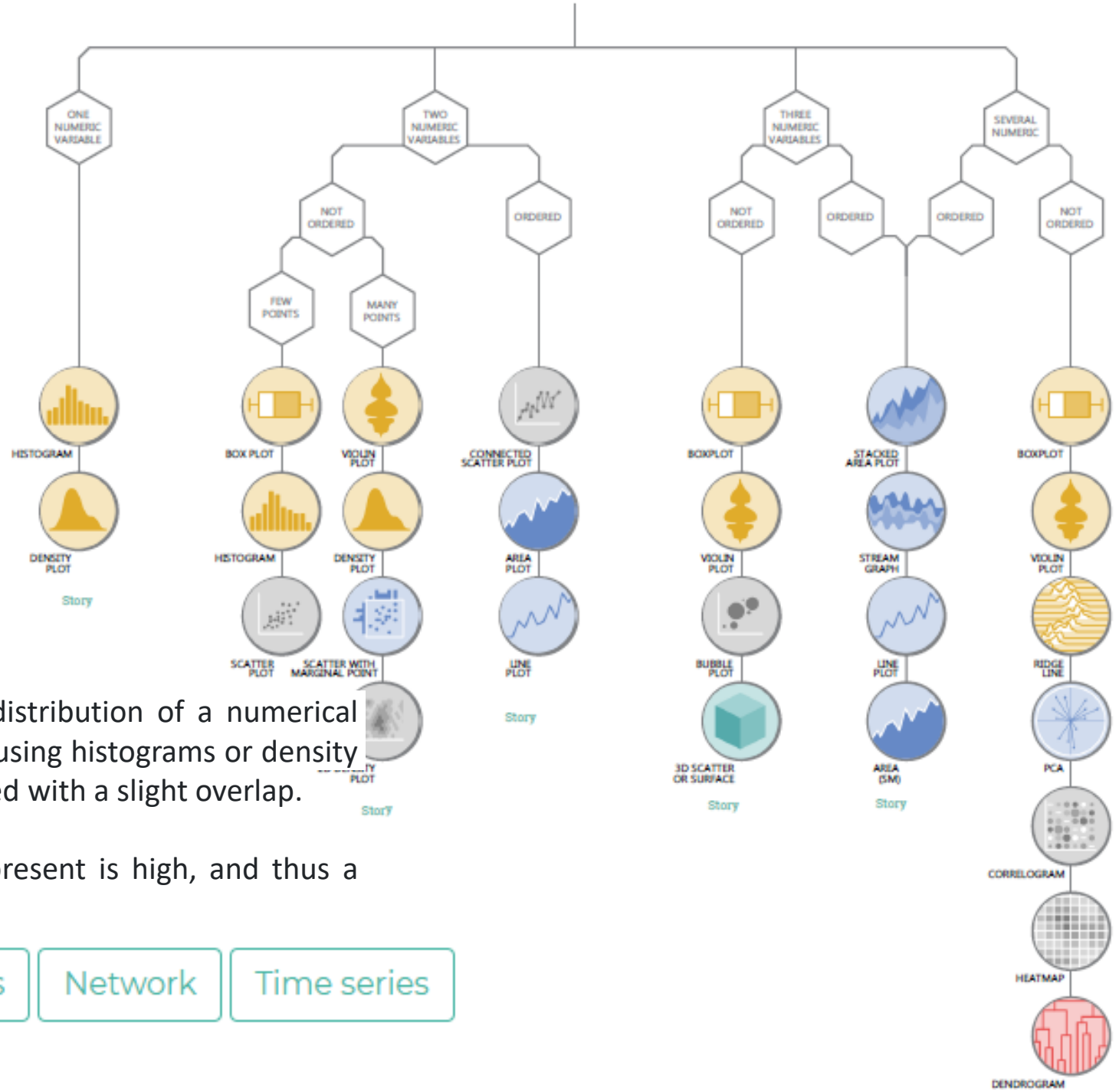
Numeric | Categoric | Num & Cat | Maps | Network | Time series

https://www.data-to-viz.com/#scatter

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Ridge Chart





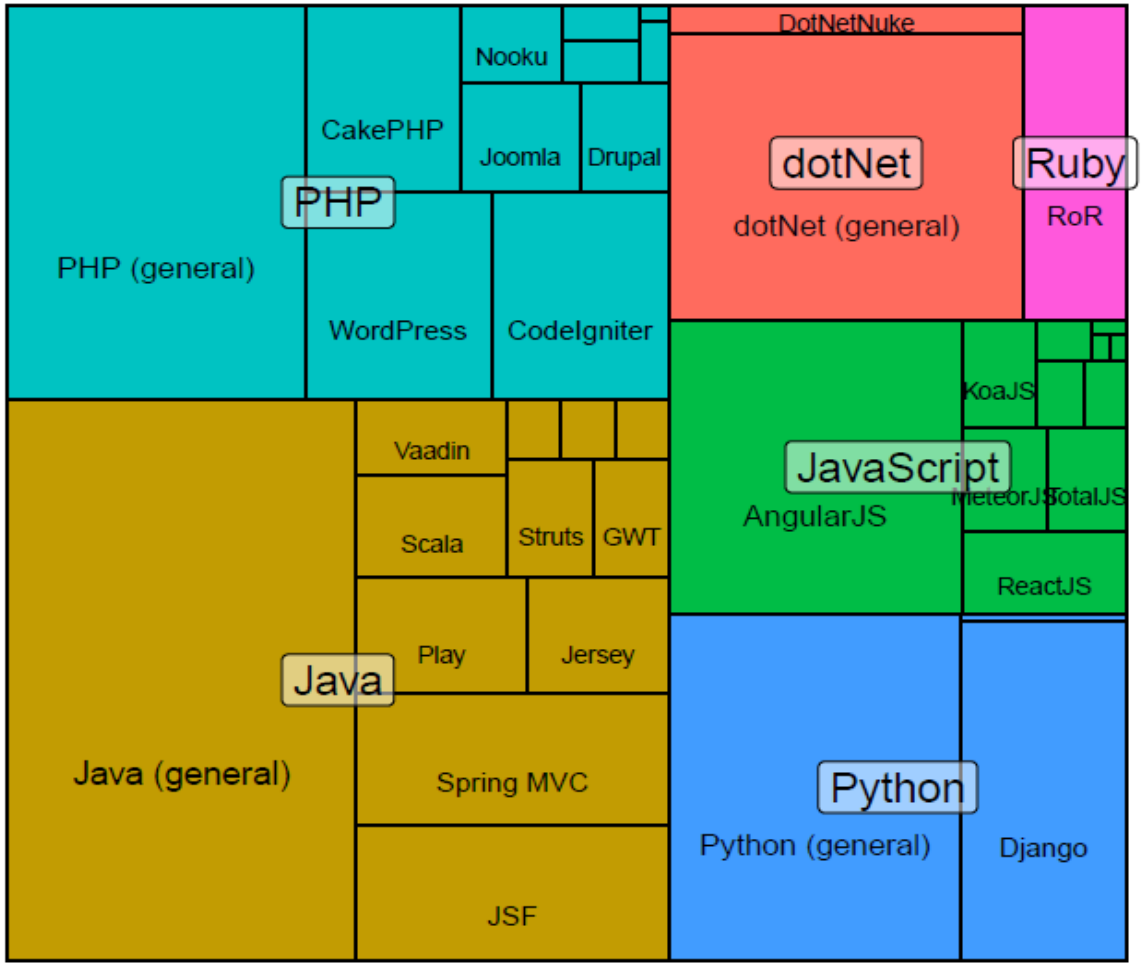
A Ridgeline plot (sometimes called Joyplot) shows the distribution of a numerical value for several groups. Distribution can be represented using histograms or density plots, all aligned to the same horizontal scale and presented with a slight overlap.

It is especially useful when the number of group to represent is high, and thus a classic window separation would take to much space.

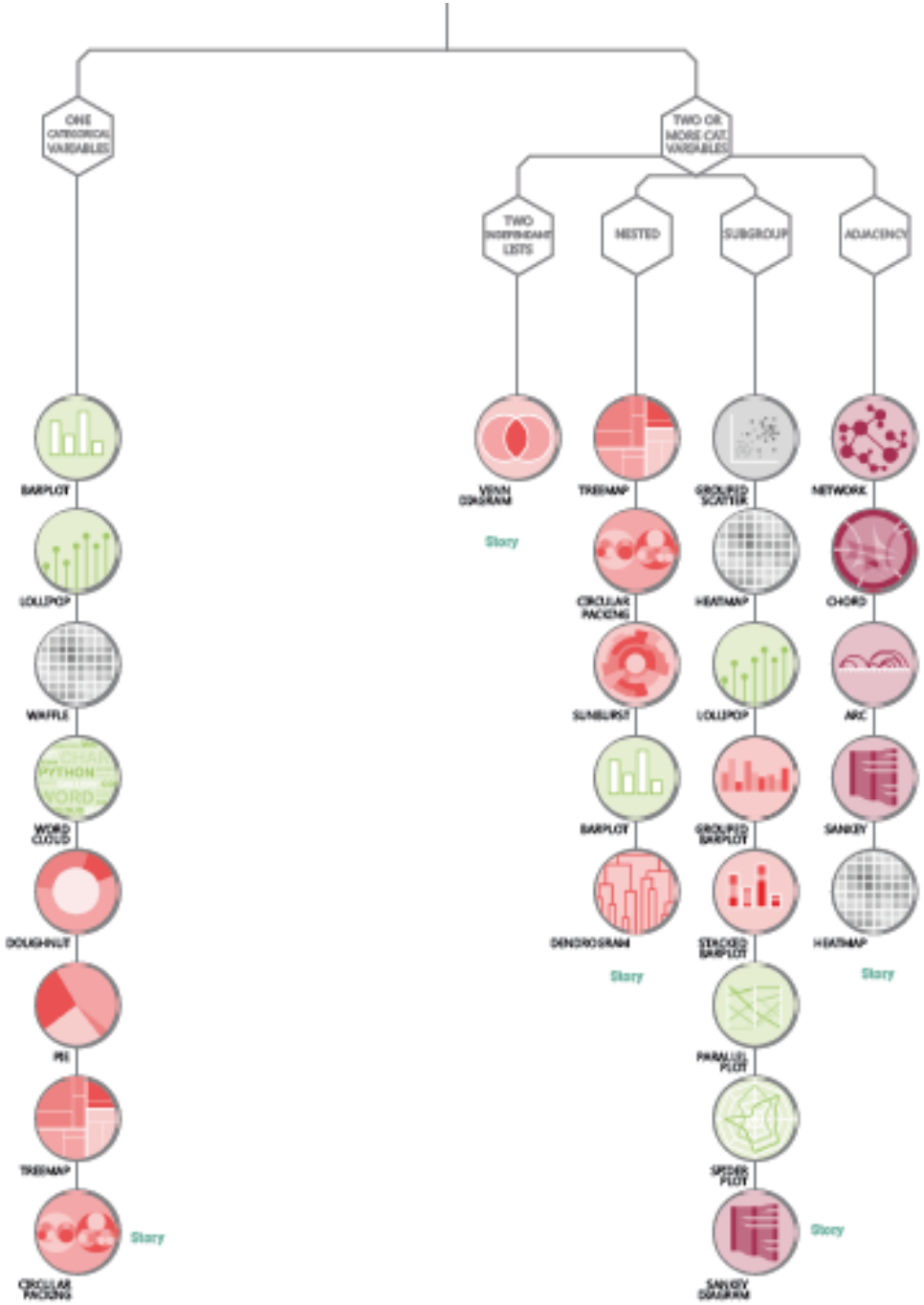
Numeric | Categoric | Num & Cat | Maps | Network | Time series

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Rectangular Treemap

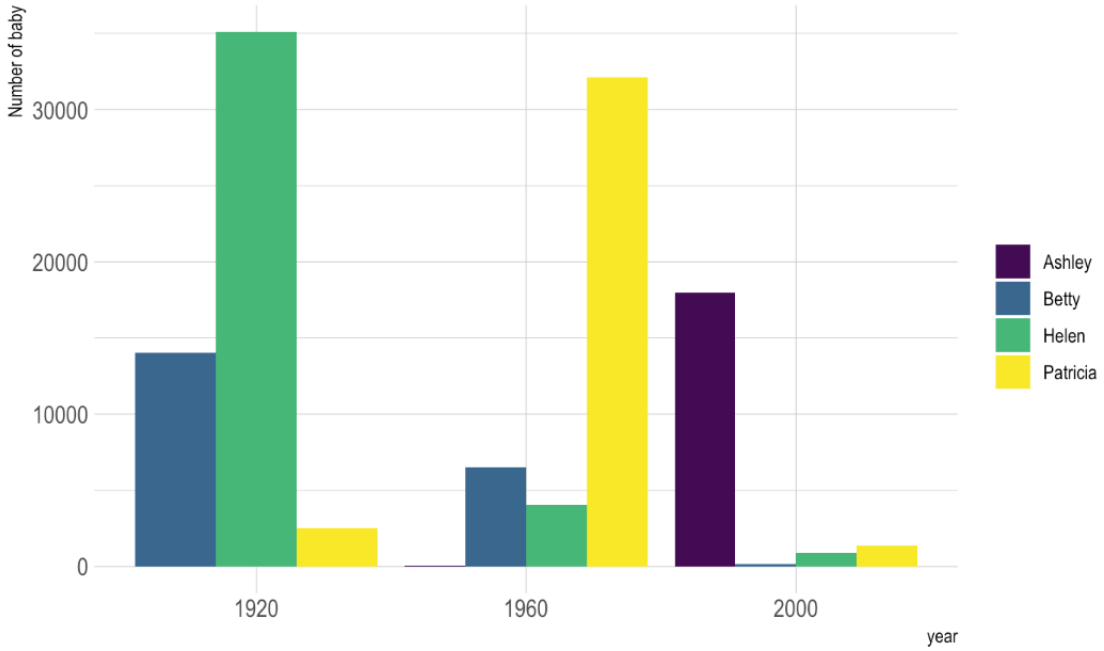


Numeric | Categoric | Num & Cat | Maps | Network | Time

https://www.data-to-viz.com/#scatter

from Data to Viz

Barplot

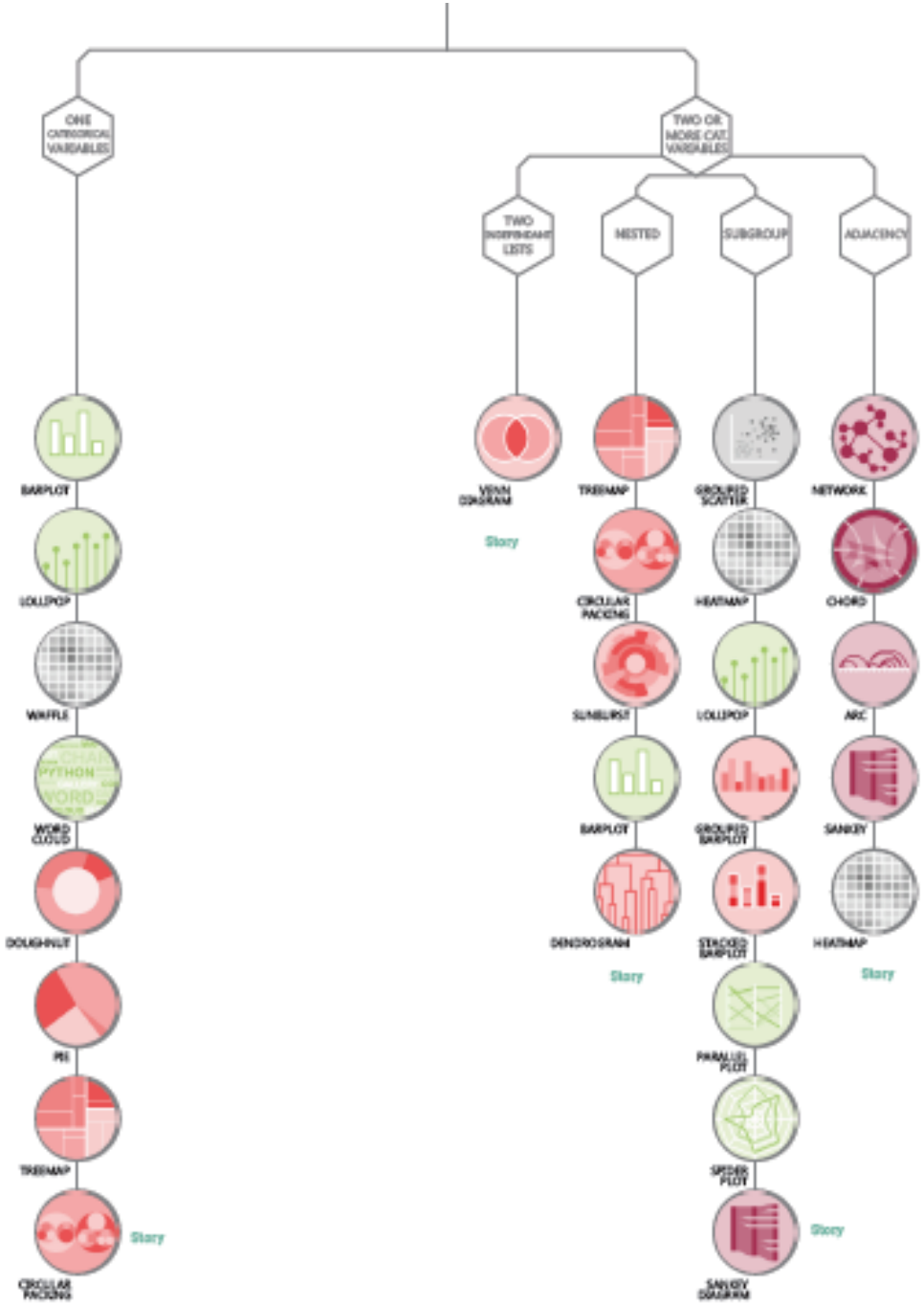


A barplot shows the relationship between a numeric and a categoric variable. Each entity of the categoric variable is represented as a bar. The size of the bar represents its numeric value.

Barplot is sometimes described as a boring way to visualize information. However it is probably the most efficient way to show this kind of data. Ordering bars and providing good annotation are often necessary.

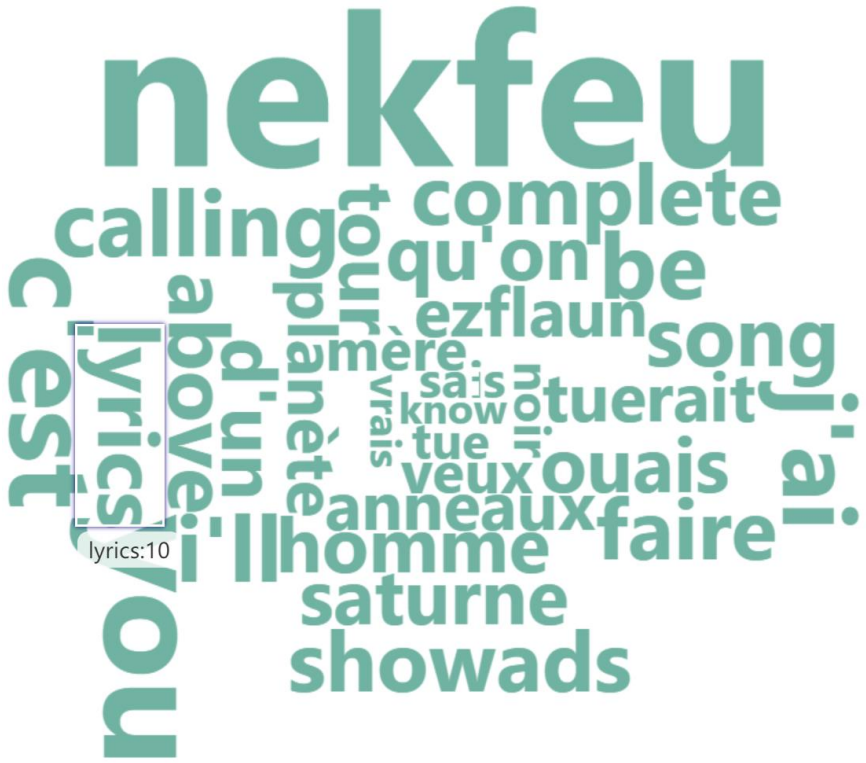
Numeric | Categoric | Num & Cat | Maps | Network | Time

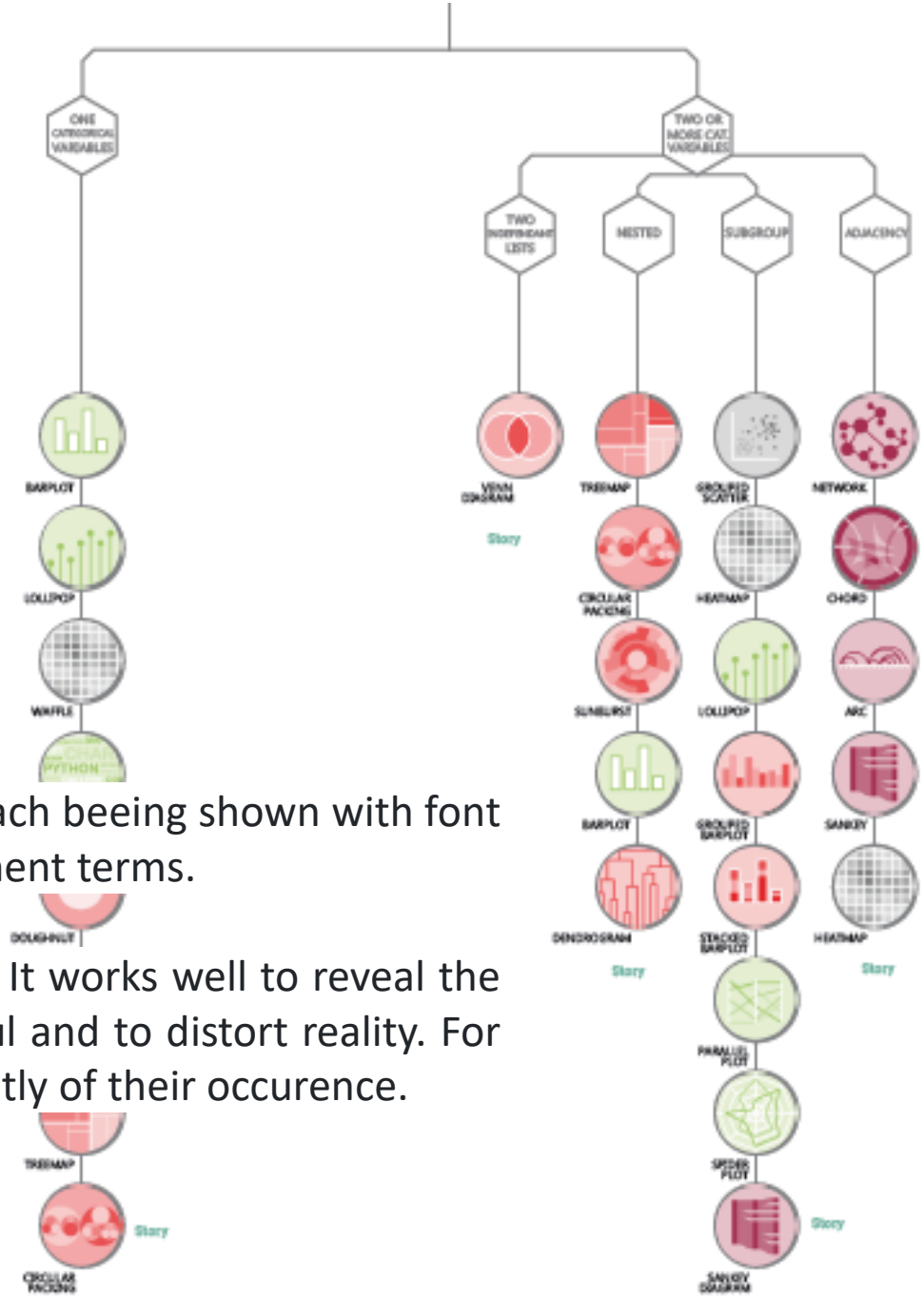
https://www.data-to-viz.com/#scatter



from Data to Viz

Wordcloud





A Wordcloud (or Tag cloud) displays a list of words, the importance of each beeing shown with font size or color. This format is useful for quickly perceiving the most prominent terms.

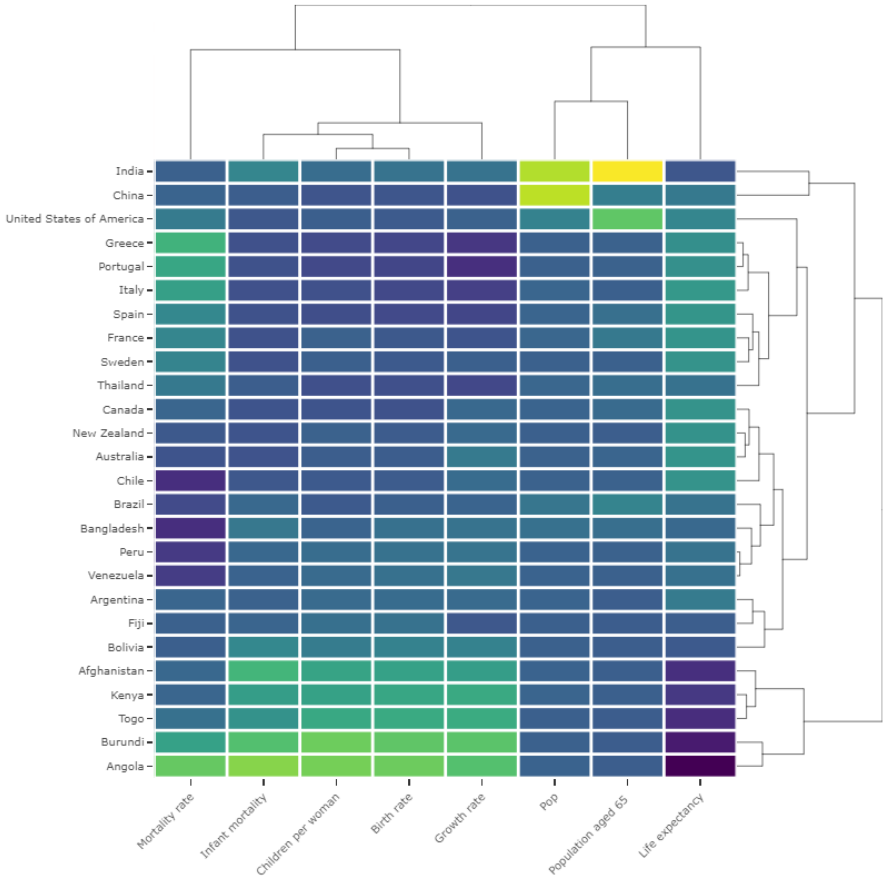
This type of visualization is well known by the public and eye-catching. It works well to reveal the essential. However, wordcloud is often criticized for not being insightful and to distort reality. For example, long words will be more prominent on the picture independently of their occurence.

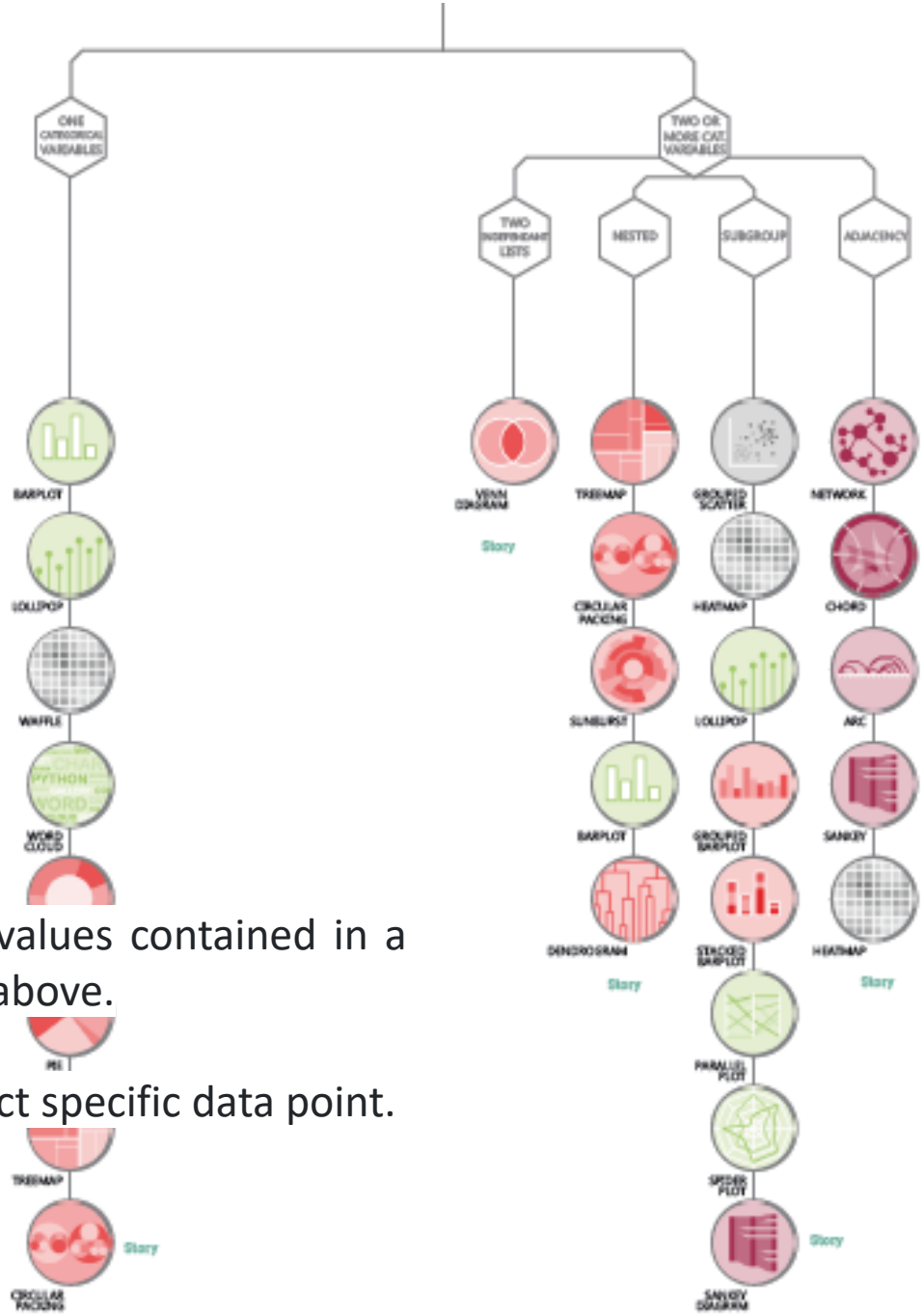


https://www.data-to-viz.com/#scatter

from Data to Viz

Heatmap





A heatmap is a graphical representation of data where the individual values contained in a matrix are represented as colors. It is a bit like looking a data table from above.

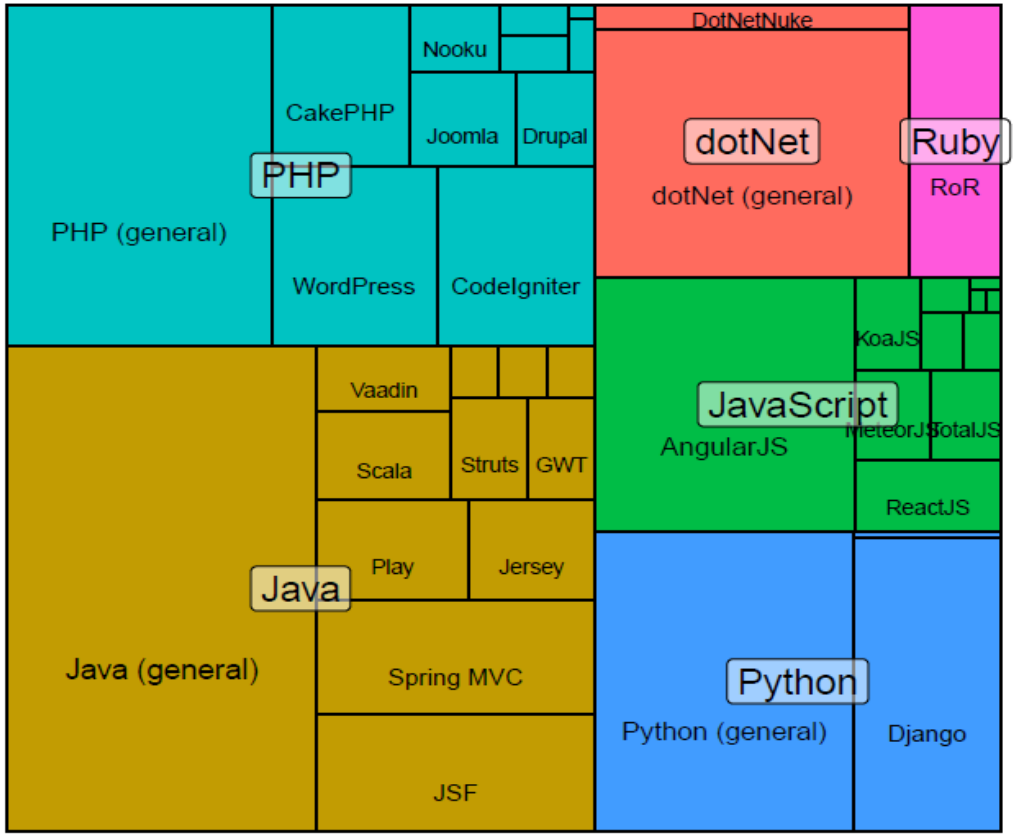
It is really useful to display a general view of numerical data, not to extract specific data point.

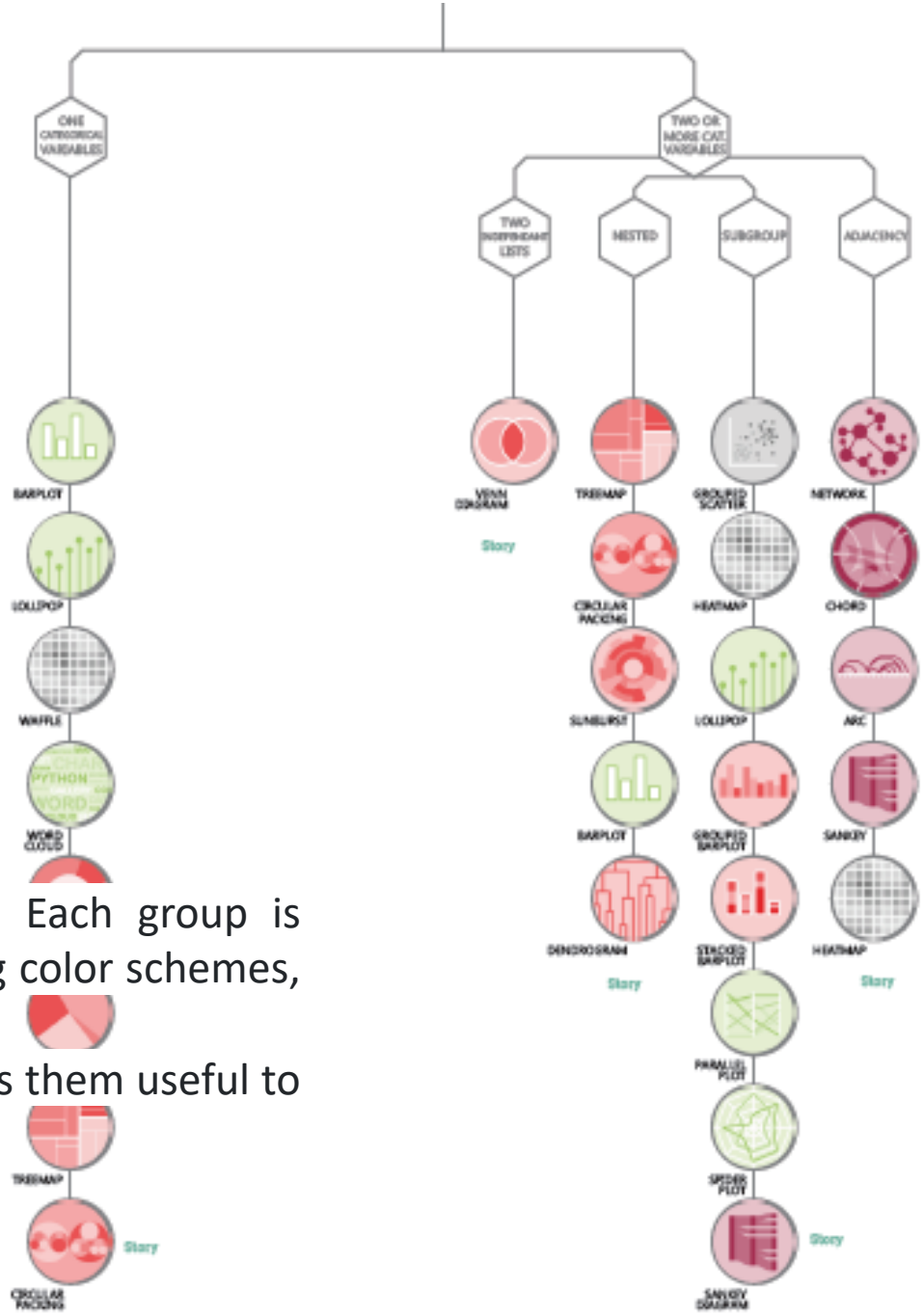


https://www.data-to-viz.com/#scatter

from Data to Viz

Rectangular Treemap





A Treemap displays hierarchical data as a set of nested rectangles. Each group is represented by a rectangle, which area is proportional to its value. Using color schemes, it is possible to represent several dimensions: groups, subgroups etc.
Treemaps have the advantage to make efficient use of space, what makes them useful to represent a big amount of data.

Numeric | Categoric | Num & Cat | Maps | Network | Time

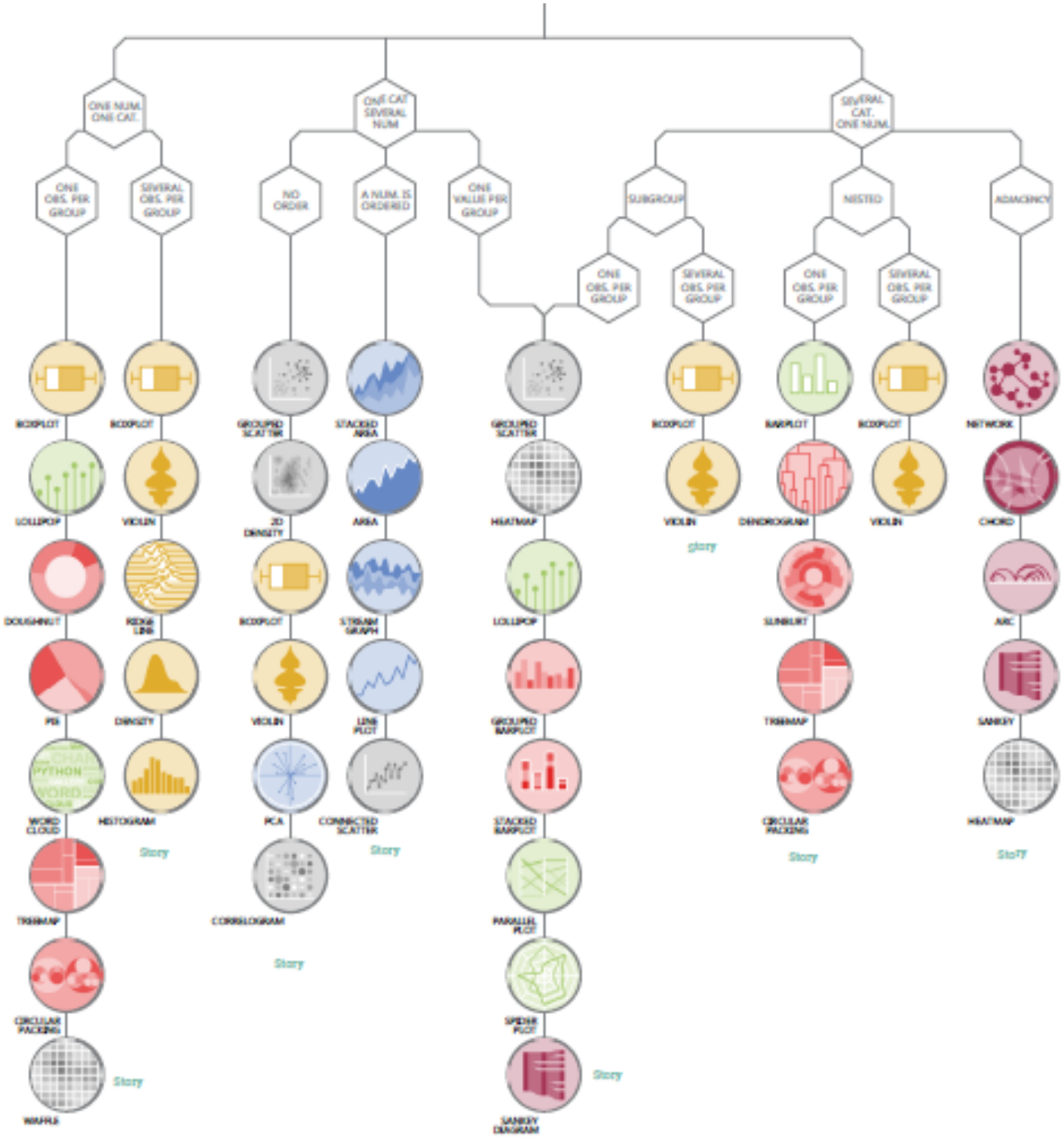
https://www.data-to-viz.com/#scatter

from Data to Viz



Source:http://lip.sourceforge.net/ctreemap.html

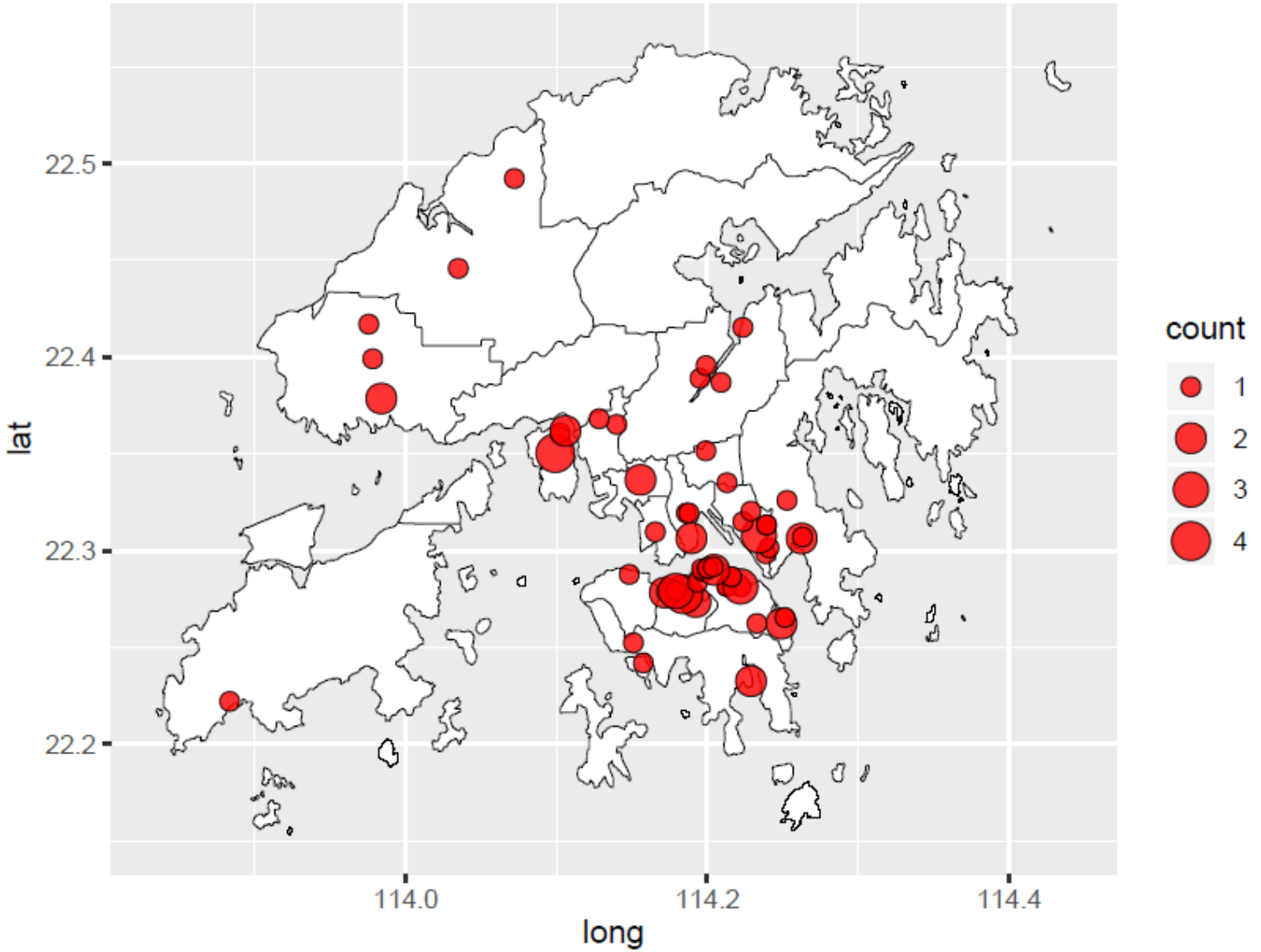
Circle Treemap



Numeric | Categoric | Num & Cat | Maps | Ne

https://www.data-to-viz.com/#scatter

from Data to Viz



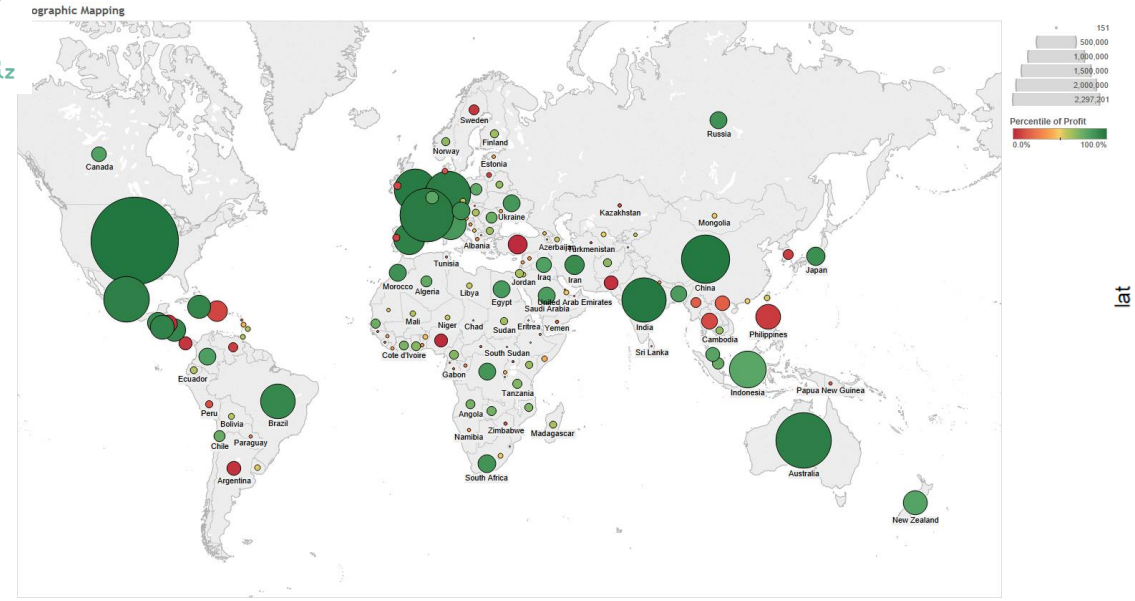


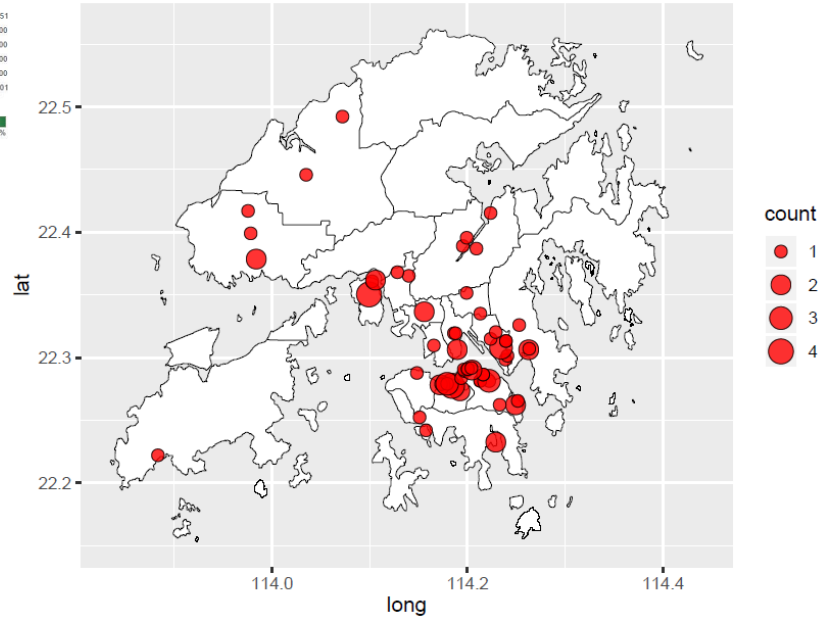
Numeric | Categoric | Num & Cat | **Maps** | Network | Time series

https://www.data-to-viz.com/#scatter

from Data to Viz

Bubble Map





A bubble map uses circles of different size to represent a numeric value on a territory.

It is possible to display a bubble per geographic coordinate, or a bubble per region (in this case the bubble is usually displayed in the baricentre of the region).

Bubble map allows to avoid the bias caused by regional area sizes in choropleth maps.

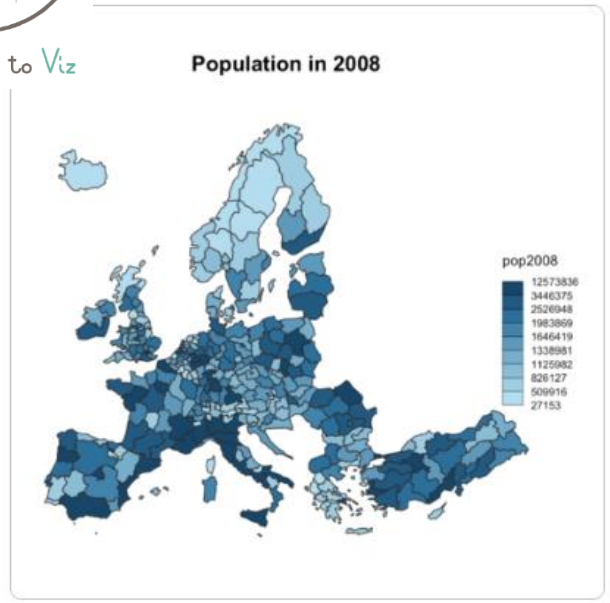
Numeric | Categoric | Num & Cat | **Maps** | Network | Time series

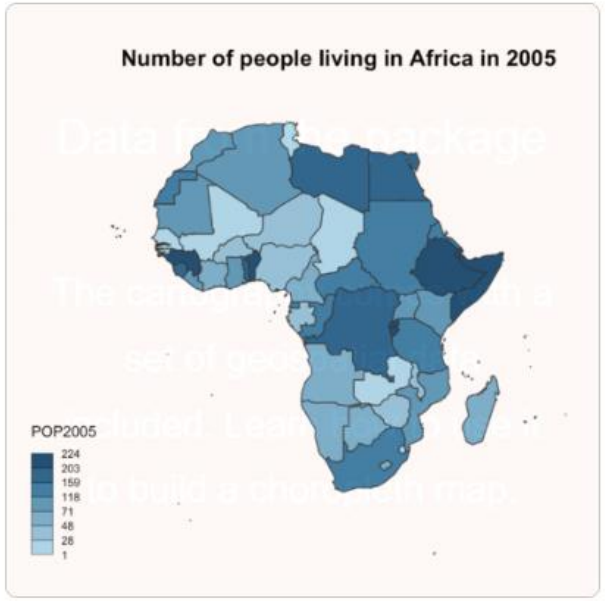
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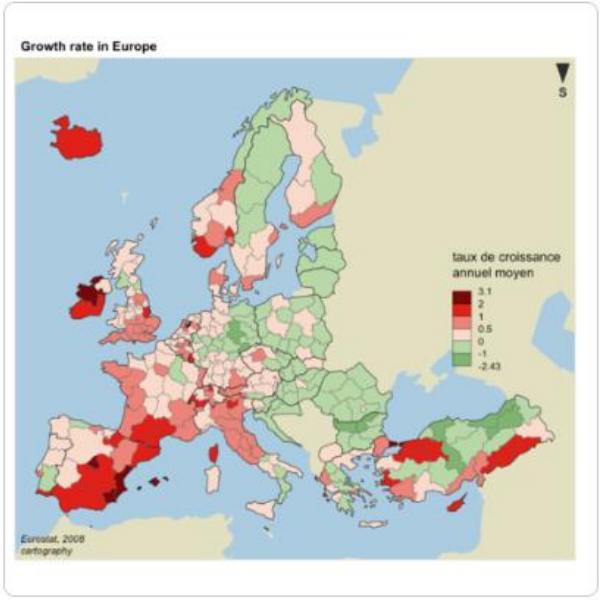


from Data to Viz

Choropleth Map







A choropleth map displays divided geographical areas or regions that are coloured in relation to a numeric variable. It allows to study how a variable evolutes along a territory. It is a powerful and widely used data visualization technique.

However, its downside is that regions with bigger sizes tend to have a bigger weight in the map interpretation, which includes a bias.

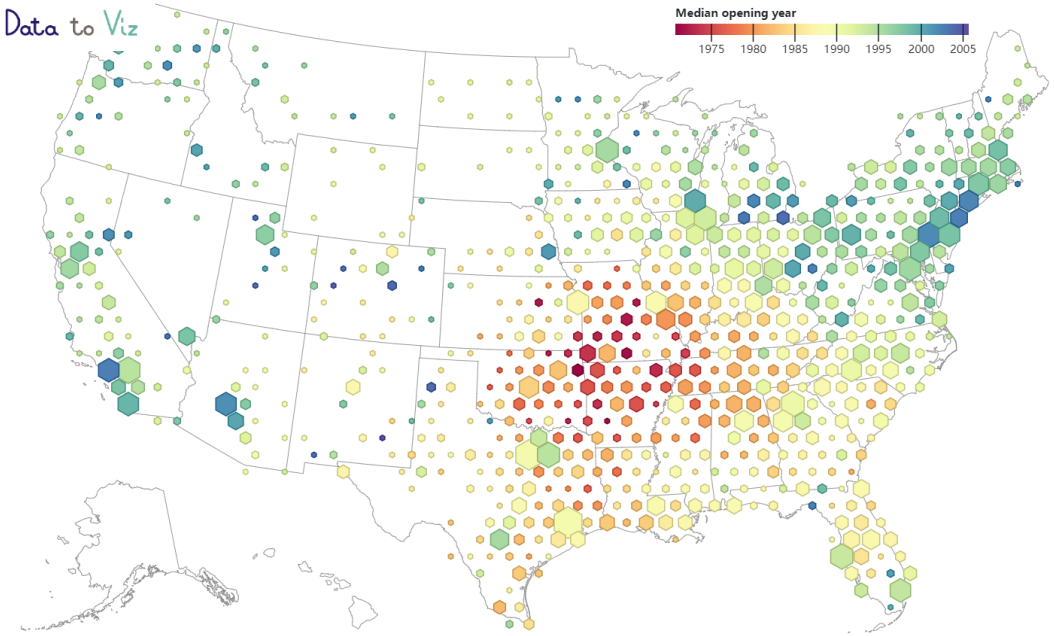
Numeric | Categoric | Num & Cat | **Maps** | Network | Time series

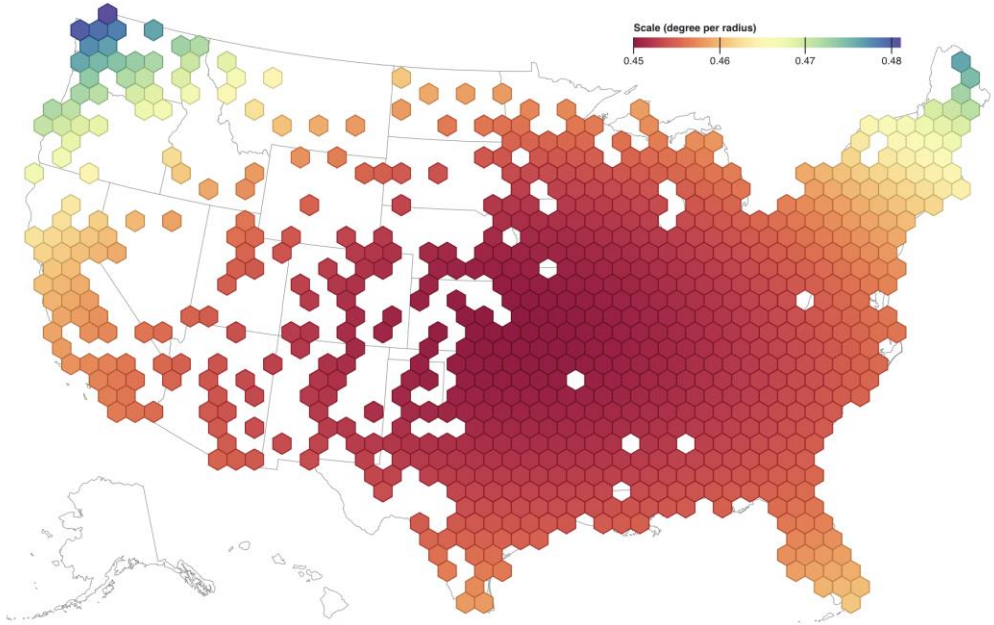
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Hexbin Map





A hexbin map splits a geographic area in a multitude of hexagons. A numeric value is attributed to each hexagone, what controls its color.

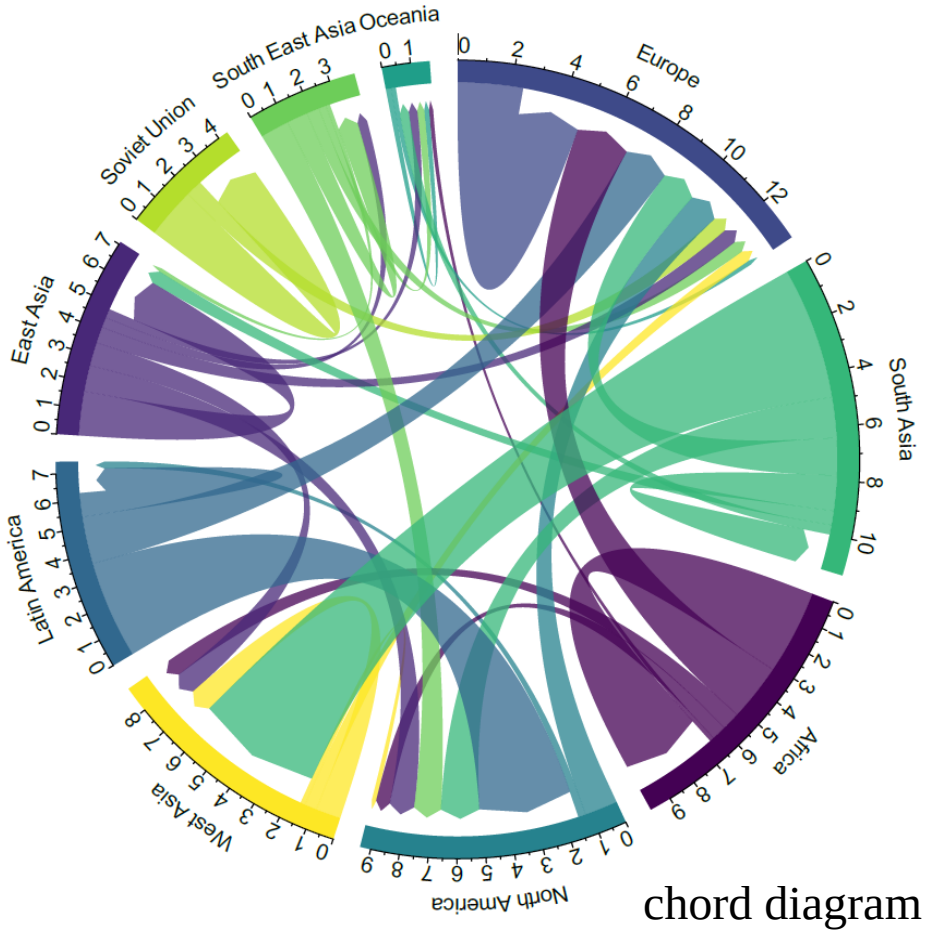
Hexbin map is a kind of choropleth map. It removes the bias introduced by the different region size in choropleth, since each region is represented by the same hexagone.

Numeric | Categoric | Num & Cat | **Maps** | Network | Time series

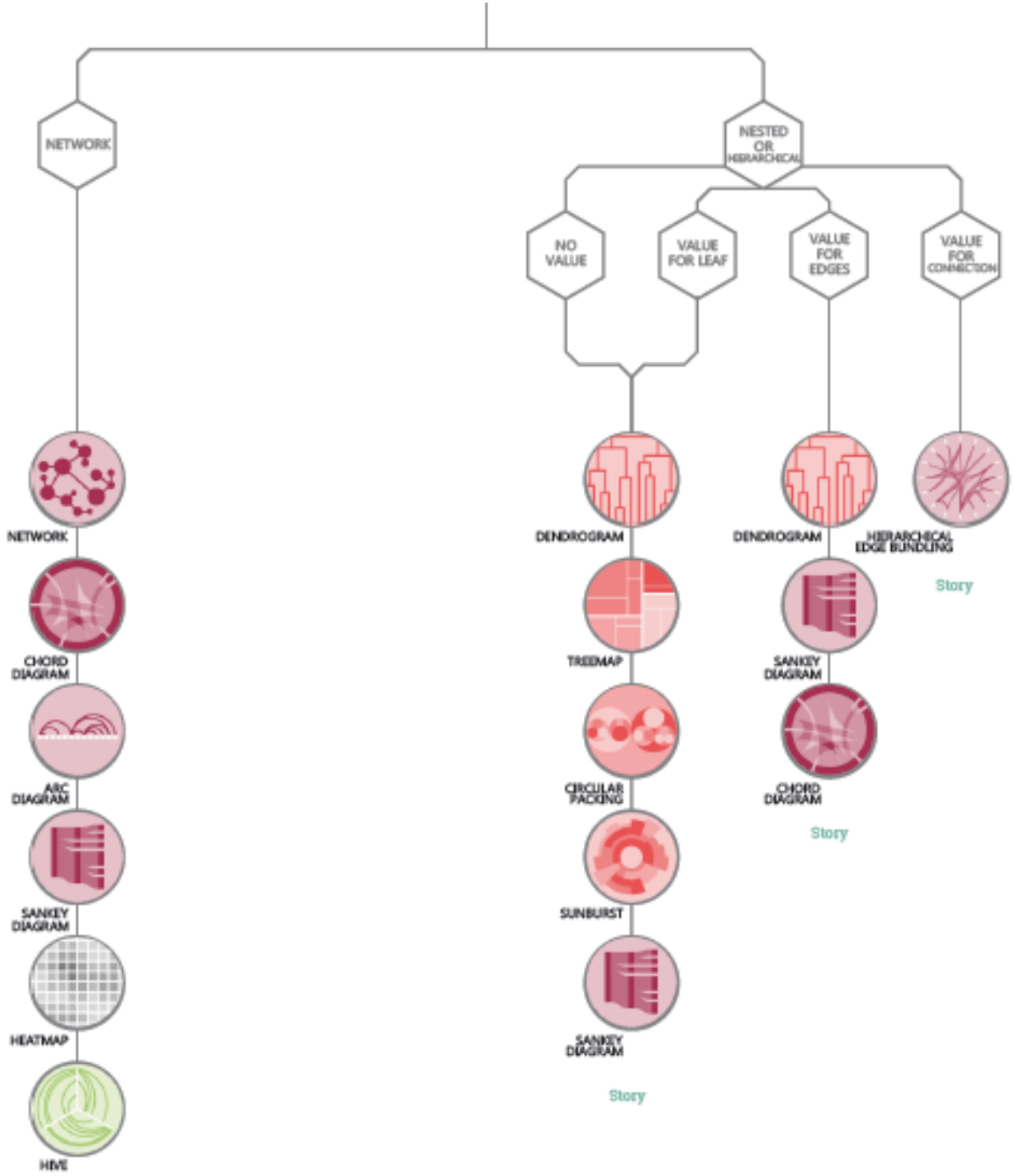
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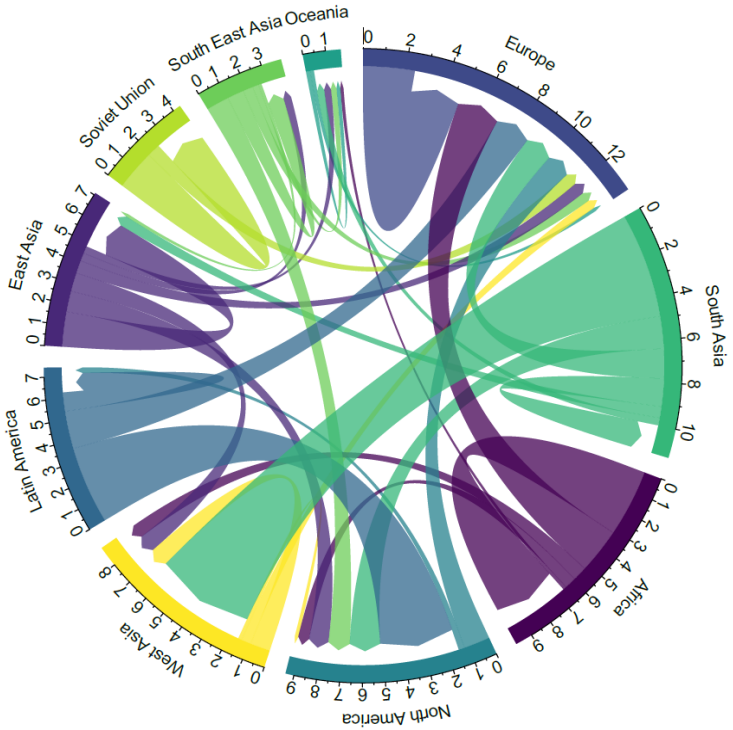
chord diagram



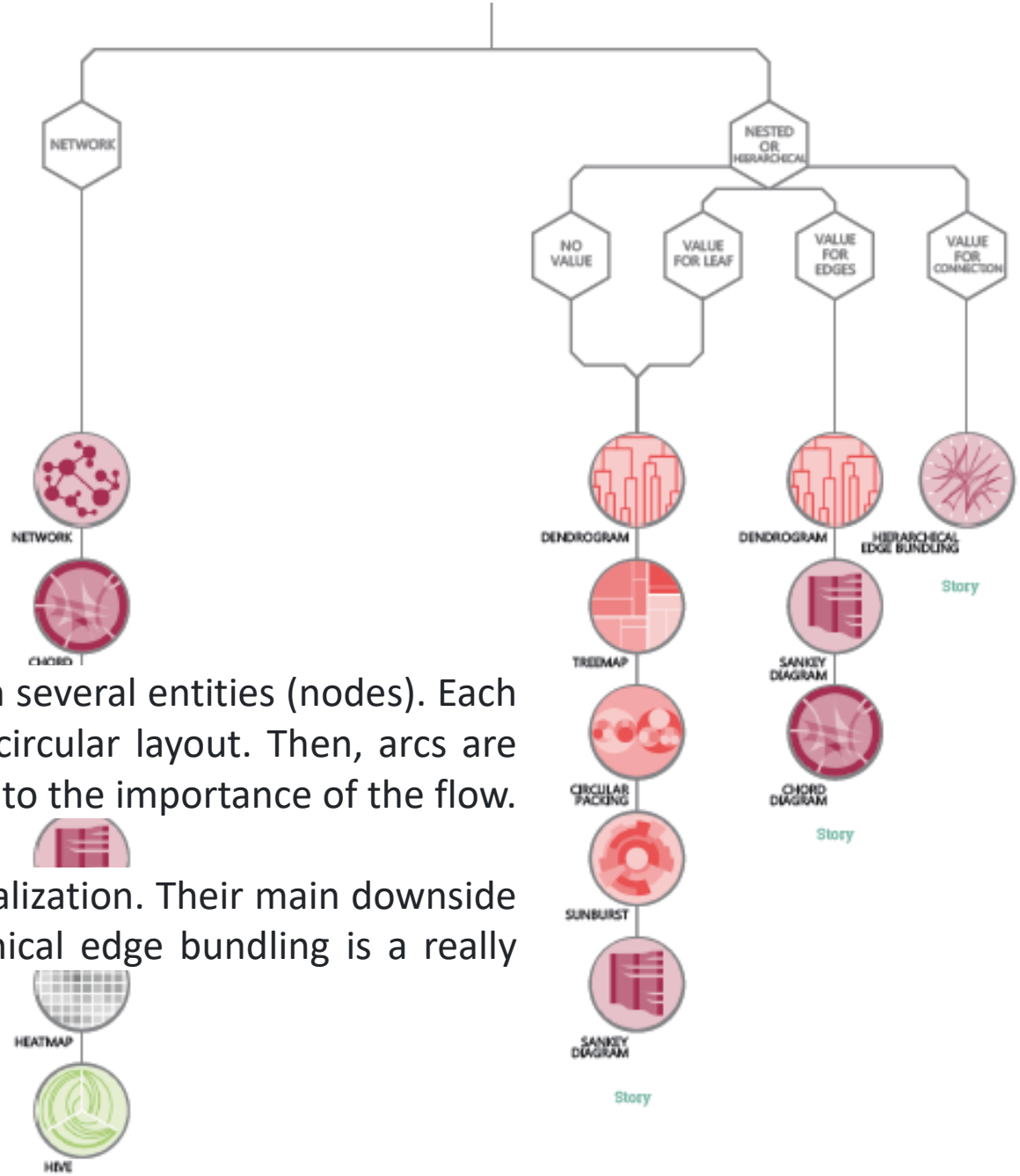
Numeric | Categoric | Num & Cat | Maps | Network

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Chord Diagram



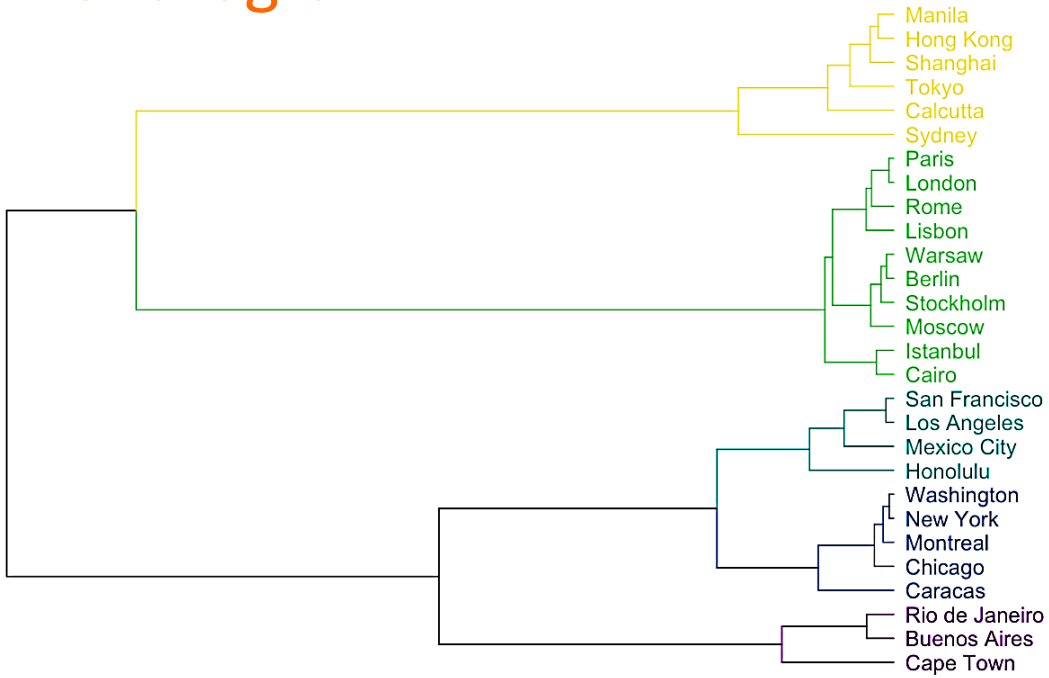
[Chord diagrams] allow to visualize flows or connections between several entities (nodes). Each entity is represented by a fragment on the outer part of the circular layout. Then, arcs are drawn between each entities. The size of the arc is proportional to the importance of the flow.

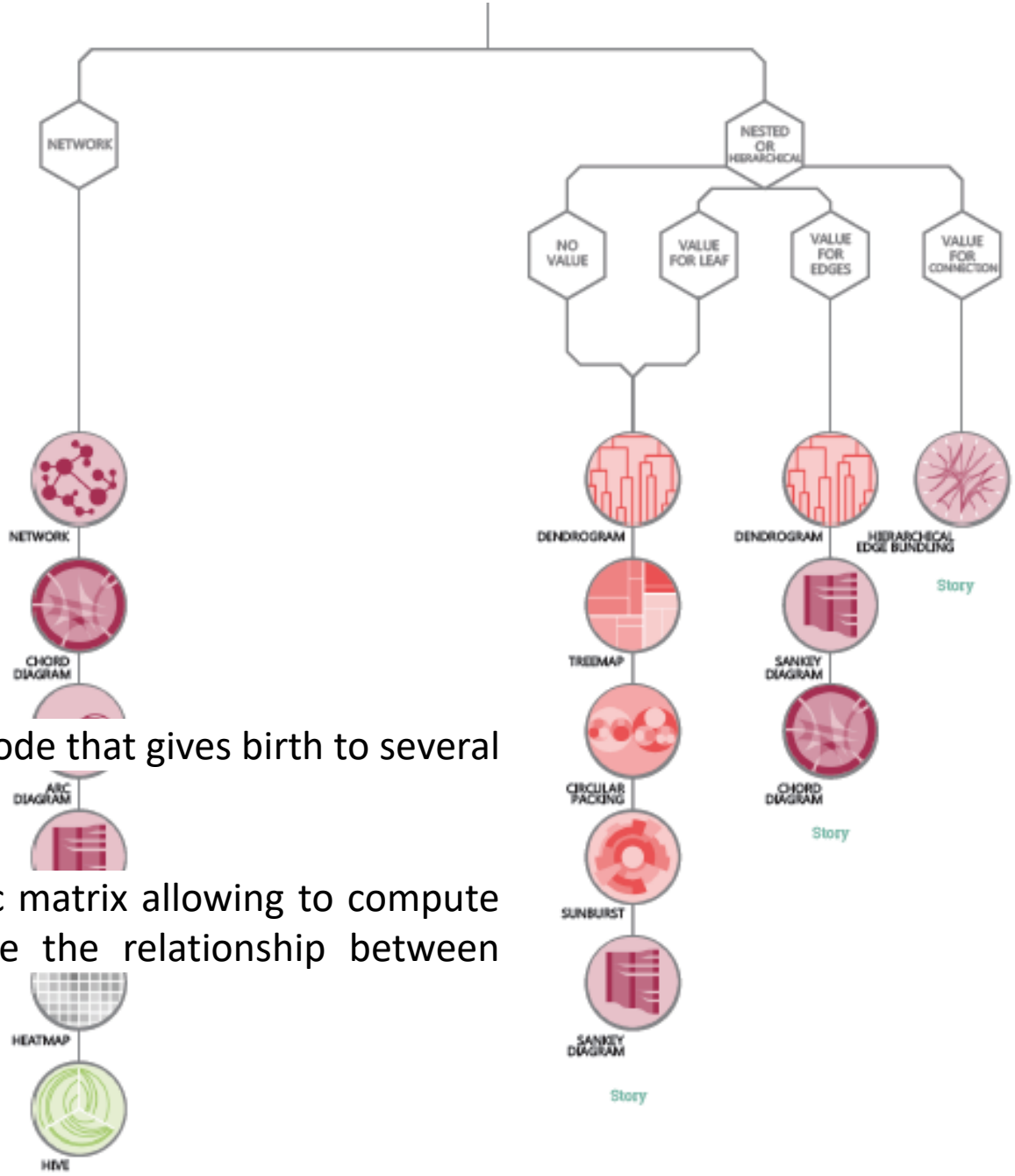
Chord diagrams are eye catching and quite popular in data visualization. Their main downside is that they becom quickly cluttered and unreadable. Hierarchical edge bundling is a really close type of graphic that allows to avoid this issue.

Numeric | Categoric | Num & Cat | Maps | **Network**

https://www.data-to-viz.com/#scatter

Dendrogram





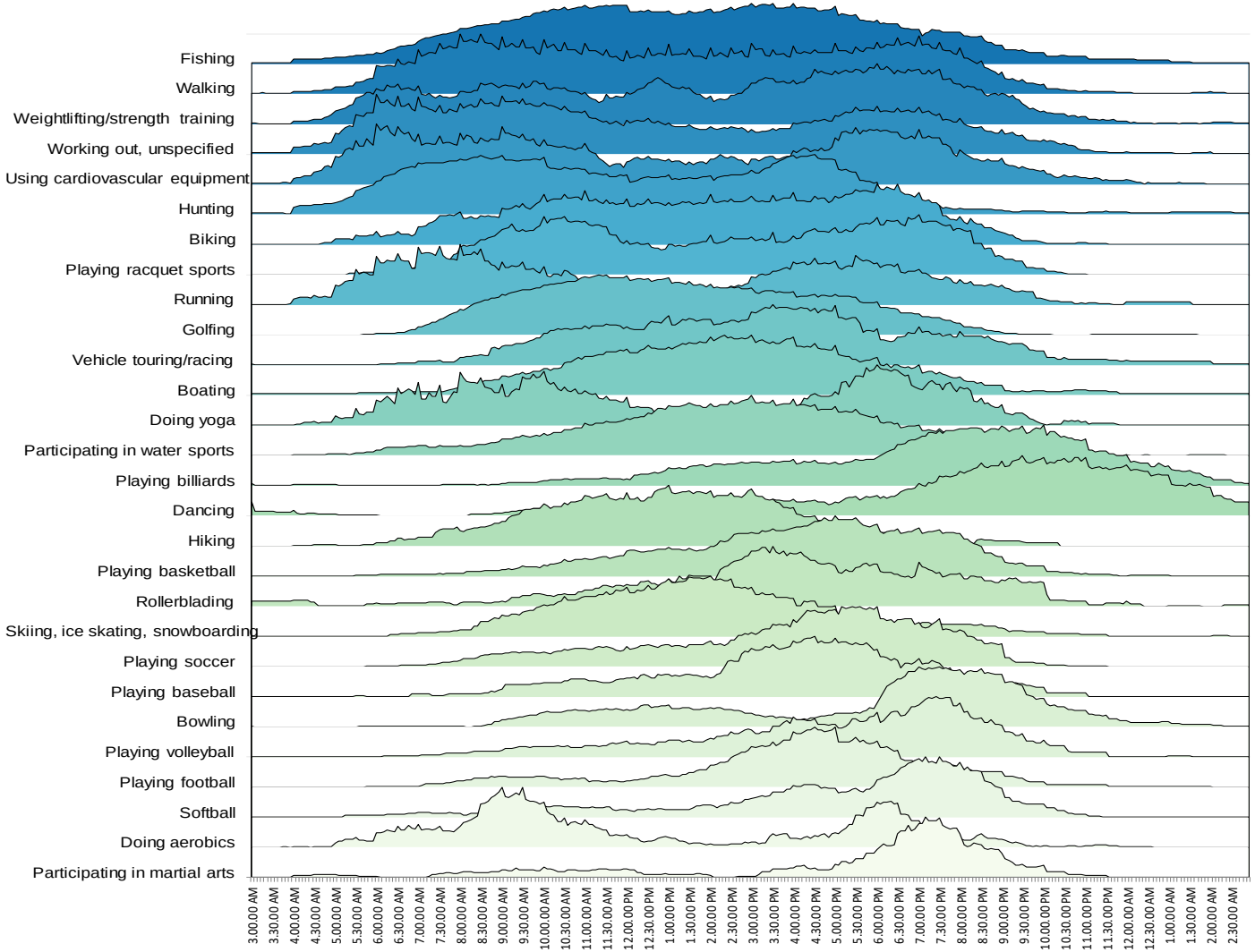
A dendrogram is a network structure. It is constituted of a root node that gives birth to several nodes connected by edges or branches.

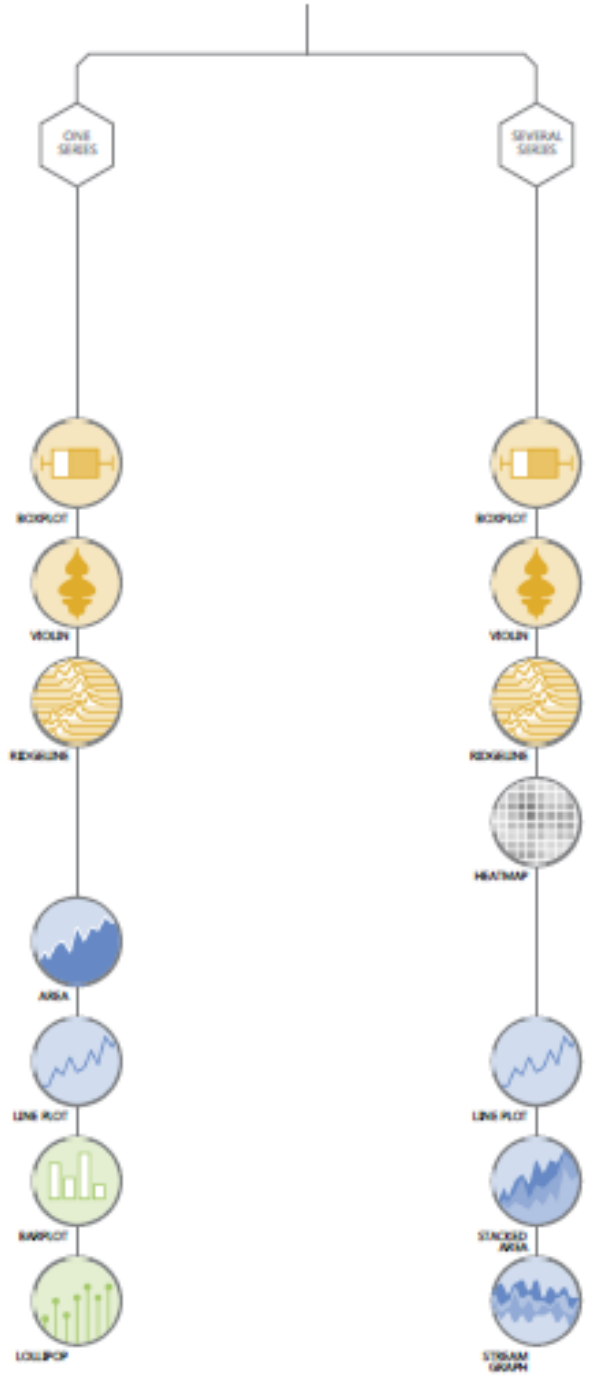
Dendrogram can be made with 2 types of dataset. i/ A numeric matrix allowing to compute distance between individuals ii/ A hierarchical dataset where the relationship between entities is provided directly.

Numeric | Categoric | Num & Cat | Maps | **Network**

https://www.data-to-viz.com/#scatter

from Data to Viz



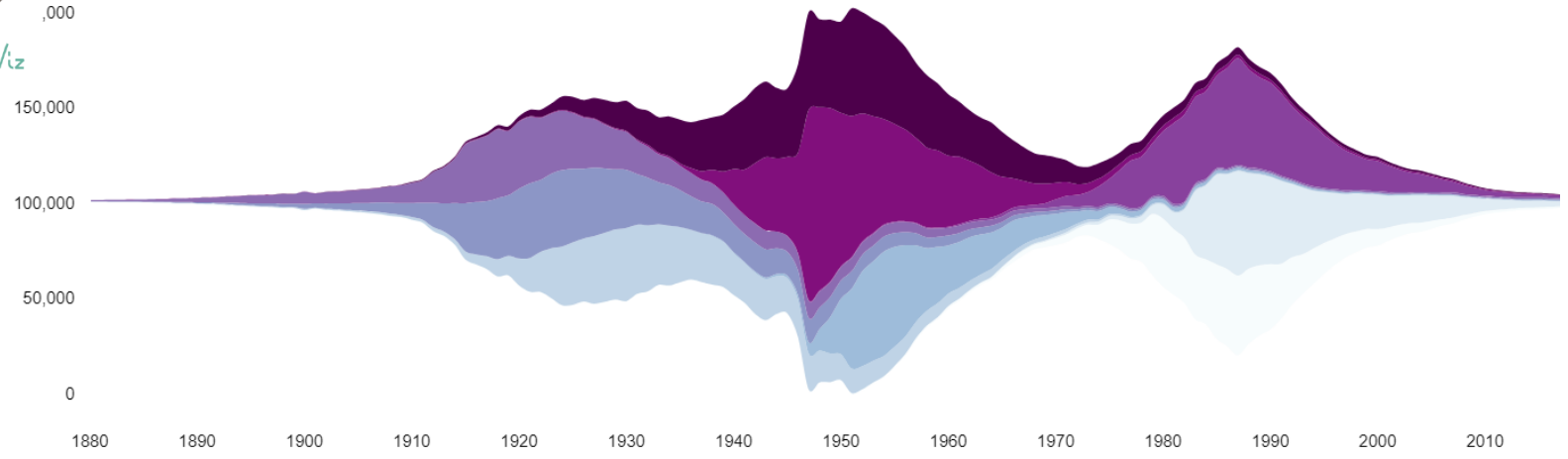


Numeric | Categoric | Num & Cat | Maps | Network | **Time series**

https://www.data-to-viz.com/#scatter

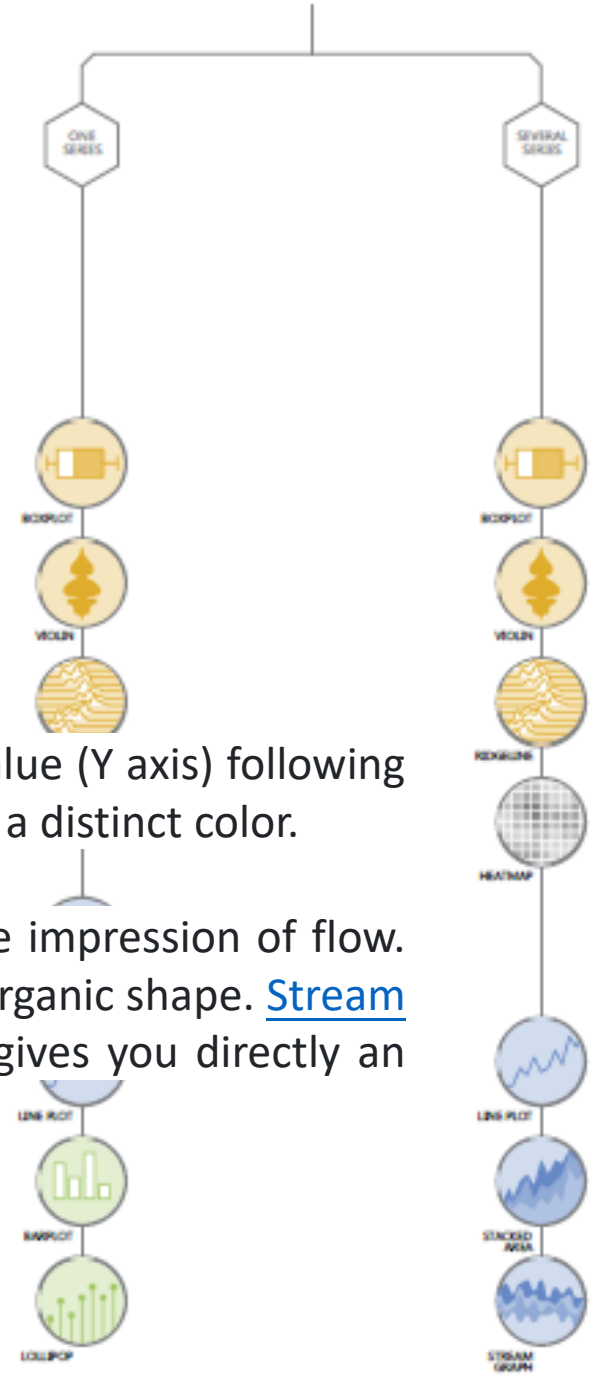
from Data to Viz

Stream graph



A Stream graph is a type of stacked area chart. It displays the evolution of a numeric value (Y axis) following another numeric value (X axis). This evolution is represented for several groups, all with a distinct color.

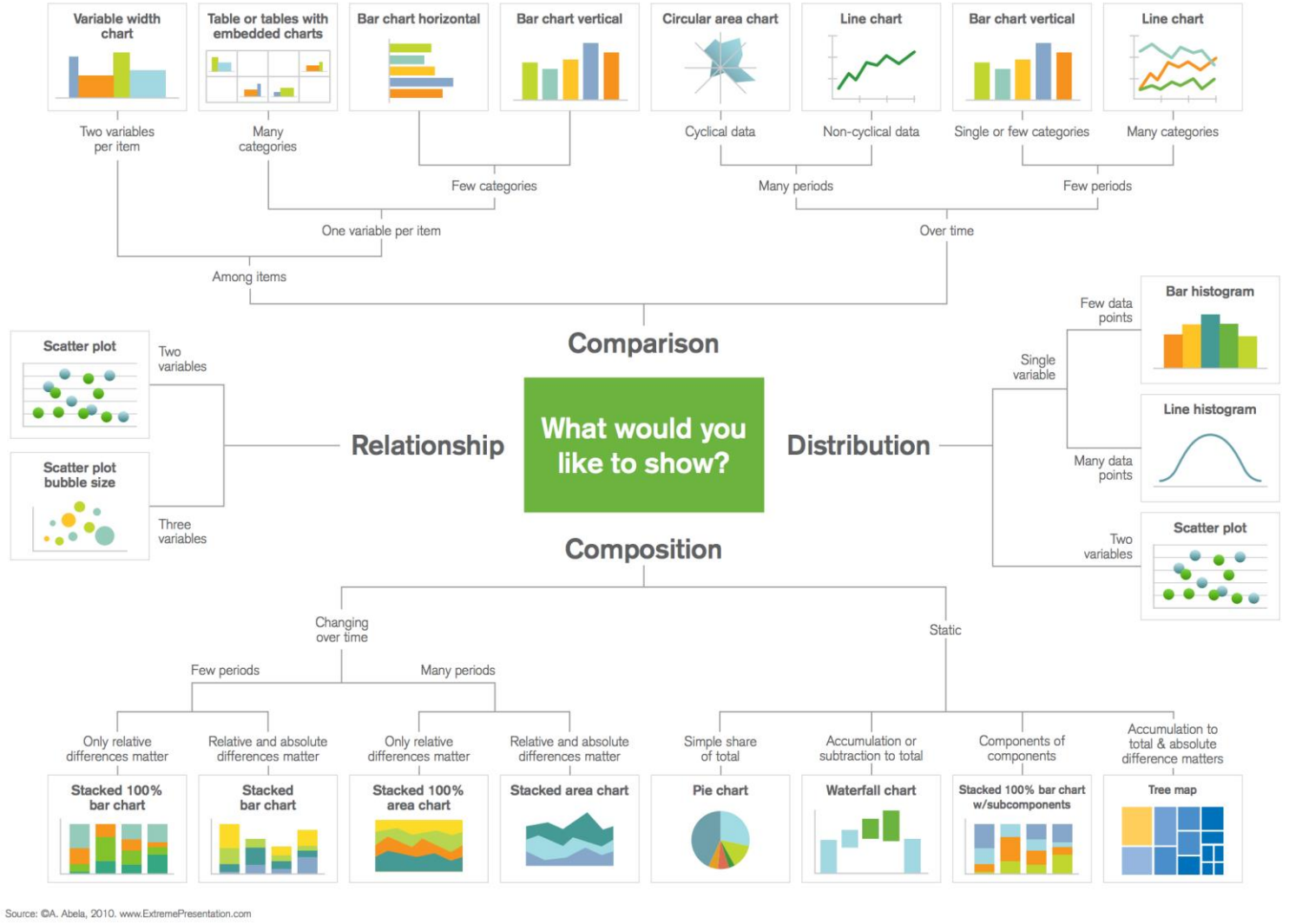
Contrary to a stacked area, there is no corner: edges are rounded what gives this nice impression of flow. Moreover, areas are usually displaced around a central axis, resulting in a flowing and organic shape. Stream chart gets really useful when displayed in an interactive mode: highlighting a group gives you directly an insight of its evolution.



Numeric | Categoric | Num & Cat | Maps | Network | **Time series**

https://www.data-to-viz.com/#scatter

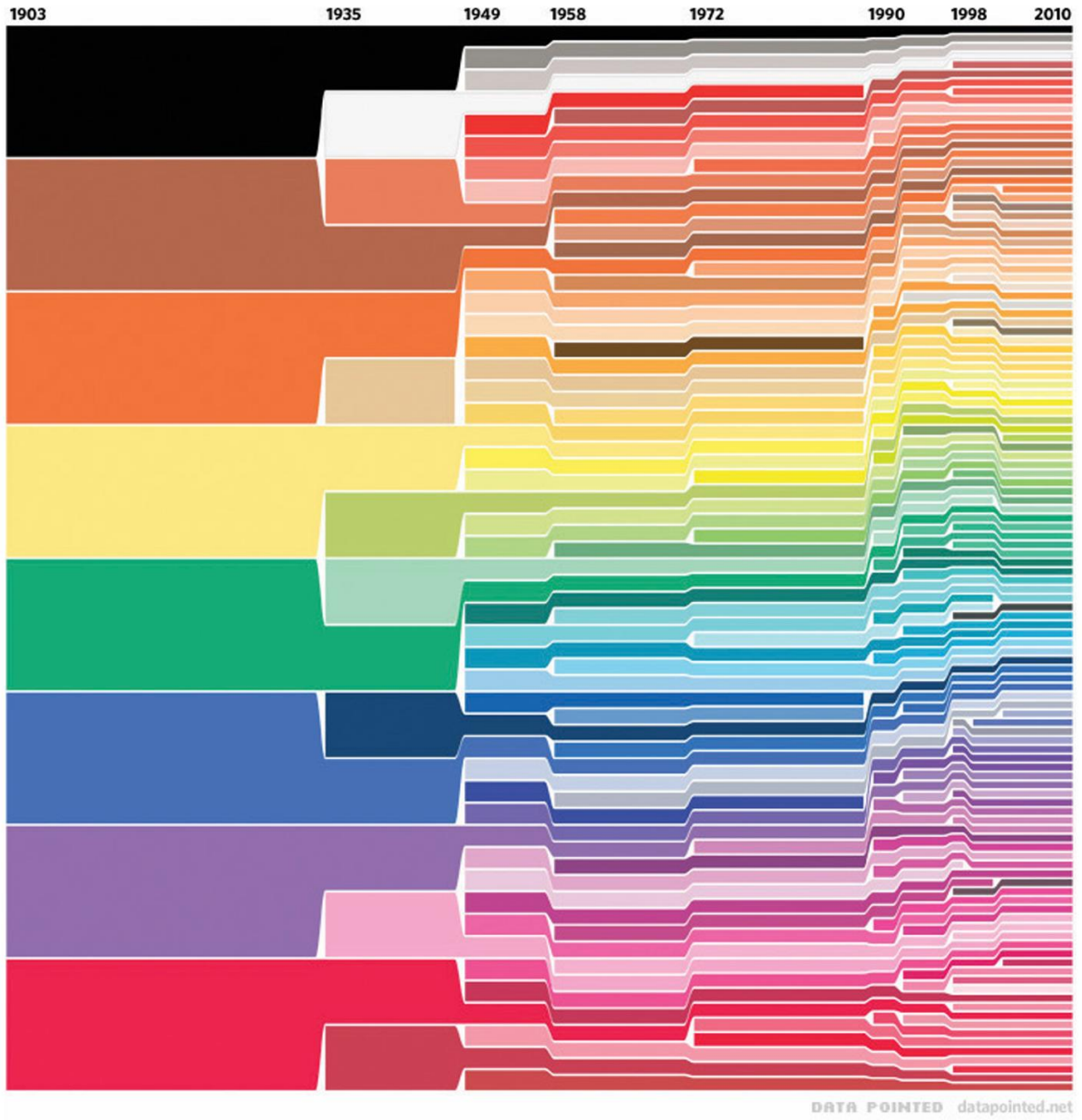
Data Visualization Selection



Source: ©A. Abela, 2010. www.ExtremePresentation.com



Color


1903
1935
1949
1958
1972
1990
1998
2010
DATA POINTED datapointed.net

Color Scale Options

Sequential

The same or similar hues are used, and saturation varies for a single metric.



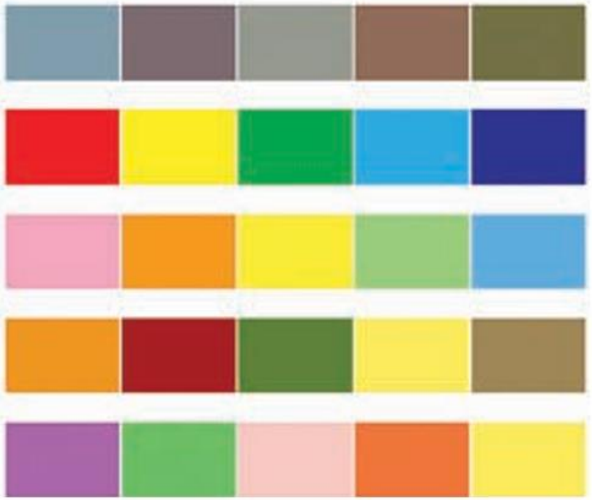
Diverging

Two hues are used to indicate a division, such as positive and negative values.



Qualitative

When data is non-numeric, contrasting colors are used for each category.



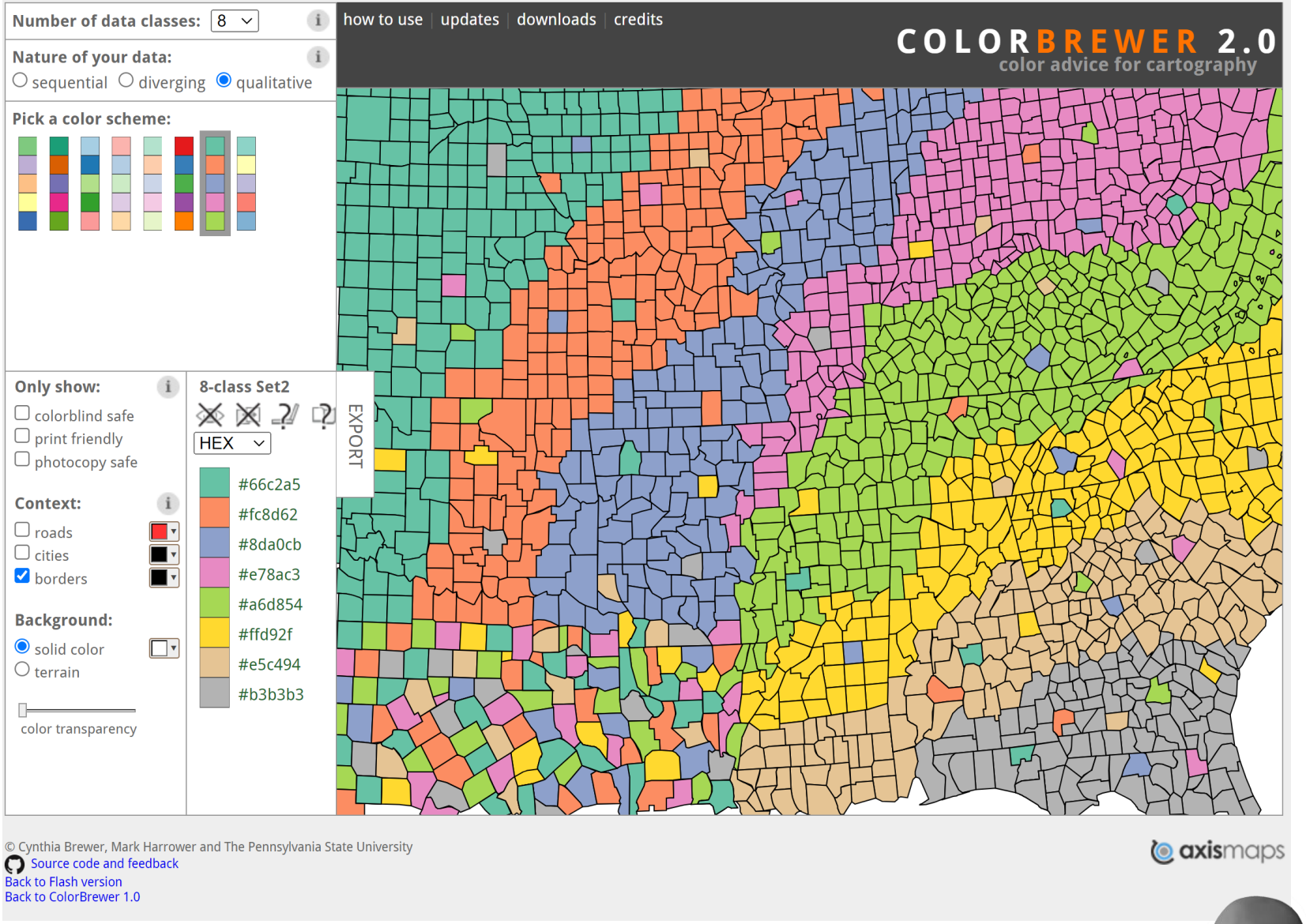


Colorbrewer

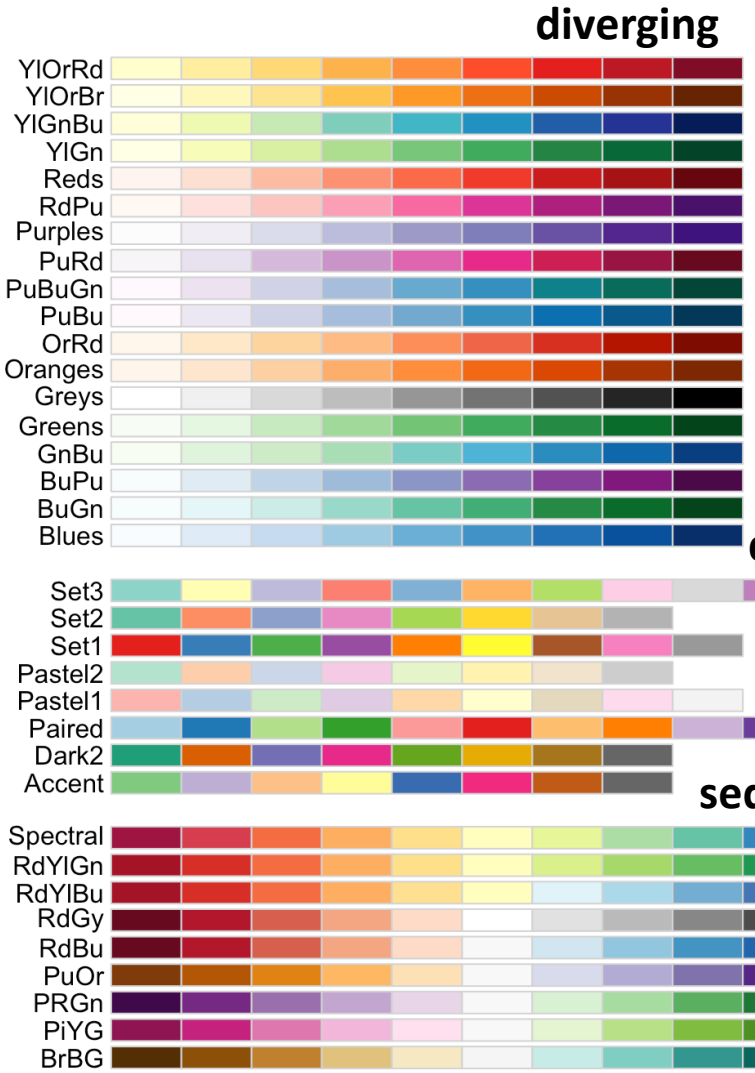
ColorBrewer is an online tool for selecting map color schemes based on palettes created by Cynthia Brewer It was launched in 2002 by Brewer, Mark Harrower, and The Pennsylvania State University.

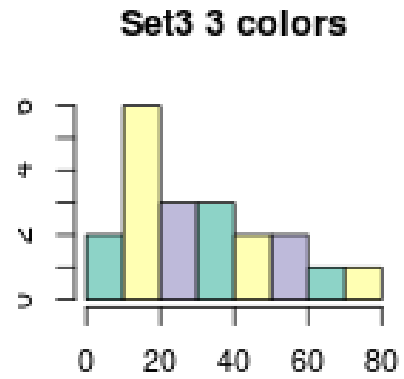
Suggested color schemes are based on data type (sequential, diverging, or qualitative). It also provides options for varied display environments, such as laptop, photocopy, and LCD projector, and colorblind safe options.

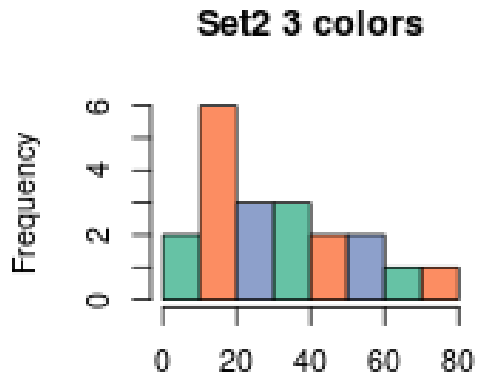
https://colorbrewer2.org

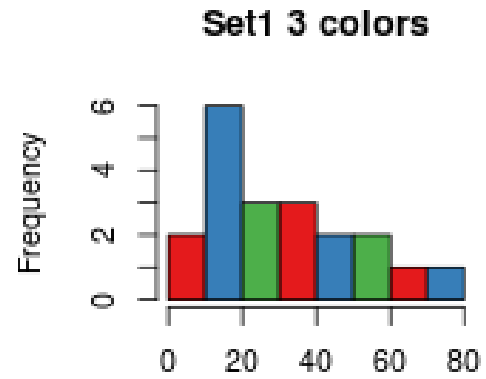


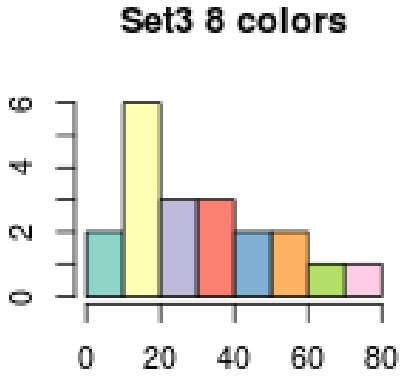
Color Palettes

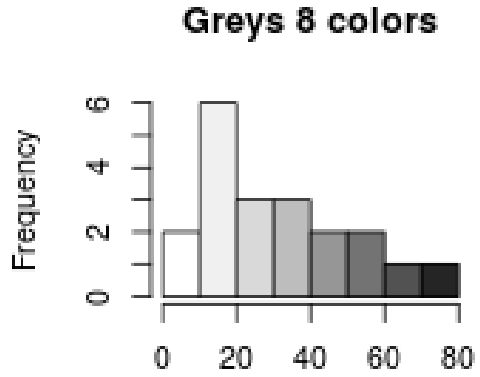


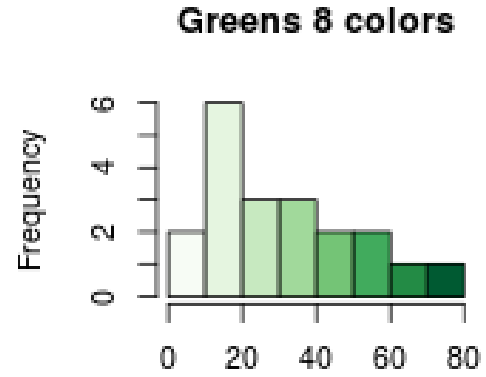




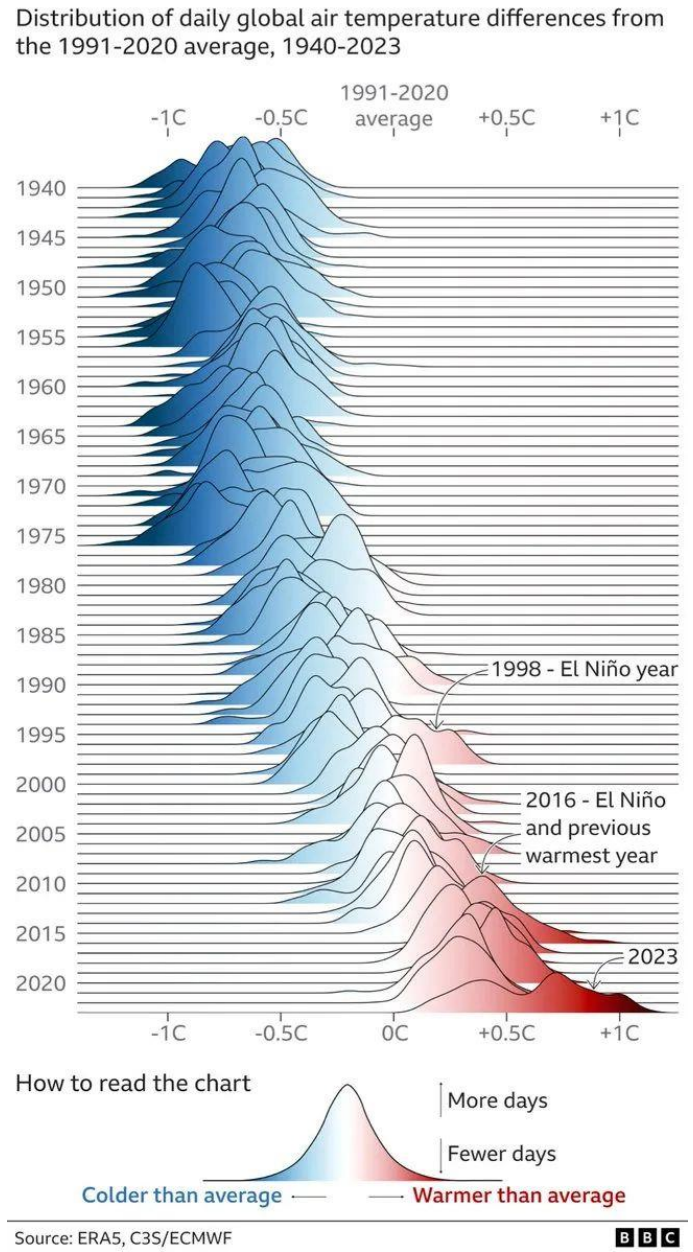








Process

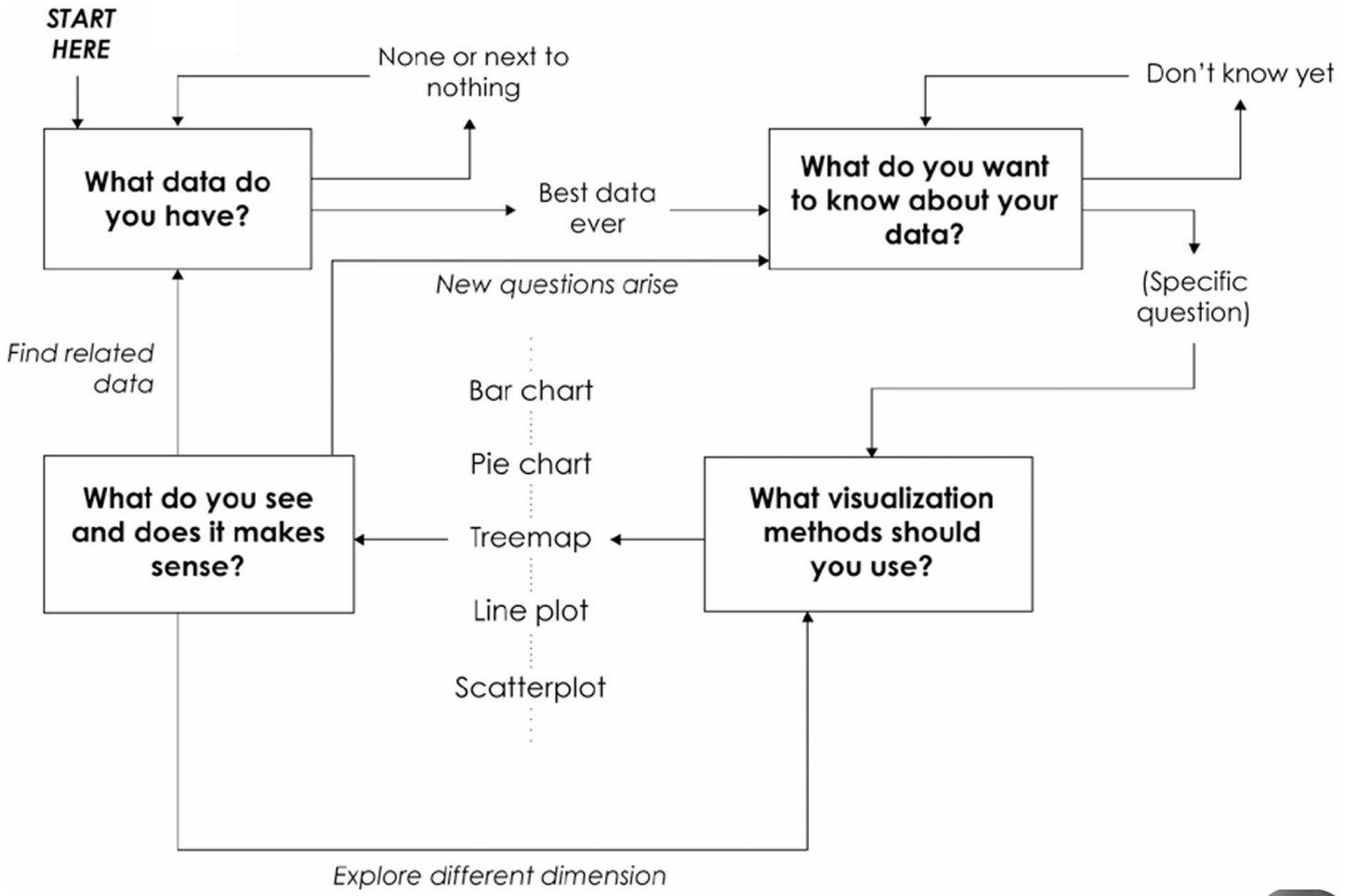

Distribution of daily global air temperature differences from the 1991-2020 average, 1940-2023
1991-2020 average
-1C
-0.5C
+0.5C
+1C
1940
1945
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
1998 - El Niño year
2016 - El Niño and previous warmest year
2023
-1C
-0.5C
0C
+0.5C
+1C
How to read the chart
More days
Fewer days
Colder than average
Warmer than average
Source: ERA5, C3S/ECMWF
BBC

Data Visualization Process

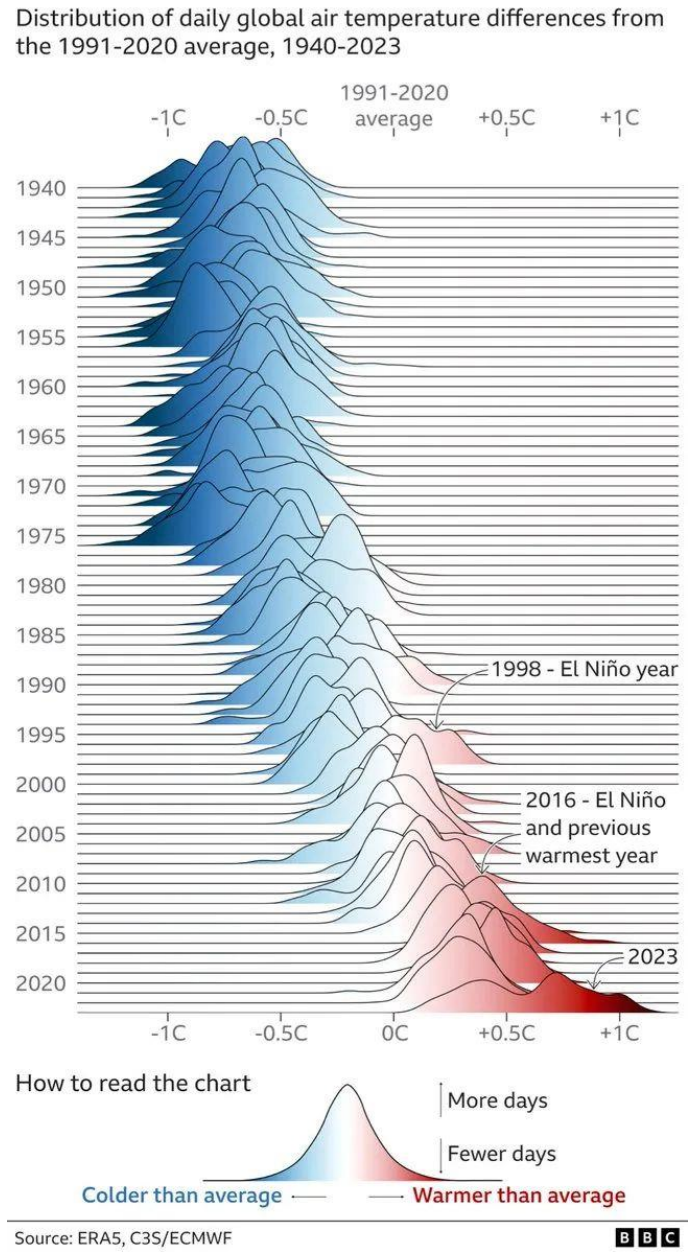
The iterative data exploration process

The specific steps you take in any analysis varies by dataset and project, but generally speaking, you should consider these four things when you explore your data visually.

- ☐ What data do you have?
- ☐ What do you want to know about your data?
- ☐ What visualization methods should you use?
- ☐ What do you see and does it makes sense?



Case Study


Distribution of daily global air temperature differences from the 1991-2020 average, 1940-2023
1991-2020 average
-1C
-0.5C
+0.5C
+1C
1940
1945
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
1998 - El Niño year
2016 - El Niño and previous warmest year
2023
-1C
-0.5C
0C
+0.5C
+1C
How to read the chart
More days
Fewer days
Colder than average
Warmer than average
Source: ERA5, C3S/ECMWF
BBC

Perspectives on the Use of English as a Business Lingua Franca in Hong Kong

Journal of Business Communication
50(3) 227–252
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DOI: 10.1177/0021943613487073
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Stephen Evans[1]

Abstract

This article presents the findings of a large-scale, multifaceted investigation into the use of English as a business lingua franca in Hong Kong's key service industries. The findings were derived from four sources: semistructured interviews with 28 Chinese professionals; four "week-in-the-life" case studies, which include an all-day office observation; analyses of telephone conference recordings and email chains; and a questionnaire survey involving more than 2,000 respondents. The evidence suggests that English, particularly its written form, plays a crucial role in business communication, although the nature and extent of its use vary according to an array of institutional and individual factors, such as a company's ownership and a professional's duties. The qualitative data illustrate the interplay between the two written (English, Chinese) and three spoken codes (English, Cantonese, Putonghua) in workplace communication, and particularly the symbiotic relationship between written English and Cantonese.

Keywords

business English, Chinese, Hong Kong, lingua franca, professional communication

Introduction

The emergence of English as the leading lingua franca of international business in the past two decades has been accompanied by a significant upsurge of scholarly interest in a range of issues relating to language choice and use in the rapidly globalizing business and professional worlds. As might be expected, much of the research in this diffuse and diverse area has been conducted in a constellation of interlocking disciplines whose principal foci of inquiry are questions of language and communication, namely,

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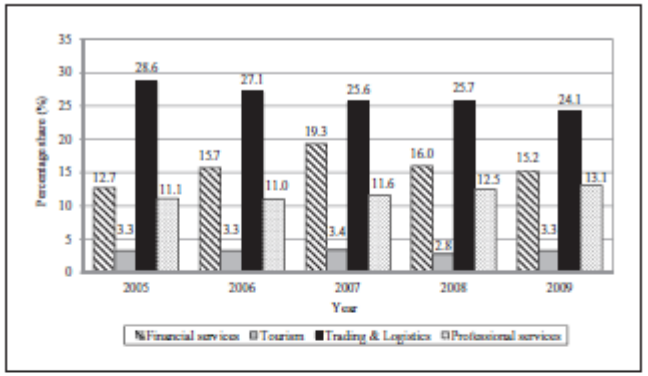


Figure 1. Percentage share of Hong Kong's GDP contributed by the four key industries (2005–2009).
Source. Census & Statistics Department (2011a).

trade and compelled the local business community to restructure the economy in favor of export-led manufacturing industries, notably textiles, clothing, electronics, plastics, and toys (Liu, 1997). By the early 1970s, the manufacturing sector accounted for 28.2% of GDP and 47% of total employment, and was largely responsible, through the ubiquitous "Made in Hong Kong" label, for the city's growing reputation as an economic powerhouse (Chiu & Lui, 2009). The second transformation was triggered by the reform and reopening of China's economy in the late 1970s, which resulted in the relocation of Hong Kong's manufacturing industries to southern China and a shift toward a service-based economy during the 1980s (Li, 2006). Such has been the scale of this restructuring in the past 30 years that manufacturing now accounts for only 1.8% of GDP and 3.5% of total employment, whereas the service sector contributes 92.6% to GDP and employs some 88.3% of Hong Kong's workforce (Hong Kong Government, 2011). The four "key" industries within the service sector are the focus of the present study, since these have been the major motors of the city's growth in recent years (Figure 1) and thus the subject of considerable government attention (Sung, 2008).

Although the study focuses on the role of English in these industries, as we shall see, this role is best appreciated if it is viewed in relation to other languages in Hong Kong's multilingual business world, notably written Chinese, Cantonese (the first language of most local professionals), and Putonghua. In the past three decades, the relative status and use of these languages have been influenced by three powerful processes: national reunification, regional integration, and economic globalization (Qin, 2007). The first process, which was initiated by the Sino-British Joint Declaration

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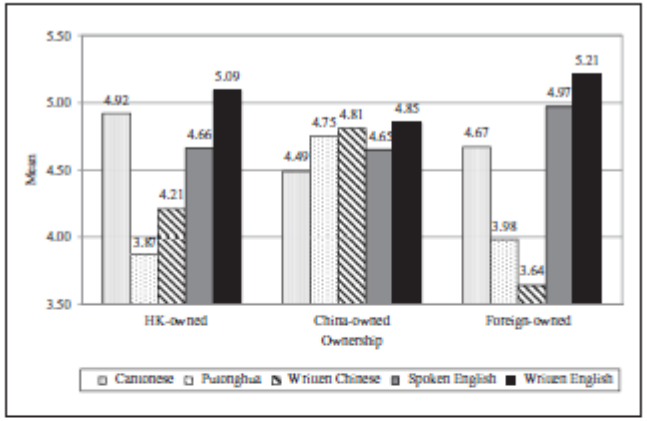


Figure 3. Perceived importance of languages by company ownership.
Note. Scale: 1 = completely unimportant, 6 = very important. Analysis of variance results showed that there are significant differences in the perceived importance of the five codes between the three categories of company ($p < .05$).

and 322 service-related professional associations. This version was completed by 1,021 participants (a return rate of around 3.6%). The survey data were analyzed using SPSS (Version 15). The results of analysis of variance are reported in Figures 3 and 4.

Just under half of the respondents (47%) were employed in the professional and business services industry, while the remainder worked in the three other "key" industries (trading and logistics, 15%; financial services, 10%; tourism, 6%) or in service-related professions that do not fall within the ambit of the four key industries (16.6%). Around a quarter of the respondents worked as civil servants (9%) or for government-related organizations (16%). Since this article focuses on the private sector, and particularly patterns of language use in Hong Kong, foreign and mainland companies, the responses of the public-sector respondents have been excluded from the analysis. The findings reported in Figures 3 and 4 are therefore based on the responses of 1,478 professionals, of whom 60% worked for Hong Kong–owned companies, 36% were employed by foreign firms, and 4% worked for Chinese enterprises. A comprehensive account of the administration and findings of the survey is presented in Evans (2010).

Findings and Discussion

This section commences by painting a broad-brush linguistic landscape of Hong Kong's service sector using ownership-based findings from the survey and then

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Evans, etal. (2013) Journal of Business Communication.

Magnitude: Dataset

Table. Percentage share of Hong Kong's GDP contributed by the four key industries (2005-2009). *Source*. Census & Statistics Department (2011a).

Group	2005	2006	2007	2008	2009
Trading & Logistics	28.6	27.1	25.6	25.7	24.1
Financial services	12.7	15.7	19.3	16	15.2
Professional services	11.1	11	11.6	12.5	13.1
Tourism	3.3	3.3	3.4	2.8	3.3

Evans, etal. (2013) Journal of Business Communication.



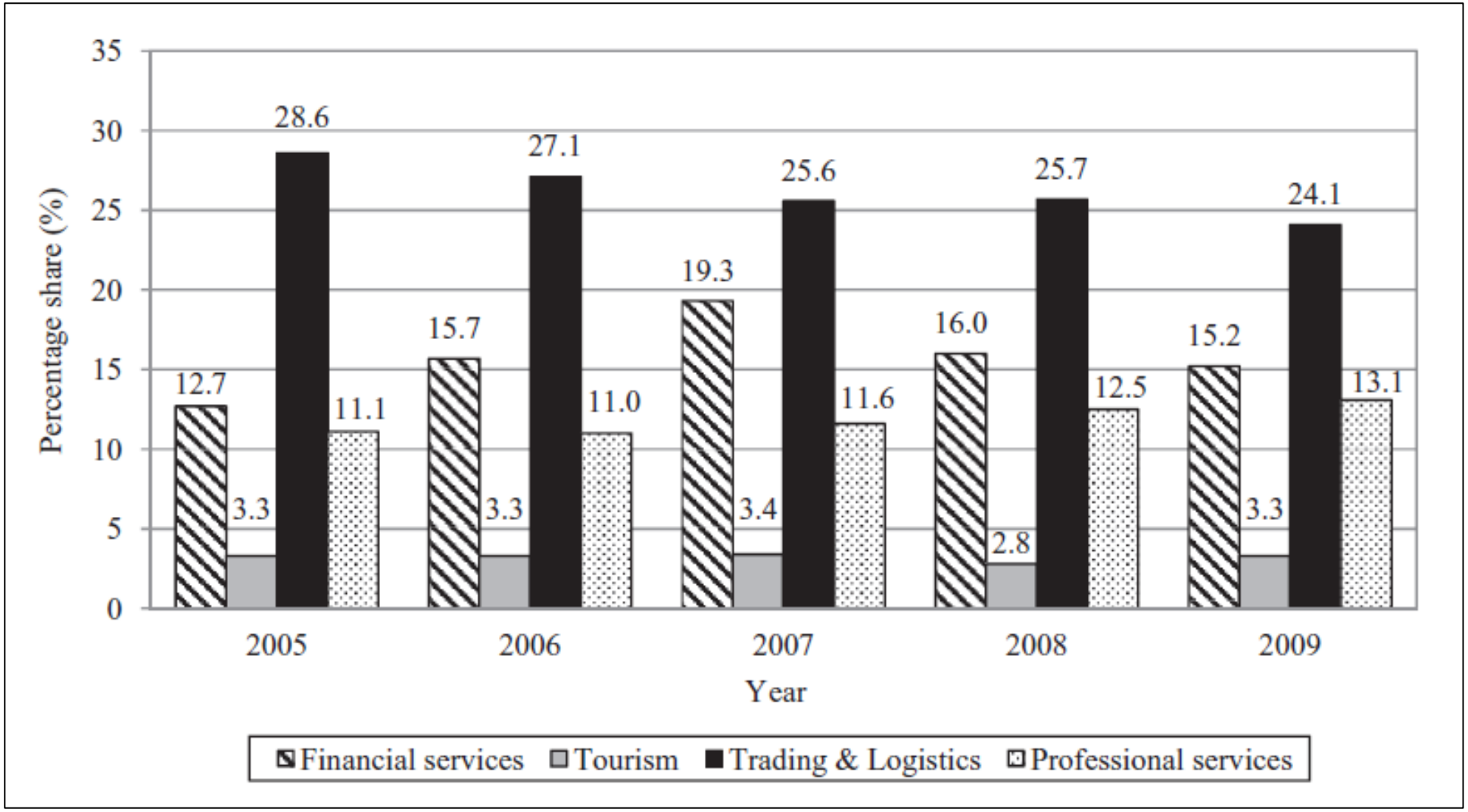
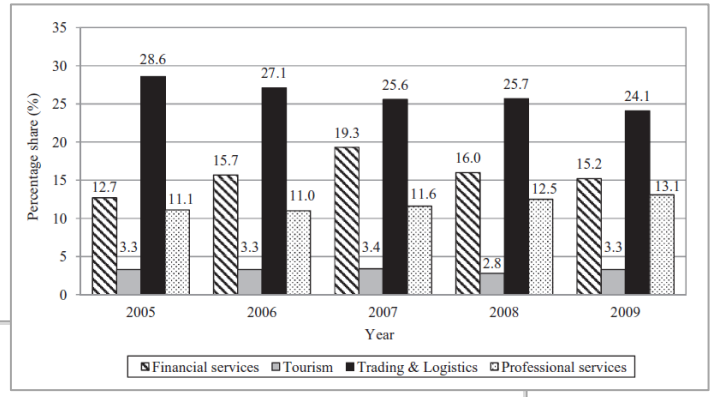
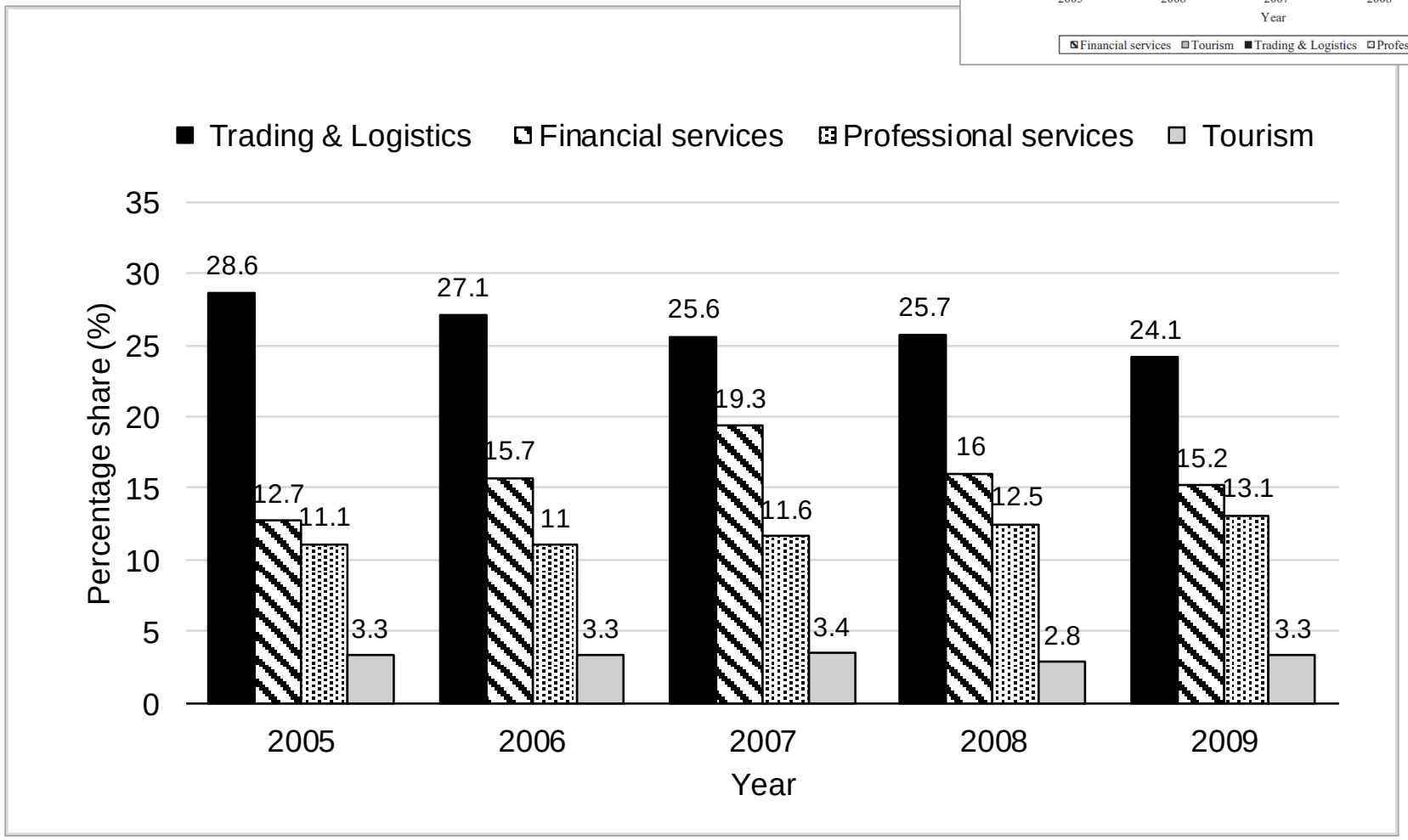
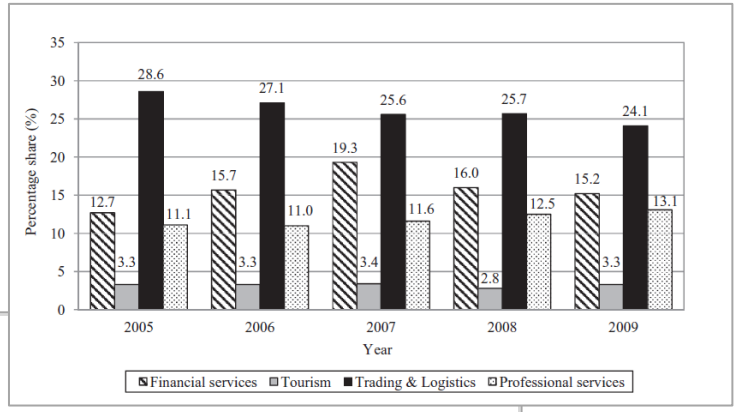


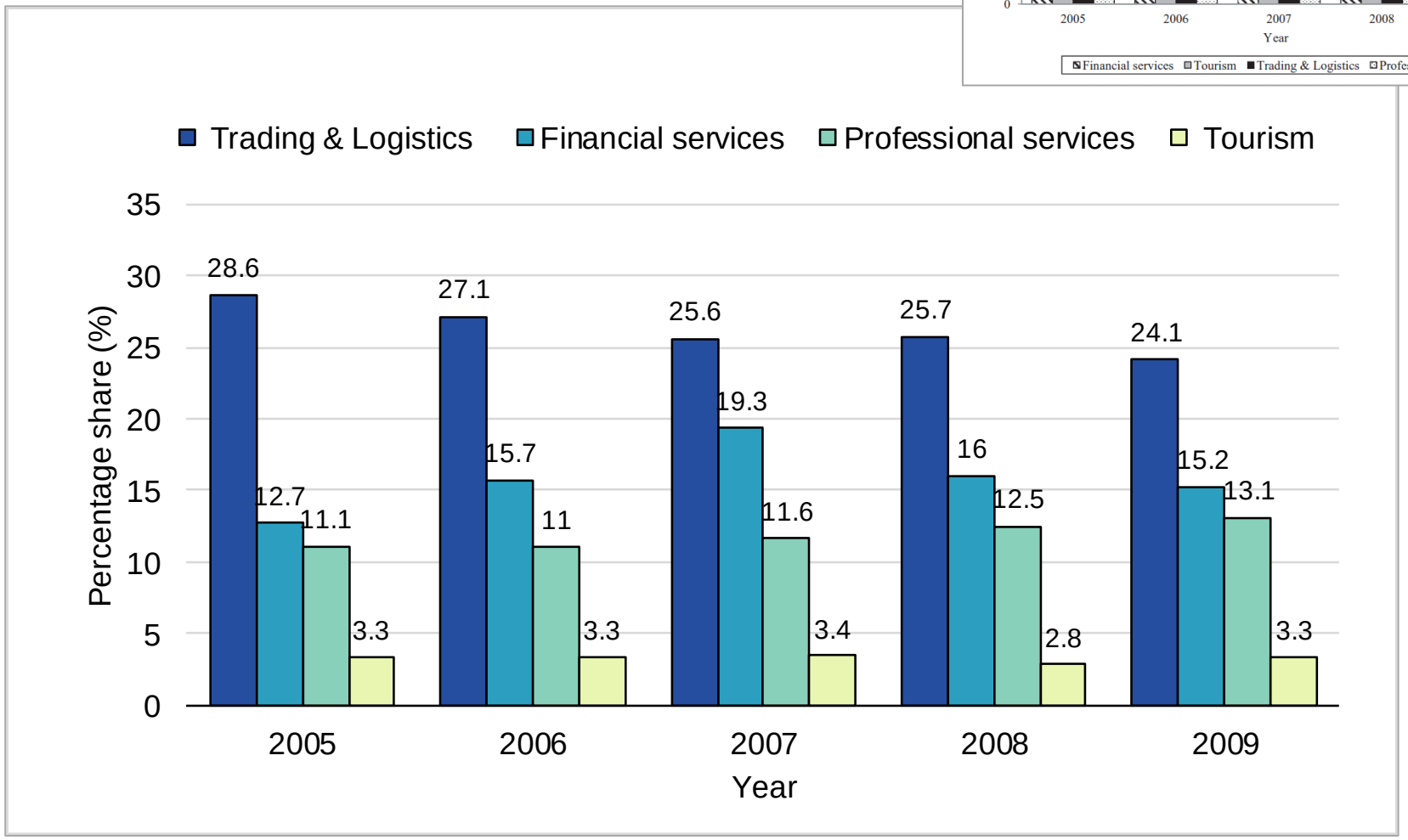
Figure 1. Percentage share of Hong Kong's GDP contributed by the four key industries (2005-2009). *Source*. Census & Statistics Department (2011a).

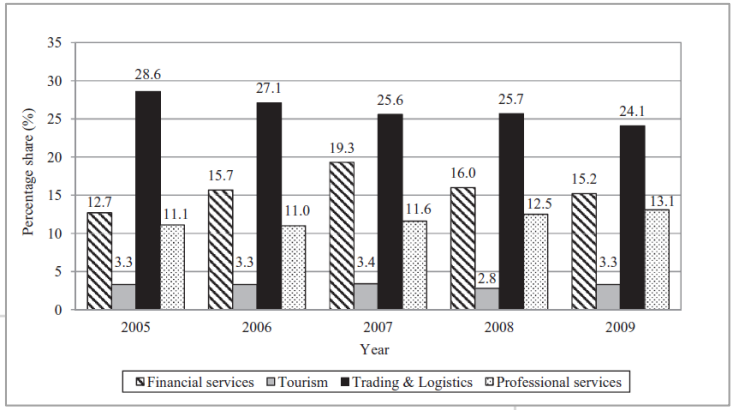
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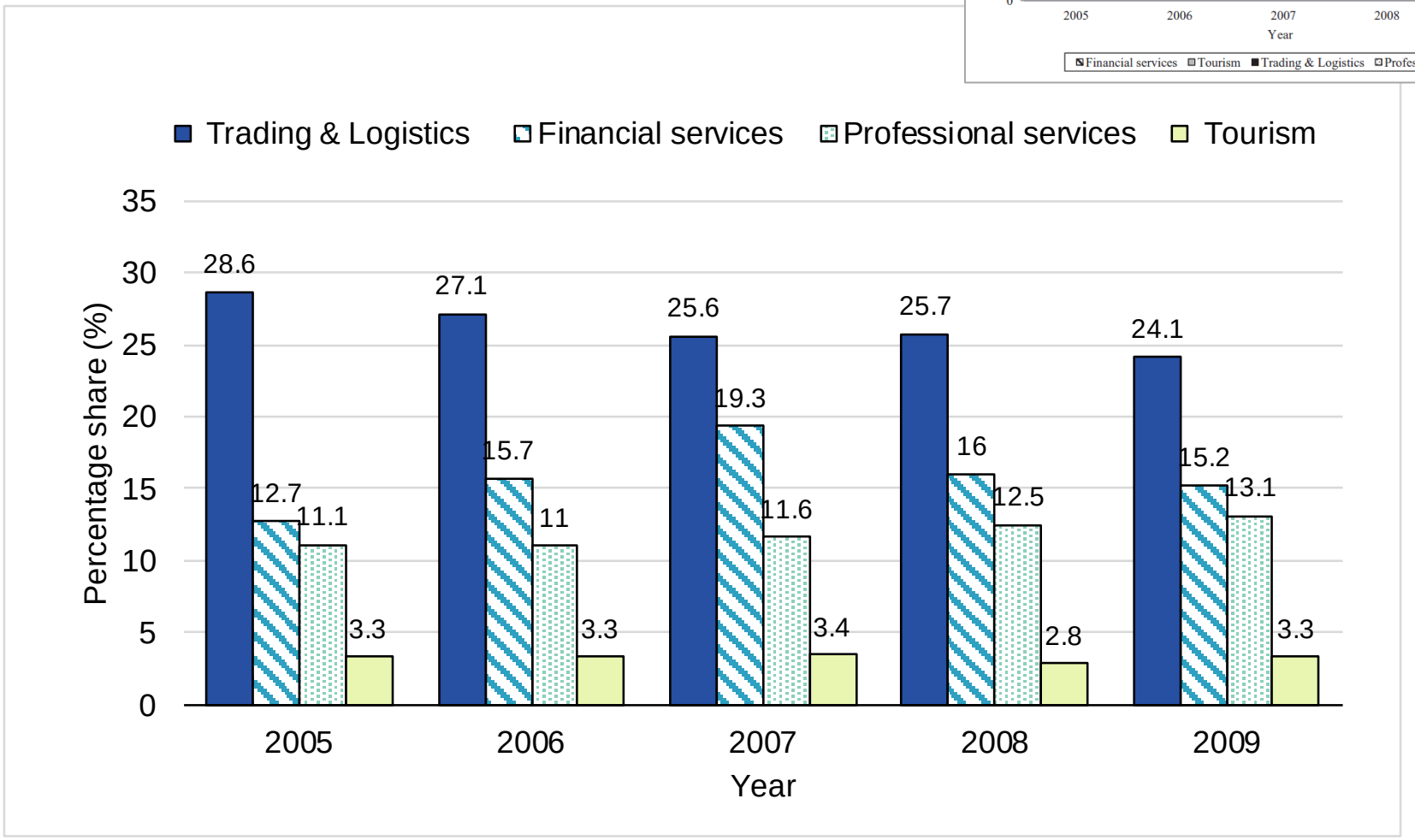




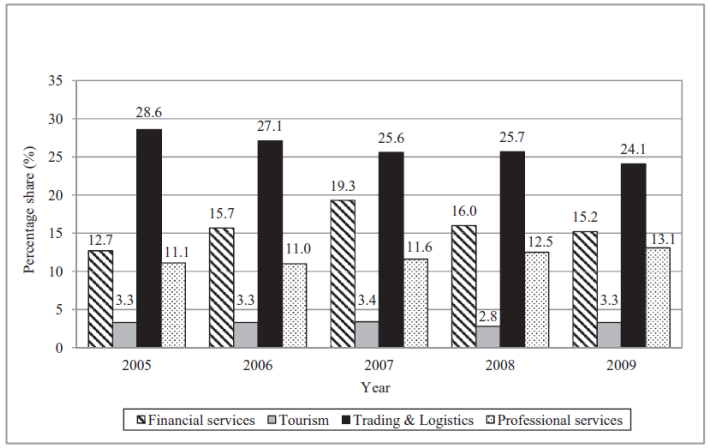

35
30
25
20
15
10
5
0
Percentage share (%)
28.6
27.1
25.6
25.7
24.1
19.3
16.0
15.7
15.2
12.7
11.1
11.0
11.6
12.5
13.1
3.3
3.4
2.8
3.3
2005
2006
2007
2008
2009
Year
Financial services
Tourism
Trading & Logistics
Professional services

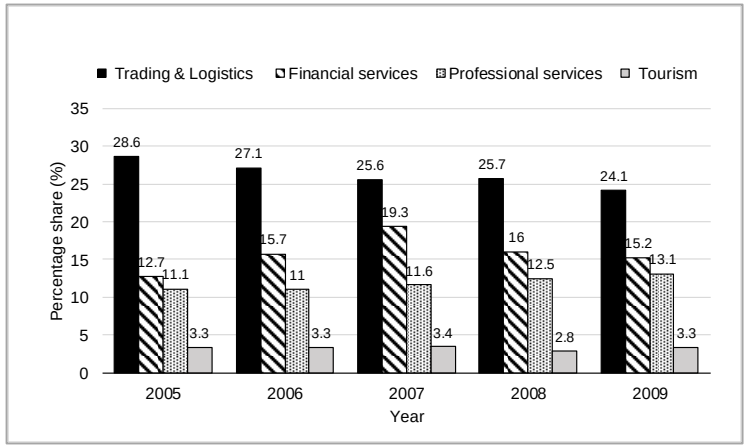

Trading & Logistics
Financial services
Professional services
Tourism
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Percentage share (%)
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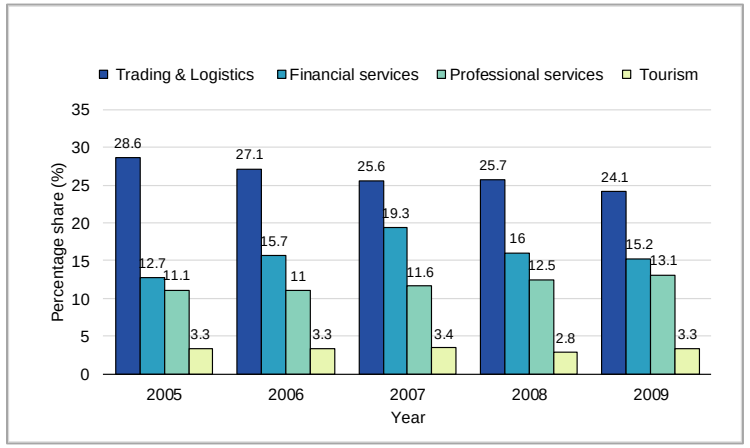

35
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2005
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Trading & Logistics
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16
12.5
2.8
24.1
15.2
13.1
3.3
2005
2006
2007
2008
2009
Year

It is important to sort the data in descending order for categorical data (except for sequential data)







Distribution: Dataset

Table Perceived importance of languages by company ownership. *Note*. Scale: 1 = *completely unimportant*, 6 = *very important*. Analysis of variance results showed that there are significant differences in the perceived importance of the five codes between the three categories of company ($p < .05$).

group	Written English	Cantonese	Spoken English	Written Chinese	Putonghua
HK-owned	4.99 ± 0.61	4.91 ± 0.59	4.69 ± 0.55	4.08 ± 0.69	3.80 ± 0.67
China-owned Ownership	4.73 ± 0.66	4.42 ± 0.58	4.66 ± 0.60	4.73 ± 0.66	4.66 ± 0.66
Foreign-owned	5.17 ± 0.50	4.70 ± 0.59	4.82 ± 0.64	3.69 ± 0.71	3.99 ± 0.67

Evans, etal. (2013) Journal of Business Communication.



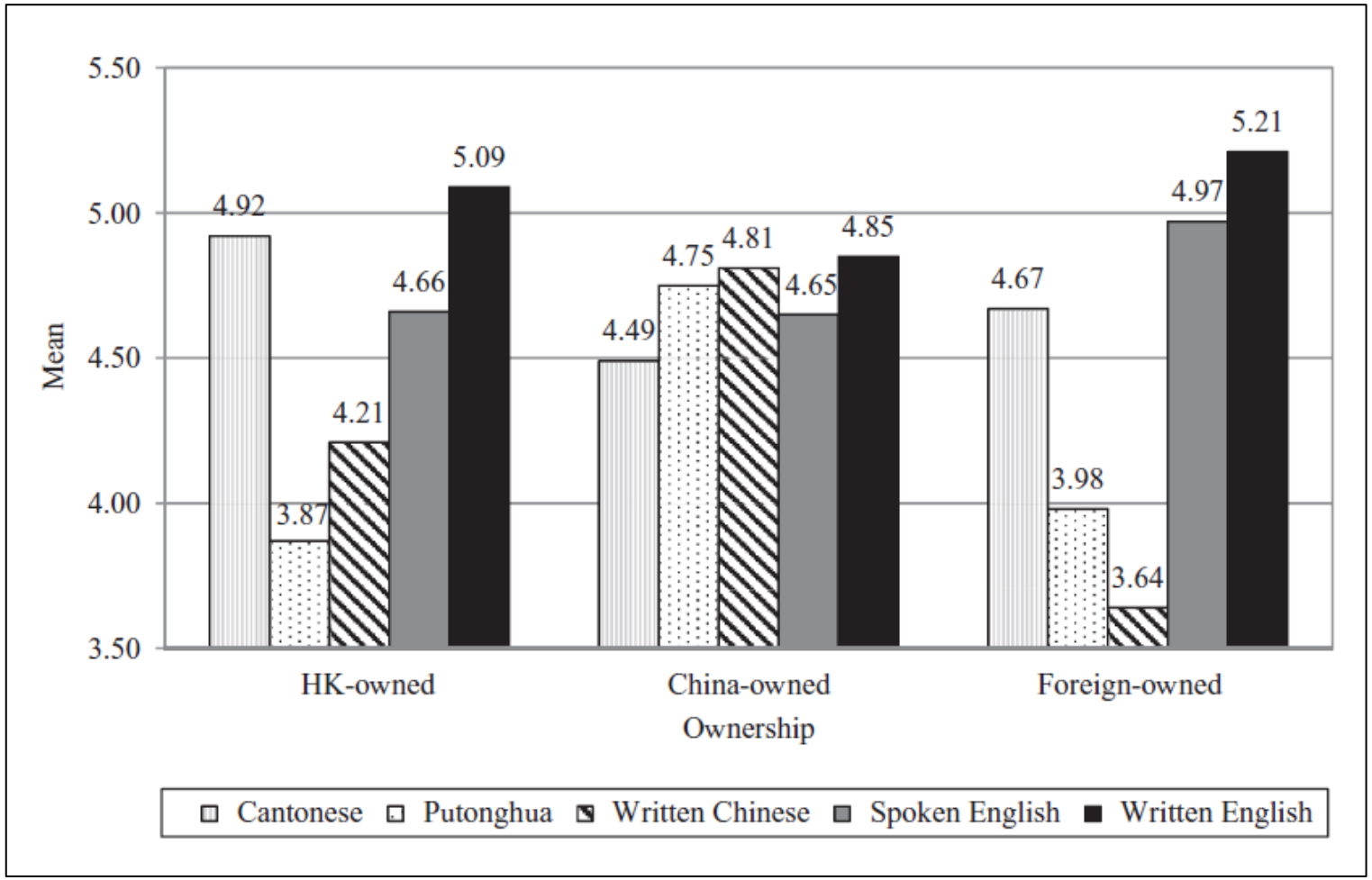
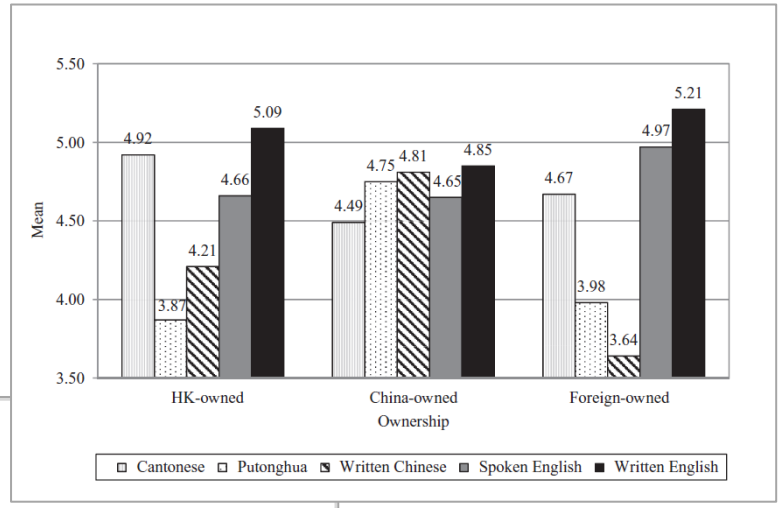
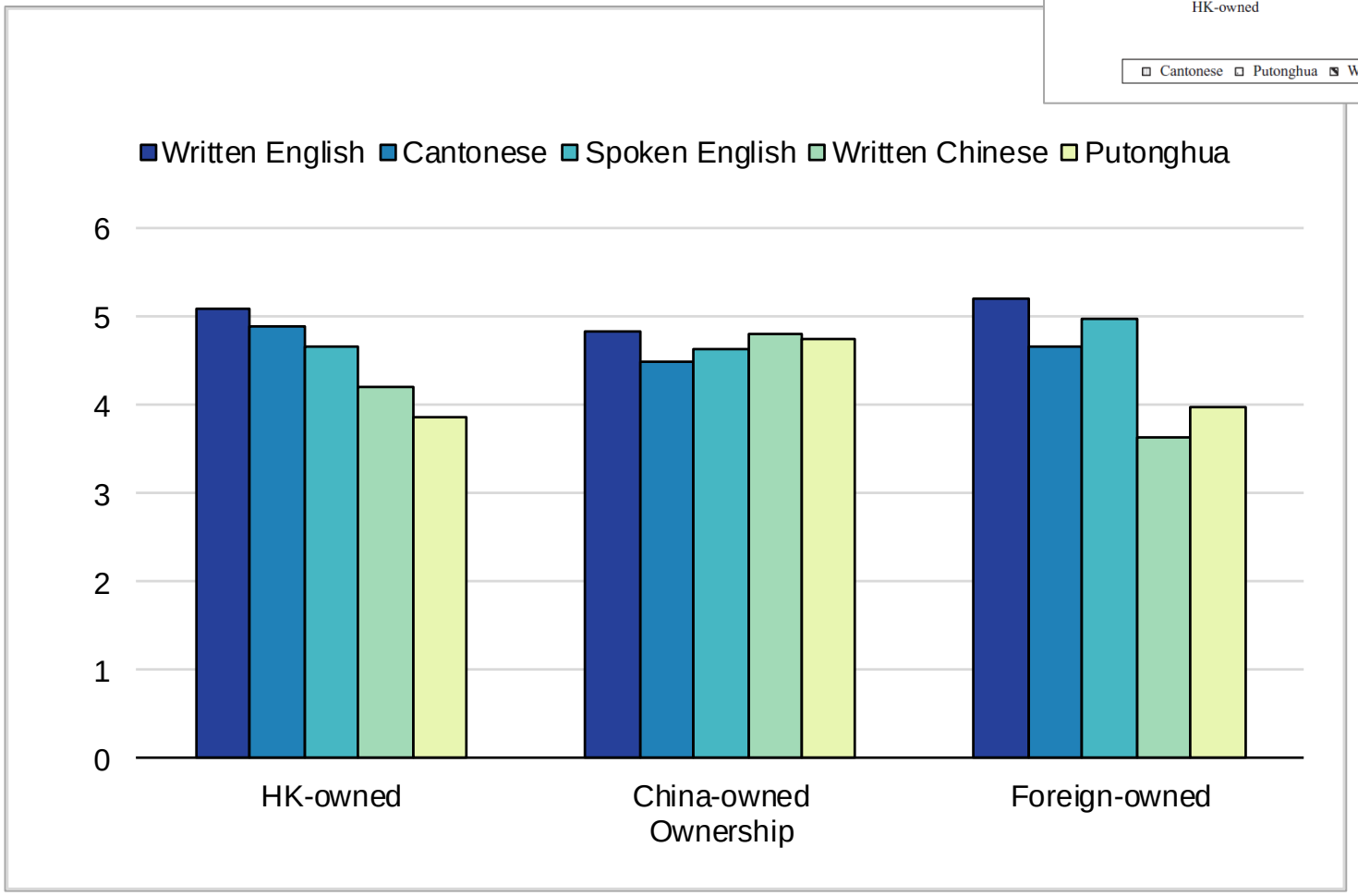


Figure 3. Perceived importance of languages by company ownership. *Note*. Scale: 1 = *completely unimportant*, 6 = *very important*. Analysis of variance results showed that there are significant differences in the perceived importance of the five codes between the three categories of company ($p < .05$).

Evans, etal. (2013) Journal of Business Communication.


5.50
5.00
4.50
4.00
3.50
Mean
4.92
3.87
4.21
4.66
5.09
4.49
4.75
4.81
4.65
4.85
4.67
3.98
3.64
4.97
5.21
HK-owned
China-owned
Foreign-owned
Ownership
Cantonese
Putonghua
Written Chinese
Spoken English
Written English


Written English
Cantonese
Spoken English
Written Chinese
Putonghua
6
5
4
3
2
1
0
HK-owned
China-owned
Foreign-owned
Ownership

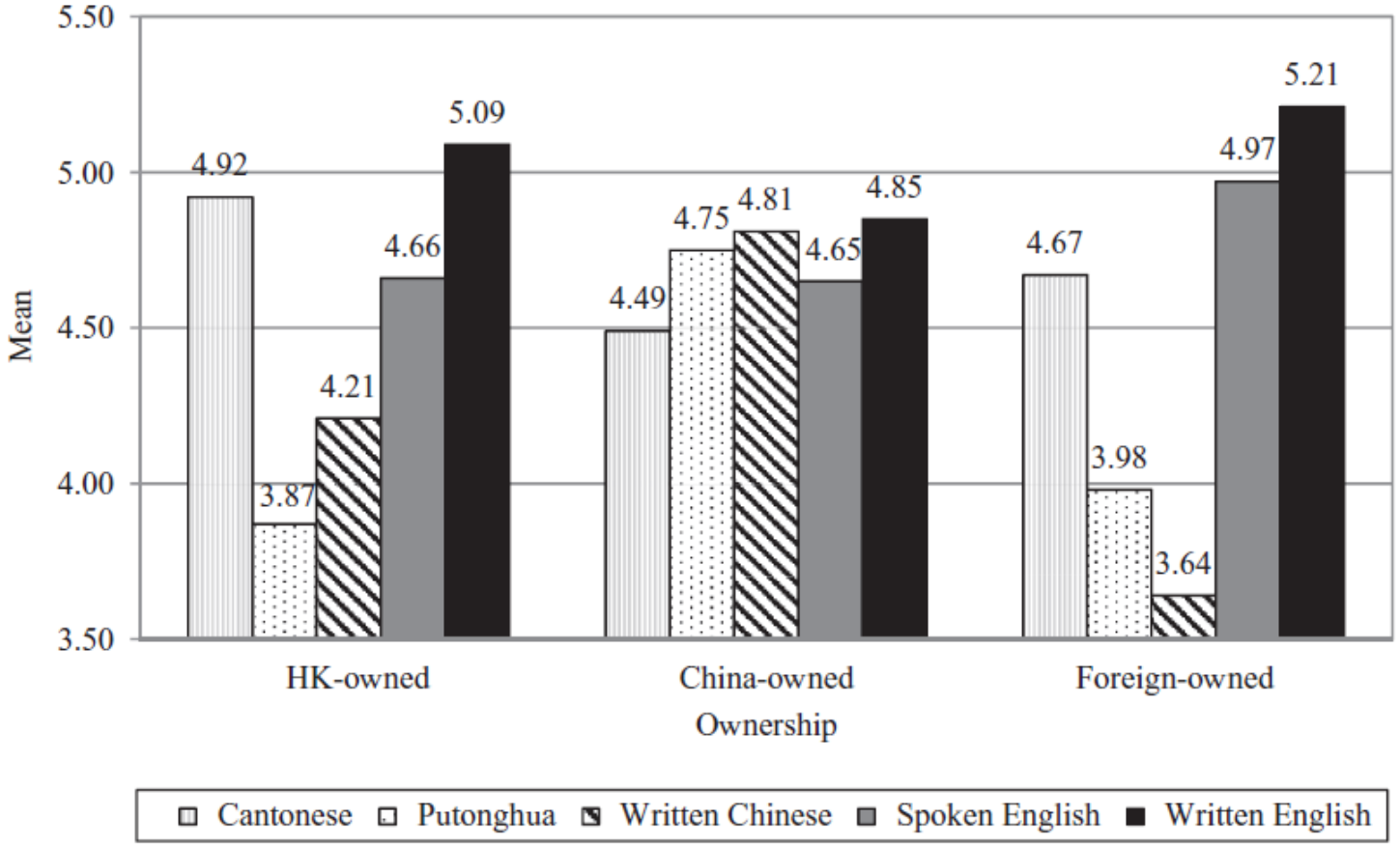
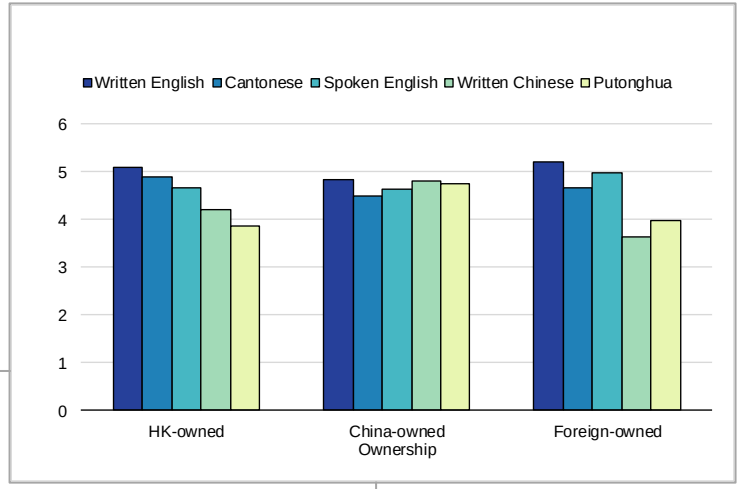
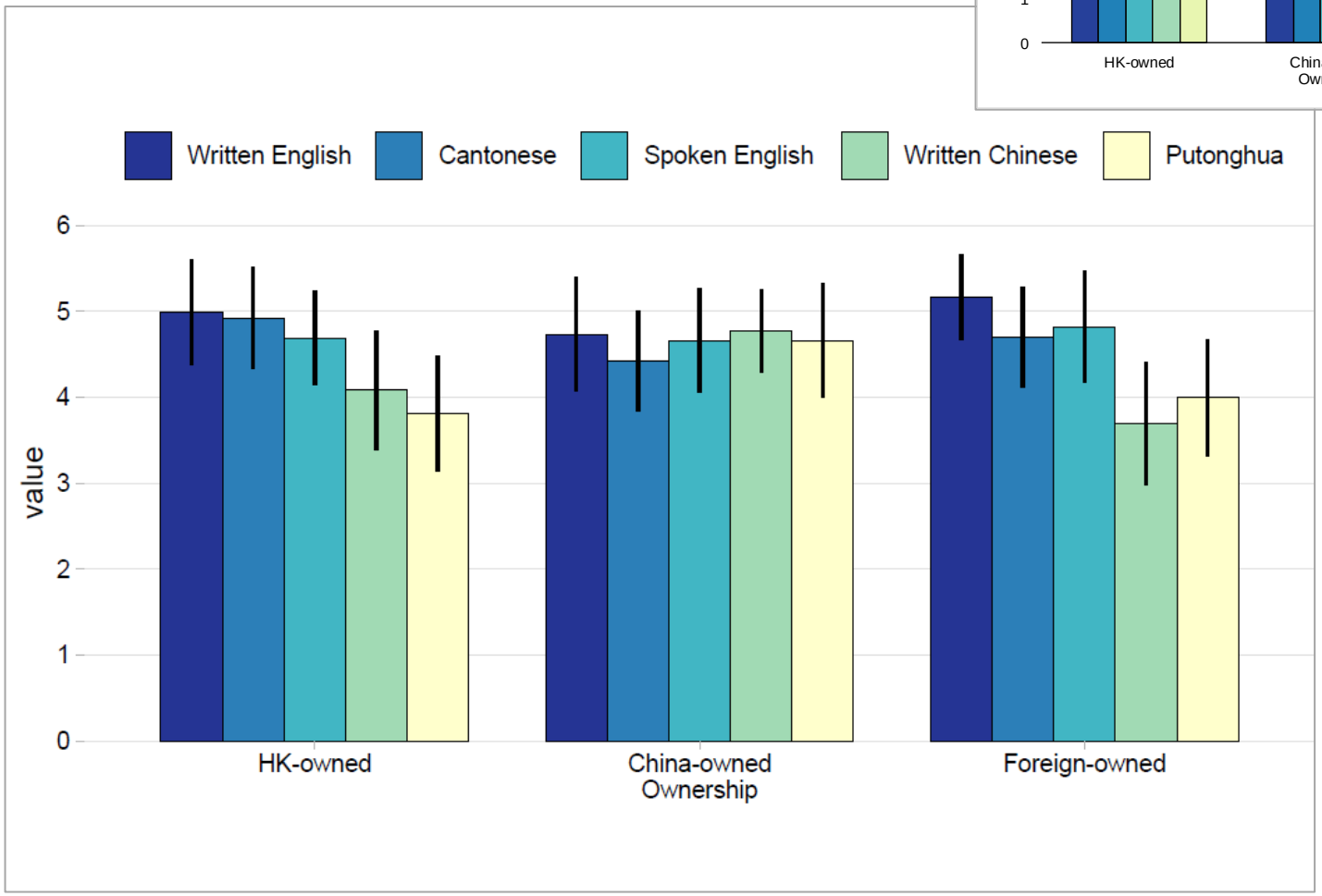


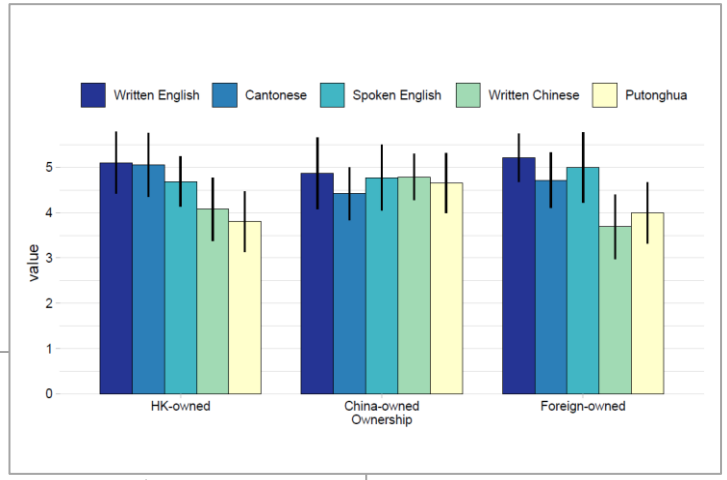
Figure 3. Perceived importance of languages by company ownership. *Note*. Scale: 1 = *completely unimportant*, 6 = *very important*. **Analysis of variance** results showed that there are significant differences in the perceived importance of the five codes between the three categories of company (***p* < .05**).

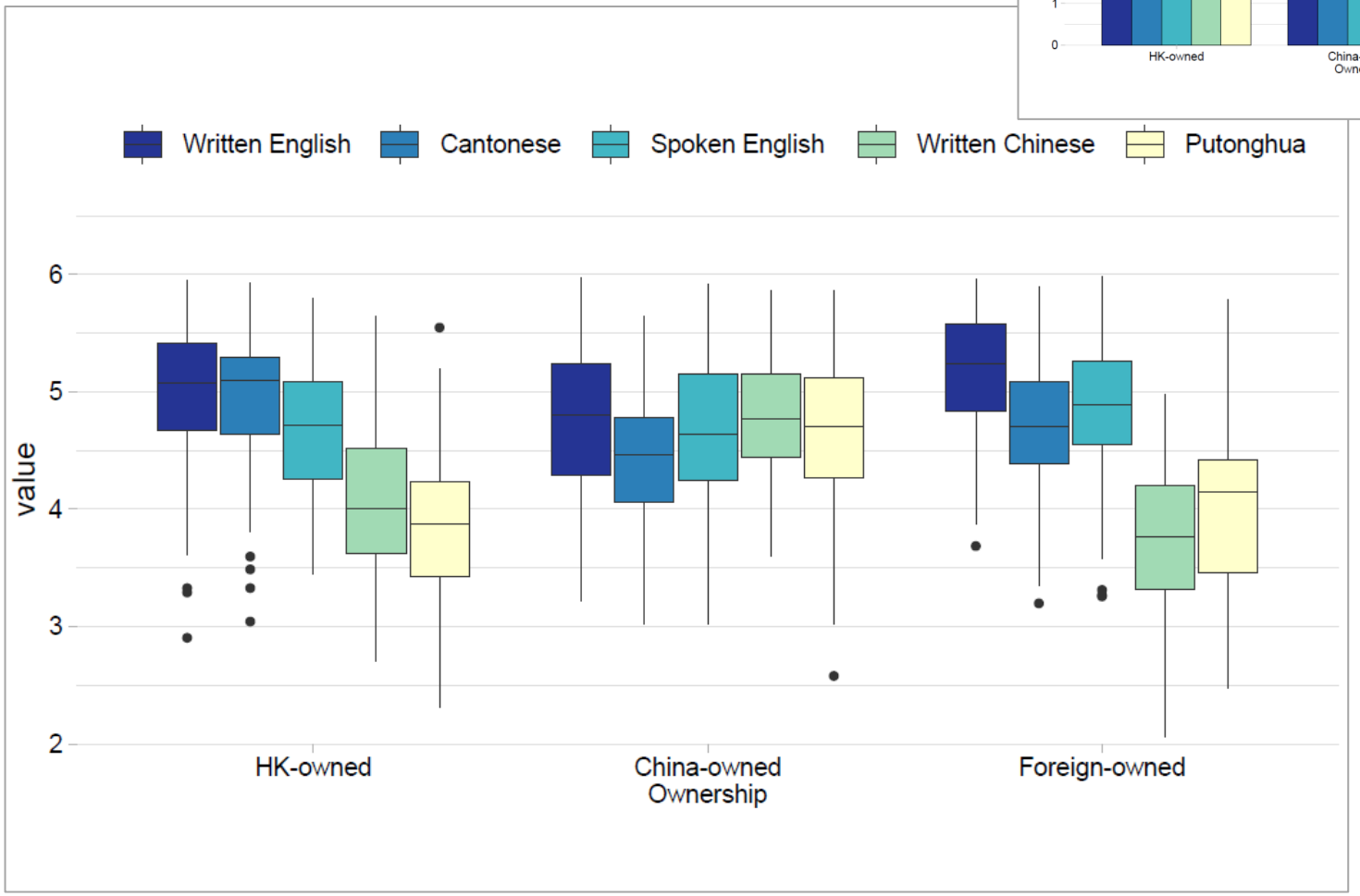
The questionnaire, which came in hard and soft forms, was completed by **2,030** professionals working in Hong Kong's service industries.





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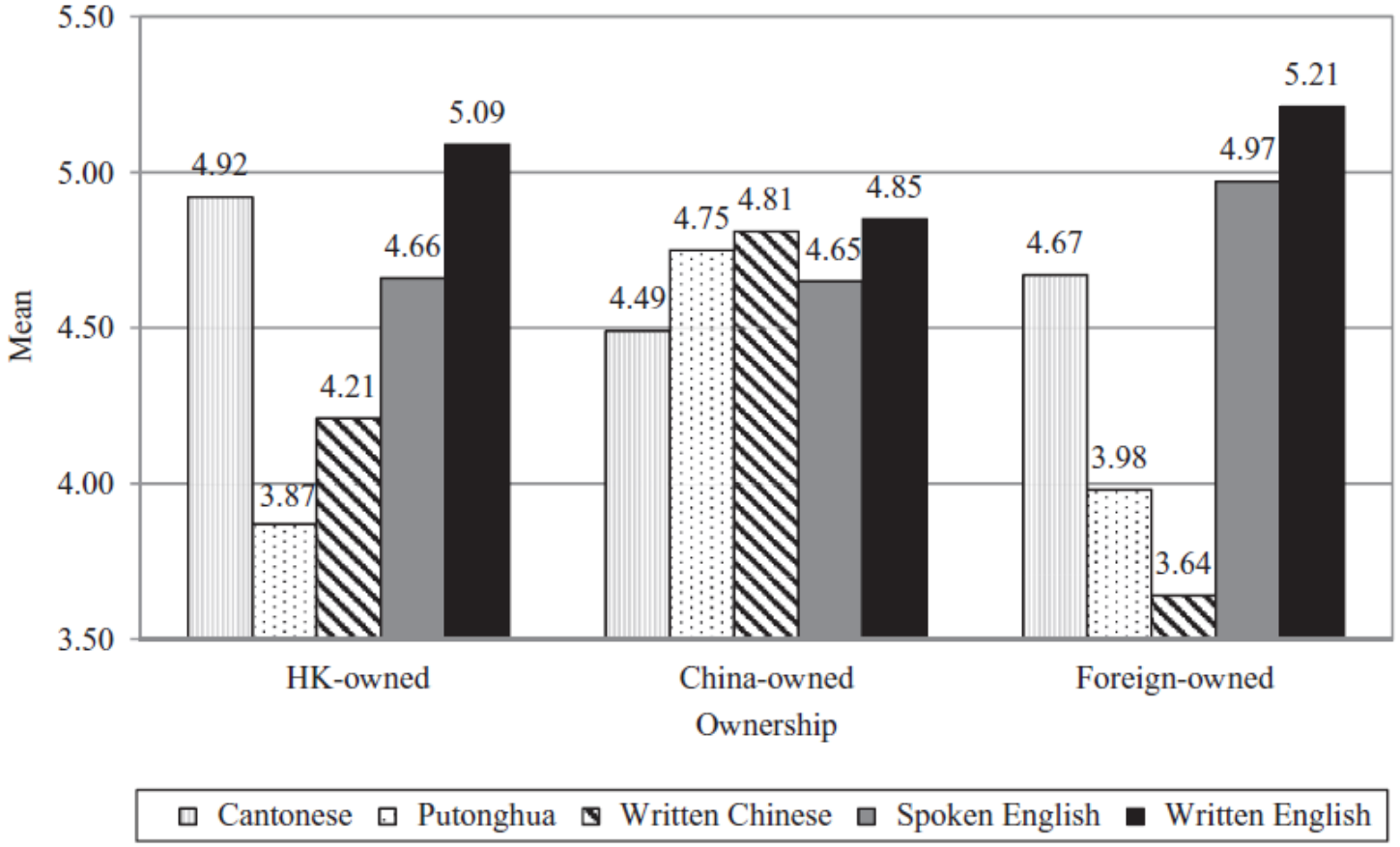
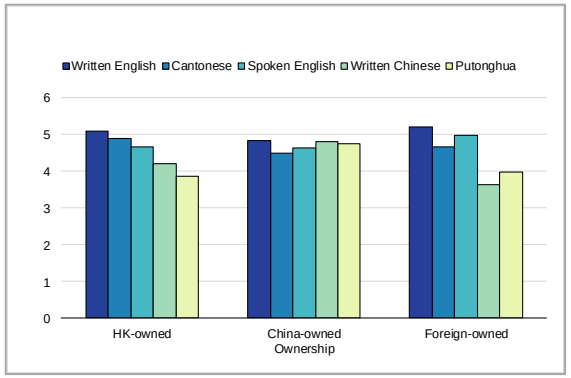
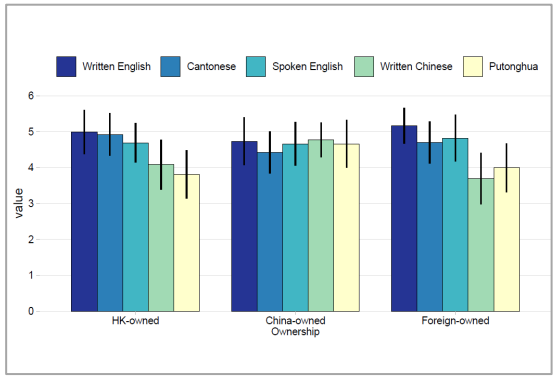
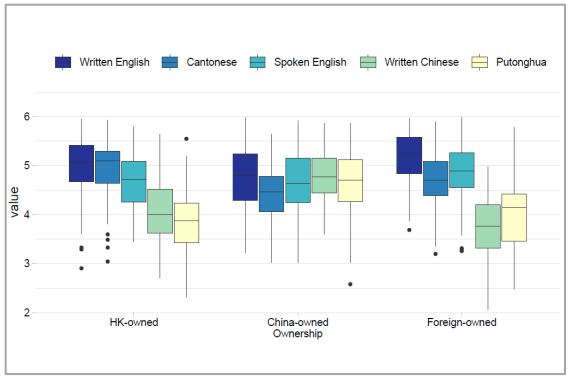
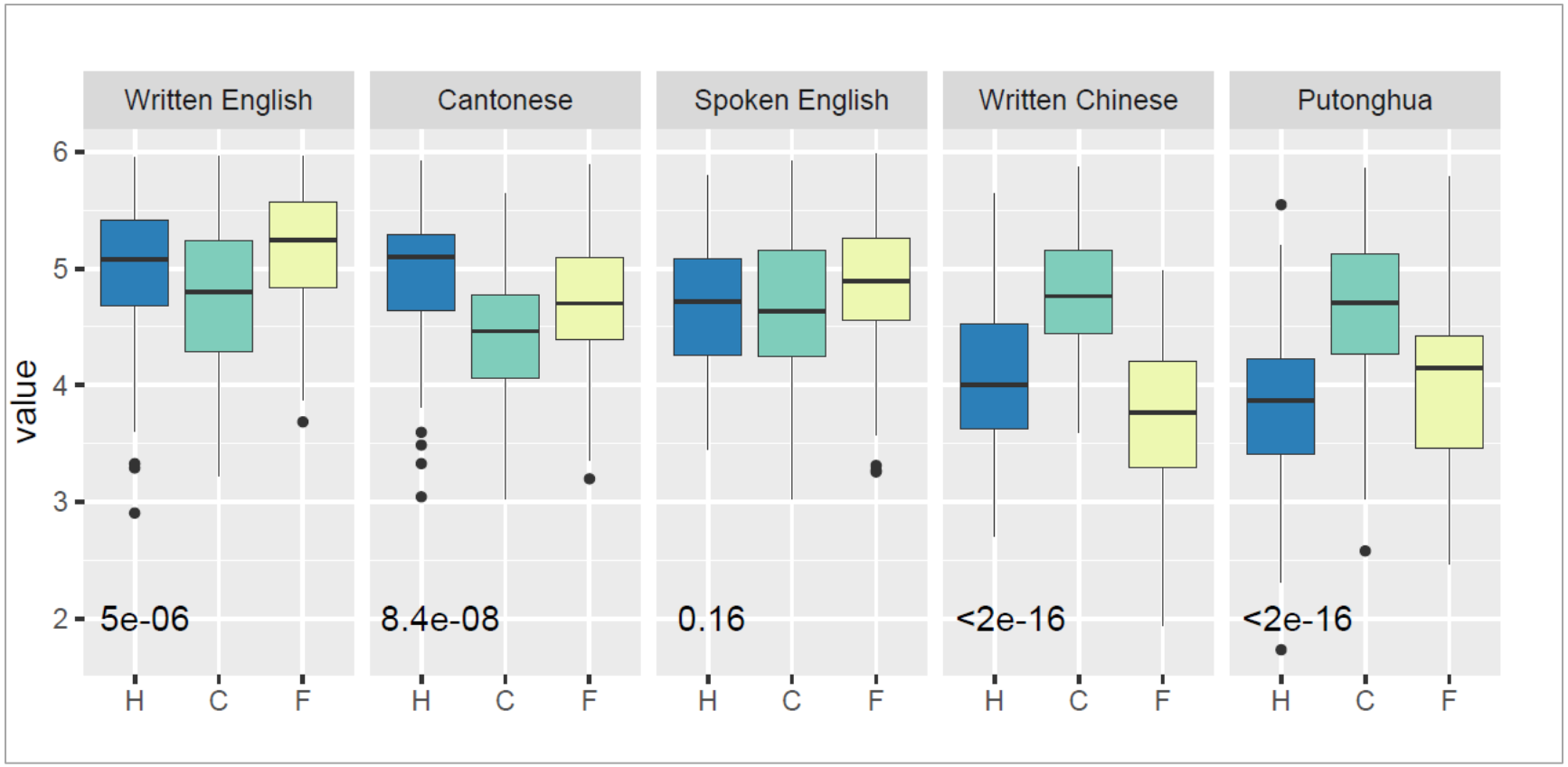


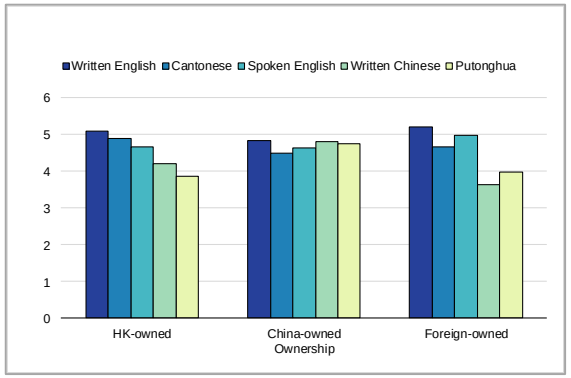
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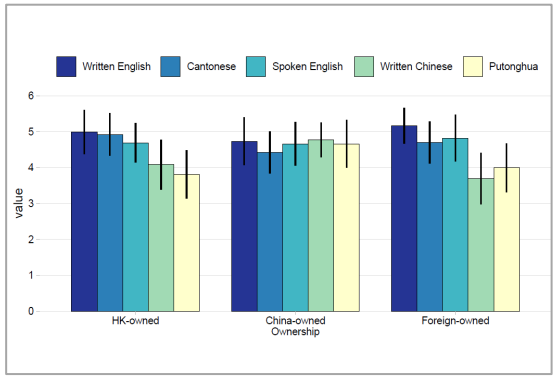

Written English
Cantonese
Spoken English
Written Chinese
Putonghua
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HK-owned
China-owned
Ownership
Foreign-owned

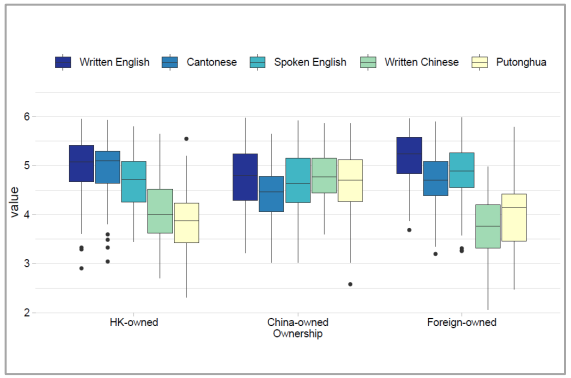

Written English
Cantonese
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value
6
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4
3
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1
0
HK-owned
China-owned
Ownership
Foreign-owned

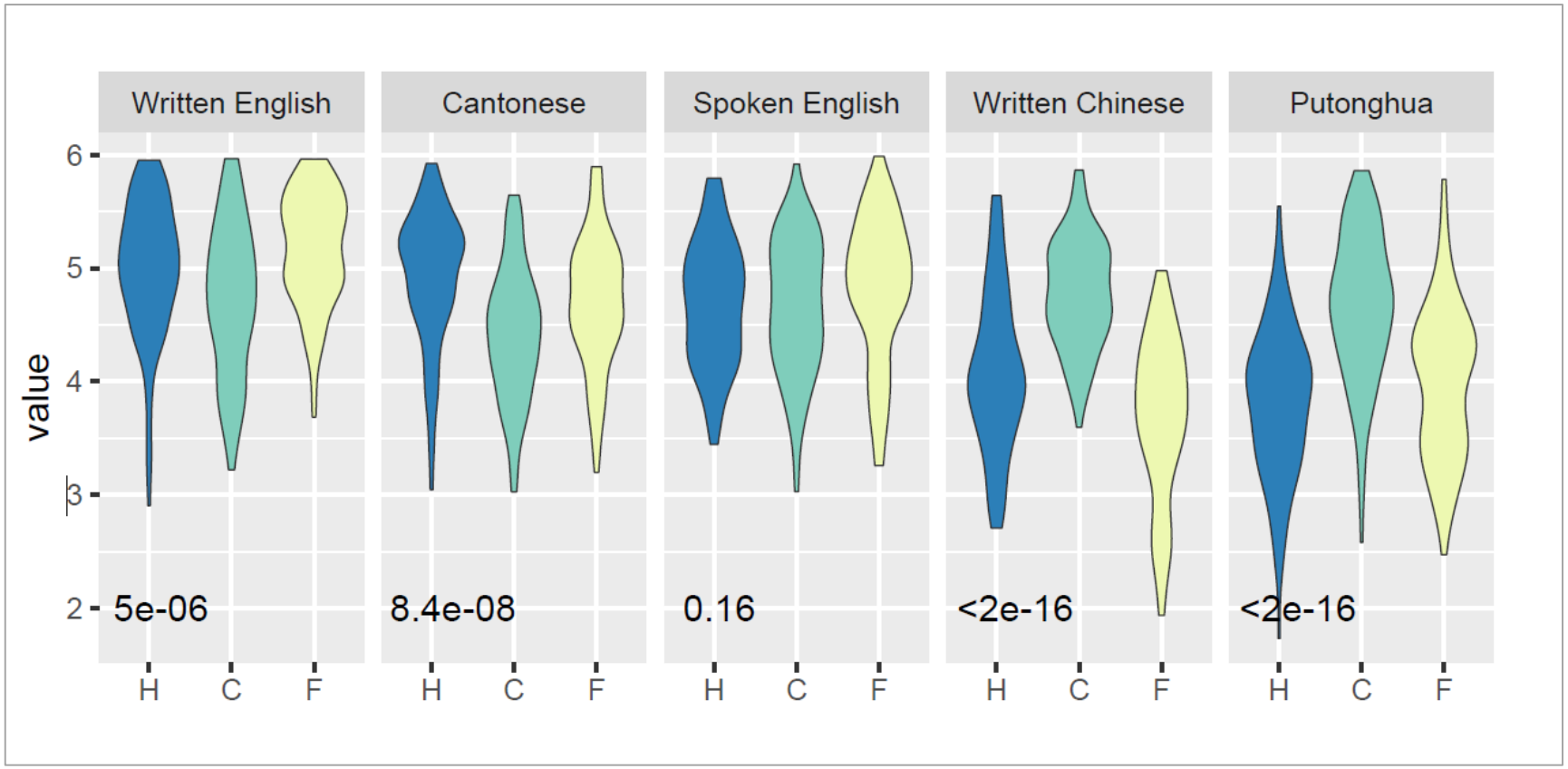

Written English
Cantonese
Spoken English
Written Chinese
Putonghua
value
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HK-owned
China-owned
Ownership
Foreign-owned


Written English
Cantonese
Spoken English
Written Chinese
Putonghua
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<2e-16
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C
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Written English
Cantonese
Spoken English
Written Chinese
Putonghua
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HK-owned
China-owned
Ownership
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Written English
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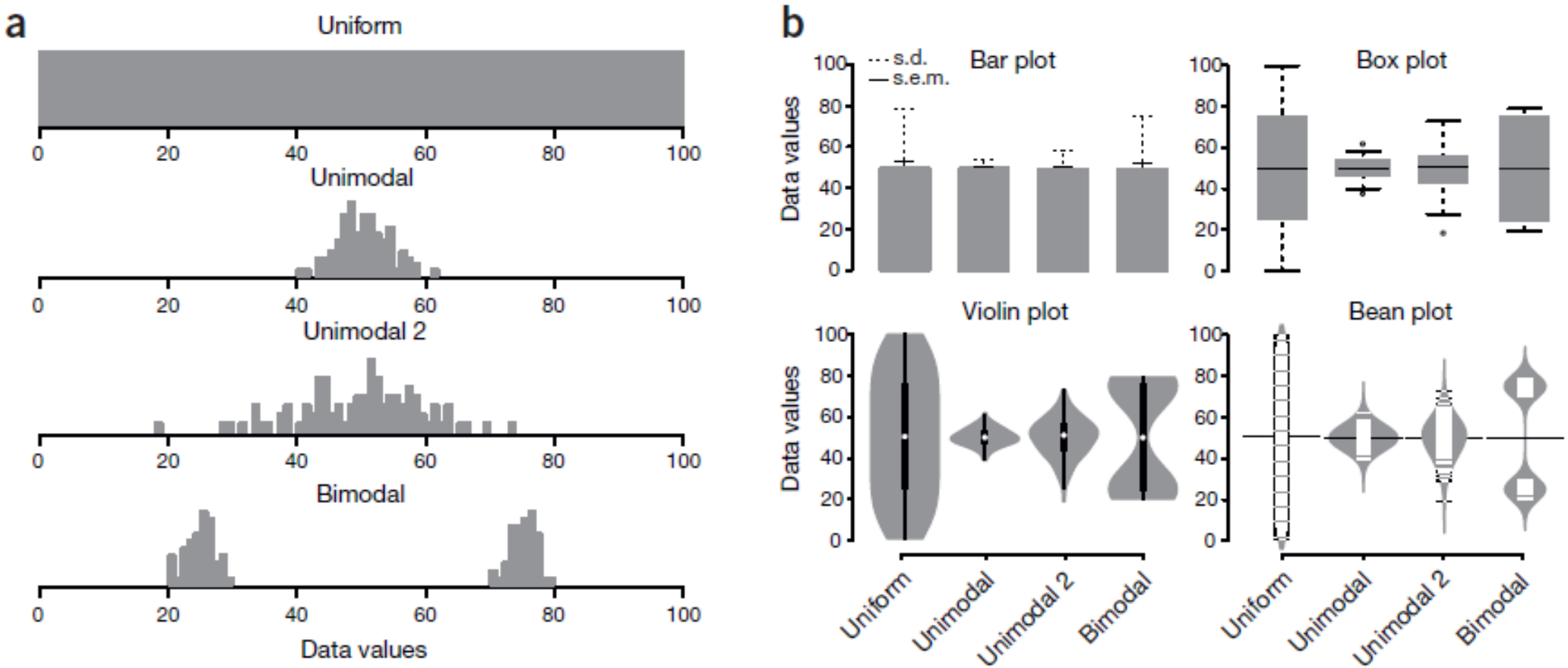
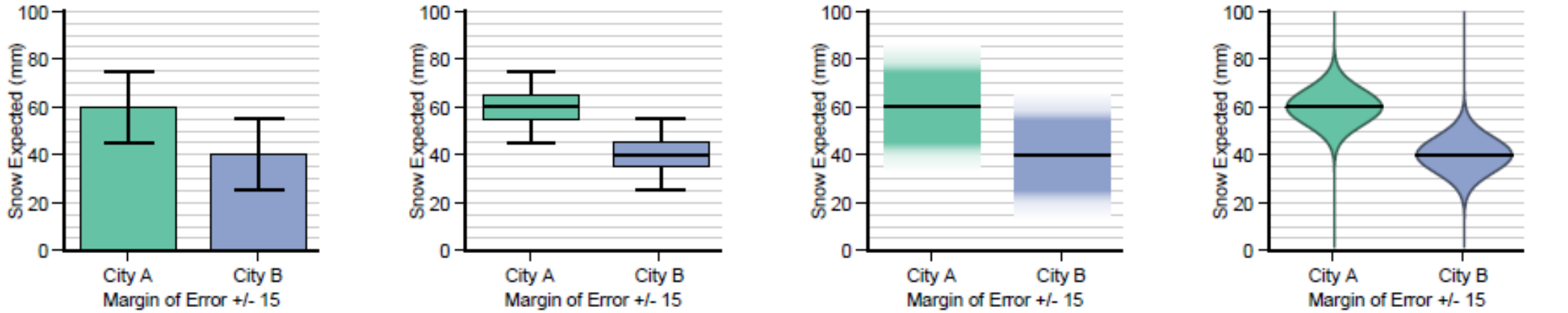


Figure 1 | Data visualization with box plots. (**a**) Hypothetical sample data sets of 100 data points each that are uniform, unimodal with one of two different variances or bimodal. Simple bar plot representations and statistical parameters may obscure such different data distributions. (**b**) Comparison of data visualization methods. Bar plots typically represent only the mean and s.d. or s.e.m. Box plots visualize the five-number summary of a data set (minimum, lower quartile, median, upper quartile and maximum). Violin and bean plots represent the actual distribution of the individual data sets.

Spitzer et al.(2014) Nature Methods, **11**, 121



(a) **Bar chart** with error bars: the height of the bars encodes the sample mean, and the whiskers encode a 95% t-confidence interval.

(b) **Modified box plot**: The whiskers are the 95% t-confidence interval, the box is a 50% t-confidence interval.

(c) **Gradient plot**: the transparency of the colored region corresponds to the cumulative density function of a t-distribution.

(d) **Violin plot**: the width of the colored region corresponds to the probability density function of a t-distribution.

Fig. 1. Four encodings for mean and error evaluated in this work. Each prioritizes a different aspect of mean and uncertainty, and results in different patterns of judgment and comprehension for tasks requiring statistical inferences.

Correll et al. (2014) IEEE transactions on visualization and computer graphics, 20, 2142.

Error bars considered harmful: Exploring alternate encodings for mean and error

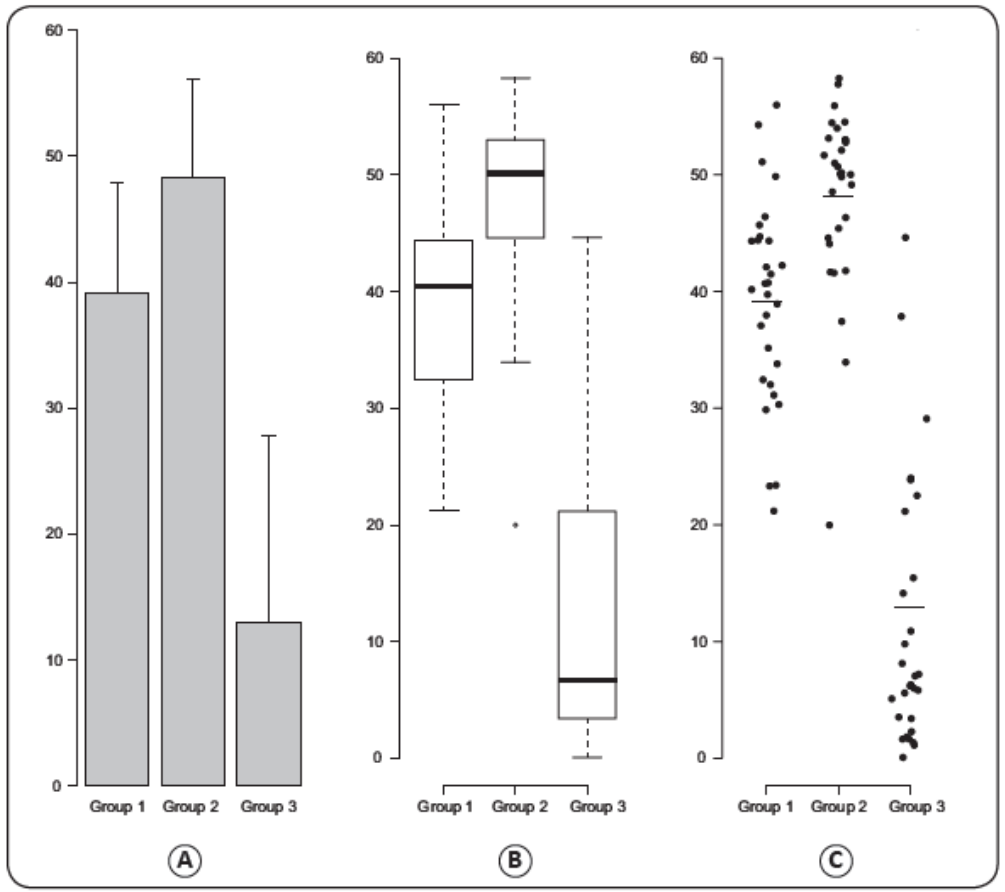


FIGURE 1 - Data are shown for three simulated samples (n = 30) from normal (Groups 1 and 2) and gamma (Group 3) distributions. A: Bar graphs with standard deviation bars are inadequate. B: Box plots adequately describe the data distribution and highlight an outlier. C: Dot plots are also adequate to describe data. The horizontal lines in this graph represent the means.

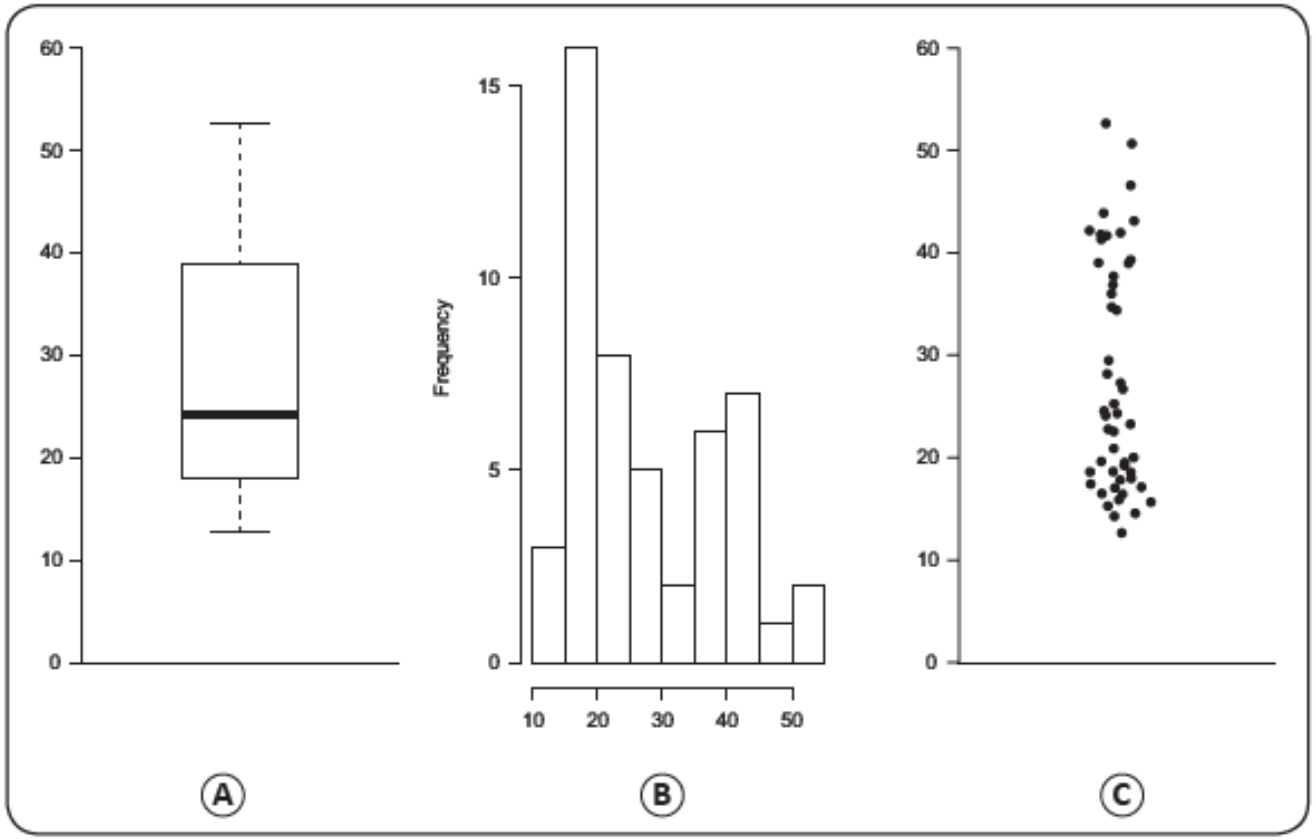
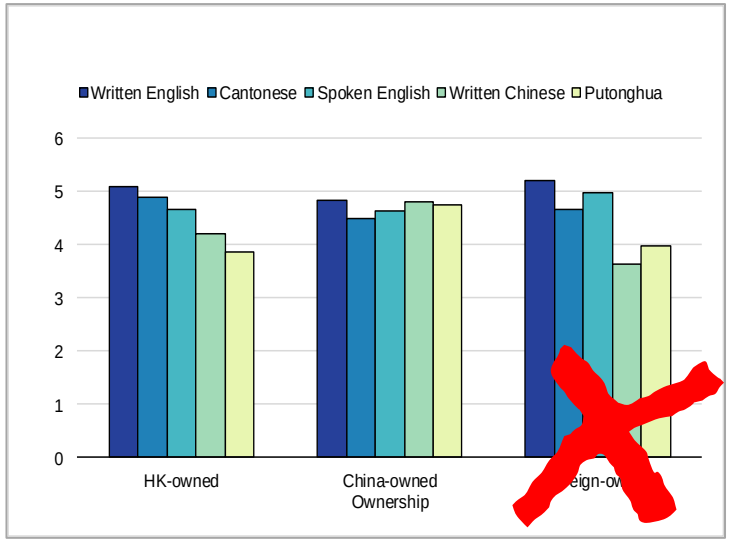


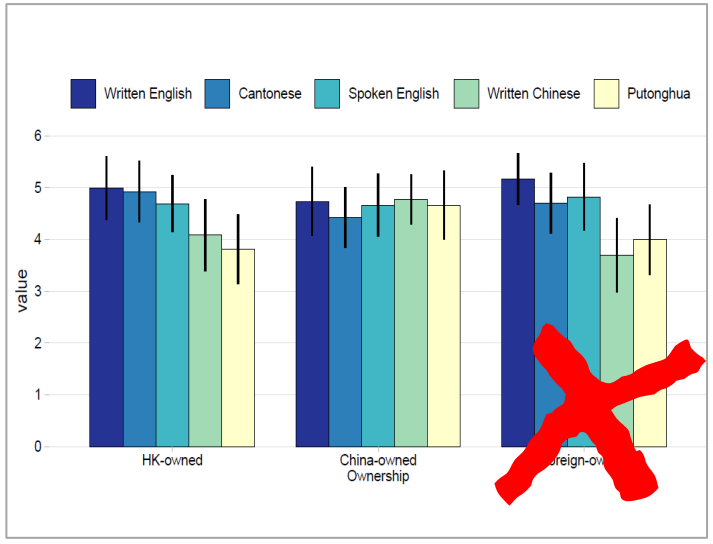
FIGURE 2 - Box plots cannot clearly describe multimodal distributions. A: Box plot for a sample from a random variable that follows a mixture of two normal distributions. The bimodality is not visible in this graph. B: A histogram for these data. The bimodality is now visible in this graph. C: A dot plot for these data. The display of two clouds of points in this figure suggests a bimodal distribution.

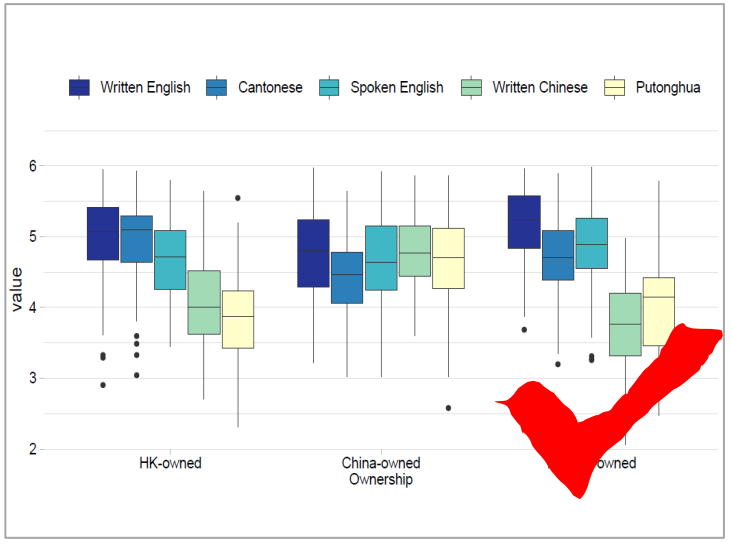
Martinez et al. (2015). Revista da Sociedade Brasileira de Medicina Tropical, 48, 494.

Description of continuous data using bar graphs: a misleading approach

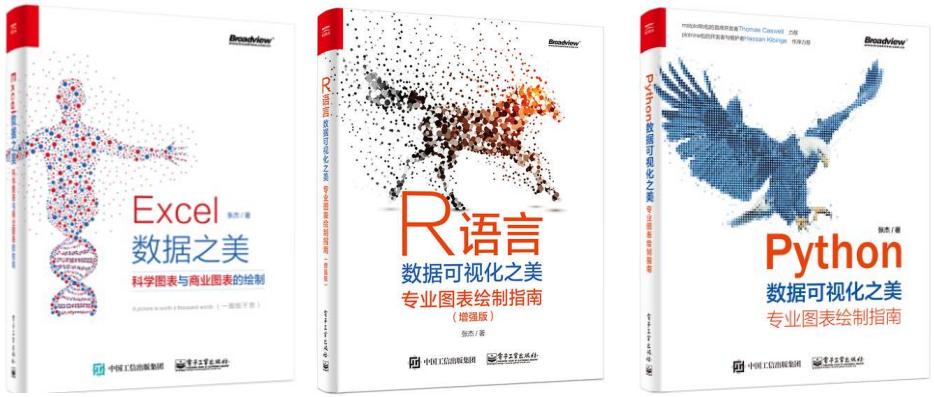
It is important to show the data error and distribution in the statistical analysis by using appropriate chart





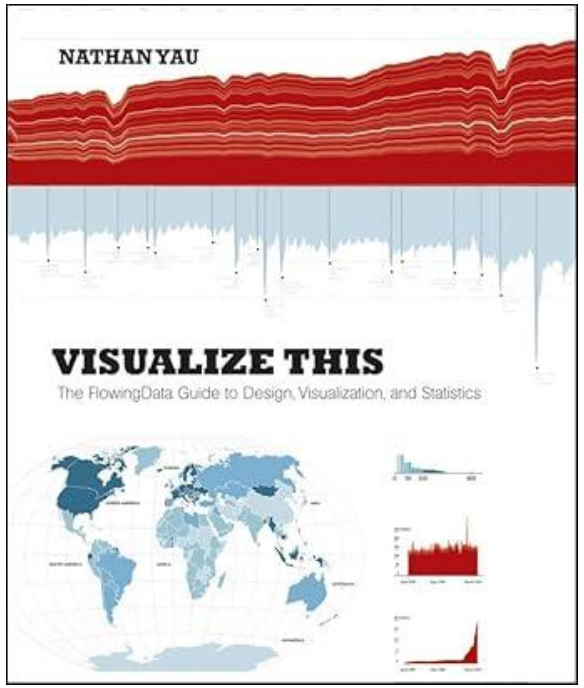


Reading Recommendation

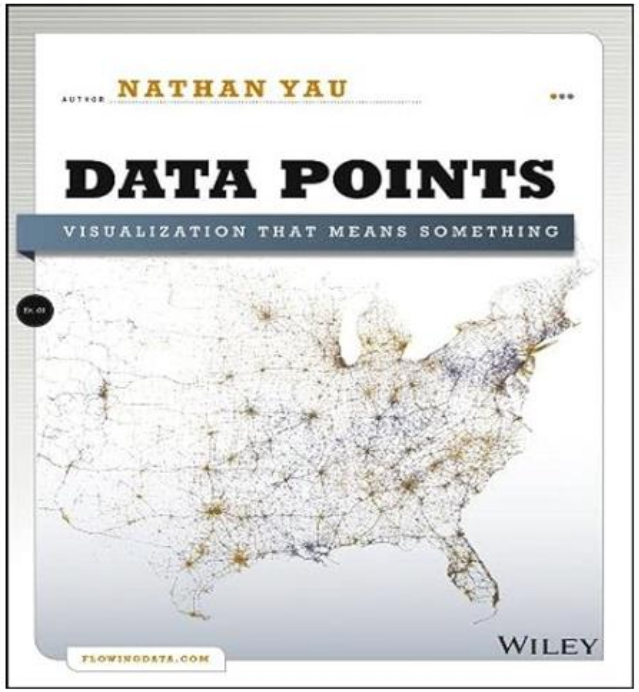


Beautiful Visualization with Excel/R/Python

Nathan Yau (Author)



Visualize This:
The FlowingData Guide to Design, Visualization, and Statistics



Data Points: Visualization That Means Something

Thanks

